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Set Up Guide for SAP Service Cloud Version 2

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1 Set Up Guide for SAP Service Cloud Version 2

This guide provides a step-by-step checklist for implementers to configure a new solution tenant for SAP Service Cloud Version 2 customers.

Activate the tenant by following the steps in the Onboarding Guide, then use the procedures in this guide to configure settings for your organization. Configure organizational structures, user accounts and roles to define access control, as well as on-screen branding and templates. The configuration procedures here are duplicated from the Solution Guide, which includes more high-level context as well as detailed usage information for each feature. This guide is intended only as a streamlined list of setup procedures for system implementers.

The configuration procedures are initially collapsed under their headings. Select the triangle at the left of the heading to expand the list of steps.

Related Information

[SAP Service Cloud Version 2 Onboarding Guide](#)

[SAP Service Cloud Version 2 Solution Guide](#)

2 Organizational Structure

An organization structure is a set of organizational units, and their hierarchical relationships, that together form the company.

As an administrator, learn to set up your organizational structure including business organizations and sub-units. The organizational structure establishes reporting lines and allows automatic work distribution.

2.1 Create Your Org Structure

Setting up an organizational structure enables automated routing of work. The organizational structure is composed of units and provides a unified representation of your company's organizational data.

To create your org structure, you must first plan your org structure. Use your existing organizational structure plan as a basis for creating the org structure in the system. When you start creating the structure, we recommend that you build from the top down; that is, start with companies and then add the sales organizations or departments, including your sales teams. Obtain all data relevant to organizational management, including managerial and functional data.

→ Tip

When editing your org structure, always enter the valid-from date for your changes.

2.1.1 Recommendations

Under  [System Settings](#) > [All Settings](#) > [Company](#) > [Organizational Structures](#) , you establish your organizational structure as a prerequisite to going live. This is also where you maintain and update your organizational structure. As you set up your org structure, we recommend that you observe the following principles:

- The uppermost node of your org structure should be an org unit defined with the [Company](#) function.

→ Tip

You must create and then edit the org unit details to select the [Company](#) function.

- The node that you create directly below the [Company](#) node should be either a [Sales Organization](#) or a [Service Organization](#). If your company uses both the sales and service functions of the solution, then your org structure will have both types of nodes below the company.

→ Tip

It's possible to setup multiple sales/service organizations in parallel organizational structures but it isn't possible to stack sales/service organizations. Because a sales organization is required to process

all corresponding sales documents (for example, leads, opportunities), and a service organization is required to process service requests. We recommend that you define only one sales organization and one service organization per org structure.

- The nodes that you create below the [Sales Organization](#) node can be for example, [Sales Units](#), and the nodes that you create below the [Service Organization](#) node can be for example, be [Service Units](#). It's also possible to create an organization, in which all units underneath a sales/service organization are not flagged as a sales/service unit.

However, it is not recommended to interrupt the setup of the sales or service function along the organizational hierarchy as application logic may not consider functional units underneath a node that is not flagged as service or sales unit.

→ Tip

Within the org structure, multiple levels of sales or service units may be defined. Service units represent the service organization and teams that process service requests. Sales units represent the sales departments or offices where sales documents are initiated. Nodes for [Sales Units](#) or [Service Units](#) can be defined below the Sales Organization, or Service Organization node, or below the nodes of other sales or service units.

- When editing your org structure, always enter the valid-from date for your changes. There's no need to explicitly activate the organizational units. They are activated based on their validity period.

⚠ Caution

During maintenance of the org structure, always set the validity of org units within the validity period of the parent org unit.

2.2 Create an Org Unit

Learn how to create an [Org Unit](#).

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Company](#) ► [Organizational Structure](#) ►.
2. Create a new org unit by choosing + ([Create](#)).
3. Enter the [Valid From](#) date (today's date, or a future date, fixed date) and the [Valid To](#) date.
You can edit the [Valid From](#) and [Valid To](#) fields in the detail view of the org unit.
4. Enter the [Address](#) details for the unit.
You can edit the [Address](#) field in the detailed view of the org unit under the [General](#) tab.
5. Select the appropriate functions for the unit.

Function	Description
Sales	The Sales function indicates that the org unit is responsible for the processing of documents, such as opportunities or sales quotes. In sales reporting, sales data is aggregated at the sales unit level. The sales organization is a specific type of sales unit, which is used to head complete areas or countries/regions. Relevance Select this function if employees assigned to this org unit will perform tasks in the areas described above or if sales data is to be aggregated at the level of this org unit.
Sales Organization	The Sales Organization function indicates that the org unit represents the top level of a sales hierarchy. Note Example For your domestic sales, you create a sales hierarchy and define the top-level org unit as a sales organization. Then below that org unit, you add additional units, defined as sales units, that represent regional sales groups and sales departments. The sales organization is also used for defining sales data in master data. • Sales data in the product and account master can be defined in relation to the sales organization in combination with the distribution channel and division. • Internal price lists can be defined based on the sales organization and distribution channels. • The combination of the sales organization, distribution channel, and division is also included in sales documents, such as sales orders or service orders. Relevance Select this function if the org unit represents the top unit of a sales hierarchy. All sales-relevant master data and pricing must include a sales organization.

Function	Description
Service	The Service function indicates that the org unit is responsible for the managing, monitoring, and execution of services, such as the handling of product inquiries, providing recommendations, and pre- and post- sales support. Customer service can be provided by a service desk or by field or in-house service teams. Relevance Select this function if employees assigned to this org unit will perform tasks in the areas described above.
Service Organization	The Service Organization function indicates that the org unit represents the overall organization responsible for customer service. Below the service organization, you can define individual service units to represent the various service and support teams and departments within the service organization. → Tip If the services provided are sold to your customers, you will also need to define a sales organization. Relevance Select this function if the org unit represents a fixed entity for service reporting that can be referenced by all service documents.
Company	In the context of organizational management, a Company represents an org unit that is financially and legally independent, that is not tied to a geographical location, and could be registered under business law. Each org structure must contain at least one org unit that is defined as a company. Saving the org unit as a company automatically creates an associated business partner which is used in the context of seller party or output form determinations. You set the default currency for the organization within the company org unit.

6. Save your entries.
7. To select the [Company](#) function for the org unit, open the unit details and make your selection in the header section.

2.3 Add Employees and Managers to Org Units

As an administrator, you can identify an employee as a manager by defining that employee as the manager of an org unit. The manager of an org unit enjoys unique access privileges and can access the data of their subordinates.

Note

Employees and managers need the Primary Indicator. An employee can be marked as main by using the primary indicator for only one org unit. An employee who is marked as main in an org unit may only be added as a secondary employee in another org unit.

Prerequisites:

- The employees you want to assign to the org unit have been created in the system.
- Alternatively, you can assign employees to org units when you create the employee data in the system.

Procedure

1. From the *Org Structure* screen, select the relevant org unit from the list.
2. On the *Employees* tab, choose **+** (*Add*) to assign employees on the *Employees* view or a manager on the *Managers* view to the org unit.
3. Select the employee or manager you want to assign to the org unit and choose **✓** (*Accept*).
4. *Save* your entries.

3 Employees

Learn how to create employees in your system.

Note

All accounts, individual customers, employees, contacts, are considered as business partners in the system. Therefore, all functions that are common to business partners are applicable to them.

Additionally, detailed information on External IDs is displayed under the [Mapping for Integration](#) section, and you can view all changes made to employees under the [Change History](#) (🔍) section.

3.1 Create Employees

You can create employee records manually or upload employee data via migration tool in the implementation project activity.

Procedure

1. Navigate to your [User Menu](#) and select ► [Settings](#) ► [All Settings](#) ► [Company](#) ► [Employees](#) ►.
2. In the worklist of the employees view, choose [Create](#) (+).
3. Enter as much personal, organizational, and contact information as you require.
4. Choose [Save](#).

3.2 Edit Employees

Edit an employee record.

Procedure

1. In the worklist of the employees view, choose the employee record.
2. In the quick view, choose [Open Details](#).
3. To delimit the validity of an employee record, enter the appropriate date in the [Valid To](#) field. From this date onward, this employee record cannot be used for new business processes.

The system automatically saves the changes you made.

3.3 Manage Attachments

As an administrator, you can upload, view, delete, and download attachments.

3.3.1 Upload Attachments From Your Computer

Here is a step-by-step guide on how to import attachments from a front-end system, such as a computer, into the system.

Procedure

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Company](#) ► [Employees](#) ►.
2. In the quick view, choose  (Open Detail View).
3. Choose  ([Attachment](#)) and then choose [Drop or Browse Files](#).
4. Select the file from your computer you want to upload and choose [Open](#).
5. From the dropdown, select the [Attachment Type](#).
6. Choose [Upload](#).

Note

You can upload multiple files at the same time.

To download attachments, select an attachments and choose  ([Download](#)).

Select an attachment and choose  ([Delete](#)) to delete an attachment.

3.3.2 Add Attachments From SAP System

Find out how to add attachments from your SAP system.

Procedure

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Company](#) ► [Employees](#) ►.

2. In the quick view, choose  (Open Detail View).
3. Choose  (*Attachment*) and then choose *Add*.
4. Choose *Add From Library*.

Note

To add a link, choose  (*Add Link*) and provide the link in the link field. You also can provide a title for this link. Save your changes.

5. Select the documents you want to add and choose *Add*.

3.3.3 Download or Delete Attachments

Here you can learn how to download and delete attachments.

To download attachments, select an attachments and click *Download* ().

Note

Links cannot be downloaded.

Select an attachment and click *Delete* () to delete an attachment.

3.4 Working Hours

An employee's scheduled working hours.

3.4.1 Add Working Day Calendar

The working days of an organizational unit within a particular company or region.

Context

Find out how you can assign the working days of an organization to an employee.

Procedure

1. Navigate to ► *User Menu* ► *System Settings* ► *All Settings* ► *Company* ► *Employees* ►.
2. Select an employee and in the quick view and choose  (Open Detail View).
3. Choose  (*Working Hours*).
4. Under the *Working Day Calendar* section, fill the *Time Zone* and *Working Day Calendar* fields from the dropdowns.
The system automatically saves this information.

3.4.2 Add Working Hours

The amount of time an employee is scheduled to spend working.

Procedure

1. Navigate to ► *User Menu* ► *System Settings* ► *All Settings* ► *Company* ► *Employees* ►.
2. Select an employee and in the quick view and choose  (Open Detail View).
3. Choose  (*Working Hours*).
4. In the *Working Hours* section, choose  (*Create*).
5. Select the *Days* and choose  (*Time*) to set the time.
6. Choose  (*Accept*).
The employees' working hours are saved.

4 Users and Control

Users and control settings allow you, as an administrator, to manage your employees, users, business roles, and security policies which in turn provide authentication and authorization services for your system.

- **Employees:** A group of people who contribute to creating goods or services in a company, based on a work contract or a contract for services. Unlike externals, internal employees are bound by instructions and obliged to adhere to the company's policies and regulations.
- **Users:** Refer to employees, business users, and other users such as partner contacts. Users are assigned to business roles and technically, access controls in your system are implemented on the business roles.
- **Business Roles:** Represents a set of authorizations to access your solution within your organization and system landscape. In the system, a business role defines a hierarchy of access controls that you can assign to application users in your organization. You can use roles to assign access to a large number of users at a time. As roles change, you need to update only the access rights within a role and the rights apply for all assigned users.
- **Security Policies:** A set of rules that defines password complexity, such as including numerical digits and password validity, like requiring a password change after a certain period of time.

Authentication and Authorization

Authentication is the process of verifying who a user is, while authorization is the process of verifying what a user will have access to.

The different authentication methods include:

- **Basic Authentication:** Basic authentication is a simple authentication method that is built into the HTTP protocol that is supported by most Web browsers and Web servers. With this authentication mechanism, the client sends the user ID and password to the server in plain text format.
- **Single Sign-On Authentication:** Single sign-on (SSO) is an authentication method that enables users to securely authenticate with multiple applications and websites by using just one set of credentials. It is a mechanism that eliminates the need for users to enter passwords for every system that they log on to.
- **Certificate-Based Authentication:** Certificate-based authentication is a cryptographic technique that enables secure identification of one computer by another across a network connection. It uses a public-key certificate. This authentication system confirms a user's or device's identity using digital certificates issued by a trusted authority such as a government agency or web server to verify its authenticity.
- **Token-Based Authentication:** Token-based authentication is a protocol which allows users to verify their identity, and in return receive a unique access token. Until the token is valid, the users can access the website or app that the token has been issued for without having to re-enter credentials each time they go back to the same webpage or app. The user retains access as long as the token remains valid. Once the user logs out or quits an app, the token becomes invalid.

For example: If you want to do single call where you intend to read case and opportunity, you call the opportunity API by passing basic auth headers (User Name and Password). If you want to make like 100s of calls, then it is best to use Auth Token. You use Auth Token call and provide details for Basic Authentication, with which you get an Access Token. Once you have the Access Token, you don't need basic authentication credential each time you want to access any APIs, instead you use bearer token. In

the bearer token, you can pass the Access Token and its validity. Until the validity expires, the authorization remains valid.

4.1 Users

Users are created for system access. Business users are created for employees to access the system through UI, whereas API users are technical users who can access the system through APIs.

In this section, we discuss about the different types of users that exist in the solution.

User Types

The different types of users in the system include:

- **Business User:** Users who have at least one business role assigned to them that governs the access to the system. These users can access the user interface of the application.
- **Technical User:** Users that have been created for a specific purpose. The different types of technical users include:
 - **Communication User** – Users created and used for integration purposes only. A communication user is used only for accessing a specific communication scenario for which they've been created. They have API access, but not UI access.
 - **Support User** – Users who are used by support teams to log on to customer tenants to access the system as part of case processing.

Note

Support teams may receive a case that requires access to the customer system. This scenario can occur if the issue can't be reproduced or resolved in internal environments. In such situations, the Cloud Access Manager (CAM) is used to generate temporary access to the corresponding customer system.

Support users aren't allowed to share these details. The CAM tool maintains a log of which user generated which support user at what date and time. So, it's always possible to link a generic support user back to the real person.

Support users follow the pattern **SAP_***.

- **Initial User** – Users who are created initially when a tenant is provisioned, and are later locked.
- **API User** – Users created and used for consuming other services through APIs. These users have API access as per the technical roles assigned. They don't have access to the UI of the application.

Note

For communication and API users, only for the first logon for the day, an event is logged in and published in the log.

- **Active Users:** It's a metric, and is defined as the number of unique users who are logging in a 30-day period.

- **Licensed Users:** It's a metric, and is defined as the total number of valid Business Users in the system.
- **Service Account User:** These users are created at a tenant Level. Displayed in the system as `SAP_SYSTEM01` is a service account user. They can't be created using HTTP APIs but are auto-created by IAM services. Some of the features of such service account users include:
 - `SAP_SYSTEM01` user is created in response to a tenant create event.
 - `SAP_SYSTEM01` user can't be used to log in to the tenant.
 - `SAP_SYSTEM01` user is used for any non-interactive user access and updates in the tenant that ensures that all access and updates are always based on a valid tenant user.
 - Each tenant contains a Service Account User (`SAP_SYSTEM01`) that is created with an all-zero UUID (00000000-0000-0000-0000-000000000000).
- **Limited User:** A user who can perform a limited number of transactions in the tenant. The total number of transactions that a limited user can perform are 1200 transactions per calendar year. To create a limited user, an administrator must choose the [Limited User](#) SKU in the [Manage SKUs](#) section while creating a business role. Once the limited user SKU is selected, you can't add any other SKU. Any user assigned to this role is counted as a limited user.
The transactions that get reported when a limited user can perform actions include:
 - Cases
 - Phone Calls
 - Emails
 - Tasks
 - Chats
 - Notes
 - Leads
 - Opportunities
 - Appointments
 - Visits
 - Attachments
 - Quotes
 - Orders

Business User List View

You can view all the business users in your system in a list format.

The following header fields are available on this page:

- **User Name:** the user name of your business user.
- **Full Name:** the full name of your business user.
- **Created On:** the calendar date on which your business user was created.
- **Valid Until:** the calendar date until which your business user is valid in the system.

Note

The user validity is independent of Employee validity. It means an employee valid until date isn't the same as the user valid until date.

Technical User List View

You can view all the technical users in your system in a list format.

The following header fields are available on this page:

- User Name: the user name of your technical user.
- User Type: the type of technical user.
- Valid From: the calendar date on which your technical user was created.
- Valid Until: the calendar date until which your technical user is valid in the system.

Note

User validity is independent of Employee validity. It means an employee valid until date isn't the same as the user valid until date.

- Security Policy: the security policy associated with the user.

4.1.1 Business User Details

View details of a business user on this page.

The *Details* tab provides the following information of a business user.

- *Employee Data*: Shows the basic employee information such as Employee Name, Employee ID, Company, Email ID, and so on.
- *User Details*: Displays the following user information.
 - *User Name*: A unique name to identify a business user. Only an administrator has the privileges to edit it.
 - *Global User ID*: A unique identifier for a user for content records.
 - *Created On*: The date on which the business user was created in the system.
 - *Valid From*: The date from which the business user is valid. An administrator can edit this date.

Note

Valid From date may not be the same as *Created On* date. *Valid From* date can be a future date. A business user is a valid user only between the interval defined by *Valid From* and *Valid Until* dates. An administrator can edit the *Valid From* date.

- *Valid Until*: The date until which the user is active in the system.
- *Password Locked*: A switch that indicates if the password is locked or not. The following are the features of the *Password Locked* switch:
 - The switch is auto-enabled after too many failed login attempts.
 - If the switch is enabled, an employee can't log on to the system from the login screen but can only log in using a certificate.
 - An administrator can disable this switch to unlock the password, however, an administrator can't lock the password.
- *Security Policy*: Displays the security policy assigned to the user. An administrator can change the assigned security policy.

- **Work Time:** Edit the time zone in which your business users work. An administrator can set the work time for business users.

Note

The **Work Time** field is for Dynamic Visit Planning functionality purposes and is used for visit planning and scheduling process. This settings won't impact the regular time zone configuration of the system.

- **SCIM Replication:** A flag that indicates if the user is locally created or SCIM replicated.
- **Reset Password:** Provides an option to reset the current password of the user. As an administrator, you can reset a user's password by clicking the **Reset Password** button, which triggers a password reset email to your user. The user can then reset the password using the **Password Reset** link in the email.

Note

- The **Password Reset** link is valid for 60 minutes and is for single password reset only.
- You can view user created and password reset email details in **Outbound Monitoring** setting.

- **Manage Certificates:** Enables an administrator to add or remove certificates for a user to enable certificate-based login for a user.

To add a new certificate for a user:

1. Choose **Manage Certificates**.
2. Upload a certificate for a user using the **Drop or Browse Files** option. Once the certificate is uploaded, it enables an user with seamless login to the system.

Note

You can upload only certificates generated from the certificate authority listed in the following note: <https://launchpad.support.sap.com/#/notes/2801396>.

To remove a certificate, select the certificate and select the delete icon under the **Actions** column.

The **Assigned Business Roles** tab lists all the business roles assigned to a particular business user. An administrator can search and add a business role. The administrator can also delete an assigned role.

Related Information

[Users \[page 19\]](#)

Users are created for system access. Business users are created for employees to access the system through UI, whereas API users are technical users who can access the system through APIs.

[Create Business Users and Assign Business Roles \[page 23\]](#)

Administrators can create business users and provide appropriate authorizations by assigning business roles.

4.1.2 Create Business Users and Assign Business Roles

Administrators can create business users and provide appropriate authorizations by assigning business roles.

Context

Business users in the system are linked to employees. Business users have business roles assigned, and the access controls in the system are implemented on business roles.

Procedure

1. Log in to the system as an administrator.
2. Navigate to [User Menu](#) on the top-right corner, and access the [System Settings](#) page.
3. Go to ► [All Settings](#) ► [Users and Control](#) ► [Users](#) ►.
4. Choose [Business Users](#) from the dropdown on the page.
5. Click the Add icon (+) on the [Business Users](#) page.
6. Search and select an employee for whom you wish to create the business user.

Note

- The employee linked to a user must have a unique Email ID.
- If a business user already exists for the employee, a message stating *The business user already exists for this employee* is displayed.

7. Once you select an employee, the [User Data](#) section is auto-populated. By default, the user name is created with the last name followed by the first name of the employee. If you wish to modify the user name, you can modify them.
8. Once you've verified all the details, click [Create](#). The business user is now created and linked to the employee. Refresh the business users' list to view your newly created business user.

Once the user is created, an email is triggered to the users. This email contains the username, a link to set the password, and the system URL. For SSO-enabled users, an SSO-enabled URL is sent in the email, using which the users can log in via SSO to the system.
9. Choose your business user, and then navigate to the [Assigned Business Roles](#) tab.
10. Add business roles to your user by using the [Search and Add](#) functionality on the screen. All the business roles that you assigned to your business user now appear in the list format on the [Assigned Business Roles](#) tab.

Note

- Ensure that your administrator has access to the role service in addition to the user service to assign roles to your business users.
- If a role assigned to your user has its [Write Access](#) set to [No Access](#) in the [Business Services](#) section, then such a user can't edit business roles.

You've now created a business user and assigned the user to a business role.

Related Information

[Business User Details \[page 21\]](#)

View details of a business user on this page.

4.1.3 Display User Metrics for SKUs

Displays the user count per SKU assignment for a tenant. Use this metric to monitor license distribution, optimize costs, and ensure compliance with your organization's software licensing agreements.

Context

This procedure helps you to view the user count for an SKU in a given tenant. SKUs are assigned to roles, and the users are assigned to those roles. Using this relationship, we determine which users are associated with each SKU.

Procedure

1. Navigate to [User Menu](#) on the top-right corner, and access the [System Settings](#) page.
2. Go to [All Settings](#) > [Users and Control](#) > [Users](#) > [All Business Users](#) page.
3. On the [All Business Users](#) page, choose the [Show Usage Metrics](#) icon.
A popup appears to display the number of business users assigned to each SKU in the tenant.

4.1.4 Technical User Details

View details of a technical users on this page.

The [Details](#) tab provides the following information of a technical user:

- [User Details](#): Displays the following user information.
 - [User Name](#): A unique name to identify a technical user. Only an administrator has the privileges to edit it.
 - [Display Name](#): A unique identifier for a user for content records.
 - [Created On](#): The date on which the technical user was created in the system.
 - [Valid Until](#): The date until which the user is active in the system. An administrator can edit this date.

- **Password Locked:** A switch that indicates if the password is locked or not. The following are the features of the **Password Locked** switch.
 - The switch is auto-enabled after too many failed login attempts.
 - An administrator can disable this switch to unlock the password, however, an administrator can't lock the password.
 - **Security Policy:** Displays the security policy assigned to the user. An administrator can change the assigned security policy.
 - **Change Password:** Provides an option to change the current password of the user. As an administrator, you can change a user's password by clicking the **Change Password** button.
 - **Manage Certificates:** Enables an administrator to add or remove certificates for a user to enable certificate-based login for a technical user.
- To add a new certificate for a technical user:
1. Choose **Manage Certificates**.
 2. Upload a certificate for a user using the **Drop or Browse Files** option. Once the certificate is uploaded, it enables an user with seamless login to the system.

Note

You can upload only certificates generated from the certificate authority listed in the following note: <https://launchpad.support.sap.com/#/notes/2801396>.

3. To remove a certificate, select the certificate and click the delete icon under the **Actions** column.

The **Assigned Technical Roles** tab lists all the technical roles assigned to a particular API user. An administrator can search and add a technical role. The administrator can also delete an assigned role.

Related Information

[Users \[page 19\]](#)

Users are created for system access. Business users are created for employees to access the system through UI, whereas API users are technical users who can access the system through APIs.

[Create Technical Users and Assign Technical Roles \[page 25\]](#)

Administrators can create API users and provide authorizations by assigning them to technical roles.

4.1.5 Create Technical Users and Assign Technical Roles

Administrators can create API users and provide authorizations by assigning them to technical roles.

Context

An API user is a type of technical user who is authorized to consume other services through APIs. These users can access the system using the APIs as per the technical roles assigned to them. They don't have access to

the UI of the application. For an API user, an Employee can't be assigned, and all the details of the user must be entered manually.

Procedure

1. Log in to the system as an administrator.
2. Navigate to the [User Menu](#) on the top-right corner, and access the [System Settings](#) page.
3. Go to ► [All Settings](#) ► [Users and Control](#) ► [Users](#) ►.
4. Choose [Technical Users](#) from the dropdown on the page.
5. Click the Add icon (+) on the [All Technical Users](#) page, and enter a [User Name](#), [Display Name](#), [Password](#), [Valid From](#), and browse to upload a certificate on the [Create API User](#) window. By default, a security policy relevant for the technical user is assigned.
6. Click [Create](#). The API user is now created. Refresh the [All Technical Users](#) page to view your newly created API user.
7. Choose your technical user, and then navigate to the [Assigned Technical Roles](#) tab.
8. Add technical roles to your user by using the [Search and Add Technical Roles](#) button on the screen. All the technical roles that you assigned to your API user now appear in a list format on the [Assigned Technical Roles](#) tab.

Note

- Ensure that your administrator has access to the role service in addition to the user service to assign roles to your technical users.
- If a role assigned to your user has its [Write Access](#) set to [No Access](#) in the [Business Services](#) section, then such a user can't edit business roles.

You've now created an API user and assigned a technical role to the user. You can now use this user to consume other services through APIs.

4.2 Roles

A business role defines the specific areas of the system the users can access when they're assigned to that role. Administrators can implement authorizations to business users by assigning business roles.

A business role represents a set of authorizations to access your solution within your organization and system landscape. A business role defines a hierarchy of access controls that you can assign to business users in your organization. As roles change, you need to update only the access rights within a role and the rights will apply for all assigned users.

Business Roles List View

Lists all the business roles in a list view. As an administrator, you can edit the following fields on this page:

- Business Role ID
- Business Role Name
- Description
- Status: You can set the status to *In Preparation*, *Active*, or *Obsolete*.
- Admin: You can activate this switch to indicate that the role is an administrator role.

Technical Roles List View

Lists all the technical roles in a list view. As an administrator, you can edit the following fields on this page:

- Technical Role ID
- Technical Role Name
- Description
- Status: You can set the status to *In Preparation*, *Active*, or *Obsolete*.
- Admin: You can activate this switch to indicate that the role is an administrator role.

4.2.1 Business Role Details

View more details of a business role on this page.

The *Details* tab provides the following information of a business role.

- *Business Services*:
 - All business services that are assigned to a business role are displayed in the *Business Services* list. As an administrator, you can add or delete business services to the business role.
 - All business services have unrestricted read and write access by default.
 - Some business services allow creation of restrictions. Such services have *Read/Write* dropdown list enabled. On setting any one of them to *Restricted*, the *Restriction Rule* link is enabled.

Note

If one role blocks write access and another role allows it, the more permissive role wins.

- An administrator can select from the available restrictions for a given service to create a restriction rule.
- You can delete invalid business services that get flagged with the following message "Some of the business services assigned to the business role are no longer valid. Please delete all the services with a warning sign to proceed further."
- Default and Mandatory Business Services: When a business role is created, a few of the services such as *Org. Units*, *Data Import and Export*, *Home Page*, *Service Catalog*, and so on, are assigned by default. These services are default (non-mandatory) business services. The respective apps are added to the

app list. These services can be removed by an administrator while either creating or editing a business role. Also, some of the business services such as [Search](#) and [Language, Region, and Currency](#) services are also assigned by default. These services and their respective apps can't be removed or disabled by an administrator. These services are mandatory business services for a role.

- [App List](#):
 - Each business service can have one or more apps associated with it. These apps define the left navigation for the users having this particular business role assigned.
 - On adding the business service to a role, the corresponding apps are added to the app list.
 - Administrators can enable or disable these apps using the switch against them.
 - You can also search for apps from the search bar. You can also filter the app list by [All Apps](#), [Generic Apps](#), and [Admin Apps](#).
 - As an administrator, you can also enable all apps at once using the [Enable All Apps](#) switch and enable all Admin Apps at once using the [Enable All Admin Apps](#) switch.

Note

The overall access to entities is controlled and governed by the [Business Services](#) section. However, access to a particular entity on the UI is managed using [Apps](#). If you want to provide restricted access to a particular entity, use the restriction rules in the [Business Services](#) section.

The [Assigned Business Users](#) tab lists all the users that have this business role assigned. An administrator can search and add users to this role. The administrator can also delete users from this role.

The [Changes](#) tab lists all the changes made to your business role. These changes include the modifications made to restriction rules for a particular service. The changes display the business service name that is affected when a restriction rule is edited. You can use the different filters available on the interface to filter and view the changes.

Related Information

[Changes](#)

4.2.2 Create Business Roles and Assign Business Users

Administrators can create business roles with appropriate authorizations and assign them to business users.

Context

Use business roles to assign access rights to multiple business users who carry out similar activities. You can also define access restrictions for a business role and in turn to the business users that are assigned to the role.

Procedure

1. Navigate to [User Menu](#) on the top-right corner, and access the [System Settings](#) page.
2. Go to [All Settings](#) > [Users and Control](#) > [Roles](#).
3. Choose [Business Roles](#) from the dropdown list.
4. Click the Add (+) icon on the [All Business Roles](#) page, and enter a business role name and description. The [Business Role ID](#) is auto-generated. You can also set this role as an admin role by enabling the [Admin](#) switch.
5. Click the Add (+) icon in the [Manage SKUs](#) section, and choose one or more SKUs from the [Select SKUs](#) list. However, if you choose Limited User SKU from the list, you can't select or add any other SKUs from the list. Also, you can't add Limited User SKU if any other SKU is already added.

Note

- In general, you can select more than one SKU from the list. You can remove SKUs that you no longer need. In such a case, the corresponding services are removed from the role. The business service is retained only if it's also part of another assigned SKU.
- If you select the limited use SKU from the [Select SKUs](#) list, a list of all business services (associated with all other existing SKUs in the tenant) is fetched and enabled, except for the Groupware services.

6. Based on the selected SKUs, a list of services is populated by default. The list of business services is filtered based on the selection of SKUs. Click the Add (+) icon in the [Business Services](#) section to open the [Add Business Services](#) popup, select all the services that are required for this role from the list of services, and then click [Add](#). You can now [Save and Activate](#) your business role.

Note

- Choosing an SKU in the [Manage SKUs](#) section is applicable for new roles only and doesn't impact the existing business roles.
- Mandatory and Non-Mandatory Services:
Some business services such as [Search](#) and [Language, Region, and Currency](#) business services are also assigned by default. These services and their respective apps can't either be removed or disabled by an administrator. These services are mandatory business services for a business role. If you try to remove these services or disable these apps while creating or editing a business role, they're added back again to the role. For an existing business role, this change has no functional impact.
By default, a list of business services such as [Org. Units](#), [Data Import and Export](#), [Home Page](#), [Service Catalog](#), and so on, is assigned. Such services are default (or non mandatory) business services. The respective apps are also added to the app list. These services can be removed by an administrator either while creating or editing a business role.
- If you set a role as an Admin role using the switch, a default list of business services necessary for the administrator role is added by default.

7. Click the Add (+) icon in the [Business Services](#) section, and select and add all the services that are required for this role from the [Add Business Services](#) popup. [Save and Activate](#) your business role.

Note

If you're editing a business role, you must explicitly use the [Save](#) option to ensure all your changes get saved on both [Business Role Details](#) and [Assigned Business Users](#) pages.

- You can add additional services for your role from the [Add Business Services](#) popup.
- Set restrictions for the business services by editing either or both [Read Access](#) and [Write Access](#) columns for a corresponding business service. By default, both these columns are set to unrestricted for a service, and the [Restriction Rule](#) is set to unassigned. Once you set [Read Access](#) and [Write Access](#) for a business service, the corresponding [Restriction Rule](#) is enabled. Click the enabled [Unassigned](#) button to set access restriction rules for your business role, based on the options listed.

Note

Access Restriction Rules: Based on the restriction types defined by each service, the following restriction rules are supported. When a business role is maintained for a user, based on supported restriction types for a service, the corresponding rules are displayed. You can select one or more rules.

Access Restriction Rules

Restriction Type	Restriction Rule	Description
EMPLOYEE	Myself	This rule provides access based on direct employee assignment
	Employees Reporting to Me	For a manager: Access is based on employee and involvement of employees reporting to a user in an org. unit (including subunits). For a non-manager: Same as Myself
	Specific Employees	This rule provides access to specific employees
ORGANIZATIONAL UNIT	My Service Organization	This rule provides access to the service organization of the employee in the organizational hierarchy
	My Sales Organization	This rule provides access to the sales organization of the employee in the organizational hierarchy
	My Organizational Unit	This rule provides access to an organizational unit of the employee in the organizational hierarchy

Restriction Type	Restriction Rule	Description
	Specific Organizational Unit	This rule provides access to specific organizational units (which the employee doesn't belong to)  Note As a prerequisite, the organizational unit must be set as the <i>Trusted Authorization Unit</i> in the account or individual customer details. The <i>Trusted Authorization Unit</i> field is enabled via adaptation. Restricting a business role for an account or individual customer service by the organizational unit only affects the account or individual where the <i>Trusted Authorization Unit</i> field is maintained. If the <i>Trusted Authorization Unit</i> is specified for an account or individual customer, the account can't be restricted by the employee, territory, or sales data.
TERRITORY	My Territory Hierarchy	This rule provides access based on the employee's territory assignment only (including sub-territories)
SALES AREA	My Sales Data	This rule provides access based on employee's sales data (sales org. + division + distribution channel)
	Sales Areas of My Sales Orgs.	This rule provides access based on all sales areas of sales organizations of employee

Restriction Type	Restriction Rule	Description
DISTRIBUTION CHAIN	My Sales Data	This rule refers to a distribution chain maintained at an employee level. Provides access based on the distribution chain. A distribution chain is a combination of Sales Org. and Distribution Channel code • A sales organization (for example, Sales Organization France) • A distribution channel code (for example, 1 = direct sales and 2 = indirect sales) Therefore, the distribution chain at the employee level has the sales data of: • Sales Organization France 1 • Sales Organization France 2
	Distribution Chains of My Sales Orgs.	If an employee is assigned under Sales Organization France and once you select this restriction, you get access for any distribution chains with Sales Organization France that is Sales Organization France *Refers to the Org. with any distribution channel. The restriction rule checks to which Sales Org. the user is assigned. Accordingly, any distribution chain with this Sales Org. is allowed (combination of the user's Sales Org. with any Distribution Channel)

Note

If one rule blocks the Write Access and another role allows it, the more permissive role takes precedence.

- The [App List](#) section, following the [Business Services](#) section, displays all the available apps for the selected business services. Select the switch against an app to make it available for the business role.

Note

The overall access to entities is controlled and governed by the [Business Services](#) section. However, access to a particular entity on the UI is managed using [Apps](#). If you want to provide restricted access to a particular entity, use the restriction rules in the [Business Services](#) section.

- Navigate to the [Assigned Users](#) tab. A list of users who have been this role assigned is displayed. If there are no business users assigned yet, the [Search and Add Users](#) button is visible.

- Click the [Search and Add Users](#) button to add business users from the quick view. Search and select users from the quick view to add them to the role.

Note

- Ensure that your administrator has access to the user service in addition to the role service to assign users to your business roles.
- You can't assign business users if [Write Access](#) is set to [No Access](#) for [User](#) service. If otherwise, the user can't edit the role but can edit the user assignments.

Results

You've now created a business role and assigned the business role to the selected list of business users.

Related Information

[Business Role Details \[page 27\]](#)

View more details of a business role on this page.

[List of Business Services \[page 35\]](#)

This document provides a list of business services available in SAP Sales Cloud Version 2 and SAP Service Cloud Version 2.

[Users \[page 19\]](#)

Users are created for system access. Business users are created for employees to access the system through UI, whereas API users are technical users who can access the system through APIs.

4.2.3 Copy Business Roles

Create copy of business roles and assign it to your business users.

Context

Administrators can now create copy of business roles using the inline copy function on the object worklist page.

Procedure

- On the [All Business Roles](#) page, choose the inline [Copy](#) option for a business role that you wish to copy. A new quick create page, [Copy Role](#) opens up.

Note

A new role ID is generated, and the role name is populated as *Copy Of <the existing business role>*, which can be edited to give a name.

2. Enter a description for the role and choose *Save and Open* to open the new business role. All the business services and the apps assigned to the role are copied as well.

Note

The new business role is set to active status by default, and if your role had administrator authorizations, the same is retained. However, none of the users assigned to the role is retained. Ensure that you assign users to your business role.

Results

Your role is now copied and ready. You can assign business users to your role.

4.2.4 Create Technical Roles and Assign Technical Users

Administrators can create technical roles with appropriate authorizations and assign them to technical API users to consume services.

Context

Use technical roles to assign access rights to multiple technical API users who carry out similar activities.

Procedure

1. Log in to the system as an administrator.
2. Navigate to *User Menu* on the top-right corner, and access the *System Settings* page.
3. Go to ► *All Settings* ► *Users and Control* ► *Roles* ▾.
4. Choose *Technical Roles* from the dropdown on the page.
5. Click the Add icon (+) on the *All Technical Roles* page, and enter a technical role name and description. The *Technical Role ID* is auto-generated. You can also set this role as an admin role by enabling the *Admin* switch.
6. Click the Add icon (+) in the *Business Services* section to open the *Add Business Services* popup, select all the services that are required for this role from the list of services, and then click *Add*. You can now *Save and Activate* your business role.

7. Set restrictions for the newly added business services by editing either or both of the [Read Access](#) and [Write Access](#) columns for a corresponding business service. By default, both these columns are set to unrestricted for a service. You can choose to impose restrictions on these services by choosing [Unrestricted](#) for [Read Access](#) and either [Unrestricted](#) or [No Access](#) for [Write Access](#).
8. Navigate to the [Assigned Technical Users](#) tab. A list of users who have been this role assigned is displayed. If there are no technical API users assigned yet, the [Search and Add Technical Users](#) button is visible.
9. Click the [Search and Add Technical Users](#) button to add technical API users from the quick view. Search and select users from the quick view to add them to the role.

Note

- Ensure that your administrator has access to the user service in addition to the role service to assign users to your technical roles.
- You can't assign technical users if [Write Access](#) is set to [No Access](#) for [User](#) service. If otherwise, the user can't edit the role but can edit the user assignments.

Results

You've now created a technical role and assigned the technical role to the selected list of technical API users.

Related Information

[Roles \[page 26\]](#)

A business role defines the specific areas of the system the users can access when they're assigned to that role. Administrators can implement authorizations to business users by assigning business roles.

4.2.5 List of Business Services

This document provides a list of business services available in SAP Sales Cloud Version 2 and SAP Service Cloud Version 2.

Business Service Name	Description	Business Service: Technical Details
Accounts	Monitor, store, and track business information about customers, prospects, and partners.	• Technical Name: <code>sap.crm.md.service.accountService</code> • UI App Name: <code>sap.crm.md.accountservice.uiapp.accountApp</code>

Business Service Name	Description	Business Service: Technical Details
Account Hierarchy	Map complex organizational structures of a business partner or a large customer with multiple subsidiaries.	• Technical Name: <code>sap.crm.md.service.accountHierarchyService</code> • UI App Name: <code>sap.crm.md.accounthierarchyService.uiapp.accountHierarchyApp</code>
Account Hierarchy Admin	Configure account hierarchy settings.	• Technical Name: <code>sap.crm.md.service.accountHierarchyService</code> • UI App Name: <code>sap.crm.md.accounthierarchyService.uiapp.accountHierarchyAdminApp</code>
Address	Add, modify, or delete addresses.	• Technical Name: <code>sap.crm.md.service.addressService</code> • UI App Name: <code>sap.crm.md.addressservice.uiapp.addressAdminApp</code>
Agent Desktop	Review customer information, create and process cases, manage interactions, and review knowledge base articles	• Technical Name: <code>sap.crm.service.agentDeskService</code> • UI App Name: <code>sap.crm.agentdeskservice.uiapp.agentDesk</code>
Agent Desktop Settings	Configure agent desktop settings.	• Technical Name: <code>sap.crm.service.agentDeskService</code> • UI App Name: <code>sap.crm.agentdeskservice.uiapp.agentDeskAdmin</code>
Alerts Service	Configure alert services to receive notifications.	Technical Name: <code>sap.crm.service.alertsService</code>
Analytics	Create analytical stories and models.	• Technical Name: <code>sap.crm.service.analyticsBusinessUserStoriesService</code> • UI App Name: <code>sap.crm.analyticsbusinessuserstorieservice.uiapp.anaStoryBusinessUser</code>

Business Service Name	Description	Business Service: Technical Details
Analytics - SAC Setup Service	Configure SAP Analytics Cloud setup.	• Technical Name: sap.crm.service.analyticsSacSPSetuService • UI App Name: sap.crm.analyticsacsacspsetupservice.uiapp.idPSetupUI
Analytics - User Story Service	View stories assigned to your user.	• Technical Name: sap.crm.service.analyticsUserStoriesService • UI App Name: sap.crm.analyticsuserstorieservice.uiapp.anaStoryAdmin
Application Log	View a summary of all errors that occurred during the processing of custom logic external hooks.	Technical Name: sap.crm.md.service.accountService
Approvals	Create approvals for streamlining your business processes.	Technical Name: sap.crm.md.service.accountService
Attachments	Choose the files that you want to upload or download.	• Technical Name: sap.crm.service.documentService • UI App Name: sap.crm.documentservice.uiapp.document
Attachments Admin	Configure your attachment settings.	• Technical Name: • UI App Name: sap.crm.documentservice.uiapp.documentAdmin
Audit Log	Review security-relevant chronological records of user actions.	• Technical Name:sap.crm.service.auditService • UI App Name: sap.crm.documentservice.uiapp.documentAdmin
Authorization Service	Configure authorization services.	Technical Name: sap.crm.service.authService
Autoflow	Automate your business processes by creating rules that can trigger various events.	• Technical Name: sap.crm.service.workflowService • UI App Name: sap.crm.workflowservice.uiapp.workflowAdmin

Business Service Name	Description	Business Service: Technical Details
Built-In Support Service	Find help, create tickets, and contact support right in your cloud application.	• Technical Name: <code>sap.crm.service.builtinSupportService</code> • UI App Name: <code>sap.crm.builtinsupportservice.uiapp.builtinSupportUi</code>
Built-In Support Service Admin	Configure support services to find help, create tickets, and contact support right in your cloud application	• Technical Name: <code>sap.crm.service.builtinSupportService</code> • UI App Name: <code>sap.crm.builtinsupportservice.uiapp.builtinSupportAdminUi</code>
Business Partners	Capture, monitor, store, and track all critical information about business partners.	• Technical Name: <code>sap.crm.md.service.businessPartnerService</code> • UI App Name: <code>sap.crm.md.businesspartnerservice.uiapp.businessPartnerApp</code>
Business Partners Admin	Configure business partner settings.	• Technical Name: <code>sap.crm.md.service.businessPartnerService</code> • UI App Name: <code>sap.crm.md.businesspartnerservice.uiapp.businessPartnerAdminApp</code>
Business Partner Relationship	Add relationships between different business partners.	• Technical Name: <code>sap.crm.md.service.businessPartnerRelationshipService</code> • UI App Name: <code>sap.crm.md.businesspartnerrelationshipservice.uiapp.businessPartnerRelationshipAdminApp</code>
Call Lists Administration	Configure settings for Call lists.	• Technical Name: <code>sap.crm.service.callListService</code> • UI App Name: <code>sap.crm.callListService.uiapp.callListAdminApp</code>

Business Service Name	Description	Business Service: Technical Details
Call Lists	Create call lists from the database of contacts, prospects, and leads. Also track and manage them.	• Technical Name: <code>sap.crm.service.callListService</code> • UI App Name: <code>sap.crm.calllistservice.uiapp.callList</code>
Case	Create cases and manage case phases and steps.	• Technical Name: <code>sap.crm.service.caseService</code> • UI App Name: <code>sap.crm.caseservice.uiapp.case</code>
Case Management	Create case types and configure case settings.	• Technical Name: <code>sap.crm.service.caseService</code> • UI App Name: <code>sap.crm.caseservice.uiapp.caseAdmin</code>
Case Type	Create and configure case types to match the workflow used by your organization.	• Technical Name: <code>sap.crm.service.caseTypeService</code> • UI App Name: <code>sap.crm.caseservice.uiapp.caseAdmin</code>
Change History	Check when and by whom an entity was modified.	• Technical Name: <code>sap.crm.service.changeHistoryService</code> • UI App Name: <code>sap.crm.service.changehistoryservice.uiapp.changeHistoryApp</code>
Change Project	Record and transport configuration changes from a source tenant to a target tenant.	• Technical Name: <code>sap.crm.service.changeProjectService</code> • UI App Name: <code>sap.crm.changeprojectservice.uiapp.changeProjectApp</code>
Chat	Chat with customers to resolve their service issues.	• Technical Name: <code>sap.crm.service.chatService</code> • UI App Name: <code>sap.crm.chatservice.uiapp.chatInteraction</code>

Business Service Name	Description	Business Service: Technical Details
Code List Restrictions	Restrict the availability of code values based on a combination of business entity, code field, and control field.	• Technical Name: <code>sap.crm.service.codeListRestrictionService</code> • UI App Name: <code>sap.crm.codelistrestrictionsservice.uiapp.codeListRestrictionsUi</code>
Competitor	Create new competitors and maintain competitor information.	• Technical Name: <code>sap.crm.md.service.competitorService</code> • UI App Name: <code>sap.crm.md.competitorservice.uiapp.competitorApp</code>
Competitor Product	Create and maintain comprehensive records of competitor products.	• Technical Name: <code>sap.crm.md.service.competitorProductService</code> • UI App Name: <code>sap.crm.md.competitorproductservice.uiapp.competitorProductAdminApp</code>
Computer Telephony Integration (CTI)	Speak with customers using partner-supplied computer telephony service.	• Technical Name: <code>sap.crm.service.ctiService</code> • UI App Name: <code>sap.crm.ctiservice.uiapp.cti</code>
Contacts	Maintain the details for contacts by creating a contact or using the relation between a contact and an account.	• Technical Name: <code>sap.crm.md.service.contactPersonService</code> • UI App Name: <code>sap.crm.md.contactpersonservice.uiapp.contactApp</code>
Contract Account	Manage contract account and their related plans.	• Technical Name: <code>sap.crm.service.contractAccountService</code> • UI App Name: <code>sap.crm.contractaccountservice.uiapp.contractaccout</code>

Business Service Name	Description	Business Service: Technical Details
Customer Insights	Analyze customer data and feedback to generate insights.	• Technical Name: sap.crm.service.insightsService • UI App Name: sap.crm.insightsservice.uiapp.insightssap.crm.insightsservice.uiapp.insightsAdminApp
Customer Insights Summary Service	Summarize customer data to enable sales teams to customize their approach	Technical Name: sap.crm.service.insightsSummaryService
Customer Returns	Create, manage, and track return orders.	• Technical Name: sap.crm.service.returnOrderService • UI App Name: sap.crm.returnorderservice.uiapp.returnOrderAdmin
Data Connector Service		• Technical Name: sap.crm.service.dataConnectorService • UI App Name: sap.crm.dataconnector.service.uiapp.dataConnectorConfigAppsap.crm.dataconnector.service.uiapp.communicationSystemsApp sap.crm.dataconnector.service.uiapp.monitoringApp
Data Import and Export	Import and export data.	• Technical Name: sap.crm.service.dataImpexService • UI App Name: sap.crm.dataimpexservice.uiapp.dataimpex
Data Depersonalization	Remove personal data of employees, individual customers, and contacts on their request.	• Technical Name: sap.crm.service.dppService • UI App Name: sap.crm.dppservice.uiapp.depersonalize
Data Disclosure	Disclose personal data of employees, individual customers, and contacts.	• Technical Name: sap.crm.service.dppService • UI App Name: sap.crm.dppservice.uiapp.disclosure

Business Service Name	Description	Business Service: Technical Details
Data Workbench	Import data from a third-party system or a csv file.	• Technical Name: <code>sap.crm.service.dataworkbenchService</code> • UI App Name: <code>sap.crm.dataworkbenchservice.uiapp.dataworkbench</code>
Digital Assistant - CX AI Toolkit Admin Service	Integrate the CX AI Toolkit from the settings page.	• Technical Name: <code>sap.crm.service.digitalAssistantCxAitkAdminService</code> • UI App Name: <code>sap.crm.digitalassistantcxaitkadminservice.uiapp.cxaiToolKitAdmin"</code>
Digital Assistant - CX AI Toolkit Service	Analyze enterprise data with trusted AI that helps you improve sales performance and enhance customer service.	• Technical Name: <code>sap.crm.service.digitalAssistantCxAitkService</code> • UI App Name: <code>sap.crm.digitalassistantcxaitkservice.uiapp.cxaiToolKit</code>
Digital Selling Workspace	Gather information and insights required for productive customer conversations	• Technical Name: <code>sap.crm.service.dswService</code> • UI App Name: <code>sap.crm.dswservice.uiapp.dswApp</code>
Document Service	Configure document services.	• Technical Name: <code>sap.crm.service.documentService</code> • UI App Name: <code>sap.crm.documentservice.uiapp.document</code>
E-mail Action	Add or remove e-mails	• Technical Name: <code>sap.crm.service.emailActionService</code> • UI App Name: <code>sap.crm.emailactionservice.uiapp.emailAction</code>
E-mail Channel	Create and manage e-mail channels used for customer interactions and case routing.	• Technical Name: <code>sap.crm.service.emailChannelService</code> • UI App Name: <code>sap.crm.emailchannelservice.uiapp.emailChannel</code>

Business Service Name	Description	Business Service: Technical Details
E-mail Monitor	Monitor inbound and outbound e-mails.	• Technical Name: sap.crm.service.emailMonitorService • UI App Name: sap.crm.emailmonitorservice.uiapp.emailMonitor
E-mail MTA Configuration	Configure a Mail Transfer Agent (MTA) to send e-mails through your company's Simple Mail Transfer Protocol (SMTP) server or to use the default MTA.	• Technical Name: sap.crm.service.emailMtaConfigService • UI App Name: sap.crm.emailmtaconfigservice.uiapp.emailMtaConfig
E-mail	Create and send e-mail messages to customers.	• Technical Name: sap.crm.service.emailService • UI App Name: sap.crm.emailservice.uiapp.email
E-mail Interaction	Track e-mail interactions.	• Technical Name: sap.crm.service.emailService • UI App Name: sap.crm.emailservice.uiapp.emailInteraction
Employees	Create employees in the system.	• Technical Name: sap.crm.md.service.employeeService • UI App Name: sap.crm.md.employeeservice.uiapp.employeeApp
Enrichment Service	Provide additional related-entity information to dependent services such as Autoflow and email templates.	Technical Name: sap.crm.service.enrichmentService
ESM Backend Integration Service	Enable an integration option with backend systems such as an S/4HANA system	• Technical Name: sap.crm.service.esmBackendIntegrationService • UI App Name: sap.crm.esmbackendintegrationservice.uiapp.integrationApp

Business Service Name	Description	Business Service: Technical Details
ESM Common Control Service		• Technical Name: sap.crm.service.esmCommonControlService • UI App Name: sap.crm.esmcommoncontrolservice.uiapp.commonControlApp
ESM Condition Service		• Technical Name: sap.crm.service.esmConditionService • UI App Name: sap.crm.esmconditionservice.uiapp.conditionApp
Event Bridge	Enable event bridge service for your users.	• Technical Name: sap.crm.service.eventBridgeService • UI App Name: sap.crm.eventbridgeservice.uiapp.eventBridgeAdmin
Extensibility Admin	Create extension fields and define the field attributes. You can also create custom validations and determinations.	• Technical Name: sap.crm.service.extensibilityService • UI App Name: sap.crm.extensibilityservice.uiapp.extensibilityAdmin
External ID Configuration		• Technical Name: sap.crm.service.externalIdAdminService • UI App Name: sap.crm.externalidadadminservice.uiapp.externalIdAdminApp
External Link		• Technical Name: sap.crm.service.externalNavigationService • UI App Name: sap.crm.externalnavigationservice.uiapp.externalNavigationApp

Business Service Name	Description	Business Service: Technical Details
Favorites	Mark frequently used objects to enable quick access in the future.	• Technical Name: sap.crm.service.favoriteService • UI App Name: sap.crm.favoriteservice.uiapp.favorite
Favorites Workspace	View all your favorites in one place.	• Technical Name: sap.crm.service.favoriteService • UI App Name: sap.crm.favoriteservice.uiapp.favoriteWorkspace
Feedback	Configure feedback service.	• Technical Name: sap.crm.service.feedbackService • UI App Name: sap.crm.feedbackservice.uiapp.feedbacksap.crm.feedbackservice.uiapp.feedbackAdmin
Financials	Create financial transactions and their related services for contract account.	• Technical Name: ap.crm.servsice.financialsService • UI App Name: sap.crm.financialsservice.uiapp.financials
Flag		Technical Name: sap.crm.service.flagService
Flexibility	Change the look and feel of the solution for all users by changing layout settings, adding mash-ups and fields, as well as adding extension fields.	• Technical Name: sap.crm.service.extensibilityService • UI App Name: sap.crm.extensibilityservice.uiapp.flex
Form	Configure the form service to work with forms.	Technical Name: sap.crm.service.formService
Functional Location	Maintain physical location of the object that requires servicing or maintenance.	• Technical Name: sap.crm.service.functionalLocationService • UI App Name: sap.crm.functionalllocationservice.uiapp.functionallLocationsap.crm.functionalllocationservice.uiapp.functionallLocationAdmin

Business Service Name	Description	Business Service: Technical Details
Global Calendar	View all appointments or visits days of an organizational unit within a particular company or region.	• Technical Name: <code>sap.crm.service.calendarService</code> • UI App Name: <code>sap.crm.calendarService.uiapp.globalCalendarApp</code>
Groupware Integration	Use SAP Sales Cloud Version 2 features from Microsoft Outlook or your Gmail inbox.	• Technical Name: <code>sap.crm.service.groupwareSsiService</code> • UI App Name: • <code>sap.crm.groupwareSsiService.uiapp.groupwareSsi</code> • <code>sap.crm.groupwareSsiService.uiapp.groupwareSsiAdmin</code>
Guided Selling	Work through the sales cycle across various phases by creating opportunities.	• Technical Name: <code>sap.crm.service.guidedSellingService</code> • UI App Name: <code>sap.crm.guidedSellingService.uiapp.guidedSelling</code>
Holiday Calendar	Add holidays of an organizational unit within a particular company or region.	• Technical Name: <code>sap.crm.service.holidayCalendarService</code> • UI App Name: <code>sap.crm.holidayCalendarService.uiapp.holidayCalendarApp</code>
Home Page	Explore the default start page that provides various card types and helps you with essential information and easy navigation.	• Technical Name: <code>sap.crm.service.homepageService</code> • UI App Name: <code>sap.crm.homepageService.uiapp.homepage</code>
Home Page Admin	Arrange the start page with information and activities based on business roles.	• Technical Name: <code>sap.crm.service.homepageService</code> • UI App Name: <code>sap.crm.homepageService.uiapp.homepageadmin</code>

Business Service Name	Description	Business Service: Technical Details
Identity and Access Management Configuration Service	Configure authentication and authorization mechanism for your users.	• Technical Name: sap.crm.service.iamConfigurationService • UI App Name: sap.crm.iamconfigurationservice.uiapp.iamConfiguration
Inbound Data Connector		Technical Name: sap.crm.service.inboundDataConnectorService
Individual Customers	Manage, monitor, store, and track business information about individual customers.	• Technical Name: sap.crm.md.service.individualCustomerService • UI App Name: sap.crm.md.individualcustomersevice.uiapp.individualCustomerApp
Installed Bases	Arrange installed items hierarchically at your customer's location.	• Technical Name: sap.crm.service.installedBaseService • UI App Name: sap.crm.installedbaseservice.uiapp.installedBase
Knowledge Base	Search for, review, and share Knowledge Base Articles.	• Technical Name: sap.crm.service.knowledgeBaseService • UI App Name: sap.crm.knowledgebaseservice.uiapp.knowledgebase
Knowledge Base Settings	Configure knowledge base settings.	• Technical Name: sap.crm.service.knowledgeBaseService • UI App Name: sap.crm.knowledgebaseservice.uiapp.knowledgebaseAdmin
Language Adaptation Service	Override the standard SAP texts and customize field label terminology to fit your business needs.	• Technical Name: sap.crm.service.languageAdaptationService • UI App Name: sap.crm.languageadaptationsevice.uiapp.languageAdaptation

Business Service Name	Description	Business Service: Technical Details
Language, Region, & Currency	Choose language, region, and currency settings	• Technical Name: sap.crm.service.i18nService • UI App Name: sap.crm.i18nservice.uiapp.i18napp
Language, Region, & Currency Admin	Configure language, region, and currency settings	• Technical Name: sap.crm.service.i18nService • UI App Name: sap.crm.i18nservice.uiapp.i18nAdmin
Lead	Capture interest in products and services, and qualify and nurture the interest to convert them to opportunities for sales.	• Technical Name: sap.crm.service.leadService • UI App Name: • sap.crm.service.leadService (<i>Lead</i>) • sap.crm.service.leadService (<i>Lead Administration</i>)
Library	Store, share files with the other users, and link files as attachments in sales and service entities.	• Technical Name: sap.crm.service.libraryService • UI App Name: • sap.crm.libraryservice.uiapp.libraryApp • sap.crm.libraryservice.uiapp.libraryAdminApp
Machine Learning Topic Analyzer Service	Use Generative AI to view trending topics.	• Technical Name: sap.crm.service.mlNlpTopicAnalyzerService • UI App Name: sap.crm.mlNlpTopicAnalyzerService.uiapp.topicAnalyzerResult
Machine Learning Management Generative AI Service	Generate account synopsis, case summary, and draft e-mail response for cases.	• Technical Name: sap.crm.service.mlScenarioManagementGenAIService • UI App Name: sap.crm.mlscenariomanagementgenaiservice.uiapp.mlScenarioMngmtGenAIAdmin

Business Service Name	Description	Business Service: Technical Details
Machine Learning Management Service	Use case text to identify sentiment, translate case text, and identify profane language.	• Technical Name: sap.crm.service.mlScenarioManagementService • UI App Name: sap.crm.mlscenariomanagementservice.uiapp.mlScenarioMngmntAdmin
Mashups	Integrate SAP Cloud solution data with data from online Web services or applications.	• Technical Name: sap.crm.service.mashupService • UI App Name: sap.crm.mashupservice.uiapp.mashupApp
Mashup Authoring	Activate preconfigured mashups and create new mashups.	• Technical Name: sap.crm.service.mashupService • UI App Name: sap.crm.mashupservice.uiapp.mashupAuthoring
Mashup Configuration	Configure, access, and use mashups.	• Technical Name: sap.crm.service.extensibilityService • UI App Name: sap.crm.extensibilityservice.uiapp.flexMashupConfigAdmin
Mashup Demo	Display mashup service consumption.	• Technical Name: sap.crm.service.mashupService • UI App Name: sap.crm.mashupservice.uiapp.mashupDemo
Messaging	Exchange text messages with customers to resolve their service issues.	• Technical Name: sap.crm.service.chatService • UI App Name: sap.crm.chat-service.uiapp.messageInteraction
Microsoft Teams Service		• Technical Name: sap.crm.service.msteamsService • UI App Name: sap.crm.msteamsservice.uiapp.msteamsadminAppsap.crm.msteamsservice.uiapp.msteamsap.crm.msteamservice.uiapp.mobile

Business Service Name	Description	Business Service: Technical Details
Mobile Admin Service	Configure mobile application for your users.	• Technical Name: <code>sap.crm.service.mobileAdminService</code> • UI App Name: <code>sap.crm.mobileadminservice.uiapp.mobile</code>
Move Process	Create move processes and their related services.	• Technical Name: <code>sap.crm.service.moveIntegrationService</code> • UI App Name: <code>sap.crm.moveintegrationservice.uiapp.moveApp</code>
Note	Store and maintain notes with rich text support for host objects like case, account, visit, and so on.	Technical Name: <code>sap.crm.service.noteService</code>
Opportunity	Configure access restrictions for Guided Selling, Pipeline Manager, Forecast Tracker, and Pipeline Flow.	Technical Name: <code>sap.crm.service.opportunityService</code>
Open Resource Discovery Service		Technical Name: <code>sap.crm.service.openResourceDiscoveryService</code>
Organizational Units	Create an element of an organizational model.	• Technical Name: <code>sap.crm.md.service.organizationalUnitService</code> • UI App Name: <code>sap.crm.md.organizationalunitsevice.uiapp.organizationalUnitApp</code>
Organizational Unit Admin	Maintain organizational units.	• Technical Name: <code>sap.crm.md.service.organizationalUnitService</code> • UI App Name: <code>sap.crm.md.organizationalunitsevice.uiapp.organizationalUnitAdminApp</code>

Business Service Name	Description	Business Service: Technical Details
Outbound Data Connector		• Technical Name: sap.crm.service.outboundDataConnectorService • UI App Name: sap.crm.outbounddataconnectorsevice.uiapp.eventMonitoringApp
Output Management	Manage activities related to the output of documents.	• Technical Name: sap.crm.service.outputManagementService • UI App Name: sap.crm.outputmanagementservice.uiapp.outputManagementApp
Page Layouts and Assignments Rules	Create and assign rules to page layouts.	• Technical Name: sap.crm.service.pageLayoutService • UI App Name: sap.crm.pagelayoutservice.uiapp.assignmentRules sap.crm.pagelayoutservice.uiapp.pageLayouts
Partners	Create and manage business information about partners.	• Technical Name: sap.crm.md.service.partnerService • UI App Name: sap.crm.md.partnerservice.uiapp.partnerApp
Partner Contacts	Create and manage business information about partner contacts.	• Technical Name: sap.crm.md.service.partnerContactService • UI App Name: sap.crm.md.partnercontactservice.uiapp.partnerContactApp
Party Processing	Configure and maintain involved parties in various business transactions.	• Technical Name: sap.crm.service.partyProcessingService • UI App Name: sap.crm.partyprocessingservice.uiapp.partyProcessingApps sap.crm.partyprocessingservice.uiapp.partySchemeApp

Business Service Name	Description	Business Service: Technical Details
Phone	Track and review phone call activities.	• Technical Name: sap.crm.service.phoneService • UI App Name: sap.crm.phoneservice.uiapp.phone
Playbook	Build a sales framework to guide your sales representatives through the entire sales cycle and help accelerate deals.	• Technical Name: sap.crm.service.salesPlaybookService • UI App Name: sap.crm.salesplaybookservice.uiapp.salesPlaybook
Premise	Manage premise and their related services.	• Technical Name: sap.crm.service.premiseService • UI App Name: sap.crm.premiseservice.uiapp.premise
Pricing	Configure pricing components and master data to enable native price calculation.	• Technical Name: sap.crm.service.pricingMasterDataService • UI App Name: • sap.crm.pricingmasterdataservice.uiapp.pricingMasterData • sap.crm.pricingmasterdataservice.uiapp.pricingMasterDataAdmin
Products	Create, view, and update products in an organizational master data.	• Technical Name: sap.crm.md.service.productService • UI App Name: sap.crm.md.productservice.uiapp.productApp
Product Groups	Create, view, and update product groups in an organizational master data.	• Technical Name: sap.crm.md.service.productGroupService • UI App Name: sap.crm.md.productgroupservice.uiapp.productGroupApp

Business Service Name	Description	Business Service: Technical Details
Product List	Configure targeted product proposals, exclusions, listings.	• Technical Name: <code>business_services.sap.crm.service.productListService</code> • UI App Name: • <code>sap.crm.productlistservice.uiapp.productListApp</code> • <code>sap.crm.productlistservice.uiapp.productListAdminApp</code>
Product Installation Point	Maintain physical location of the object that requires servicing or maintenance.	• Technical Name: <code>sap.crm.service.productInstallPointService</code> • UI App Name: <code>sap.crm.productinstallpointservice.uiapp.productInstallPointsap.crm.productinstallpointservice.uiapp.productInstallPointAdmin</code>
Qualtrics XM Discover Service	Create and configure feedback flow.	• Technical Name: <code>sap.crm.service.qualtricsXmDiscoverService</code> • UI App Name: <code>sap.crm.qualtricsxmdiscoverservice.uiapp.xmDiscoverAppLauncher</code>
Question Bank	Store survey questions in the question bank for easy reuse in multiple surveys.	• Technical Name: <code>sap.crm.service.questionBankService</code> • UI App Name: <code>sap.crm.questionbankservice.uiapp.questionBankApp</code>
Read Access Logging Service	Monitor and record access to sensitive personal data.	• Technical Name: <code>sap.crm.md.service.raLAppService</code> • UI App Name: <code>sap.crm.md.raLAppservice.uiapp.raLApp</code>
Registered Products	Use products that are associated with a specific customer and generally has a serial number.	• Technical Name: <code>sap.crm.service.registeredProductService</code> • UI App Name: <code>sap.crm.registeredproductservice.uiapp.registeredProduct</code>

Business Service Name	Description	Business Service: Technical Details
Roles	Configure different user roles in the system to access specific services.	• Technical Name: sap.crm.service.roleService • UI App Name: sap.crm.roleservice.uiapp.roleAdmin
Campaign	Engage with customers and prospects to promote products and services.	• Technical Name: sap.crm.service.campaignService • UI App Name: sap.crm.service.campaignService
Sales Order	Record and view sales interactions related to sales orders.	• Technical Name: sap.crm.service.salesOrderService • UI App Name: • sap.crm.salesorderservice.uiapp.salesOrderAdmin • sap.crm.salesorderservice.uiapp.salesOrderApp
Sales Quote	Create and manage sales quotations for business partners.	• Technical Name: sap.crm.service.salesQuoteService • UI App Name: • sap.crm.salesquoteservice.uiapp.salesQuoteApp • salesquoteservice.uiapp.salesQuoteAdminApp
SAML IdP Service	Configure SAML IdP services for your users.	• Technical Name: sap.crm.service.saml2idpService • UI App Name: sap.crm.saml2idpservice.uiapp.idpConfigAdmin
Search	Search for entity details across the solution.	• Technical Name: sap.crm.service.searchService • UI App Name: sap.crm.searchservice.uiapp.searchApp
Search Admin	Manage global search settings.	• Technical Name: sap.crm.service.searchService • UI App Name: sap.crm.searchservice.uiapp.searchAdmin

Business Service Name	Description	Business Service: Technical Details
Search Referring Service	Maintain the attributes of referring entities up to date	Technical Name: <code>sap.crm.service.searchReferringService</code>
Security Policies	Create security policies for your organization	• Technical Name: <code>sap.crm.service.securityPolicyService</code> • UI App Name: <code>sap.crm.securitypolicy.service.uiapp.securityPolicyAdmin</code>
Service Category Catalog	Configure service catalog classifications.	• Technical Name: <code>sap.crm.service.serviceCategoryService</code> • UI App Name: <code>sap.crm.servicecategoryservice.uiapp.serviceCatalogClassification</code>
Service Category Catalog Settings	Create and manage service category catalogs.	• Technical Name: <code>sap.crm.service.serviceCategoryService</code> • UI App Name: <code>sap.crm.servicecategoryservice.uiapp.serviceCatalogAdmin</code>
Service Category Path		• Technical Name: <code>sap.crm.service.serviceCategoryService</code> • UI App Name: <code>sap.crm.servicecategoryservice.uiapp.serviceCategoryPath</code>
Service Category Catalog Selection		• Technical Name: <code>sap.crm.service.serviceCategoryService</code> • UI App Name: <code>sap.crm.servicecategoryservice.uiapp.serviceCatalogSelection</code>
Service Level	Create, view, and maintain service level agreements.	• Technical Name: <code>sap.crm.service.serviceLevelService</code> • UI App Name: <code>sap.crm.servicelevelservice.uiapp.serviceLevel</code>

Business Service Name	Description	Business Service: Technical Details
Service Orders	Replicate service orders to external systems.	• Technical Name: sap.crm.service.s4hcServiceOrderService (cloud) • sap.crm.service.s4hopServiceOrderService(On-prem mashup) • UI App Name: sap.crm.s4hcserviceorderservice.uiapp.externalReplica (cloud) • sap.crm.s4hopserviceorderservice.uiapp.s4hopServiceOrder(On-prem mashup)
Service Quotation	Create and manage service quotations for business partners.	• Technical Name: sap.crm.service.s4hcServiceQuotationService • UI App Name: sap.crm.s4hcservicequotationservice.uiapp.s4hcServiceQuotation • sap.crm.servicequotationservice.uiapp.serviceQuotationAdmin
Settings	Configure dashboard settings for your system.	• Technical Name: sap.crm.service.adminService • UI App Name: sap.crm.adminservice.uiapp.admin
Settings	Configure administration settings for your system.	• Technical Name: sap.crm.service.adminService • UI App Name: sap.crm.adminservice.uiapp.adminconsole
Shell Service	Adjust the interface to optimize it for interaction.	Technical Name: sap.crm.service.shellService
Surveys - Design Time	Create surveys that sales representatives can use to collect business-related data.	• Technical Name: sap.crm.service.surveyDesignService • UI App Name: sap.crm.surveydesignservice.uiapp.survey

Business Service Name	Description	Business Service: Technical Details
Surveys - runtime	Answer Surveys and help collect important information for deriving insights.	• Technical Name: sap.crm.service.surveyService • UI App Name: sap.crm.surveydesignservice.uiapp.survey
Surveys Administration	Configure your settings for surveys.	• Technical Name: sap.crm.service.surveyDesignService • UI App Name: sap.crm.surveydesignservice.uiapp.surveyAdminApp
Target Group	Segment customers and prospects to form the right target audience for sales campaigns.	• Technical Name: • sap.crm.service.targetGroupService • sap.crm.service.targetGroupMemberService • UI App Name: sap.crm.service.targetGroupService
Tasks	Create and manage your tasks associated with entities such as leads, opportunities, and quotes, and so on.	• Technical Name:sap.crm.service.taskService • UI App Name: • sap.crm.taskservice.uiapp.taskApp • sap.crm.taskservice.uiapp.taskAdmin • sap.crm.taskservice.uiapp.taskTile • sap.crm.taskservice.uiapp.taskManagerApp
Templates	Create and edit templates.	• Technical Name: sap.crm.service.templateService • UI App Name: sap.crm.templateservice.uiapp.template

Business Service Name	Description	Business Service: Technical Details
Template Settings	Configure template settings.	• Technical Name: sap.crm.service.templateService • UI App Name: sap.crm.templateservice.uiapp.templateAdmin
Timeline	View timeline events for your apps.	• Technical Name: sap.crm.service.timelineService • UI App Name: sap.crm.timelineservice.uiapp.timeline
Timeline Settings	Configure timeline for your apps.	• Technical Name: sap.crm.service.timelineService • UI App Name: sap.crm.timelineservice.uiapp.timelineAdmin
User Settings	Configure user settings.	• Technical Name: sap.crm.service.userService • UI App Name: sap.crm.userservice.uiapp.userAdmin
Utilities Admin Settings	Configure utilities admin settings.	• Technical Name: sap.crm.service.commonIntegrationService • UI App Name: sap.crm.commonintegrationservice.uiapp.commonadminui
Value Mapping		• Technical Name: sap.crm.service.valueMappingService • UI App Name: sap.crm.valuemappingservice.uiapp.valuemappingApp
Warranty	Assign a warranty to a registered product or installation point, and determine its coverage in a ticket.	• Technical Name: sap.crm.service.warrantyService • UI App Name: sap.crm.warrantyservice.uiapp.warranty

Business Service Name	Description	Business Service: Technical Details
Where-Used Service	Get a list of content, within a system, where objects such as extensions fields, code lists, and so on is used or referenced.	• Technical Name: <code>sap.crm.service.whereUsedService</code> • UI App Name: <code>sap.crm.whereusedservice.uiapp.whereUsed</code>
Working Day Calendar	Add working days of an organizational unit within a particular company or region.	• Technical Name: <code>sap.crm.service.workingDayCalendarService</code> • UI App Name: <code>sap.crm.workingdaycalendar.service.uiapp.workingDayCalendarApp</code>

4.3 Security Policies

A security policy is a set of rules that defines password complexity, such as including numerical digits and password validity, like requiring a password change after a certain period of time.

You can define multiple security policies because work areas or departments of a company may have different password security requirements.

As an administrator, you can manage the security level, if desired, by editing and enhancing the security policy, for example, by changing the complexity and validity for all passwords, in accordance with your company's security requirements.

4.3.1 Security Policy Features

Lists and describes the features of a security policy.

The security policy features are classified as:

- System password-related features
 - Enable Password Login - if enabled, allows users to log in to the system with a password.
 - Set Default - if enabled is set as the default security policy for all the users.
- System password complexity-related features
 - Minimum Number of Characters - defines the minimum number of characters in a valid password.
 - Minimum Number of Lowercase Alphabets - defines the minimum number of lowercase alphabets in a valid password.
 - Minimum Number of Uppercase Alphabets - defines the minimum number of uppercase alphabets in a valid password.

- Minimum Number of Digits - defines the number of numerals in a password.
- Minimum Number of Special Characters - defines the minimum number of special characters in a valid password.
- System password validity-related features
 - Allowed Number of Incorrect Password Attempts - defines the number of successive login failures before a password is locked. A message is displayed on the login screen once the password is locked.
 - Password History - defines the number of passwords to store in the memory to compare with the new productive password set by the user.
 - Maximum Password Validity (in Days) - defines the maximum number of days a password is valid for.
 - Minimum Waiting Time Before Changing Password (in Days) - defines the minimum number of days a user must wait before changing the existing password to a new password.
 - Unused Productive Password Validity (in Days) - defines the validity of the productive password after changing from the initial password.

4.3.2 Create Security Policies

You can define multiple security policies as different work areas or departments in a company can have different password security requirements.

Context

As an administrator, you can increase the security level by editing and enhancing a security policy. For example, by changing the complexity and validity for all passwords, in accordance with your company's security requirements.

Procedure

1. Log in to the system as an administrator.
2. Navigate to [User Menu](#) on the top-right corner, and access the [System Settings](#) page.
3. Go to [All Settings](#) [Users and Control](#) [Security Policies](#).
4. Click the add icon (+) on the [Security Policies](#) page.
5. Enter a [Policy ID](#), [Policy Name](#), and [Description](#) for the security policy. You can further choose to set this policy as the default using the [Set Default](#) switch.
6. Enhance security complexity for your policy by entering inputs for: [Minimum Number of Characters](#), [Minimum Number of Lowercase Alphabets](#), [Minimum Number of Uppercase Alphabets](#), [Minimum Number of Digits](#), [Minimum Number of Special Characters](#), [Allowed Number of Incorrect Password Attempts](#), [Password History](#), [Maximum Password Validity \(in Days\)](#), [Minimum Password Change Waiting Time \(in Days\)](#), and then choose [Save](#).
7. Refresh the [Security Policies](#) page to view your newly created security policy. All the security policies that you've created appear in the list format on the [All Policies](#) page.

You've now created a security policy.

4.3.3 Edit Security Policy

Administrators can edit security policies.

Procedure

1. Log in to the system as an administrator.
2. Navigate to [User Menu](#) on the top-right corner, and access the [System Settings](#) page.
3. Go to ► [All Settings](#) ► [Users and Control](#) ► [Security Policies](#) ►.
4. Click an existing policy ID to view the details. The [Security Policy](#) quick view opens.
5. Hover the mouse over the fields that you want to edit. Click on pencil icon next to the field to edit the field.

Results

Once edited, the changes are saved automatically.

4.4 Identity Provider Configuration

Identity provider refers to an entity that manages identity information and provides authentication services to trusted service providers.

Every user type must authenticate to the system for regular browser-based front-end access, as well as for electronic data exchange, such as business-to-business communication. The system doesn't support anonymous access. We recommend using Single Sign-On (SSO) for basic security. To protect accounts further, configure the identity provider (IdP) of the SSO solution to provide enhanced security.

4.4.1 Configure Identity Provider

Administrators can configure identity provider to enable Single-Sign-On (SSO) for the system.

Enable Single Sign-On for the System

To enable SSO for a tenant, you've to connect the identity provider to the tenant. To do so, download the metadata of your system, upload the tenant metadata XML file to your identity provider (IdP), and then download the configured metadata from your identity provider. Afterwards on your system, choose the identity provider, and create the identity provider to start using the system.

Download the System Metadata

Download the tenant metadata XML and further upload it to IdP to configure it.

1. Log in to the system as an administrator.
2. Navigate to the [User Menu](#) on the top-right corner, and access the [System Settings](#) page.
3. Go to ► [All Settings](#) ► [Users and Control](#) ► [IdP Configuration](#) ► page.
4. Choose [Download Metadata](#). The tenant metadata XML file is downloaded to your local drive.

Note

If your IdP certificate is due for renewal in the next 20 days, the message "Click the [Renew and Download Metadata](#) button before the certificate validate expiration date to avoid disruptions to your SSO users" appears. In addition to the message, the certificate validity period is displayed as well.

If you intend to renew and reconfigure your identity provider, click the [Renew and Download Metadata](#) option to download and further reconfigure the identity provider. The [Renew and Download Metadata](#) option extends the validity of your certificate and stops your SSO users from logging into the system until you've reconfigured your IdP with the renewed metadata file. Use the downloaded service-provider metadata file to configure your IdP.

Configure System Metadata in Your Identity Provider (IdP)

Upload the system metadata XML file to IdP, configure it, and then download the configured IdP metadata file.

Note

Use this procedure to configure authentication services provided by SAP Cloud Identity Services. You can follow similar steps to configure other IdPs as well.

1. Log on to the [SAP Cloud Identity Services](#) as an administrator. You can also use an external identity provider.
2. On the SAP Cloud Identity Services screen, navigate to ► [Applications & Resources](#) ► [Applications](#) ►, and choose [Add](#) to create a new application for your system.
3. In the [Add Application](#) pop-up window, enter a name, and choose [Save](#). A new application page opens.
4. Under ► [Trust](#) ► [Single Sign-On](#) ►, configure the following settings:
 - Choose [Type](#), and select SAML 2.0

- Choose [SAML 2.0 Configuration](#), and upload the downloaded tenant metadata XML from your local drive.
 - Choose [Subject Name Identifier](#), and configure the attribute, which the application uses to identify the users.
5. Navigate to [Tenant Settings](#), and choose [SAML 2.0 Configuration](#) to open a new screen.
 6. Choose [Download Metadata File](#) to download the IdP metadata file.

Create a Trusted Identity Provider and Upload IdP Metadata to Your System

Navigate to your tenant to create a trusted IdP and choose a default IdP.

1. Log in to your system as an administrator.
2. Navigate to the [User Menu](#) on the top-right corner, and access the [System Settings](#) page.
3. Go to [All Settings](#) > [Users and Control](#) > [Configure IdP](#).
4. Choose the add (+) icon under the [Trusted Identity Provider](#) section.
5. Under the [Upload IdP Metadata](#) quick view, enter an [Alias](#), and upload the configured metadata XML file (downloaded from your IdP) by either dropping or browsing to upload from the system.
6. Choose [Save](#) to save your configuration.
7. Activate your configuration by enabling the [Set Active](#) switch. To make it a default option, turn on the [Set Default](#) switch.

You've now configured Single Sign-On for your system.

Related Information

[Access Admin Console](#)

4.5 System for Cross-Domain Identity Management (SCIM) API

System for Cross-domain Identity Management (SCIM) specification is designed to efficiently manage user identities in cloud-based applications and services.

The specification suite builds upon the existing schema and deployments, placing specific emphasis on simplicity of development and integration, while applying existing authentication, authorization, and privacy models. It intends to reduce the cost and complexity of user management operations by providing a common user schema and extension model, as well as binding documents to provide patterns for exchanging this schema using standard protocols.

Getting Started with the SAP Sales and Service Cloud V2 SCIM API

The SAP Sales Cloud and SAP Service Cloud Version 2 SCIM API is an API based on the SCIM (System for Cross-domain Identity Management) specification. It's used as an interface between SAP Cloud Identity

Services (IAS) and SAP Sales and Service Cloud V2 solutions to transfer user information as part of user management. In SAP Sales and Service Cloud V2, the SCIM API is used to provision Users and User Groups.

- [User Service](#) 
- [User Group Service](#) 

SAP Sales and Service Cloud V2 have a standalone user and roles management. To efficiently manage the users and user groups in V2 in an integrated scenario, SCIM API is implemented.

→ Remember

- The incoming data from the Source (SCIM API) is considered as master data and overwrites any User attributes already maintained in the SAP Sales and Service Cloud Version 2 system.
- Each group has 1:1 mapping with the business role in SAP Sales and Service Cloud Version 2.
- SCIM API is applicable for management of business users but not to integration users (the users that are created using communication systems).
- Username is the unique attribute and is used to resolve any conflicts.
- SAP Sales and Service Cloud Version 2 don't support hard-deletion or soft-deletion of a user. The user status is set to Obsolete when a Delete request is received.
- When a create user request is received with an existing Username, a 409 – CONFLICT response is raised, and the user can be updated with a PUT request.
- When a Create request is received with an existing deleted user (Status: Obsolete), the same user is reactivated with the same ID.
- When a create Group request is received with an existing Display Name (Role Name), the request isn't rejected but is linked to the existing role. If the role isn't existing, a new Role is created with Default Services assigned.
- When you use a PUT request to update the Members list, the system removes the role from any existing users in SAP Sales and Service Cloud Version 2 who aren't included in the updated list.
- If it's required to add or remove roles for specific users, PATCH operation on the Group Resource can be used to modify the members.
- Whenever a role is created in SAP Sales and Service Cloud Version 2, it's automatically available as a User Group when the groups' endpoint is accessed.
- The default security policy assigned: **S_BUSINESS_USER_WITHOUT_PASSWORD**.

Set Up Integration with Cloud Identity Services (IAS) and SAP Identity Provisioning Services (IPS)

1. Create a Technical API User.

You can consume the SCIM APIs using the Technical User (API User). To create a technical user:

1. On the SAP Sales Cloud and SAP Service Cloud Version 2, navigate to the [User Menu](#) on the top-right corner, and access the [System Settings](#) page.
2. Go to [All Settings](#)  [Users and Control](#)  [Roles](#) .
3. Select [Technical Roles](#) from the dropdown, and create a new technical role as an admin role. Assign the [User](#), [Role](#), [Employee](#), and [Admin Service](#) business services.
4. Save and activate the role.

5. Navigate to ► [All Settings](#) ► [Users and Control](#) ► [Users](#) ►.
6. Select [Technical Users](#) from the dropdown, create a new technical user, and assign the previously created technical role to this user.

Note

- Ensure that your [User Name](#) doesn't contain a space and a period and also ensure you choose a strong password that meets the security policy guidelines.
- Choose [Refresh](#) once you assign a technical user to the role.

2. Create an Application in IAS. Create an SAML 2.0 application following the procedure at [Create SAML 2.0 Application](#).

1. Sign in to the administration console for SAP Cloud Identity Services. Navigate to ► [Applications and Resources](#) ► [Applications](#) ► tile and choose [Create](#).
2. Enter or choose the following information while you create an application, and choose [+ Create](#).

Field	Information
Display Name	Required field. The display name of the application appears on the logon and registration pages. Ensure you enter a display name that helps you identify your system.
Home URL	Enter the SAP Sales Cloud V2 or SAP Service Cloud V2 tenant URL.
Organization ID	Global. Auto-filled.
Protocol Type	Select the SAML 2.0 radio-button.

3. Set Up Single Sign-On on the SAP Sales Cloud and SAP Service Cloud Version 2 system.
 1. On the SAP Sales Cloud and SAP Service Cloud Version 2 system, navigate to the ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Users and Control](#) ► [Identity Provider Configuration](#) ► page.
 2. Choose [Download Metadata](#) to download the service provider metadata from the SAP Sales Cloud and SAP Service Cloud Version 2 system.
 3. Upload the metadata to SAP IAS.
Follow the steps listed at: [Make a New Trust Configuration](#).
 1. On the SAP Cloud Identity Services, navigate to ► [Applications and Resources](#) ► [Applications](#) ► tile. Select your application on the left.
 2. Choose the [Trust](#) tab.
 3. Under [Single Sign-On](#), choose [SAML 2.0 Configuration](#).
 4. Upload the service provider metadata XML file that you downloaded from the SAP Sales Cloud and SAP Service Cloud Version 2 system. Choose [Load from File](#), and upload the metadata file.
 4. Download the configured metadata from SAP IAS.
Follow the steps listed at: [Tenant SAML 2.0 Configurations](#).
 1. On the SAP Cloud Identity Services, navigate to ► [Applications and Resources](#) ► [Tenant Settings](#) ► [Single Sign-On](#) ►.
 2. Choose the [Download Metadata File](#) to download the metadata file.
 5. To re-upload the downloaded metadata from SAP IAS to SAP Sales Cloud and SAP Service Cloud Version 2.

Follow the steps listed at: [Configure Identity Provider \[page 62\]](#).

1. On the SAP Sales Cloud and SAP Service Cloud Version 2 system, navigate to the [Identity Provider Configuration](#) page, choose **+** ([Add](#)), enter an alias, and choose [Upload Metadata](#).
2. In the [Upload IdP Metadata](#) quick view, enter an [Alias](#), enable the [Active](#) toggle, and upload the configured metadata XML file (downloaded from your IdP).

Note

The alias name can be the URL of your IdP service.

3. Leave the [Client ID](#) and [Client Secret](#) blank, and choose [Save](#).
4. Set Up SAP IAS as Source.

Follow the steps listed in [Identity Authentication](#).

1. On [SAP Cloud Identity Services](#), navigate to **Identity Provisioning** > [Source Systems](#) , and choose [Add](#).
2. Enter or choose the following information on the [Details](#) tab.
 1. For [Type](#), enter **HTTP** input.
 2. Choose **Identity Authentication** from the [Type](#) dropdown.
 3. For [System Name](#), enter a system name with which you can identify your source system; here it's, the IAS service. For example, **IAS for SAP Sales and Service Cloud V2**.
 4. For [Description](#), enter a description for system identification. You can enter the same description as the system name with additional information that helps you identify the system.
3. Navigate to the [Transformations](#) tab, and complete the following steps:
 1. Copy the code syntax under [Default transformation for SCIM API version 2](#) section from: [Identity Authentication](#).
 2. On the [Transformations](#) tab, go to **Switch to JSON Editor** > [Edit](#) > [Paste](#) .
 3. Choose [Save](#), and enter a description to confirm your transformation changes.
4. Navigate to the [Properties](#) tab, enter the properties as listed in the [Mandatory Properties](#) section in [SAP Sales Cloud and SAP Service Cloud](#), and choose [Save](#).
Enter the IAS tenant URL under the [URL](#).
5. Navigate to the [Outbound Certificates](#) tab, generate, and download the certificate.

5. Set Up Sales Cloud and Service Cloud V2 as Target.

Follow the steps listed at: [SAP Sales Cloud and SAP Service Cloud](#).

1. On [SAP Cloud Identity Services](#), navigate to **Identity Provisioning** > [Target Systems](#) , and choose [Add](#).
2. Enter or choose the following information on the [Details](#) tab.
 1. Choose **SAP Sales and Service Cloud** from the [Type](#) dropdown.
 2. In the [System Name](#) field, enter a system name with which you can identify your target system. For example, **Service3 VLAB**.
 3. In the [Description](#) field, enter a description for system identification. You can enter the same description as the system name with additional information that helps you identify the system.
 4. Select the source system that was created previously from the [Source Systems](#) dropdown.
3. Navigate to the [Transformations](#) tab, and complete the following steps:
 1. Copy the code syntax under [Default transformation for connector version 4](#) section from: [SAP Sales Cloud and SAP Service Cloud](#).
 2. On the [Transformations](#) tab, go to **Switch to JSON Editor** > [Paste](#) .
 3. Choose [Save](#), and enter a description to confirm your transformation changes.

4. Navigate to the [Properties](#) tab, enter the properties as listed in the [Mandatory Properties](#) table, and choose [Save](#).

Mandatory Properties

Property Name	Description & Value
Type	Enter: HTTP
URL	SAP Sales Cloud and SAP Service Cloud Version 2 system URL.
ProxyType	Enter: Internet
Authentication	Enter your authentication method: ClientCertificateAuthentication
c4c.support.patch.operation	By default, it's PUT. Modify if you want PATCH operation. This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to true, PATCH operations are used to update users and groups in the target system. • If set to false, PUT operations are used to update users and groups in the target system. Default value: false
ips.delete.threshold.groups	Enter a numeric value. Use this property to control the number of groups to be deleted in a target system by defining a threshold. This property prevents you from accidentally deleting a huge number of groups, for example by adding a filter or condition.
ips.delete.threshold.users	Enter a numeric value. Use this property to control the number of users to be deleted in a target system by defining a threshold. This property prevents you from accidentally deleting a huge number of users, for example by adding a filter or condition.

Note

An exhaustive list of properties and their descriptions is available at: [SAP Sales Cloud and SAP Service Cloud](#).

5. Navigate to the [Outbound Certificates](#) tab, generate, and download the certificate.
6. Authentication Set Up in SAP IAS.
 1. On [SAP Cloud Identity Services](#), go to ► [Users and Authorization](#) ► [Administrators](#) ► [+Add](#) ► [System](#) ►.

1. Enter a display name for *Name*, enable all configurations for the user, and choose *Save*.
2. Select the system user from the left-hand pane, and go to ► *Configure System Authentication* ► *Certificate* ► *Configure Certificate* ►, and choose *Upload Certificate* to upload the source system certificate downloaded during source setup, and choose *Save*.
2. Go to SAP Sales Cloud and SAP Service Cloud Version 2 system, navigate to ► *User Menu* ► *System Settings* ► *All Settings* ► *Users and Control* ► *Users* ►, and choose the API user that you created previously and upload the certificate downloaded from the target system.

Note

Ensure that the certificate that you upload is of **.cer** extension type only.

You've now successfully configured the SCIM API on your system.

7. Verify the Set Up.
 1. To verify that your SCIM configuration is successful, create or edit a user on SAP Cloud Identity Services. For a selected user, ensure that the following fields are updated.
 - User Type: Public
 - Valid From
 - Valid To
 - Login Name
 - Employee Number
 2. On the SAP Cloud Identity Services, navigate to ► *Users and Authorization* ► *User Management* ►, and edit the fields for a user. Open the *Groups* tab and assign a group to the user.
 3. On the SAP Cloud Identity Services, navigate to ► *Identity Provisioning* ► *Source Systems* ►, select the system on the left pane, and choose *Jobs*. In the *Job Type* list, choose *Run Now*.
Once the job is successfully run, the changes made on the SAP Cloud Identity Services reflect on the SAP Sales Cloud and SAP Service Cloud Version 2 system.

Authentication Mechanism

SCIM API supports the following authentication mechanisms:

- Basic Authentication
- Certificate Authentication
- Bearer Token Authentication

Monitoring Tools

SAP Sales Cloud and SAP Service Cloud Version 2 doesn't provide any monitoring tools for the direct usage of SCIM APIs. As a user, you must monitor the replications using SCIM APIs.

4.6 Settings

This page lists all the identity and access management settings.

Cross-Site Request Forgery Settings

Cross-Site Request Forgery (CSRF) is a web-security vulnerability that allows an attacker to induce users to perform actions that they don't intend to perform. CSRF attacks exploit the trust a Web application has in an authenticated user. To prevent such attacks, CSRF settings are provided.

Configure Cross-Site Request Forgery Settings

1. Log in to the system as an administrator.
2. Navigate to the [User Menu](#) on the right-top corner, and access the [System Settings](#) page.
3. Go to ► [All Settings](#) ► [Users and Control](#) ► [Settings](#) ► [Cross-Site Request Forgery Settings](#) ►.
4. Select one of the modes: [Strict](#), [None](#), or [Lax](#) from the dropdown for third-party integration.

ⓘ Note

- Lax: This mode allows cookies to be sent for top-level navigation only.
- None: This mode disables SameSite-based protection. The website can use its own CSRF protection mechanisms.
- Strict: This mode disables cookies being sent to all third-party websites. Cookies are sent only if the domain is the same as the path for which the cookie has been set.

ⓘ Note

By default, the [Strict](#) mode is enabled for third-party integrations. However, for existing customers, this change doesn't affect the existing integrations in the current release. In a future release, the existing integrations can be affected.

5 Data Import and Export

This document describes how you can use Data Import and Export to import data into your solution and export data from your solution.

You can use Data Import and Export to import the following:

- Data from a legacy system into SAP Service Cloud Version 2. For example, you can transfer all the existing data from any Cloud CRM system to SAP Service Cloud Version 2.
- Operational data from third-party systems into SAP Service Cloud Version 2.
- Data that is manually maintained in a CSV file (data file) into SAP Service Cloud Version 2. For example, customer and product information collected from a campaign and maintained in a data file.

As an administrator, you are expected to understand the following:

- Your business requirement and knowledge of data imported to SAP Service Cloud Version 2.
- Sequence in which the entities must be imported. For information on each of the entities, see entity documentation.

Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Data Import and Export](#) ► [Import and Export Data](#) ►.

You can use the filters to view all tasks, your tasks, and application tasks. Application tasks are import and export tasks triggered from the respective entity workspaces.

Once a task is triggered, it goes through the following stages:

- **New** indicates that a new task is triggered.
- **In Process** indicates that the task is being processed.
- **Completed** indicates that the task is processed without any errors.
- **Completed With Errors** indicates that the task is completed but with errors. Error files are generated and the task moves to Completed With Errors. Errors generated, if any, are available for download. The error file specifies the error message along with the specific root causes for the error. For example, if data validation failed due to an invalid character and incorrect ID, an error message Data Validation Failed: Account name contains an invalid character, Account ID is incorrect is displayed. If there are multiple errors the errors are displayed as comma separated values. Verify the file and import it again.
- **Failed** indicates that the task failed because of technical errors. Technical errors could be due to incorrect file format or invalid data type for any of the fields in the import file. The error count for tasks in failed status will be zero.

ⓘ Note

Tasks older than 30 days are automatically removed from Monitor, irrespective of task status.

In cases where you want to retain information beyond one month (such as external keys used for an entity during import), we recommend you to maintain local copies of such information.

You can view all tasks triggered in your tenant.

Prerequisites

Type	Task
Technical	Template files can be downloaded from the system. To check if the file is in an acceptable format, do the following: 1. Open the CSV file in Notepad. 2. Click  File > Save As >  3. In the Save As dialog, choose Encoding as UTF-8. Keep the file extension as .CSV. 4. Once the file is saved, check if the content is displayed correctly.
Functional	You understand the business requirement of the data to be imported. For each entity you want to import data into, you understand the dependency within the entity section of the respective entity document.

5.1 Create Data Using Import

You can create new data and import it to the system.

Templates are available by default for each entity type for administrators who have a business understanding of the fields that need to be mapped between systems. This ensures consistency in mapping across all users.

Typical tasks of an administrator include:

- Downloading templates for various entities and nodes
- Preparing data to an accepted format
- Importing data

→ Tip

It is recommended to download the template from the system for every import create instead of using a locally saved template version.

Administrator data, such as created by, created on, updated by, and updated on, for the data being imported will be based on the user performing the task and the time of import. Existing administrator data cannot be retained for historical data imports or migrated data imported through the [Data Import and Export](#) functionality.

5.1.1 Download Template

You can download templates to prepare data for importing.

Procedure

1. Navigate to **User Menu** > **System Settings** > **All Settings** > **Data Import and Export** > **Import and Export Data**.
2. Select **Download Template**.
3. Select the entity for which you want to import the data.
4. Select the **File Type** between **Comma Separated Values (.csv)** and **Excel Workbook (.xlsx)**.
5. Turn on the **Complete Entity** switch to import the root entity and all its subentities. Disable it if only the root entity or root entity with selected subentities is to be imported.

When **Complete Entity** is selected, the system downloads a zip file with multiple csv and xlsx template files (one file each for the root entity and individual subentities).

When **Complete Entity** is deselected, you can select subentities based on your import requirement along with root entity template, considering that only root entity is to be imported or a root entity with selected child entities is to be imported.

- If no subentity is selected, a single csv/excel template file of the root entity is made available for download.
- If certain subentities are selected, a zip file with multiple csv/xlsx template files (one file each for root entity and every selected subentity) is available for download.

6. Select **Download Template**.

Note

The template is downloaded in the same language as of the user's login language.

7. Update the downloaded template with data to be imported.

Note

Only comma and semicolon are supported as value separators.

5.1.2 Prepare Data

Data that is either downloaded from a legacy system, a third-party system, or manually maintained in a data file, it must be cleansed to ensure that it's in as per the Data Import and Export template.

For example, you must ensure that all the technical IDs are maintained based on the dependency, and also check if the data is as per the Data Import and Export template.

Instructions

- The login language while importing data and the template language used while creating data using import must be the same.
- For date fields, follow these instructions:
 - Use format yyyy-mm-dd. For example, 2015-11-25.
 - To import a timestamp field, use format yyyy-mm-ddThh:mm:ssZ. For example, 2021-12-31T23:59:59Z. Enter the UTC time.

Note

Date and currency fields get auto converted in excel when entering the data, ensure that the format is in the expected format after data entry by changing the data type of the cell to Text/General.

- If you have a Boolean entry, ensure that the value is entered either as true or false in small case.

Accepted Values

- true
- True
- TRUE

- false
- False
- FALSE

Note

Any blank boolean field is treated as undefined.

- Columns with numbers that are treated as text.
Data of a few columns that display numbers is treated as text. Refer to the code list in the templates for fields that have a specific list of values.

→ Remember

Use a notepad application to modify the CSV data file before importing. This action avoids any format changes to the CSV data file.

When using any excel application to modify the CSV data file, make sure to import the **.csv** file using **UTF-8** character encoding.

Only comma and semicolon are supported as value separators.

- In addition to the import data fields, you must also provide the following:
 - An external_key in the root entity file
 - A parent_external_key in the root entity file for hierarchical entity structure
 - A parent_external_key in the subentity files

Add the parent record's `external_key` as `parent_external_key` for child records to define the parent-child relationship between records.

- For hierarchy entities, ensure that a value is maintained in the `external_key` column for all records, and the `external_key` value must be unique for each record in the import file.
- The `external_key` and `parent_external_key` fields are string type fields used by data impex for processing the respective import tasks. It doesn't reflect in the entity records created in the system via import. Data impex doesn't store these field values as they're used only for a particular import task.

catalogId	parentId	displayId	imageUrl	name	description	language-Code	external_key	parent_external_key
-----------	----------	-----------	----------	------	-------------	---------------	--------------	---------------------

In this example, `external_key` is a unique non-null value.

- Even if subentities aren't being used, you must include the template files for those subentities during import.
- Reordering of the column names is supported for templates and export files, however, with the following restrictions:
 - Reordering is supported only for imports triggered from the [Data Import and Export](#) user interface and not by other UIs.
 - Reordering isn't supported by decision tables and ID-mapping functionalities.
 - The column position for Data Import and Export custom attributes such as `external_key`, `parent_external_key`, `to_be_deleted`) must not be modified. Their position must remain the same as in the downloaded template/export file.
 - If a file contains `parent_external_key` column, ensure that the position of the column remains as the last column then even after re-ordering.

The primary key (ID/Display ID/Technical ID) of the respective entity is different from the `external_key` and `parent_external_key` fields. The primary key (ID/Display ID/Technical ID) of the entity is generated when the entity records imported are created in the tenant. Hence, when new records are created during import, this field has to be left blank in the template.

Administrator data like created by, created on, updated by, and updated on for the data being imported is based on the user performing the task and the time of import. Existing administrator data can't be retained for historical data imports or migrated data imported through Data Import and Export.

Note

To import only the root entity, use a single csv/excel file.

To import a complete entity or a root entity along with select subentities, use a .zip file that includes multiple csv/excel files

Don't create a zip file by zipping a folder that contains the entity files. To prepare a zip file, select all csv/excel files (except the codelist files) to include in the .zip file, right-click, and create a zipped folder. Don't create a zipped folder directly from the parent folder that was downloaded.

Extension field label names must not contain either a "," (comma) or ";" (semicolon).

Attachments present in the template can't be imported via Data Import and Export. However, include the csv/excel blank template in the zip file to import the complete entity.

→ Tip

It's recommended to import only a maximum of 25,000 records in an Excel file or 50,000 records in a csv file.

5.1.3 Create Data

You can create new data and import it to the system.

Procedure

1. Navigate to ► *User Menu* ► *System Settings* ► *All Settings* ► *Data Import and Export* ► *Import and Export Data* .
2. Select ► *Import* ► *Create* .
3. Select *Complete Entity* if you want to create the root entity and all its subentities or select *Individual Entities* if you want to import only the root entity or root entity with selected subentities.
4. Turn on the *Initial Load* switch to relax checks and reduce errors during initial migration.
5. Turn on the *Simulation Mode* switch to simulate and view errors.

ⓘ Note

Simulation mode performs a basic validation to identify file upload errors. Errors related to hierarchical relationships aren't validated in simulation mode.

6. Select the *File Type* between *Comma Separated Values (.csv)* and *Excel Workbook (.xlsx)*.

→ Remember

Use that latest template file for create.

7. Select the *File Type* between *Comma Separated Values (.csv)* and *Excel Workbook (.xlsx)*.
8. Browse for the *.zip prepared data file you want to upload.

ⓘ Note

Once you upload the file, an option to *Preview Sample Data* appears, using which you can view the preview of the first 10 entries in your uploaded file.

ⓘ Note

Maintaining an external key is mandatory for import of the complete entity.

9. Click *Import Data*.

Results

The system validates the first 10 entries in the import file for data type and file type errors. Errors, if any, are displayed in the [Import and Export Data](#) screen. You can't import the file without resolving the errors. The system displays a message on the status of the import.

Note

If the error message reads "Operation failed. Please contact the support team", try importing the file again. If the issue persists, raise a support ticket.

5.2 Prepare Data for Import Update

You can update existing data records and import them to the system.

To update existing data records, you must first export the data from the system.

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Data Import and Export](#) ► [Import and Export Data](#) .
2. Select [Export](#).
3. Select the entity for which you want to update the data.
4. Select the subentities and fields from [Entity and field selection](#).
5. Select the [File Type](#) between [Comma Separated Values \(.csv\)](#) and [Excel Workbook \(.xlsx\)](#).
6. Select [Apply Filter](#) to filter based on the values available for the respective entities. Select [Define Conditions](#) to add specific filter conditions and [Add Condition Group](#) to add more conditions.
7. Select entities that need to be downloaded as part of the export file.
8. Select [Export Data](#).

The system displays the status of the export in the [Import and Export Data](#) screen.

→ Remember

Use that latest export file for update. If an old export file is used, existing data will be overwritten considering update file data as the latest data. Data which is not to be updated can be left unchanged in the complete entity file. Update is only supported for the complete entity or root entity. If only certain subentities need an update, modify only those entities and upload the complete entity file without altering the other data. It is recommended to export only a maximum of 50,000 records in a single file.

Add a column with header **to_be_deleted** column in the export file and set the value as true to delete subnode entities. If the column value is blank or false, updated data, if any, would be considered. Administrator data like created by, created on, updated by, and updated on for the data being imported will be based on the user performing the task and the time of import. Existing administrator data cannot be retained for historical data imports or migrated data imported through Data Import and Export.

5.2.1 Update Data

Once the exported file is updated with the necessary information, you can import it back into the system.

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Data Import and Export](#) ► [Import and Export Data](#) .
2. Select ► [Import](#) ► [Update](#) .

Note

The [Editable](#) column in the [Field_Definitions](#) tab of the downloaded excel file lists whether a particular field is updatable or not. Update only those fields which have value [True](#).

Note

For hierarchy entities, ensure that the `external_key` value is maintained for all records, and the `external_key` value must be unique for each record.

3. Select the entity for which you want to update the data.
4. Select [Complete Entity](#) if you want to import the root entity and all its subentities or select [Root Entity](#) if you want to import only the root entity.

Note

Updating the root entity is currently available only for Opportunities.

Note

If you want to update the `noted_HTML_Content` before importing the file, then you must add a column called `noted_HTML_Content` and then populate the content.

5. Select [Ignore](#) or [Update](#) to ignore or overwrite blank values, respectively.

Note

Reordering of the column names is supported for templates and export files, however, with the following restrictions:

- Reordering is supported only for imports triggered from the [Data Import and Export](#) UI and not from other UIs.
- Reordering isn't supported by decision tables and ID-mapping functionalities.
- The column position for Data Import and Export custom attributes such as `external_key`, `parent_external_key`, `to_be_deleted` must not be modified. Their position must remain the same as in the downloaded template or the export file.
- If a file contains `parent_external_key` or `to_be_deleted` columns, ensure that the positions of these columns remain as the last column then even after re-ordering.

Extension field label names must not contain either a "," (comma) or ";" (semicolon).

6. Turn on the [Initial Load](#) switch to relax checks and reduce errors during initial migration.
7. Turn on the [Simulation Mode](#) switch to simulate and view errors.
8. Select the [File Type](#) between [Comma-Separated Values \(.csv\)](#) and [Excel Workbook \(.xlsx\)](#).

9. Browse for the *.zip prepared data file you want to upload.

📘 Note

Once you upload the file, an option to [Preview Sample Data](#) appears, using which you can view the preview of the first 10 entries in your uploaded file.

10. Click [Import Data](#).

The system validates the first 10 entries in the import file for data type and file type errors. Errors, if any, are displayed in the [Import and Export Data](#) screen. You can't import the file without resolving the errors. The system displays a message on the status of the import.

📘 Note

If the error message reads "Operation failed. Please contact the support team," try importing the data again. If the issue persists, raise a support ticket.

5.3 Prepare Data for Import Delete

You can delete existing data records from the system.

To delete existing data records, you must first export the data from the system.

1. Navigate to your [User Menu](#) and select ► [System Settings](#) ► [All Settings](#) ► [Data Import and Export](#) ► [Import and Export Data](#) ►.
2. Select [Export](#).
3. Select the entity for which you want to delete the data.
4. Select the subentities and fields from [Entity and field selection](#) and the root node.
5. Select the [File Type](#) between [Comma Separated Values \(.csv\)](#) and [Excel Workbook \(.xlsx\)](#).
6. Select [Apply Filter](#) to filter records that are to be deleted based on the values available for the respective entities.
Select [Define Conditions](#) to add specific filter conditions and [Add Condition Group](#) to add more conditions.
7. Select [Export Data](#).
The system displays the status of the export in the [Import and Export Data](#) screen.

→ Remember

Use that latest export file for delete. Ensure that the file contains records that need to be deleted from the system. Other records, if any, can be removed from the file.

5.3.1 Delete Data

You can delete existing data records using Data Import and Export.

1. Navigate to [User Menu](#) and select ► [System Settings](#) ► [All Settings](#) ► [Data Import and Export](#) ► [Import](#) ► [Delete](#) ►.

2. Select the entity.
3. Select the file type.

Note

Reordering of the column names is supported for templates and export files, however, with the following restrictions:

- Reordering is supported only for imports triggered from the [Data Import and Export](#) user interface and not by other UIs.
- Reordering isn't supported by decision tables and ID-mapping functionalities.
- The column position for Data Import and Export custom attributes such as `external_key`, `parent_external_key`, `to_be_deleted` must not be modified. Their position must remain the same as in the downloaded template/export file.
- If a file contains `parent_external_key` or `to_be_deleted` columns, ensure that the positions of these columns remain as the last column then even after re-ordering.

Note

For hierarchy entities, ensure that the `external_key` value is maintained for all records, and the `external_key` value must be unique for each record.

4. Upload the export file that contains records that you want to delete.

Note

Once you upload the file, an option to [Preview Sample Data](#) appears, using which you can view the preview of the first 10 entries in your uploaded file.

Note

You can only delete the complete entity. Upload the root entity file.

5. Choose [Import Data](#).

The system validates the first 10 entries in the import file for data type and file type errors. Errors, if any, are displayed in the Import and Export Data screen. You can't import the file without resolving the errors. The system displays a message on the status of the import.

Note

If the error message reads "Operation failed. Please contact the support team", try again. If the issue persists, raise a support ticket.

5.4 Export Data

You can export data from your solution.

Procedure

1. Navigate to your *User Menu* and select ► *System Settings* ► *All Settings* ► *Data Import and Export* ► *Import and Export Data* ►.
2. Select *Export*.
3. Select the entity for which you want to export the data.
4. Select the subentities and fields from *Entity and field selection*.

Note

The attribute notelid_HTML_content is removed from the export file for different entities that contained it.

5. Select the *File Type* between *Comma Separated Values (.csv)* and *Excel Workbook (.xlsx)*.
6. Select *Apply Filter* to filter based on the values available for the respective entities. For administrative data, you can further narrow down your results using the date filter.
 - a. Select *Define Conditions* to add specific filter conditions.
 - b. Select *Add Condition Group* to add more conditions.
7. Select entities that need to be downloaded as part of the export file.

Note

The *Editable* column in the *Field_Definitions* tab of the downloaded excel file lists whether a particular field is updatable or not.

8. Select *Export Data*.

The system displays the status of the export in the *Import and Export Data* screen.

Note

You can export only a maximum of 50,000 records per export task in a single file.

5.5 ID Mapping

If an external key is not maintained or retained previously, you can maintain external keys against the Technical IDs or UUIDs of existing records using the template file that contains a common Technical ID and external key columns for all entities. For example, records imported via Data Import and Export before ID mapping activation or via other sources.

ID Mapping is currently available for the following entities:

- Accounts
- Competitors
- Competitor Products
- Contacts
- Employees
- Individual Customers
- Organizational Unit
- Products
- Product Groups

1. Navigate to ► [User Menu](#) ► [Settings](#) ► [All Settings](#) ► [Data Import and Export](#) ► [ID Mapping](#) ►.
2. Select [Download Template](#) to download the ID mapping template.

5.5.1 Configure ID Mapping

As an administrator, you can activate ID mapping for an entity. Once you activate ID mapping for an entity, the external keys used in the Import Create mode file for the entity are retained and can be reused instead of Technical IDs to make subsequent updates and to cross reference in other entities via Import Update mode.

Procedure

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Data Import and Export](#) ► [ID Mapping](#) ► [Configuration](#) ►.
2. Turn on the [Enable ID Mapping](#) switch to enable ID mapping for the respective entities.

After activating ID mapping for an entity, the external keys used in the Import Create mode file for that entity will be retained and can be reused instead of Technical IDs and for cross references in other entities.

For example, if ID mapping is activated for the Accounts entity, external key of Accounts can be used while importing opportunities instead of Account Technical UUIDs.

5.5.2 Create ID Mapping Using Import

You can create a new ID Mapping for existing entity records and use these IDs for subsequent updates or as cross references in related entities.

Procedure

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Data Import and Export](#) ► [ID Mapping](#) ► [ID Mapping](#) .
2. Select [Download Template](#) and choose the file type.
Update the template file with the external key and UUID of the entities.
3. Select ► [Import](#) ► [Create Records](#) .
4. Select the entity and the file type.
5. Upload the completed ID-Mapping template.
6. Choose [Import ID Mapping](#).

5.5.3 Update ID Mapping Using Import

You can update existing ID Mapping information using import.

Procedure

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Data Import and Export](#) ► [ID Mapping](#) ► [ID Mapping](#) .
2. Choose [Export](#).
3. Select the entity and the file type.
4. Choose [Export ID Mapping](#)

Update the external keys as required. Add a column with the header to_be_updated in the export file and set the value as true to update ID mappings.

To remove an ID mapping, add a column with the header to_be_updated in the export file, set the value as true, and clear the external key.

5. Choose ► [Import](#) ► [Update Records](#) .
6. Select the entity and the file type.
7. Upload the updated ID Mapping file.
8. Choose [Import ID Mapping](#).

5.5.4 Export ID Mapping

You can export existing ID mapping information.

Procedure

1. Navigate to **User Menu** > **System Settings** > **All Settings** > **Data Import and Export** > **ID Mapping** > **ID Mapping**.
2. Choose **Export**.
3. Select the entity and the file type.
4. Choose **Export ID Mapping**.

5.6 Entity-Specific Information

This section describes the entity-specific information available in Data Import and Export.

Data Import and Export supports the following entities:

Supported Entities

Entity	Import Create	Import Update - Complete Entity	Import Update - Root Entity	Initial Load	Simulation Mode	Export
Accounts	Yes	Yes	Yes	No	No	Yes
Cases	Yes	Yes	No	No	Yes	Yes
Chat	Yes	Yes	No	Yes	Yes	Yes
Competitors	Yes	Yes	No	No	Yes	Yes
Competitor Products	Yes	Yes	No	No	Yes	Yes
Contacts	Yes	Yes	Yes	No	No	Yes
Email	Yes	No	No	No	No	Yes
Employees	Yes	Yes	Yes	No	No	Yes
Exchange Rate	Yes	No	No	No	No	Yes
Functional Location	Yes	Yes	No	No	Yes	Yes
Individual Customers	Yes	Yes	Yes	No	No	Yes
Installed Base	Yes	Yes	No	No	Yes	Yes

Entity	Import Create	Import Update - Complete Entity	Import Update - Root Entity	Initial Load	Simulation Mode	Export
Organizational Unit	Yes	Yes	No	No	Yes*	Yes
Phone Call	Yes	Yes	No	Yes	Yes	Yes
Products	Yes	Yes	No	No	No	Yes
Product Groups	Yes	No	No	No	Yes*	Yes
Product Installation Point	Yes	Yes	No	No	Yes	Yes
Registered Products	Yes	Yes	No	No	Yes	Yes
Service Catalog	Yes	Yes	No	No	Yes	Yes
Service Category	Yes	Yes	No	No	Yes*	Yes
Task	Yes	Yes	No	No	No	Yes
User	Yes	Yes	No	No	No	Yes
Warranty	Yes	Yes	No	No	Yes	Yes

* indicates that the entity is a hierarchical entity.

[Account \[page 85\]](#)

Account management capabilities offer a holistic view of the customer, and allow you to capture, monitor, store and track all critical business information about customers, prospects and partners.

[Account Hierarchy \[page 94\]](#)

Account hierarchy is a collection of accounts linked to each other through a parent-child relationship.

[Case Overview and Structure \[page 96\]](#)

Overview of the Case entity and the structure in the solution.

[Competitors \[page 101\]](#)

Use Competitors to create new competitors and maintain competitor information.

[Competitor Products \[page 103\]](#)

Competitor Products enables you to create and maintain comprehensive records of competitor offerings. Leverage this data to effectively compare and strategically position your own products.

[Contact Person \[page 104\]](#)

Use contacts to represent the relationship between contacts and accounts. Contacts can be maintained only for corporate accounts.

[Emails \[page 108\]](#)

Learn about how you can leverage the email functionality of your system to manage email channels, maintain templates as per different scenarios and automatically generate cases that are sent to service agents for processing.

[Employee \[page 108\]](#)

Employee Master Data capabilities offer a holistic view of internal employees (a business partner person with valid employee type "INTERNAL").

[Functional Location \[page 113\]](#)

[Individual Customer \[page 118\]](#)

Individual Customers represent persons with whom business is done.

[Installed Base \[page 125\]](#)

Installed base is a hierarchical arrangement of items installed at your customer's location.

[Organizational Unit \[page 128\]](#)

Organizational unit containing a hierarchy that represents the hierarchical structure of the organization.

[Product \[page 135\]](#)

A product is the item offered for sale and it can be a service or a physical item. A product can be of real or virtual form. Each product is made at a cost and sold at a price and the price depends on the quality, market, the marketing, and the targeted segment.

[Product Group \[page 139\]](#)

A product group is a group of products categorized based on the shared attributes like usage, features, production processes and so on. A product group can also be the market or customer segment in which the products are sold or the prices at which they're offered.

[Product Installation Point \[page 140\]](#)

An installation point is the physical location of the object that requires servicing or maintenance based on the service or maintenance plan for that object.

[Service Categories Overview and Structure \[page 144\]](#)

Overview of the Service Category entity and the structure in the solution.

[Service Catalogs \[page 146\]](#)

Overview of the Service Catalog entity and the structure in the solution.

[Registered Products \[page 149\]](#)

A registered product is an instance of a product that is associated with a specific customer and generally has a unique serial number. The registered product information allows the service agent to identify the customer product and any associated warranties or service entitlements

[Warranty \[page 153\]](#)

Warranty is assigned to a registered product or installation point, and you can determine its coverage in a ticket.

[Users \[page 154\]](#)

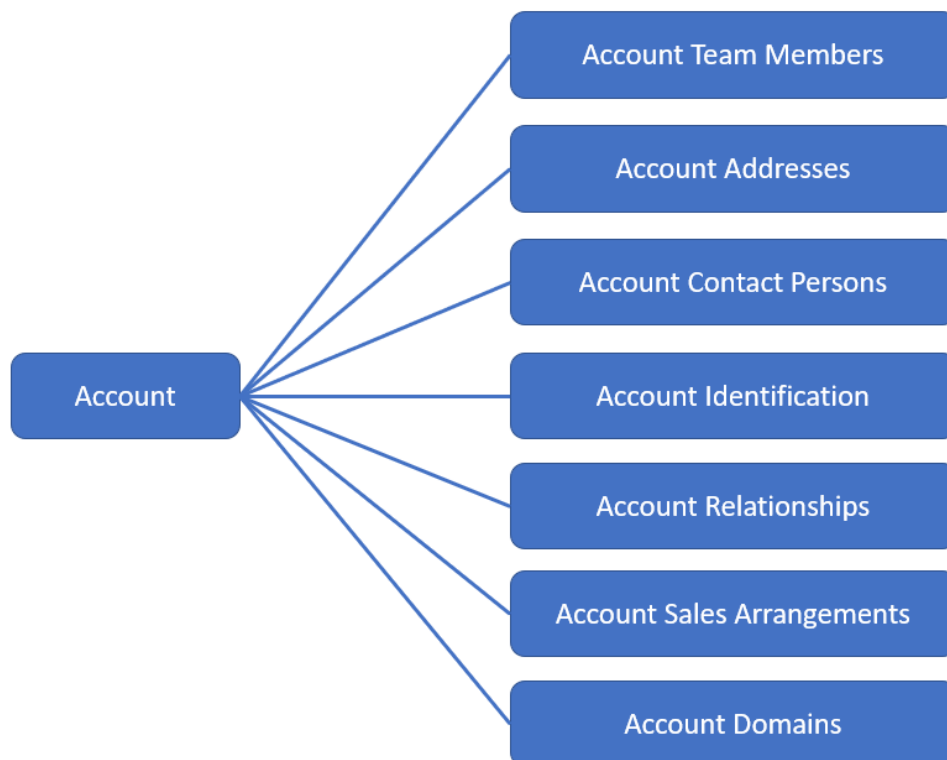
Users in the system refers to employees, business users and other users such as partner contacts.

Users are assigned to business roles and access controls in the system are implemented on business roles.

5.6.1 Account

Account management capabilities offer a holistic view of the customer, and allow you to capture, monitor, store and track all critical business information about customers, prospects and partners.

Use this information to focus on your most profitable customers, maintain satisfaction and loyalty and consistent interaction across all channels.



5.6.1.1 Account Dependencies

Dependencies on Other Entities

Entity	Description
Contact Person	Used in Account Contact Persons to assign a Contact to Corporate account(s)
Employee	Used in Account Team Members to assign internal and external employees to the account
Organizational Unit	Used in Account Team to assign the employees fo a specific Sales Organization to Corporate account(s) Used in Account Sales Data to assign a Sales Organization to the account

Entity	Description
Account Relationships	The related Business Partner, and in case it is provided, the related Sales Organization must exist before creating a relationship to the account

Dependencies within the Entity

Entity	Description
Account Addresses	The default address can either be maintained in Account or Account Addresses but not in both places simultaneously
Account Contact Persons	Default address maintained at Account or Account Addresses must exist to fill workplace address fields

5.6.1.2 Account Fields

Download the template from [Data Import and Export](#) for the field level information.

5.6.1.2.1 Account

The Account header entity contains all central information of a corporate account.

It contains information about the account name, main address with communication data like email, ABC classification and blocking information. Use the address fields in the Account header entity in case only the default account address is to be maintained. It is highly recommended to fill the Country field even if you do not provide full address details.

Template Header	Data Type	Notes
external_key	STRING	
Technical ID	STRING	
Account ID	STRING	If this is not provided a number from a number range is drawn. Display ID of Account
Prospect	BOOLEAN	A prospect is someone who may become a customer.

Template Header	Data Type	Notes
Role	STRING	This field is mandatory Use CRM000 for customer Or BUP002 for prospect Or any specific Z-code
ABC Classification	STRING	ABC classification of a customer
Nielsen ID	STRING	
Status	STRING	ACTIVE will be determined by the system if not explicitly specified.
Natural Person	BOOLEAN	
Source Lead Technical ID	STRING	
Blocking Reasons_Billing Block	STRING	
Blocking Reasons_Delivery Block	STRING	
Blocking Reasons_Order Block	STRING	
Blocking Reasons_Sales Support Block	BOOLEAN	
Name	STRING	This field is mandatory Name of the account holder
Additional Name	STRING	
Additional Name 2	STRING	
Additional Name 3	STRING	
Industry	STRING	Primary business activity of the company
Legal Form	STRING	
Contact Permission	STRING	
D-U-N-S	STRING	
Owner Technical ID	STRING	Technical ID of the Account Owner (Employee) part of the Account Team
Main Address_Country/Region	STRING	The country of the account main address
Main Address_State_State	STRING	The region (federal state, county) of the address
Main Address_Postal Code	STRING	Postal code for the street address
Main Address_City	STRING	City of the address
Main Address_Street	STRING	Street
Main Address_House Number	STRING	
Main Address_County	STRING	
Main Address_District	STRING	District of the address

Template Header	Data Type	Notes
Main Address_Address Line 1	STRING	
Main Address_Address Line 2	STRING	
Main Address_Address Line 4	STRING	
Main Address_Address Line 5	STRING	
Main Address_P.O. Box	STRING	
Main Address_P.O. Box Address	BOOLEAN	
Main Address_Latitude	NUMBER	
Main Address_Longitude	NUMBER	
Main Address_Internal Geo Location	BOOLEAN	
Main Address_Language	STRING	
Main Address_Best Reached By	STRING	
Main Communication Data_Email	STRING	Email address
Main Communication Data_Website	STRING	
Main Communication Data_Fax	STRING	
Main Communication Data_Phone	STRING	
Main Communication Data_Mobile	STRING	

5.6.1.2.2 Account Team Members

The Account Team entity allows access to the account team. Here you can assign employees with a specific role to the corporate account.

Template Header	Data Type	Notes
parent_external_key	STRING	
Employee Technical ID	STRING	This field is mandatory Technical ID of Employee
Employee ID	STRING	Employee display ID can be used to reference an employee alternatively to Employee Technical ID. (Either of the two can be used.)
Role	STRING	This field is mandatory The type of responsibility the employee has for the account.
Sales Organization Technical ID	STRING	Technical ID of the Sales Organization. It is required for Sales Area dependent Account Team.

Template Header	Data Type	Notes
Distribution Channel	STRING	
division	STRING	
Main	BOOLEAN	Main indicator for given party role, sales area and validity period
Valid From	DATETIME	Start date of the validity of the account
Valid To	DATETIME	End date of the validity of the account

5.6.1.2.3 Account Addresses

The Account Addresses entity offers access and maintenance of the corporate accounts addresses.

The address includes postal address details as well as communication data details like email or phone number. You can specify for an individual address whether it should be the main address. Use the Account Addresses entity only in case of multiple addresses. If you want to maintain only the main address, then use the Account entity.

Template Header	Data Type	Notes
parent_external_key	STRING	
Main	BOOLEAN	Specification of main address
Country/Region	STRING	The country/region of the address
State_State	STRING	The region (federal state, county) of the address
City	STRING	City or district of the address
Street	STRING	Street of the address
House Number	STRING	
Postal Code	STRING	Postal code for the street address
County	STRING	
District	STRING	District of the address
Address Line 1	STRING	
Address Line 2	STRING	
Address Line 4	STRING	
Address Line 5	STRING	
P.O. Box	STRING	
P.O. Box Address	BOOLEAN	
Latitude	NUMBER	
Longitude	NUMBER	

Template Header	Data Type	Notes
Internal Geo Location	BOOLEAN	
Language	STRING	
Best Reached By	STRING	
Email	STRING	Email address
Website	STRING	
Fax	STRING	
Phone	STRING	
Mobile	STRING	

5.6.1.2.4 Account Contacts Persons

The Account Contact Persons entity offers access to the corporate accounts contact data.

With this entity you can assign contacts to an account and add business address details like communication data email, phone or mobile number. The relationship to the account will always be established based on the main address of the account.

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Contact Technical ID	STRING	This field is mandatory
Contact ID	STRING	The Contact display ID can be used to reference an contact person alternatively to Contact Technical ID. (Either of the two can be used.)
Main	STRING	Specifies that the assigned Contact person is the main contact for this Account.

5.6.1.2.5 Account Identification

The Account Identification entity offers access to the corporate accounts identification data.

Template Header	Data Type	Notes
parent_external_key	STRING	

Template Header	Data Type	Notes
ID Type	STRING	This field is mandatory A code representation of the type of an identification number
ID Number	STRING	This field is mandatory The identification number itself
Valid From	DATETIME	
Valid To	DATETIME	
Institution Responsible	STRING	Name of the institution (government agency, registry office), company (Dun & Bradstreet), an organization (UN) that issued the identification number
Entry Date	DATETIME	
Country/Region	STRING	
State_State	STRING	

5.6.1.2.6 Account Relationships

The Account Relationships entity offers access to the corporate accounts relationship data.

Relationships from the account can be established to other business partners by providing the Business Partner Technical ID and the Relationship Role, which classifies the relationship.

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Business Partner Relationship Role	STRING	This field is mandatory A coded representation of the type of Relationship.
Business Partner Technical ID	STRING	This field is mandatory
Business Partner ID	STRING	The Business Partner ID can be used to reference an Business Partner alternatively to Business Partner Technical ID. (Either of the two can be used.)
Sales Organization Technical ID	STRING	For Sales Area dependent relationships the Sales Organization Technical ID must be provided.
Distribution Channel	STRING	
Division	STRING	

Template Header	Data Type	Notes
Main Relationship	BOOLEAN	Indicates whether the relationship for the provided Business Partner Relationship Role and the Sales Area is the main relationship.

5.6.1.2.7 Account Sales Arrangements

The Account Sales Data entity allows access to the corporate accounts sales data.

Template Header	Data Type	Notes
parent_external_key	STRING	
Sales Organization Technical ID	STRING	This field is mandatory Technical ID of the Sales Organization.
Sales Organization ID	STRING	
Distribution Channel	STRING	If you provide a distribution channel you need to provide Sales Organization ID
Division	STRING	If you provide a division you need to provide distribution channel and Sales Organization ID
Marked for Deletion	BOOLEAN	Use this to mark a Sales Area for deletion. Required in integrated environments where physical deletion of sales arrangements is not allowed
Sales Office Technical ID	STRING	
Sales Office ID	STRING	
Sales Group Technical ID	STRING	
Sales Group ID	STRING	
Delivery Priority	STRING	
Complete Delivery	BOOLEAN	
Currency	STRING	
Customer Group	STRING	
Price List	STRING	
Price Group	STRING	
Cash Discount Terms	STRING	
Incoterms_Incoterms	STRING	
Incoterms_Incoterms Location	STRING	
Blocking Reasons_Billing Block	STRING	

Template Header	Data Type	Notes
Blocking Reasons_Delivery Block	STRING	
Blocking Reasons_Order Block	STRING	
Blocking Reasons_Sales Support Block	BOOLEAN	

5.6.1.2.8 Account Domains

The Account Domains entity offers access and maintenance of the corporate accounts' (web) domains.

You can assign additional (web) domains to an account apart from the address of the website that is maintained. As address maintenance only allows one website per address, this domain information enables multi-domain assignment to an account.

This information can then be used to identify accounts by domain information from Email conversations if multiple domains are in use.

Template Header	Data Type	Notes
parent_external_key	STRING	
Domain	STRING	This field is mandatory. Enter the domain of the account.

5.6.2 Account Hierarchy

Account hierarchy is a collection of accounts linked to each other through a parent-child relationship.

Use account hierarchy to:

- Map complex organizational structures of a business partner (for example, buying group, co-operative or chain of retail outlets).
- Map the complex organizational structure of a large customer with multiple levels of subsidiaries.

5.6.2.1 Account Hierarchy Fields

Download the template from [Data Import and Export](#) for the field level information.

5.6.2.1.1 Account Hierarchy

This is the main entity file that contains all the account hierarchies, sub hierarchies, and their linkage.

Account Hierarchy

Template Header	Data Type	Notes
external_key	STRING	
Technical ID	STRING	This is a mandatory field.
Sales Organization Technical ID	STRING	
Distribution Channel Code	STRING	
Division Code	STRING	
Parent Account Technical ID	STRING	This field can be left blank for the top account.
Parent Account Sales Organization ID	STRING	This field is not mandatory but is used in combination with Distribution Channel and Division. Hence, all three fields must be maintained.
Parent Account Distribution Channel Code	STRING	This field is not mandatory but is used in combination with Sales Organization ID and Division. Hence, all three fields must be maintained.
Parent Account Division Code	STRING	This field is not mandatory but is used in combination with Sales organization ID and Distribution Channel. Hence, all three fields must be maintained.
Is Rebate Relevant	BOOLEAN	
Is Pricing Determination Relevant	BOOLEAN	
Usage Type Code	STRING	This is a mandatory field. The usage type should be same for the entire hierarchy.

5.6.2.1.2 Account Hierarchy External IDs

The sub entity file contains the External IDs of the account hierarchies and the source system information.

External IDs

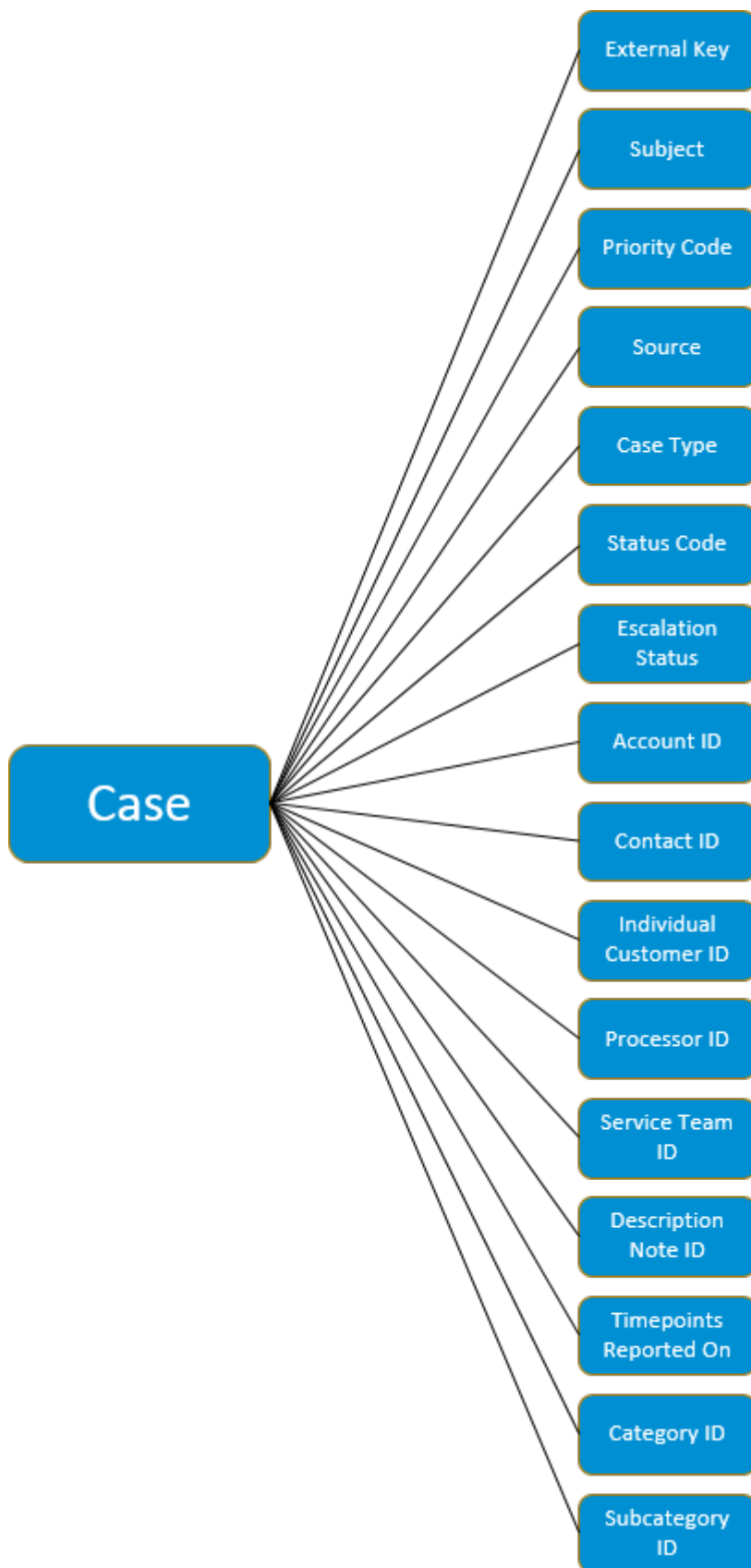
Template Header	Data Type	Notes
external_key	STRING	
External ID	STRING	This is the ID in the source system.
Business System Technical ID	STRING	This is the UUID of the business system.
Business System ID	STRING	This is the ID of the business system.

Template Header	Data Type	Notes
ID Type	STRING	ID Type is constant for account hierarchy. The field value is sap.oitc.985
Main	BOOLEAN	If there are multiple External IDs, you can use this field to mark an external ID as main or default.

5.6.3 Case Overview and Structure

Overview of the Case entity and the structure in the solution.

Cases are a record of a request for some type of service or support from an account, customer, or employee. This record allows you to track interactions with the service requester. Cases also record details like how much time has passed since the case was created, what actions were taken to resolve the issue, priority, associated products, or warranties, and much more. Cases are sometimes referred to as service requests.



5.6.3.1 Case Dependencies

Case has dependencies on the following entities.

Entity	Notes
Hypernotes	Hypernotes are used for the case description and notes content.
Attachments	

5.6.3.2 Supported Operations for Case

Data import and export supports the following operations for the case entity.

Complete Entity Import	Insert	Update	Delete
Yes	Yes	Yes	No

5.6.3.3 Case Fields

Download the template from Data Import and Export for the field level information.

5.6.3.3.1 Case

Template Header	Data Type	Notes
external_key	STRING	
Subject	STRING	
Priority Code	STRING	
Source	STRING	
Case Type	STRING	
Status Code	STRING	
Escalation Status	STRING	
Language	STRING	
Account_Account ID	STRING	
Contact_Contact ID	STRING	

Template Header	Data Type	Notes
Individual Customer_Individual Customer ID	STRING	
Processor_Processor ID	STRING	
Service Team_Service Team ID	STRING	
Supplier_Supplier ID	STRING	
Employee_Employee ID	STRING	
Supplier Contact_Supplier Contact ID	STRING	
Bill-To Account_Bill-To Account ID	STRING	
Bill-To Individual Customer_Bill-To Individual Customer ID	STRING	
Bill-To Supplier_Bill-To Supplier ID	STRING	
Ship-To Account_Ship-To Account ID	STRING	
Ship-to Individual Customer_Ship-To Individual Customer ID	STRING	
Ship-To Supplier_Ship-To Supplier ID	STRING	
Payer Account_Payer Account ID	STRING	
Payer Individual Customer_Payer Individual Customer ID	STRING	
Payer Supplier_Payer Supplier ID	STRING	
Description_Note ID_HTML_CONTENT	STRING	
Description_Note ID	STRING	
TimePoints_Reported on	DATETIME	
Incident Category Level 1_Category Level 1 ID	STRING	
Incident Category Level 1_Category Level 1 Display ID	STRING	
Category Level 2_Category Level 2 ID	STRING	
Category Level 2_Category Level 2 Display ID	STRING	
Category Level 3_Category Level 3 ID	STRING	
Category Level 3_Category Level 3 Display ID	STRING	
Category Level 4_Category Level 4 ID	STRING	
Category Level 4_Category Level 4 Display ID	STRING	
Company_Company ID	STRING	
Irrelevant Indicator	BOOLEAN	

5.6.3.3.2 Case Approvers

Template Header	Data Type	Notes
parent_external_key	STRING	
Approver ID	STRING	

5.6.3.3.3 Case Functional Locations

Template Header	Data Type	Notes
parent_external_key	STRING	
Functional Location ID	STRING	

5.6.3.3.4 Case Installed Bases

Template Header	Data Type	Notes
parent_external_key	STRING	
Installed Base ID	STRING	

5.6.3.3.5 Case Notes

Template Header	Data Type	Notes
parent_external_key	STRING	
Note ID_HTML_CONTENT	STRING	
Note ID_NOTE_TYPE	STRING	
Note ID	STRING	

5.6.3.3.6 Case Product Install Points

Template Header	Data Type	Notes
parent_external_key	STRING	
Product Install Point ID	STRING	

5.6.3.3.7 Case Registered Products

Template Header	Data Type	Notes
parent_external_key	STRING	
Registered Product ID	STRING	

5.6.4 Competitors

Use Competitors to create new competitors and maintain competitor information.

5.6.4.1 Competitor Dependencies

Download the template from Data Import and Export for the field level information.

5.6.4.1.1 Competitors

The Competitor header entity contains all information about the competitor.

Template Header	Data Type	Notes
external_key	STRING	
Competitor ID	STRING	Value must be in alphanumeric format.
Status	STRING	Value must be in code format.
Name	STRING	This is a mandatory field.
Additional Name	STRING	
Additional Name 2	STRING	

Template Header	Data Type	Notes
Additional Name 3	STRING	
Default Address_Country/Region	STRING	
Default Address_Region_Country/Region	STRING	Value must be in code format.
Default Address_Region_State	STRING	Value must be in code format.
Default Address_Region_Postal Code	STRING	Value must be in code format.
Default Address_Region_City	STRING	
Default Address_Street	STRING	
Default Address_House Number	STRING	
Default Address_County	STRING	
Default Address_District	STRING	
Default Address_Address Line 1	STRING	
Default Address_Address Line 2	STRING	
Default Address_Address Line 4	STRING	
Default Address_Address Line 5	STRING	
Default Address_P.O. Box	STRING	
Default Address_P.O. Box Address	BOOLEAN	
Default Address_P.O. Box Postal Code	STRING	
Default Address_Company Postal Code	STRING	
Default Address_P.O. Box City	STRING	
Default Address_P.O. Box Country/Region	STRING	Value must be in code format.
Default Address_P.O. Box State	STRING	Value must be in code format.
Default Address_Latitude	NUMBER	
Default Address_Longitude	NUMBER	
Default Address_Language	STRING	Value must be in code format.
Default Address_Best Reached By	STRING	Value must be in code format.
Default Address_Different City	STRING	
Default Address_Additional House Number	STRING	
Default Address_C/O	STRING	
Default Communication_Email	STRING	Value must be in email format.
Default Communication_Website	STRING	Value must be in URL format.
Default Communication_Fax	STRING	
Default Communication_Phone	STRING	
Default Communication_Mobile	STRING	

5.6.4.1.2 Competitor External IDs

Template Header	Data Type	Notes
parent_external_key	STRING	
External ID	STRING	
Technical ID of Communication System	STRING	
Communication System ID	STRING	
ID Type	STRING	
Main	BOOLEAN	

5.6.5 Competitor Products

Competitor Products enables you to create and maintain comprehensive records of competitor offerings. Leverage this data to effectively compare and strategically position your own products.

5.6.5.1 Competitor Product Dependencies

Dependencies on other Entities

Entity	Description
Product	Used in competitor product root to assign a product to competitor product.
Competitor	Used in competitor product to assign competitor to competitor product.

5.6.5.2 Competitor Product Fields

Download the template from Data Import and Export for the field level information.

5.6.5.2.1 Competitor Product

Template Header	Data Type	Notes
external_key	STRING	
Technical ID	STRING	
Competitor Product ID	STRING	
Status Code	STRING	
Competitor Technical ID	STRING	
Competitor ID	STRING	
Competitor Product Name	STRING	
Top Selling	BOOLEAN	
List Price/ Currency_Price	NUMBER	
List Price/ Currency_Currency	STRING	
Unit of Measure Code	STRING	
Product Technical ID	STRING	
Product ID	STRING	
Comparison Code	STRING	

5.6.6 Contact Person

Use contacts to represent the relationship between contacts and accounts. Contacts can be maintained only for corporate accounts.

Contact Persons are individuals you are having direct interactions with for your business to be successful. Contacts mostly have a relationship with a corporate account and they may be involved in business processes like activities, orders, opportunities, and so on. Contact Person Relationships can only be maintained for corporate accounts whereas in general a contact can also exist without any account assignment yet. With the contact entity and its dedicated sub-entities you can access and modify all central and related information of a contact (master data).



5.6.6.1 Contact Person Dependencies

Dependencies on Other Entities

Entity	Description
Account	Used in Contact Person (Root) and (Is) Contact Person For entity to assign a Contact to existing Corporate Account(s)
Account Address	Used as basis and prerequisite for the contacts business address based communication and workplace data

Dependencies within the Entity

Entity	Description
(Is) Contact Person For	The contacts main account relationship data can either be maintained in Contact Person (Root) or (Is) Contact Person For entity not in both places simultaneously. Recommendation is to maintain this Business Address based details like communication data email, phone or mobile number for the main account at the Contact Person (Root) entity whereas non-main-account information needs to be handled via the specific sub-entity.

5.6.6.2 Contact Person Fields

Download the template from [Data Import and Export](#) for the field level information.

5.6.6.2.1 Contact Person

The Contact Person entity allows you access to all central information for a contact person such as Contact Name and assignment of the contact to the main account (including the business address details of that relationship). The assignment to an account is mandatory in case you want to maintain any workplace details like "Email", "Phone", "Mobile", "Fax", "Building", "Floor", "Room", etc. based on the business address. Use the address fields in the Contact Person header entity in case only the workplace address to the main account is to be maintained.

Template Header	Data Type	Notes
external_key	STRING	
Technical ID	STRING	
Contact ID	STRING	Human-readable identifier of a contact person. The Contact ID is internally determined by the system via number range if not explicitly specified.
Last Name	STRING	This field is mandatory. Family name of a contact person
First Name	STRING	Given name of a contact person
Middle Name	STRING	
Date of Birth	DATETIME	
Gender	STRING	
Title	STRING	
Academic Title	STRING	
Language	STRING	
Status	STRING	ACTIVE will be determined by the system if not explicitly specified.
Account Technical ID	STRING	UUID of the main account to which the contact is assigned to
Account ID	STRING	Display ID of the main account to which the contact is assigned to (Read Only)
Department	STRING	
Department From Business Card	STRING	
Job Title	STRING	Description of the function of the Contact Person within the main accounts company
Function	STRING	
VIP Contact	STRING	
Building	STRING	Building maintained for the assignment to the main account
Floor	STRING	
Room	STRING	
Phone	STRING	Conventional phone number for the contact person's workplace with main account
Mobile	STRING	Mobile phone number for the contact person's workplace with main account
Email	STRING	Email address for the contact person's workplace with main account

Template Header	Data Type	Notes
Best Reached By	STRING	
Contact Permission	STRING	

5.6.6.2.2 Contact Person for

The 'Contact Person Is Contact Person For' sub-entity offers access to the contact account relationship data. With this entity you can assign contacts to accounts and add business address details like communication data email, phone or mobile number. The relationship to the account will always be established based on the main address of the account. This specific entity has to be used in case a contact person serves as contact for multiple accounts and the referred account is not the main one. The related information for the contacts' main account can also be maintained at the Contact Person (root) entity.

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Account Technical ID	STRING	This field is mandatory for the sub-entity UUID of the account to which the contact is assigned to
Account ID	STRING	
Main Contact	BOOLEAN	Indicates whether the account is the contact person's main account (in case of multiple account assignments for the contact)
Function	STRING	
Department	STRING	
VIP Contact	STRING	
Department From Business Card	STRING	
Job Title	STRING	Description of the function of the contact person within the assigned accounts company
Building	STRING	Building maintained for the assignment to the assigned account
Floor	STRING	
Room	STRING	
Phone	STRING	Conventional phone number for the contact person's workplace with assigned account

Template Header	Data Type	Notes
Mobile	STRING	Mobile phone number for the contact person's workplace with main account
Email	STRING	Email address for the contact person's workplace with main account
Best Reached By	STRING	

5.6.7 Emails

Learn about how you can leverage the email functionality of your system to manage email channels, maintain templates as per different scenarios and automatically generate cases that are sent to service agents for processing.

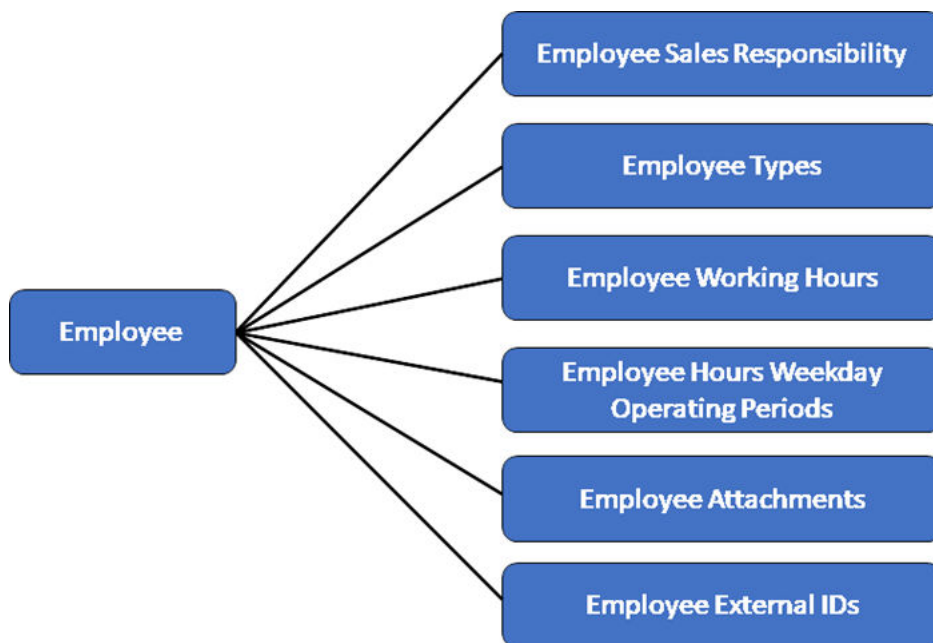
5.6.7.1 Email Dependencies

5.6.7.2 Email Fields

5.6.8 Employee

Employee Master Data capabilities offer a holistic view of internal employees (a business partner person with valid employee type "INTERNAL").

It allows you to maintain all relevant data for your internal employees such as basic data (name, workplace address, communication data), organizational data (manager assignment, department), sales responsibility data, working hours information, and so on.



5.6.8.1 Employee Dependencies

Dependencies on Other Entities

Entity	Description
Organizational Unit	Used in Employee Sales Responsibility to assign the employees to a specific Sales Organization and OrgUnit

Dependencies within the Entity

Entity	Description
Employee Type	The employee type address can be maintained in Employee
Employee Working Hours Weekday Operating Period	The Employee Working Hours Weekday Operating Period can be maintained in Employee

5.6.8.2 Employee Fields

Download the template from [Data Import and Export](#) for the field level information.

5.6.8.2.1 Employee

The Employee header entity contains all central information of a employee. It contains information about the account name and workplace address with communication data like email.

Template Header	Data Type	Notes
external_key	STRING	
Technical ID	STRING	
Business Partner ID	STRING	A proprietary identifier for a business partner; alternative identifier for the employee
Employee ID	STRING	This field is mandatory An identifier for an Employee
Title	STRING	
Academic Title	STRING	
First Name	STRING	Given name of a person
Middle Name	STRING	
Last Name	STRING	This field is mandatory Family name of a person
Additional Last Name	STRING	
Date of Birth	DATETIME	The date on which the employee was born.
Nationality	STRING	
Marital Status	STRING	
Gender	STRING	
Language	STRING	
Workplace Address_Street	STRING	Street of the address
Workplace Address_House Number	STRING	House number of the address
Workplace Address_City	STRING	City of the address
Workplace Address_Postal Code	STRING	Postal code for the street address
Workplace Address_Country/Region	STRING	The country/region of the address
Workplace Address_State_State	STRING	The state of the address

Template Header	Data Type	Notes
Workplace Address_District	STRING	The district of the address
Workplace Address_Latitude	NUMBER	
Workplace Address_Longitude	NUMBER	
Workplace Address_Building	STRING	
Workplace Address_Floor	STRING	
Workplace Address_Room	STRING	
Workplace Address_Phone	STRING	
Workplace Address_Mobile	STRING	
Workplace Address_Email	STRING	Email address
Workplace Address_Address Line 1	STRING	
Workplace Address_Address Line 2	STRING	
Workplace Address_Address Line 4	STRING	
Workplace Address_Address Line 5	STRING	

5.6.8.2.2 Employee Sales Responsibility

The Employee Sales Responsibility entity allows the employee to maintain the sales data.

Template Header	Data Type	Notes
parent_external_key	STRING	
Sales Organization Technical ID	STRING	
Distribution Channel	STRING	A coded representation of a distribution channel
Division	STRING	A coded representation of a division
Default Sales Responsibility	BOOLEAN	An indicator to mark one Employee Sales Responsibility-record as default for the employee

5.6.8.2.3 Employee Types

The Employee Types entity offers the possibility to maintain the type of the employee (internal or external) and the validity.

Template Header	Data Type	Notes
parent_external_key	STRING	

Template Header	Data Type	Notes
Internal Employee	BOOLEAN	This field is mandatory
Valid From	DATETIME	This field is mandatory
Valid To	DATETIME	This field is mandatory

5.6.8.2.4 Employee External IDs

The Employee External IDs entity displays the external ids for employee records.

Template Header	Data Type	Notes
parent_external_key	STRING	
External ID	STRING	
Business System Technical ID	STRING	
Business System ID	STRING	
ID Type	STRING	
Main	BOOLEAN	

5.6.8.2.5 Employee Working Hours

The Employee Working Hours entity offers access to the working hours data.

Template Header	Data Type	Notes
parent_external_key	STRING	
Type	STRING	A coded representation of a working hours type
Valid From	DATETIME	The period from which the operating hours are valid
Valid To	DATETIME	The period till which the operating hours are valid
Time Zone	STRING	Time zone in which the operating hours are specified
Working Day Calendar	STRING	Calendar with general working days in a week and public holidays

5.6.8.2.6 Employee Working Hours Operating Periods

The Employee Working Hours Operating Periods entity offers access to the operating period data. With this entity you can assign working hours operating period to working hours.

Template Header	Data Type	Notes
parent_external_key	STRING	
Weekday	STRING	
Start Time	DATETIME	An active time period of a day in a day program
End Time	DATETIME	An active time period of a day in a day program

5.6.9 Functional Location

5.6.9.1 Functional Location Dependencies

Dependencies on Other Entities

Entity	Description
Account	Used as Customer, Ship To, Bill To, Owner and Payer
Individual Customer	Used as Customer, Ship To, Bill To, Owner and Payer
Contact	Used as Main contact of Customer, Ship To, Bill To, Owner and Payer Accounts
Employee	Used as Service Technician and Employee Responsible
Organizational Unit	Used as Responsible Planner, Service Technician Team, Sales Org
Warranty	

5.6.9.2 Functional Location Fields

Download the template from Data Import and Export for the field level information.

5.6.9.2.1 Functional Location

Template Header	String Type	Notes
external_key	STRING	
Valid From	DATETIME	
Valid To	DATETIME	
Account_Contact UUID	STRING	
Account_Account UUID	STRING	
Bill-to Account_Bill-To Contact UUID	STRING	
Bill-to Account_Bill-To Account UUID	STRING	
Ship-to Account_Ship-To Contact UUID	STRING	
Ship-to Account_Ship-To Account UUID	STRING	
Payer Account_Payer Contact UUID	STRING	
Payer Account_Payer Account UUID	STRING	
Owner Account_Owner Contact UUID	STRING	
Owner Account_Owner Account UUID	STRING	
Individual Customer_Individual Customer UUID	STRING	
Bill-To Individual Customer_Bill-To Individual Customer UUID	STRING	
Ship-to Individual Customer_Ship-To Individual Customer UUID	STRING	
Payer Individual Customer_Payer Individual Customer UUID	STRING	
Owner Individual Customer_Owner Individual Customer UUID	STRING	
Service Technician Team_Service Technician Team UUID	STRING	
Responsible Planner_Responsible Planner UUID	STRING	
Service Technician_Service Technician UUID	STRING	
Employee Responsible_Employee Responsible UUID	STRING	
Warranty_Warranty UUID	STRING	
Warranty_Warranty End Date	DATETIME	
Warranty_Warranty Start Date	DATETIME	
Status	STRING	
ID	STRING	
Address_Postal Address_Building	STRING	
Address_Postal Address_City	STRING	
Address_Postal Address_Country/Region Code	STRING	

Template Header	String Type	Notes
Address_Postal Address_State Code	STRING	
Address_Postal Address_County	STRING	
Address_Postal Address_District	STRING	
Address_Postal Address_Floor	STRING	
Address_Postal Address_House Number	STRING	
Address_Postal Address_Additional House Number	STRING	
Address_Postal Address_Room	STRING	
Address_Postal Address_Street	STRING	
Address_Postal Address_Street Postal Code	STRING	
Address_Postal Address_Time Zone	STRING	
Address_Postal Address_Street Prefix	STRING	
Address_Postal Address_Street Suffix	STRING	
Address_Postal Address_Additional Street Prefix	STRING	
Address_Postal Address_Additional Street Suffix	STRING	
Address_Postal Address_C/O	STRING	
Address_Postal Address_P.O. Box City	STRING	
Address_Postal Address_P.O. Box Country/Region	STRING	
Address_Postal Address_P.O. Box	STRING	
Address_Postal Address_P.O. Box Postal Code	STRING	
Address_Postal Address_P.O. Box State	STRING	
Address_Geographical Location_Latitude	STRING	
Address_Geographical Location_Longitude	STRING	
Sales Office_Sales Office UUID	STRING	
Sales Group_Sales Group UUID	STRING	
Sales Organization_Sales Organization UUID	STRING	
Division Code	STRING	
Distribution Channel Code	STRING	
Type	STRING	
Parent Registered Product_Parent Registered Product UUID	STRING	
Parent Installation Point_Parent Installation Point UUID	STRING	
Parent Installed Base_Parent Installed Base UUID	STRING	
Extensions_asdD	STRING	
Extensions_z_bool_search_ref	BOOLEAN	

5.6.9.2.2 Functional Location Customer Information

Template Header	String Type	Notes
parent_external_key	STRING	
content	STRING	
languageCode	STRING	

5.6.9.2.3 Functional Location Descriptions

Template Header	String Type	Notes
parent_external_key	STRING	
content	STRING	
languageCode	STRING	

5.6.9.2.4 Functional Location External IDs

Template Header	String Type	Notes
parent_external_key	STRING	
id	STRING	
External ID	STRING	
Communication System ID	STRING	
Communication System UUID	STRING	
Type	STRING	
Default	BOOLEAN	

5.6.9.2.5 Functional Location Notes

Template Header	String Type	Notes
parent_external_key	STRING	
content	STRING	

Template Header	String Type	Notes
languageCode	STRING	

5.6.9.2.6 Functional Location Account Custom Parties

Template Header	String Type	Notes
parent_external_key	STRING	
Role Code	STRING	
Contact UUID	STRING	
Account UUID	STRING	

5.6.9.2.7 Functional Location Employee Custom Parties

Template Header	String Type	Notes
parent_external_key	STRING	
Role Code	STRING	
Employee UUID	STRING	

5.6.9.2.8 Functional Location_Individual Customer Custom Parties

Template Header	String Type	Notes
parent_external_key	STRING	
Role Code	STRING	
Individual Customer UUID	STRING	

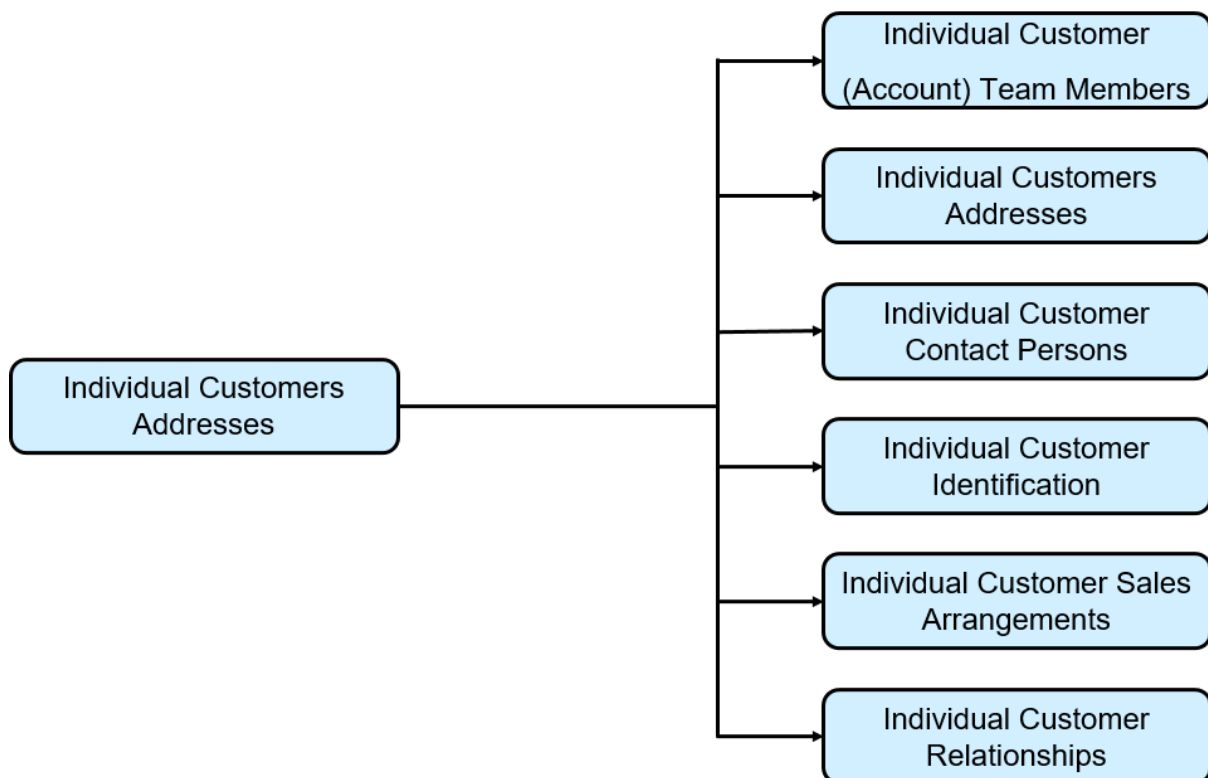
5.6.9.2.9 Functional Location Organization Custom Parties

Template Header	String Type	Notes
parent_external_key	STRING	
Role Code	STRING	
Organization UUID	STRING	

5.6.10 Individual Customer

Individual Customers represent persons with whom business is done.

Similar to account management capabilities for accounts with this entity a holistic view of the Individual Customer is offered, that allows you to capture, monitor, store and track all critical business information about customers and prospects. The Individual Customer thereby directly represents a person with whom your business is done, whereas an account is always an organization. Use this information to focus on your most profitable individual customers, maintain satisfaction and loyalty and consistent interaction across all channels.



5.6.10.1 Individual Customer Dependencies

Dependencies on Other Entities

Entity	Description
Contact Person	Used in Individual Customer Contact Persons entity to assign a Contact to individual customers
Employee	Used in Individual Customer (Root) and Individual Customer Account Team Members to assign internal and external employees e.g. as owner to the Individual Customer
Organizational Unit	Used in Individual Customer Account Team Members and Individual Customer Sales Data to assign data relevant for a specific Sales Organization to the Individual Customer

Dependencies within the Entity

Entity	Description
Individual Customer Addresses	The default address can either be maintained in the Individual Customer (Root) or Individual Customer Addresses entity but not in both places simultaneously.
Individual Customer Account Team Members	The owner for the individual customer can either be maintained in the Individual Customer (Root) or Individual Customer Account Team Members entity but not in both places simultaneously.

5.6.10.2 Individual Customer Fields

Download the template from [Data Import and Export](#) for the field level information.

5.6.10.2.1 Individual Customer

The Individual Customer header entity contains all central information of an individual customer. It contains information about the customers name, main address with communication data like email, ABC classification and blocking information. Use the address fields in the individual customer header entity in case only the

default address is to be maintained. It is recommended to fill the Country field even if you do not provide full address details.

Template Header	Data Type	Notes
external_key	STRING	
Technical ID	STRING	
Customer ID	STRING	Human-readable identifier of an Individual Customer. The Customer ID is internally determined by the system via number range if not explicitly specified
Prospect	BOOLEAN	A prospect is someone who may become a customer.
Role	STRING	This field is mandatory Use CRM000 for customer Or BUP002 for prospect Or any specific Z-code
ABC Classification	STRING	ABC classification of a customer
Status	STRING	ACTIVE will be determined by the system if not explicitly specified.
Blocking Reasons_Billing Block	STRING	
Blocking Reasons_Delivery Block	STRING	
Blocking Reasons_Order Block	STRING	
Blocking Reasons_Sales Support Block	BOOLEAN	
Date of Birth	DATETIME	The date of birth
Language	STRING	
Nationality	STRING	Identifies the nation to which the customer belongs
Gender	STRING	
Title	STRING	A form of address, for example, Mr. or Ms.
Academic Title	STRING	Identifies the academic degree
First Name	STRING	Given name of an individual customer
Middle Name	STRING	
Last Name	STRING	This field is mandatory Family name of an individual customer
Contact Permission	STRING	An indicator that defines whether it is allowed to contact the customer
Owner Technical ID	STRING	Technical ID of the Account Owner (Employee) as part of the Individual Customer (Account) Team

Template Header	Data Type	Notes
Main Address_Country/Region	STRING	The country of the customer's address
Main Address_State_State	STRING	The region (federal state, county) of the customer's address
Main Address_Postal Code	STRING	The postal code for the customer's street address
Main Address_City	STRING	The city of the customer's address
Main Address_Street	STRING	The street of the customer's address
Main Address_House Number	STRING	The house number of the customer's address
Main Address_County	STRING	
Main Address_District	STRING	District of the address
Main Address_Address Line 1	STRING	
Main Address_Address Line 2	STRING	
Main Address_Address Line 4	STRING	
Main Address_Address Line 5	STRING	
Main Address_Latitude	NUMBER	
Main Address_Longitude	NUMBER	
Main Address_Best Reached By	STRING	
Main Communication Data_Email	STRING	
Main Communication Data_Website	STRING	
Main Communication Data_Fax	STRING	
Main Communication Data_Phone	STRING	Phone number of the customer
Main Communication Data_Mobile	STRING	

5.6.10.2.2 Individual Customer Account Team Members

The Individual Customer (Account) Team Members entity allows access to the Individual Customer Team. Here you can assign employees with a specific party role to the individual customer. These assignments can even be done sales organization dependent.

Template Header	Data Type	Notes
parent_external_key	STRING	
Employee Technical ID	STRING	This field is mandatory Technical ID of the assigned Employee
Employee ID	STRING	

Template Header	Data Type	Notes
Role	STRING	This field is mandatory The type of responsibility the employee has for the business partner.
Sales Organization Technical ID	STRING	Technical ID of the Sales Organization. It is required for Sales Area dependent Individual Customer Team.
Distribution Channel	STRING	
Division	STRING	
Main	BOOLEAN	Main indicator for given party role, sales area and validity period
Valid From	DATETIME	Start date of the validity of the customer team assignment of the employee
Valid To	DATETIME	End date of the validity of the customer team assignment of the employee

5.6.10.2.3 Individual Customer Addresses

The Individual Customer Addresses entity offers access and maintenance of the individual customer addresses. The address includes postal address details as well as communication data details like email or phone number. You can specify for a dedicated address whether it should be the main address for the individual customer. Use the Individual Customer Addresses entity in case of multiple addresses. If you want to maintain/change only the main address, then use the Individual Customer header entity.

Template Header	Data Type	Notes
parent_external_key	STRING	
Main	BOOLEAN	Specification of main address
Country/Region	STRING	Country/region of the customer's address
State_State	STRING	The region (federal state, county) of the customer's address
City	STRING	City of customer's address
Street	STRING	Street of customer's address
House Number	STRING	House number of customer's address
Postal Code	STRING	Postal code of the customer's street address
County	STRING	
District	STRING	District of customer's address
Address Line 1	STRING	

Template Header	Data Type	Notes
Address Line 2	STRING	
Address Line 4	STRING	
Address Line 5	STRING	
Latitude	NUMBER	
Longitude	NUMBER	
Best Reached By	STRING	
E-mail	STRING	Email address
Website	STRING	
Fax	STRING	
Phone	STRING	
Mobile	STRING	

5.6.10.2.4 Individual Customer Identifications

The Individual Customer Identification entity offers access to the individual customers identification data.

Template Header	Data Type	Notes
parent_external_key	STRING	
ID Type	STRING	This field is mandatory A code representation of the type of an identification number
ID Number	STRING	This field is mandatory The identification number itself
Valid From	DATETIME	
Valid To	DATETIME	
Institution Responsible	STRING	Name of the institution (government agency, registry office), company (Dun & Bradstreet), an organization (UN) that issued the identification number
Entry Date	DATETIME	
Country/Region	STRING	
State_State	STRING	

5.6.10.2.5 Individual Customer Sales Arrangements

The Individual Customer Sales Data entity allows access to the individual customers sales data.

Template Header	Data Type	Notes
parent_external_key	STRING	
Sales Organization Technical ID	STRING	This field is mandatory Technical ID of the Sales Organization.
Sales Organization ID	STRING	
Distribution Channel	STRING	
Division	STRING	
Marked for Deletion	BOOLEAN	Use this to mark a Sales Area for deletion. Required in integrated environments where physical deletion of sales arrangements is not allowed
Sales Office Technical ID	STRING	
Sales Office ID	STRING	
Sales Group Technical ID	STRING	
Sales Group ID	STRING	
Delivery Priority	STRING	
Complete Delivery	BOOLEAN	
Currency	STRING	
Customer Group	STRING	
Price List	STRING	
Price Group	STRING	
Cash Discount Terms	STRING	
Incoterms_Incoterms	STRING	
Incoterms_Incoterms Location	STRING	
Blocking Reasons_Billing Block	STRING	
Blocking Reasons_Delivery Block	STRING	
Blocking Reasons_Order Block	STRING	
Blocking Reasons_Sales Support Block	BOOLEAN	

5.6.10.2.6 Individual Customer Relationships

The Individual Customer Relationships allows establish relationships between business partners.

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	This field is mandatory
Business Partner Relationship Role	STRING	
Business Partner Technical ID	STRING	
Business Partner ID	STRING	
Sales Organization Technical ID	STRING	
Sales Organization ID	STRING	
Sales Group Technical ID	STRING	
Distribution Channel	STRING	
Division	STRING	
Main Relationship	STRING	

5.6.11 Installed Base

Installed base is a hierarchical arrangement of items installed at your customer's location.

5.6.11.1 Installed Base Dependencies

Dependencies on Other Entities

Entity	Description
Account	Used as Customer, Ship To, Bill To, Owner and Payer
Individual Customer	Used as Customer, Ship To, Bill To, Owner and Payer
Contact	Used as Main contact of Customer, Ship To, Bill To, Owner and Payer Accounts
Employee	Used as Service Technician and Employee Responsible
Organizational Unit	Used as Responsible Planner, Service Technician Team, Sales Org

5.6.11.2 Installed Base Fields

Download the template from Data Import and Export for the field level information.

5.6.11.2.1 Installed Base

Template Header	String Type	Notes
external_key	STRING	
Account_Contact UUID	STRING	
Account_Account UUID	STRING	
Bill-to Account_Bill-To Contact UUID	STRING	
Bill-to Account_Bill-To Account UUID	STRING	
Ship-to Account_Ship-To Contact UUID	STRING	
Ship-to Account_Ship-To Account UUID	STRING	
Individual Customer_Individual Customer UUID	STRING	
Bill-To Individual Customer_Bill-To Individual Customer UUID	STRING	
Ship-to Individual Customer_Ship-To Individual Customer UUID	STRING	
Status	STRING	
ID	STRING	
Address_Postal Address_Building	STRING	
Address_Postal Address_City	STRING	
Address_Postal Address_Country/Region Code	STRING	
Address_Postal Address_State Code	STRING	
Address_Postal Address_County	STRING	
Address_Postal Address_District	STRING	
Address_Postal Address_Floor	STRING	
Address_Postal Address_House Number	STRING	
Address_Postal Address_Additional House Number	STRING	
Address_Postal Address_Room	STRING	
Address_Postal Address_Street	STRING	
Address_Postal Address_Street Postal Code	STRING	
Address_Postal Address_Time Zone	STRING	
Address_Postal Address_Street Prefix	STRING	

Template Header	String Type	Notes
Address_Postal Address_Street Suffix	STRING	
Address_Postal Address_Additional Street Prefix	STRING	
Address_Postal Address_Additional Street Suffix	STRING	
Address_Postal Address_C/O	STRING	
Address_Postal Address_P.O. Box City	STRING	
Address_Postal Address_P.O. Box Country/Region	STRING	
Address_Postal Address_P.O. Box	STRING	
Address_Postal Address_P.O. Box Postal Code	STRING	
Address_Postal Address_P.O. Box State	STRING	
Address_Geographical Location_Latitude	STRING	
Address_Geographical Location_Longitude	STRING	
Payer Account_Payer Contact UUID	STRING	
Payer Account_Payer Account UUID	STRING	
Payer Individual Customer_Payer Individual Customer UUID	STRING	
Owner Account_Owner Contact UUID	STRING	
Owner Account_Owner Account UUID	STRING	
Owner Individual Customer_Owner Individual Customer UUID	STRING	
Service Technician Team_Service Technician Team UUID	STRING	
Responsible Planner_Responsible Planner UUID	STRING	
Service Technician_Service Technician UUID	STRING	
Employee Responsible_Employee Responsible UUID	STRING	

5.6.11.2.2 Installed Base Descriptions

Template Header	Data Type	Notes
parent_external_key	STRING	
content	STRING	
languageCode	STRING	

5.6.11.2.3 Installed Base Notes

Template Header	Data Type	Notes
parent_external_key	STRING	
content	STRING	
languageCode	STRING	

5.6.12 Organizational Unit

Organizational unit containing a hierarchy that represents the hierarchical structure of the organization.

5.6.12.1 Organizational Unit Dependencies

Dependencies on Other Entities

Entity	Description
Organizational Unit	Assign a parent org unit
Employee Assignments	You can assign an employee to the org unit

5.6.12.2 Organizational Unit Fields

Download the template from [Data Import and Export](#) for the field level information.

5.6.12.2.1 Organizational Unit

Note

Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.

Template Header	Data Type	Notes
external_key	STRING	
Technical ID	STRING	
Valid From	DATETIME	If not provided, defaulted to current date
Valid To	DATETIME	If not provided, defaulted to the maximum end date, 31-December-9999
Organizational Unit ID	STRING	This field is mandatory
Name	STRING	
Status	STRING	
Current Functions_Company	BOOLEAN	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Current Functions_Sales Organization	BOOLEAN	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Current Functions_Service Organization	BOOLEAN	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Current Functions_Sales Office	BOOLEAN	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Current Functions_Sales Group	BOOLEAN	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Current Functions_Sales	BOOLEAN	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Current Functions_Service	BOOLEAN	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Current Functions_Currency	STRING	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Manager_Manager Technical ID	STRING	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.

Template Header	Data Type	Notes
Parent Unit_Parent Unit Technical ID	STRING	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Address_Country/Region	STRING	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Address__State	STRING	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Address__State	STRING	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Address_Postal Code	STRING	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Address_City	STRING	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Address_Street	STRING	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Address_House Number	STRING	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Address_County	STRING	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Address_District	STRING	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Address_Address Line 1	STRING	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Address_Address Line 2	STRING	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.

Template Header	Data Type	Notes
Address_Address Line 4	STRING	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Address_Address Line 5	STRING	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Address_Latitude	NUMBER	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Address_Longitude	NUMBER	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Communication Data_Email	STRING	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Communication Data_Website	STRING	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Communication Data_Fax	STRING	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Communication Data_Phone	STRING	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.
Communication Data_Mobile	STRING	Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.

5.6.12.2.2 Organizational Unit Employee Assignments

Note

Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Valid From	DATETIME	
Valid To	DATETIME	
Employee Technical ID	STRING	The employee should be present in the system. This field or the employee display ID is mandatory. The employee must be valid during the period given in the Valid From and Valid To.
Role	STRING	

5.6.12.2.3 Organizational Unit External IDs

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
External ID	STRING	
Business System Technical ID	STRING	
Business System ID	STRING	
ID Type	STRING	
Main	BOOLEAN	

5.6.12.2.4 Organizational Unit Functions

Note

Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	

Template Header	Data Type	Notes
Valid From	DATETIME	
Valid To	DATETIME	
Company	BOOLEAN	
Sales Organization	BOOLEAN	
Service Organization	BOOLEAN	
Sales Office	BOOLEAN	
Sales Group	BOOLEAN	
Sales	BOOLEAN	
Service	BOOLEAN	
Currency	STRING	Recommended to maintain a currency either for the company or for the Sales Organization

5.6.12.2.5 Organizational Unit Name and Address

Note

Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Valid From	DATETIME	
Valid To	DATETIME	
Name	STRING	Recommended to maintain this field
Address Technical ID	STRING	
Country/Region	STRING	
_Country/Region	STRING	
_State	STRING	
City	STRING	
Street	STRING	
House Number	STRING	
Postal Code	STRING	
County	STRING	

Template Header	Data Type	Notes
District	STRING	
Address Line 1	STRING	
Address Line 2	STRING	
Address Line 4	STRING	
Address Line 5	STRING	
Latitude	NUMBER	
Longitude	NUMBER	
Email	STRING	
Website	STRING	
Fax	STRING	
Phone	STRING	
Mobile	STRING	

5.6.12.2.6 Organizational Unit Parent Units

Note

Do not upload the root field information simultaneously with the sub node fields as it will lead to inconsistencies. Please use only root fields or subnode fields.

Template Header	Data Types	Notes
parent_external_key	STRING	
Technical ID	STRING	
Valid From	DATETIME	
Valid To	DATETIME	
Parent Unit Technical ID	STRING	Recommended to maintain a valid Parent Unit Technical ID or a Parent Unit Display ID
Reporting Line	BOOLEAN	There can be only one reporting line parent org unit at the same point in time.

5.6.12.2.7 Organizational Unit Allowed Sales Areas

Template Header	Data Types	Notes
parent_external_key	STRING	
Technical ID	STRING	
Distribution Channel	STRING	
Division	STRING	

5.6.13 Product

A product is the item offered for sale and it can be a service or a physical item. A product can be of real or virtual form. Each product is made at a cost and sold at a price and the price depends on the quality, market, the marketing, and the targeted segment.

5.6.13.1 Product Dependencies

Dependencies on Other Entities

Entity	Description
Product Group	Used in product root to assign a product group to product(s)
Organizational Unit	Used in product sales distribution chains to assign sales data to product(s)

5.6.13.2 Product Fields

Download the template from Data Import and Export for the field level information.

5.6.13.2.1 Product

Template Header	Data Type	Notes
external_key	STRING	
Technical ID	STRING	
Product ID	STRING	Mandatory field
Status Code	STRING	Mandatory field. If no value is provided, then the default will be In_PREPARATION
Unit of Measure Code	STRING	Mandatory field
Name	STRING	Mandatory field
Description	STRING	
Product Group Technical ID	STRING	Technical ID of the product group created in the system.
Product Group ID	STRING	Display ID of the product group created in the system.

5.6.13.2.2 Product Descriptions

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Description	STRING	A user-defined language-dependent text that provides more information about the product. Maximum length: 5000
Language Code	STRING	

5.6.13.2.3 Product External IDs

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
External ID	STRING	
Business System Technical ID	STRING	Technical GUID of the Communication System.
Business System ID	STRING	Display ID of the Communication System.

Template Header	Data Type	Notes
ID Type	STRING	
Main	BOOLEAN	

5.6.13.2.4 Product Names

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Name	STRING	Definition of product names in multiple languages. Maximum length: 255
Language Code	STRING	

5.6.13.2.5 Product Sales Distribution Chains

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Sales Organization Technical ID	STRING	Organizational Unit Technical ID which is created in the system. This is a mandatory field.
Distribution Channel Code	STRING	This is a mandatory field.
Sales Unit Of Measure Code	STRING	
Minimum Order Quantity_Quantity	NUMBER	
Minimum Order Quantity_Unit of Measure Code	STRING	Minimum Order Quantity Unit of Measure Code must be equal to the field Unit of Measure Code of the entity Product. If it is not maintained during the creation of the corresponding Product Sales Distribution Chains entry, then it defaults to the Unit of Measure Code of the Product.
Sales Status Code	STRING	Sales status code that is defined in administrator settings > sales status schema

5.6.13.2.5.1 Customer Part Numbers

Customer Part Numbers is a sub-entity of product sales distribution chain.

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Customer Technical ID *	STRING	This is a mandatory field.
Customer ID	STRING	
Customer Part Number *	STRING	This is a mandatory field.
Description	STRING	

5.6.13.2.6 Product Unit of Measure Conversion

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Quantity	NUMBER	
Unit of Measure Code	STRING	
Corresponding Quantity	NUMBER	
Corresponding Unit of Measure Code	STRING	

5.6.13.2.7 Product Unit of Measure

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Unit of Measure Code	STRING	

5.6.13.2.7.1 Global Trade Item Numbers

Global Trade Item Numbers is a sub-entity of product units of measure.

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Global Trade Item Number	STRING	
Main	BOOLEAN	

5.6.14 Product Group

A product group is a group of products categorized based on the shared attributes like usage, features, production processes and so on. A product group can also be the market or customer segment in which the products are sold or the prices at which they're offered.

For the field level information of product group, download the product group template from ► [User Menu](#)
► [System Settings](#) ► [Monitoring](#) ► [Data Import and Export](#) ►.

Template Header	Data Type	Notes
parent_external_key	STRING	
external_key	STRING	
Technical ID	STRING	
Product Group ID	STRING	Mandatory field
Parent Product Group Technical ID		Technical ID of the parent product group created in the system.
Name	STRING	Mandatory field
Parent Product Group ID	STRING	Display ID of the parent product group created in the system.

5.6.14.1 Product Group External IDs

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
External ID	STRING	
Business System Technical ID	STRING	

Template Header	Data Type	Notes
Business System ID	STRING	
ID Type	STRING	
Main	BOOLEAN	

5.6.14.2 Product Group Names

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Name	STRING	Definition of product group names in multiple languages. Maximum length: 255
Language Code	STRING	

5.6.15 Product Installation Point

An installation point is the physical location of the object that requires servicing or maintenance based on the service or maintenance plan for that object.

5.6.15.1 Product Installation Point Dependencies

Dependencies on Other Entities

Entity	Description
Account	Used as Customer, Ship To, Bill To, Owner and Payer
Individual Customer	Used as Customer, Ship To, Bill To, Owner and Payer
Contact	Used as Main contact of Customer, Ship To, Bill To, Owner and Payer Accounts
Employee	Used as Service Technician and Employee Responsible
Organizational Unit	Used as Responsible Planner, Service Technician Team, Sales Org
Product	

5.6.15.2 Product Installation Point Fields

Download the template from Data Import and Export for the field level information.

5.6.15.2.1 Product Installation Point

Template Header	Data Type	Notes
external_key	STRING	
Reference Product_Product UUID	STRING	
Type	STRING	
Quantity_Quantity	NUMBER	
Quantity_Unit of Measure	STRING	
Account_Contact UUID	STRING	
Account_Account UUID	STRING	
Bill-to Account_Bill-To Contact UUID	STRING	
Bill-to Account_Bill-To Account UUID	STRING	
Ship-to Account_Ship-To Contact UUID	STRING	
Ship-to Account_Ship-To Account UUID	STRING	
Payer Account_Payer Contact UUID	STRING	
Payer Account_Payer Account UUID	STRING	
Owner Account_Owner Contact UUID	STRING	
Owner Account_Owner Account UUID	STRING	
Individual Customer_Individual Customer UUID	STRING	
Bill-To Individual Customer_Bill-To Individual Customer UUID	STRING	
Ship-to Individual Customer_Ship-To Individual Customer UUID	STRING	
Payer Individual Customer_Payer Individual Customer UUID	STRING	
Owner Individual Customer_Owner Individual Customer UUID	STRING	
Service Technician Team_Service Technician Team UUID	STRING	
Responsible Planner_Responsible Planner UUID	STRING	
Service Technician_Service Technician UUID	STRING	
Employee Responsible_Employee Responsible UUID	STRING	

Template Header	Data Type	Notes
Status	STRING	
ID	STRING	
Address_Postal Address_Building	STRING	
Address_Postal Address_City	STRING	
Address_Postal Address_Country/Region Code	STRING	
Address_Postal Address_State Code	STRING	
Address_Postal Address_County	STRING	
Address_Postal Address_District	STRING	
Address_Postal Address_Floor	STRING	
Address_Postal Address_House Number	STRING	
Address_Postal Address_Additional House Number	STRING	
Address_Postal Address_Room	STRING	
Address_Postal Address_Street	STRING	
Address_Postal Address_Street Postal Code	STRING	
Address_Postal Address_Time Zone	STRING	
Address_Postal Address_Street Prefix	STRING	
Address_Postal Address_Street Suffix	STRING	
Address_Postal Address_Additional Street Prefix	STRING	
Address_Postal Address_Additional Street Suffix	STRING	
Address_Postal Address_C/O	STRING	
Address_Postal Address_P.O. Box City	STRING	
Address_Postal Address_P.O. Box Country/Region	STRING	
Address_Postal Address_P.O. Box	STRING	
Address_Postal Address_P.O. Box Postal Code	STRING	
Address_Postal Address_P.O. Box State	STRING	
Address_Geographical Location_Latitude	STRING	
Address_Geographical Location_Longitude	STRING	
Parent Registered Product_Parent Registered Product UUID	STRING	
Parent Installation Point_Parent Installation Point UUID	STRING	
Parent Installed Base_Parent Installed Base UUID	STRING	

5.6.15.2.2 Product Installation Point External IDs

Template Header	Data Type	Notes
parent_external_key	STRING	
id	STRING	
External ID	STRING	
Communication System ID	STRING	
Communication System UUID	STRING	
Type	STRING	
Default	BOOLEAN	

5.6.15.2.3 Product Installation Point Customer Information

Template Header	Data Type	Notes
parent_external_key	STRING	
content	STRING	
languageCode	STRING	

5.6.15.2.4 Product Installation Point Notes

Template Header	Data Type	Notes
parent_external_key	STRING	
content	STRING	
languageCode	STRING	

5.6.15.2.5 Product Installation Point Descriptions

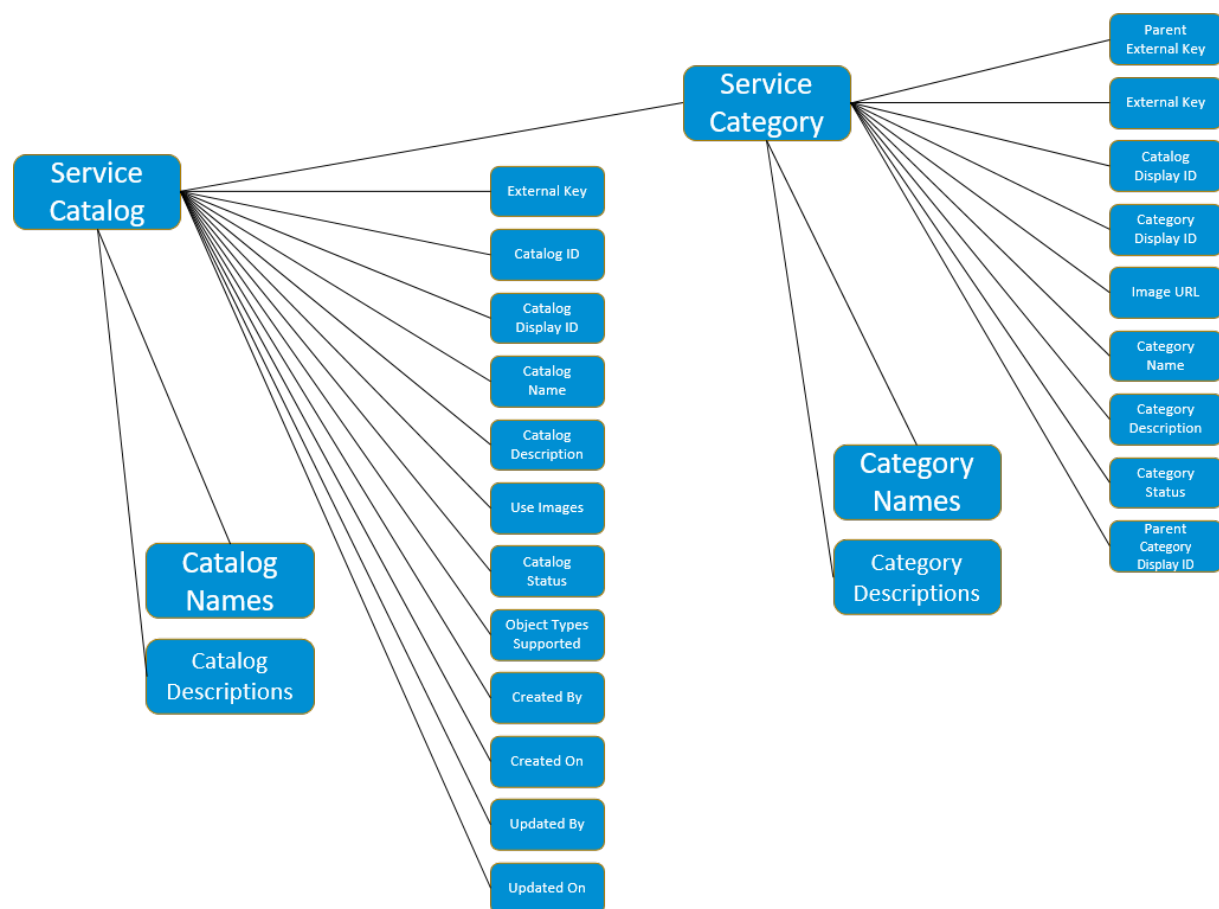
Template Header	Data Type	Notes
parent_external_key	STRING	
content	STRING	

Template Header	Data Type	Notes
languageCode	STRING	

5.6.16 Service Categories Overview and Structure

Overview of the Service Category entity and the structure in the solution.

Creating and defining categories helps build valuable knowledge about the customer and also collects data that can be analyzed to understand business requirements and market demands.



5.6.16.1 Service Categories Dependencies

Dependencies on Other Entities

Entity	Description
Service Catalog	Each service category must be associated with a service catalog

Dependencies within the Entity

Entity	Description
Category Names	Provide name values in different languages
Category Descriptions	Provide descriptions in different languages

5.6.16.2 Service Category Fields

Download the template from [Data Import and Export](#) for the field level information.

5.6.16.2.1 Service Category

Enter each category in a separate row. The **Name** and **Description** values entered in this root file are stored for English. Use the **Category Names** and **Category Descriptions** files to enter translations for these values in additional languages.

Template Header	Data Type	Notes
parent_external_key	STRING	Enter the external key of the parent category. Leave blank for top-level categories.
external_key	STRING	
Catalog Display ID	STRING	required
Category Display ID	STRING	required
Image URL	STRING	

Template Header	Data Type	Notes
Category Name	STRING	required
Category Description	STRING	
Category Status	STRING	Enter status codes as provided in category status file
Parent Category Display ID	STRING	Enter the display ID of the parent category. Leave blank for top-level categories.

5.6.16.2.2 Category Names

Use Category Names to enter name values for different languages. Prepare a file for all category names to be translated into different languages and enter each language translation in a separate row.

Template Header	Data Type	Notes
parent_external_key	STRING	Enter the external key of the category to be translated
content	STRING	Enter the translated name here
languageCode	STRING	Use language codes provided in the Language-Code file

5.6.16.2.3 Category Descriptions

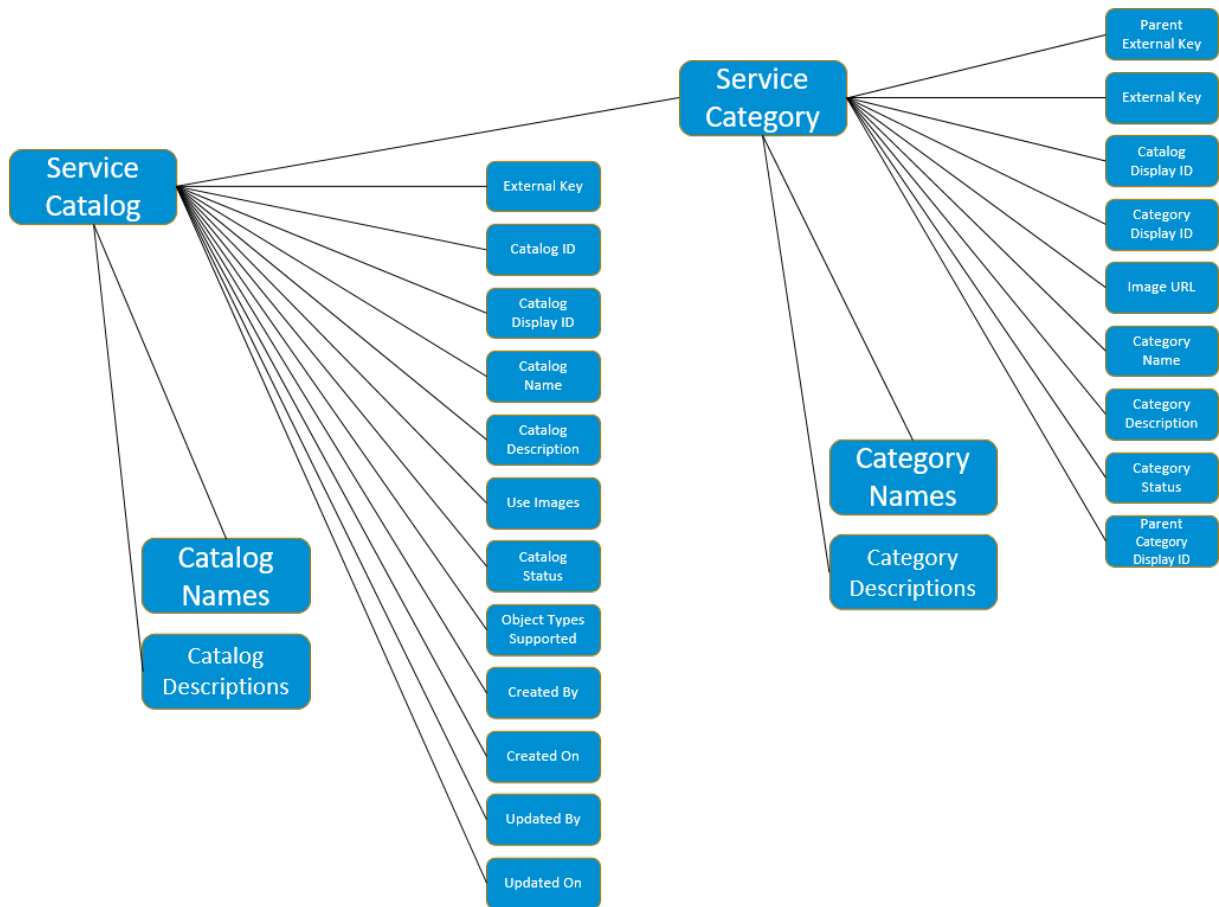
Use Category Descriptions to enter description values for different languages. Prepare a file for all category descriptions to be translated in different languages and enter each language translation in a separate row.

Template Header	Data Type	Notes
parent_external_key	STRING	Enter the external key of the category to be translated
content	STRING	Enter the translated description here
languageCode	STRING	Use language codes provided in the Language-Code file

5.6.17 Service Catalogs

Overview of the Service Catalog entity and the structure in the solution.

Service catalogs are collections of service categories. Each service category must be assigned to a service catalog.



5.6.17.1 Service Catalog Dependencies

Dependencies within the Entity

Entity	Description
Catalog Names	Provide name values in different languages
Catalog Descriptions	Provide descriptions in different languages

5.6.17.2 Service Catalog Fields

Download the template from [Data Import and Export](#) for the field level information.

5.6.17.2.1 Service Catalog

Enter each Catalog to be imported in a separate row. The **Name** and **Description** values entered in this root file are stored for English. Use the **Catalog Names** and **Catalog Descriptions** files to enter translations for these values in additional languages.

Template Header	Data Type	Notes
external_key	STRING	
Catalog ID		required - unique technical ID for the catalog
Catalog Display ID	STRING	required
Catalog Name	STRING	required
Catalog Description	STRING	
Use Images	BOOLEAN	
Catalog Status	STRING	Enter status codes provided in the catalog status file
Object Types Supported	ARRAY	• Case: 2886 • Phone: 86 • Chat: 1976 • Email: 39 If the catalog is used for all object types, enter "2886","86","1976","39" in the field
Catalog Administration Data_Created By	STRING	
Catalog Administration Data_Created On	DATETIME	
Catalog Administration Data_Updated By	STRING	
Catalog Administration Data_Updated On	DATETIME	

5.6.17.2.2 Catalog Names

Use Catalog Names to enter name values for different languages. Prepare a file for all catalog names to be translated into different languages and enter each language translation in a separate row.

Template Header	Data Type	Notes
parent_external_key	STRING	Enter the external key of the catalog to be translated
content	STRING	Enter the translated name here
languageCode	STRING	Use language codes provided in the Language-Code file

5.6.17.2.3 Catalog Descriptions

Use Catalog Descriptions to enter description values for different languages. Prepare a file for all catalog descriptions to be translated in different languages and enter each language translation in a separate row.

Template Header	Data Type	Notes
parent_external_key	STRING	Enter the external key of the catalog to be translated
content	STRING	Enter the translated description here
languageCode	STRING	Use language codes provided in the Language-Code file

5.6.18 Registered Products

A registered product is an instance of a product that is associated with a specific customer and generally has a unique serial number. The registered product information allows the service agent to identify the customer product and any associated warranties or service entitlements

5.6.18.1 Registered Product Dependencies

Dependencies on Other Entities

Entity	Description
Account	Used as Customer, Ship To, Bill To, Owner and Payer
Individual Customer	Used as Customer, Ship To, Bill To, Owner and Payer
Contact	Used as Main contact of Customer, Ship To, Bill To, Owner and Payer Accounts
Employee	Used as Service Technician and Employee Responsible
Organizational Unit	Used as Responsible Planner, Service Technician Team, Sales Org
Product	
Warranty	

5.6.18.2 Registered Product Fields

Download the template from Data Import and Export for the field level information.

5.6.18.2.1 Registered Product

Template Header	Data Type	Notes
external_key	STRING	
ID	STRING	
Status	STRING	
Account_Contact UUID	STRING	
Account_Account UUID	STRING	
Individual Customer_Individual Customer UUID	STRING	
Ship-to Account_Ship-To Contact UUID	STRING	
Ship-to Account_Ship-To Account UUID	STRING	
Ship-to Individual Customer_Ship-To Individual Customer UUID	STRING	
Bill-to Account_Bill-To Contact UUID	STRING	
Bill-to Account_Bill-To Account UUID	STRING	
Bill-to Individual Customer_Bill-To Individual Customer UUID	STRING	
Payer Account_Payer Contact UUID	STRING	
Payer Account_Payer Account UUID	STRING	
Payer Individual Customer_Payer Individual Customer UUID	STRING	
Owner Account_Owner Contact UUID	STRING	
Owner Account_Owner Account UUID	STRING	
Owner Individual Customer_Owner Individual Customer UUID	STRING	
Service Technician Team_Service Technician Team UUID	STRING	
Responsible Planner_Responsible Planner UUID	STRING	
Service Technician_Service Technician UUID	STRING	
Employee Responsible_Employee Responsible UUID	STRING	
Address_Postal Address_Building	STRING	
Address_Postal Address_City	STRING	

Template Header	Data Type	Notes
Address_Postal Address_Country/Region Code	STRING	
Address_Postal Address_State Code	STRING	
Address_Postal Address_County	STRING	
Address_Postal Address_District	STRING	
Address_Postal Address_Floor	STRING	
Address_Postal Address_House Number	STRING	
Address_Postal Address_Additional House Number	STRING	
Address_Postal Address_Room	STRING	
Address_Postal Address_Street	STRING	
Address_Postal Address_Street Postal Code	STRING	
Address_Postal Address_Time Zone	STRING	
Address_Postal Address_Street Prefix	STRING	
Address_Postal Address_Street Suffix	STRING	
Address_Postal Address_Additional Street Prefix	STRING	
Address_Postal Address_Additional Street Suffix	STRING	
Address_Postal Address_C/O	STRING	
Address_Postal Address_P.O. Box City	STRING	
Address_Postal Address_P.O. Box Country/Region	STRING	
Address_Postal Address_P.O. Box	STRING	
Address_Postal Address_P.O. Box Postal Code	STRING	
Address_Postal Address_P.O. Box State	STRING	
Address_Geographical Location_Latitude	STRING	
Address_Geographical Location_Longitude	STRING	
Registered Product category_Registered Product Category Code	STRING	
Reference Product_Product UUID	STRING	
Parent Registered Product_Parent Registered Product UUID	STRING	
Parent Installation Point_Parent Installation Point UUID	STRING	
Parent Installed Base_Parent Installed Base UUID	STRING	
Reference Date	DATETIME	
Serial ID	STRING	
Warranty_Warranty UUID	STRING	
Warranty_Warranty End Date	DATETIME	
Warranty_Warranty Start Date	DATETIME	

Template Header	Data Type	Notes
Sales Office_Sales Office UUID	STRING	
Sales Group_Sales Group UUID	STRING	
Sales Organization_Sales Organization UUID	STRING	
Division Code	STRING	
Distribution Channel Code	STRING	

5.6.18.2.2 Registered Product External IDs

Template Header	Data Type	Notes
parent_external_key	STRING	
id	STRING	
External ID	STRING	
Communication System ID	STRING	
Communication System UUID	STRING	
Type	STRING	
Default	BOOLEAN	

5.6.18.2.3 Registered Product Descriptions

Template Header	Data Type	Notes
parent_external_key	STRING	
content	STRING	
languageCode	STRING	

5.6.18.2.4 Registered Product Notes

Template Header	Data Type	Notes
parent_external_key	STRING	
content	STRING	

Template Header	Data Type	Notes
languageCode	STRING	

5.6.18.2.5 Registered Product Customer Information

Template Header	Data Type	Notes
parent_external_key	STRING	
content	STRING	
languageCode	STRING	

5.6.19 Warranty

Warranty is assigned to a registered product or installation point, and you can determine its coverage in a ticket.

5.6.19.1 Warranty Fields

Download the template from Data Import and Export for the field level information.

5.6.19.1.1 Warranty

Template Header	Data Type	Notes
external_key	STRING	
Status	STRING	
Duration	DATETIME	

5.6.19.1.2 Warranty External IDs

Template Header	Data Type	Notes
parent_external_key	STRING	
id	STRING	
External ID	STRING	
Communication System ID	STRING	
Communication System UUID	STRING	
Type	STRING	
Default	BOOLEAN	

5.6.19.1.3 Warranty Descriptions

Template Header	Data Type	Notes
parent_external_key	STRING	
content	STRING	
languageCode	STRING	

5.6.20 Users

Users in the system refers to employees, business users and other users such as partner contacts. Users are assigned to business roles and access controls in the system are implemented on business roles.

Each business user must actively possess at least one assigned business role that governs their access to the system.

5.6.20.1 User Fields

Download the template from [Data Import and Export](#) for field level information.

5.6.20.1.1 Users

The Users header entity contains all central information of a user.

Template Header	Data Type	Notes
external_key	STRING	Unique field used by data import and export for processing the respective import tasks
Technical ID	STRING	Technical reference ID for user based on integration scenarios
System Specific ID	STRING	
User Name	STRING	Name of the user
Status	STRING	Status of the user
Given Name	STRING	First name of the user
Family Name	STRING	Last name of the user
User Type	STRING	
Employee ID	STRING	
Security Policy_Technical ID	STRING	
Valid From	DATETIME	
Valid Until	DATETIME	
Admin Data_Created By	STRING	
Admin Data_Created On	DATETIME	
Admin Data_Changed By	STRING	
Admin Data_Changed On	DATETIME	

5.6.20.1.2 User Roles

The User Roles entity contains all the information of a user role.

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Role ID	STRING	

6 Configure Agent Desktop

Administrators can configure agent desktop in [Settings](#).

Find [Settings](#) in the user profile menu at the top of the screen.

→ Tip

Expand a collapsible section heading to view the configuration steps.

6.1 Configure General Settings

Enable automatic activity creation and open agent desktop automatically when a service agent accepts an incoming interaction.

Context

You can configure the solution to automatically create activity entities for either incoming or outbound interactions, or for both. This screen is also where you enable agent desktop to open automatically upon accepting an incoming interaction.

Procedure

1. Go to your user profile and select ► [System Settings](#) ► [Agent Desktop](#) ► [General Settings](#) ►.
2. Select the checkbox to enable one or more of these options:
 - [Create Activity from Interaction - Inbound](#) Create an activity record for each incoming interaction automatically.
 - [Create Activity from Interaction - Outbound](#) Create an activity record for each outgoing interaction automatically.
 - [Open Agent Desktop](#) Open the agent desktop when an agent accepts an incoming interaction.
3. Save your changes.

6.2 Configure Communication Provider

Configure communication system provider and widget size.

Prerequisites

A third-party communication system integrated with your solution. The communication system must be available as a widget or embedded frame inside the agent desktop interface. For more details, see the integration information in the related links that follow the steps.

Note

A third-party communication system provider is required for communication channels. Contracting a communication system provider, including all costs incurred by using the provider services, is your responsibility. These contracts and charges aren't part of SAP Service Cloud Version 2. The availability of the inbound and outbound call functionality depends on the services that your provider supports.

Restriction

The maximum number of communications providers is 5.

Procedure

1. Go to your user profile and select [System Settings](#) > [Agent Desktop](#) > [Live Interaction Widget Settings](#).
2. Under [Configure Widget and Provider](#), enter the required provider details:
 - [Provider Name](#)
 - [Provider ID](#)
 - [Provider URL](#) - **URL for the communication system widget**

3. Add channels. Select the add icon (+) at the top of the table to add a new channel.

Set the parameters depending on the channel type:

Channel Parameters

Type	ID	Identify Customer By
Phone	(unique channel ID)	Email
Chat	(unique channel ID)	Phone
SMS	(unique channel ID)	Phone
(Social Media Service)	(unique channel ID)	Email

4. Activate channel. Set the [Active](#) switch for the channel entry to on.

Interactions on a channel are only possible if the channel is active.

You can configure multiple channels of each type for your solution. For example: different phone numbers or SMS short codes with your communication provider for each support team can be configured as separate channels with unique IDs.

5. Save your changes.

6.3 Configure Timeline

Configure the events that create a timeline update for different entities and conditions.

Context

Different entities can subscribe to timeline events so that the event generates a timeline entry for that entity.

Procedure

1. Go to your user profile and select [System Settings](#) > [Timeline](#) > [Timeline Configuration](#).
2. Select [Event View](#) to list all events. Expand an event to view the event status.

Check the [Timeline Event Title](#) column to see how the event appears in the timeline with placeholders for entities and parties.
3. Select [Subscriber View](#) to see events grouped by subscribing entity. Expand the subscriber to view the subscribed events.
4. Toggle event visibility in the timeline in the [Show on Timeline](#) column.
5. Save your changes.

6.4 Configure Knowledge Base Provider

Configure your knowledge base provider to integrate with the native widget. Widget functions include: recommended articles based on linked case or registered product, article search, a top trending article list, copy article URL, and send article as email.

Prerequisites

Create a communication system for your knowledge base provider.

For knowledge base providers, the **outbound** protocol is **HTTPS**.

Find a link to complete details for setting up a communication system in the **Related Information** section.

Procedure

1. Go to your user profile and select ► [System Settings](#) ► [Integration](#) ► [Communication Configuration](#) ►.
2. Search for and select one of the following :
 - [Knowledge Base Provider: NICE CXone Expert \(Mindtouch\)](#)
 - [Knowledge Base Provider: SAP Build Work Zone](#)
 - [Knowledge Base Provider: Generic](#) - all other knowledge base providers
3. Select the [Copy](#) (📄) button in the provider details screen you opened.
4. Edit the name of your new communication configuration as desired.
5. Set the two API paths under [HTTP Data](#):
 - Retrieve articles based on ...
 - Get Top Trending Articles from ...
6. Set the [Active](#) switch to on and save the configuration.

Next Steps

Additional configuration is required if your knowledge base provider is [NICE CXone Expert \(Mindtouch\)](#).

1. Go to your user profile and select ► [System Settings](#) ► [Knowledge Base](#) ► [Provider Settings](#) ►.
2. Select [NICE CXone Expert \(Mindtouch\)](#) under [Knowledgebase Providers](#) and enter these required parameters:
 - [User](#) - user name for account
 - [Key](#)
 - [Secret](#)

Additional configuration is also required if your Knowledge base provider is SAP Build Work Zone. See the link in the **Related Information** section.

Related Information

[Create Communication Systems](#)
[Create Communication Configurations](#)
[Configure SAP Build Work Zone Native KB Widget Integration](#)

6.5 Configure Inbound Email Channels to Create Cases

Email messages routed to your solution can automatically generate cases that are sent to service agents for processing. You, as an administrator can add the inbound email addresses to use for case creation.

Prerequisites

- Decide which email addresses to use for customer support. There is currently a limit of 20 channels per tenant.
- In your organization's mail server, configure email forwarding from each support address to the technical address of your SAP Service Cloud Version 2 tenant. Find the tenant technical address on the email channel settings page.

Context

When an email message is routed to the solution, the system creates a case for the corresponding email channel according to the settings you configure for the channel. In addition, the solution can then route cases to a team or agent using routing rules you specify.

Procedure

1. Go to your user profile and select ► [System Settings](#) ► [Email](#) ► [Channel Settings](#) ►.
2. Add a channel for each inbound support address you wish to use. Select the create icon (+) at the top of the [All Email Channels](#) list to create a new channel.
3. Enter the support email address for this channel and enter a display name for the channel.

You can also enter an address to send a blind carbon copy (BCC) of outbound messages you send as replies to this channel. Some organizations use this secondary address as another way to track support responses.

4. Choose [Case](#) for [Object](#).

When you choose [Case](#), the [Type](#) and [Default Account](#) fields appear.

5. Choose the case type, and the default account for this channel.

All cases created from this channel use the case type you select here. The default account is assigned to the case if the solution is unable to determine a unique account or customer based on the **from** address on the email message.

6. Select [Save](#) to save the new email channel.

The technical email address for your tenant appears on this screen once you save the channel.

7. Select [Activate](#) to set the channel to [Active](#) status.

A two-step verification process is required before the channel is activated. A message is sent to the channel email address with a verification code. A second message is sent from the verification service with a link you use to activate the channel.

Email channels only create cases from incoming messages when the channel status is set to [Active](#) and the forwarding rules are in place on your organization's mail server.

Once a channel is in [Active](#) status, you can edit all attributes of the channel, **except** the channel email address. For example, you can edit the channel name, display name, object, case type, or default account.

Related Information

[Configure Case Routing to Employee \[page 194\]](#)

You, as an administrator, can set up rules that route cases to employees automatically.

[Configure Case Routing to Team \[page 195\]](#)

You, as an administrator, can set up rules that route cases to service teams automatically.

6.6 Create Email Templates

Email templates streamline email messaging with preset text, graphics, signatures, and attachments. Create templates for service responses, signatures, branding, and campaigns by uploading an HTML-based layout file.

Prerequisites

Prepare an HTML template layout. You can include technical placeholders to fill in data from the object record when the template is applied in the context of an associated object. See the steps following for how to find a list of placeholders.

Context

You can create email templates based on HTML layouts you create in your preferred HTML editor. Upload the HTML layout file and include any standard attachments for the template. You can also use placeholders to automatically pull data from associated objects in the system. For example: you can automatically fill in the customer name and ticket ID for a service response.

Procedure

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Email](#) ► [Templates](#) .
2. Add a new template. Choose + [\(Create\)](#) at the top of the [Templates](#) list to create a new template.
3. Enter a name for your new template.
4. Select a [Template Type](#):

- **Response:** use for service responses from your support teams.
- **Signature:** can include placeholders for employee information.
- **Branding:** create a custom brand-themed email template.
- **Campaign:** for newsletters and surveys.
- **Approval Response Template:** create a response template for approvals.
- **Approval Template:** create an approval request template.

The required fields for the template information vary depending on the type of template you select. Some types are only available for email channels. You can enter a description for response templates and campaign templates.

5. Select an [Object Type](#).

- Account
- Contact
- Individual Customer
- Case

Object type selection is limited based on the template type. Not all objects are available for every template type.

6. Select a [Channel Type](#) if necessary.

The channel type defaults to [Email](#) for most template types.

7. Set up the formatting for your template.

Depending on the type of template and object you selected in the previous steps, some or all of the following tabs are available for working with template formatting:

- **[Viewer](#):** Shows a preview of the template as it appears in the email editor. Placeholders are shown where relevant data is filled in once the template is applied.
- **[Documents](#):** Upload or change the HTML file defining the template layout here. Use any HTML editor you prefer to create your template layout.
- **[Attachments](#):** Upload any documents you want to include as standard attachments with email messages created from this template.
- **[Placeholder](#):** Displays a list of available data fields associated with the object type you selected for your template. Consult this list when creating your HTML layout file to insert placeholders for object data. Placeholders are formatted as the technical field name surrounded by double curly braces. For example: `{{familyName}}`, `{{givenName}}`.

Note

While creating or editing the HTML approval template, to add the source document link, use [Insert link](#) and enter `{{uiLink}}` in the [URL](#) field, and enter a name for the source document in the [Text](#) field. When approval email notification is received by the intended approvers, the approver

can access the source document by clicking on the link in the email with manual login and then navigate to the source document approval step.

8. Save your changes when finished.

7 Configure Timeline

Configure the events that create a timeline update for different entities and conditions.

Context

Different entities can subscribe to timeline events so that the event generates a timeline entry for that entity.

Procedure

1. Go to your user profile and select ► [System Settings](#) ► [Timeline](#) ► [Timeline Configuration](#) ►.
2. Select [Event View](#) to list all events. Expand an event to view the event status.
Check the [Timeline Event Title](#) column to see how the event appears in the timeline with placeholders for entities and parties.
3. Select [Subscriber View](#) to see events grouped by subscribing entity. Expand the subscriber to view the subscribed events.
4. Toggle event visibility in the timeline in the [Show on Timeline](#) column.
5. Save your changes.

8 Configure Knowledge Base Provider

Configure your knowledge base provider to integrate with the native widget. Widget functions include: recommended articles based on linked case or registered product, article search, a top trending article list, copy article URL, and send article as email.

Prerequisites

Create a communication system for your knowledge base provider.

For knowledge base providers, the **outbound** protocol is **HTTPS**.

Find a link to complete details for setting up a communication system in the **Related Information** section.

Procedure

1. Go to your user profile and select ► [System Settings](#) ► [Integration](#) ► [Communication Configuration](#) ►.
2. Search for and select one of the following :
 - [Knowledge Base Provider: NICE CXone Expert \(Mindtouch\)](#)
 - [Knowledge Base Provider: SAP Build Work Zone](#)
 - [Knowledge Base Provider: Generic](#) - all other knowledge base providers
3. Select the [Copy](#) (📄) button in the provider details screen you opened.
4. Edit the name of your new communication configuration as desired.
5. Set the two API paths under [HTTP Data](#):
 - Retrieve articles based on ...
 - Get Top Trending Articles from ...
6. Set the [Active](#) switch to on and save the configuration.

Next Steps

Additional configuration is required if your knowledge base provider is [NICE CXone Expert \(Mindtouch\)](#).

1. Go to your user profile and select ► [System Settings](#) ► [Knowledge Base](#) ► [Provider Settings](#) ►.
2. Select [NICE CXone Expert \(Mindtouch\)](#) under [Knowledgebase Providers](#) and enter these required parameters:
 - [User](#) - user name for account

- [Key](#)
- [Secret](#)

Additional configuration is also required if your Knowledge base provider is SAP Build Work Zone. See the link in the **Related Information** section.

Related Information

[Create Communication Systems](#)

[Create Communication Configurations](#)

[Configure SAP Build Work Zone Native KB Widget Integration](#)

9 Integrate External SAP Systems into Agent Desktop with Mashups

Use the screen adaptation function of Agent Desktop and the mashup capabilities built into SAP Service Cloud Version 2 to integrate external system functions into Agent Desktop. The example that follows integrates SAP S/4HANA or SAP S/4HANA Cloud Public Edition Service Management to create service orders and service quotation. As part of the standard integration, only service order and service quotation is supported.

Overview

Incorporate a back-office application, such as SAP S/4HANA Cloud Public Edition Service Management or SAP S/4HANA, into the front-office interface of SAP Service Cloud Version 2 Agent Desktop using a mashup..

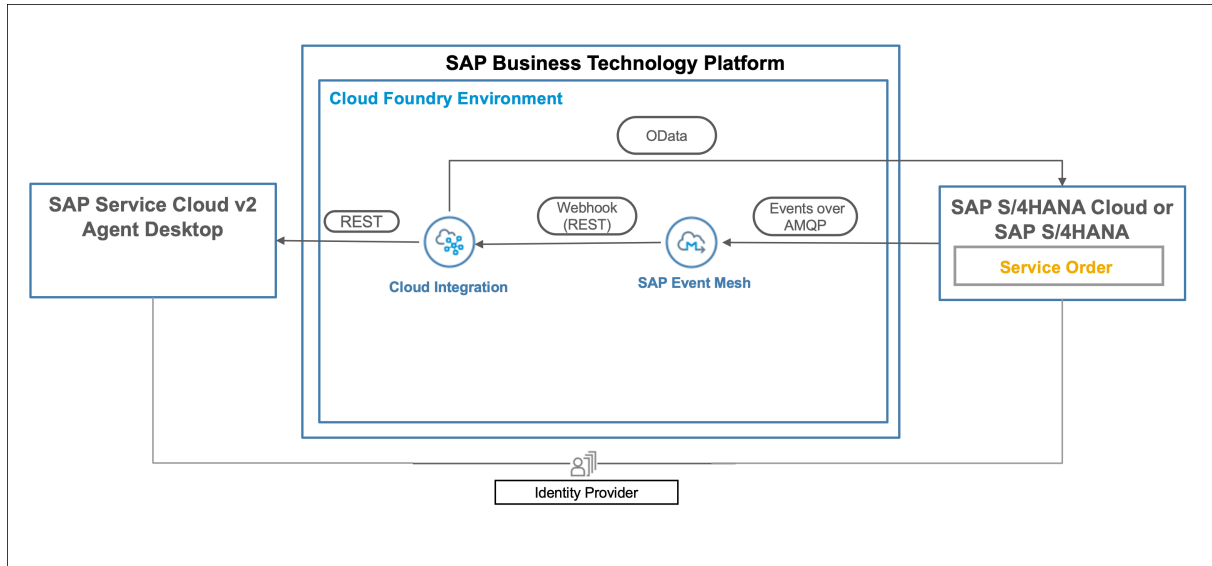
Prerequisites

You require the following software components for this example integration:

- SAP Service Cloud Version 2
- SAP Agent Desktop
- SAP S/4HANA or SAP S/4HANA Cloud Public Edition Service Management
- SAP Event Mesh
- SAP Cloud Integration

Architecture and Design

The following diagram illustrates this integration architecture:



Related Information

[SAP Event Mesh documentation](#)

[SAP Cloud Integration documentation](#)

9.1 Configure SAP Event Mesh

Configure SAP Event Mesh for integration between SAP S/4HANA or SAP S/4HANA Cloud Public Edition and SAP Service Cloud Version 2, Agent Desktop.

Context

Log on as an administrator to configure this integration.

Procedure

1. Go to the Cloud Cockpit of your BTP sub-account.
2. Navigate to the [Instance and Subscriptions](#) option.
3. Under [Instances](#), create a new [Service Instance](#) of the [Event Mesh](#) service, with a **Default** plan.

Provide the JSON parameters for the instance following this example:

Sample Code

```
{
  "options": {
    "management": true,
    "messagingrest": true,
    "messaging": true
  },
  "rules": {
    "topicRules": {
      "publishFilter": [
        "${namespace}/*"
      ],
      "subscribeFilter": [
        "${namespace}/*"
      ]
    },
    "queueRules": {
      "publishFilter": [
        "${namespace}/*"
      ],
      "subscribeFilter": [
        "${namespace}/*"
      ]
    }
  },
  "version": "1.1.0",
  "emname": "<Unique Instance name>",
  "namespace": "<Unique namespace value>"
}
```

Make a note of the [Namespace](#) for this service instance. You need it later for configuring the [Enterprise Event Enablement](#) in SAP S/4HANA Cloud.

4. Create a [Service Key](#) for this newly created event mesh instance.

Make a note of the **Client ID** and **Secret** of this key under the `amqp10ws` protocol. You need it later for configuring the [Communication System](#) in SAP S/4HANA Cloud.

5. Launch the [Event Mesh Administrator](#) user interface to create a subscription to the **Event Mesh** application.
6. Choose the service instance you created previously.

Find the service instance listed under [Message Clients](#).

7. Go to the [Queues](#) tab, and create a new queue. Prefix the queue name with the namespace you created and noted previously.

Once the queue is created, you can create subscriptions to the SAP S/4HANA Cloud outbound topics corresponding to service order events in the SAP S/4HANA Cloud solution.

8. Add subscriptions to your queue: select the queue you created in the previous step, and go to  [Actions](#)  [Create Subscriptions](#) .

For events you want to relay to the agent desktop timeline, create subscriptions using the following pattern:

<Message Client Namespace>**ce**<S4 Hana outbound topic name>

For example: s4toc4/v2/srvo/**ce**/sap/s4/beh/serviceorder/v1/ServiceOrder/Completed/v1

Other events supported currently are:

- /sap/s4/beh/serviceorder/v1/ServiceOrder/Created/v1
 - /sap/s4/beh/serviceorder/v1/ServiceOrder/Released/v1
9. Create a webhook in the event mesh instance. Go to the instance [WebHooks](#) tab. Create a new webhook and point it to the integration flow endpoint by providing the required authentication.

Related Information

[SAP Event Mesh documentation](#)

9.2 Configure SAP S/4HANA Cloud

Configure SAP S/4HANA Cloud for integration with SAP Service Cloud Version 2, Agent Desktop.

Procedure

1. Log on to your SAP S/4HANA Cloud solution and go to [Communication Management](#).
2. Create a new communication arrangement for the communication scenario: SAP_COM_0092 (Enterprise Eventing Integration).

Note

You also need to create two more communication arrangements once you finish this one.

3. Enter the value you noted for the event mesh instance namespace under [Additional Properties](#).
4. Enter a unique value for the property: [Channel](#).

Make a note of the channel property value. You need it later to configure the channel binding.

5. Create a [Communication System](#) with the host name set to your event mesh instance.

For example: **enterprise-messaging-messaging-gateway.cfapps.sap.hana.ondemand.com:443**

6. Enter the authentication and token endpoint in the **OAuth 2.0** settings. Use the **Service Key** token endpoint value that you created previously.
7. Enter the OAuth 2.0 client ID and client secret corresponding to the service key amqp protocol under [Outbound communication](#).

8. Enter your mesh instance with the `amqp` protocol under *Outbound services*.

For example: `https://enterprise-messaging-messaging-gateway.cfapps.sap.hana.ondemand.com:443/protocols/amqp10ws`

9. Activate the communication arrangement.

If you can't activate the communication arrangement, check and correct your entries made in the previous steps. Continue this procedure when the communication arrangement status is active.

10. Go to *Enterprise Event Enablement* *Configure Channel Binding* and search for the communication arrangement, using the value you entered for the **Channel** property.

Verify that the communication arrangement status is **Active**. If the status is anything other than **Active**, try reactivating the communication arrangement.

11. Go to the channel details and create *Outbound Topics* corresponding to the service order events that need to be relayed to the Agent Desktop timeline.

The topics configured in this channel must be the same as the subscriptions you created in the event mesh instance.

Events supported currently are:

- `/sap/s4/beh/serviceorder/v1/ServiceOrder/Created/v1`
- `/sap/s4/beh/serviceorder/v1/ServiceOrder/Released/v1`
- `/sap/s4/beh/serviceorder/v1/ServiceOrder/Completed/v1`

12. Go to **Administration** **Security** **Maintain Protection Allowlists**. In the *Clickjacking Protection* tab, click on the **+** icon and add the *Trusted Host Name*, *Schema*, and *Port* for your SAP Service Cloud tenant.

This step allows trusted access for SAP Service Cloud and Agent Desktop to your S/4 HANA Cloud tenant.

13. Create a new communication arrangement for the communication scenario: `SAP_COM_0350` (Service Order OData Integration).

This communication arrangement invokes the **Service Order OData API** from the integration flow. The log in credentials for the inbound user configured in this communication arrangement must be maintained in the integration flow.

14. Create a Communication Arrangement for the Communication Scenario: `SAP_COM_0008` (Business Partner, Customer, and Supplier Integration).

This communication arrangement invokes the **Business Partner OData API** from the integration flow.

9.3 Configure SAP S/4HANA

Configure SAP S/4HANA for integration with SAP Service Cloud Version 2, Agent Desktop.

Using the enterprise event enablement feature of SAP S/4HANA, events can be exchanged between SAP S/4HANA and event mesh. In order to enable SAP S/4HANA systems for enterprise messaging you need to create channels and maintain outbound event topic bindings for the channel. For more information, see [Creating, Configuring and Managing Channels](#)

9.4 Configure SAP Cloud Integration

Configure and deploy the SAP Cloud Integration flow (iFlow) between SAP Service Cloud Version 2, Agent Desktop, and SAP S/4HANA or SAP S/4HANA Cloud Public Edition.

9.4.1 Configure iFlow Parameters for SAP S/4HANA Cloud to SAP Service Cloud Service Order Integration

Prerequisites

As a prerequisite, go to ► [Operations View](#) ► [Security Material](#) ► [Add](#) ► [User Credentials](#) ► and deploy the following:

- User credentials to connect to SAP S/4HANA Cloud Public Edition system
- User credentials to connect to SAP Service Cloud Version 2.

Configure the iFlow Parameters

Configure the iFlow parameters for the SAP S/4HANA Cloud Public Edition to SAP Service Cloud integration.

- Configure Sender

Parameter	Description
Sender	Select S4_HANA_Cloud
Adapter Type	HTTPS
Address	<Webhook configured in event mesh>
Authorization	User Role
User Role	ESBMessaging.send

- Configure Receiver
 - S4_HANA_Cloud

Parameter	Description
Adapter Type	HTTP
S4 host URL	<SAP S/4HANA Cloud host URL>

Parameter	Description
Proxy Type	Internet
Credential Name	<User credential created for connecting to SAP S/4HANA Cloud>
User Role	ESBMessaging.send
Timeout (in ms)	60000

- Service_Cloud_V2

Parameter	Description
Adapter Type	HTTP
C4C host URL	<SAP Service Cloud V2 host URL>
Credential Name	<User credential created for connecting to SAP Service Cloud V2>
Timeout (in ms)	60000

- More configurations

Parameter	Description
Type	All Parameters
Agent Desktop Add-on	• False: For standalone scenario • True: For side by side scenario
Receiver Communication	Default communication system of Service Cloud version 2
Sender Communication	Communication system of Service Cloud version 2 which is configured for the communication configuration Integrate Service Entities with SAP S/4HANA Cloud

9.4.2 Configure iFlow Parameters for SAP S/4HANA to SAP Service Cloud Service Order Integration

Prerequisites

As a prerequisite, go to ► [Operations View](#) ► [Security Material](#) ► [Add](#) ► [User Credentials](#) ► and deploy the following:

- User credentials to connect to SAP S/4HANA system
- User credentials to connect to SAP Service Cloud Version 2.

Configure the iFlow Parameters

Configure the iFlow parameters for the SAP S/4HANA to SAP Service Cloud integration.

- Configure Sender

Parameter	Description
Sender	Select S4
Adapter Type	HTTPS
Address	<Webhook configured in event mesh>
Authorization	User Role
User Role	ESBMessaging.send

- Configure Receiver

- S4_HANA

Parameter	Description
Adapter Type	HTTP
S4 host URL	<SAP S/4HANA host URL>
Proxy Type	On-premise
Credential Name	<User credential created for connecting to SAP S/4HANA>
User Role	ESBMessaging.send
Timeout (in ms)	60000

- Service_Cloud_V2

Parameter	Description
Adapter Type	HTTP
C4C host URL	<SAP Service Cloud V2 host URL>
Credential Name	<User credential created for connecting to SAP Service Cloud V2>
Timeout (in ms)	60000

- More configurations

Parameter	Description
Type	All Parameters
Agent Desktop Add-on	• False: For standalone scenario • True: For side by side scenario
Receiver Communication	Default communication system of Service Cloud version 2

Parameter	Description
Sender Communication	Communication system of Service Cloud version 2 which is configured for the communication configuration Integrate Service Entities with SAP S/4HANA Cloud

9.5 Configure SAP Service Cloud

Configure SAP Service Cloud, Service Agent Desktop for integration with SAP S/4HANA Cloud.

Prerequisites

Master data replication of **Business Partner (BP)** in SAP S/4HANA or SAP S/4HANA Cloud Public Edition to **Customer** in SAP Service Cloud.

For this integration, every customer record in SAP Service Cloud requires a corresponding remote business partner in S/4HANA Cloud Service Management.

Context

⚠ Restriction

Individual customers aren't supported currently.

Procedure

1. Review Section 4 of the guide: **Setting Up Opportunity-to-Order with SAP Cloud for Customer**.
Find a link to this guide in the Related Links section.
2. Follow the procedure in section 4 to set up master data integration for business partners (BP).

Related Information

[Setting Up Opportunity-to-Order with SAP Cloud for Customer](#) 

9.6 Configure Service Agent Desktop

Configure SAP Service Cloud, Agent Desktop for integration with SAP S/4HANA or SAP S/4HANA Cloud Public Edition Service Management.

Context

An inbound communication configuration linking the Event Bridge service to the Inbound entity is delivered with the connector-service. Create the following artifacts for this inbound communication configuration:

- Communication System
- Communication Configuration

Procedure

1. Go to your user profile menu and choose ► [System Settings](#) ► [Integration](#) ► [Communication Configuration](#) .
2. Choose [Integrate Service Entities with SAP S/4HANA Cloud](#).
3. Select [Copy](#) icon.
4. Create a new communication configuration for the communication system you created .
5. Activate the communication configuration you created.

You don't need an outbound configuration for this integration.

9.7 Configure Mashup: Create Scenario

Configure a create scenario mashup in SAP Service Cloud Version 2 for integration with SAP S/4HANA or SAP S/4HANA Cloud.

Prerequisites

- The S/4HANA or S/4HANA Cloud URLs that are included in the mashup need to be added to a list of allowed URLs in SAP Service Cloud Version 2. You can do this by navigating to your user profile on the top-right corner of your screen and click ► [Settings](#) ► [All Settings](#) ► [General](#) ► [Content Security Policy Setting](#) and scroll to the section [Frame Source](#) and add the S/4HANA Cloud hostname URLs.

❖ Example

If the Service Order Create URL is `https://my123456.s4hana.ondemand.com/ui?sap-ushell-config=headerless#ServiceOrder-create` then, add `https://my123456.s4hana.ondemand.com/` in the [Frame Source](#).

Similarly, add the login URL of S/4HANA Cloud system in the Frame Source.

- Assign the customer service manager role `SAP_BR_CUSTOMER_SERVICE_MGR` to the user in S/4HANA.
- Assign the business catalog `SAP_CRM_BC_SERV_PROCCORDER_PC` to the user in S/4HANA Cloud Service Management.

For further details on assigning business catalogs to S/4HANA Cloud users, see the related links section following the steps.

Context

Procedure

1. Go to your user profile menu and choose [System Settings](#) [Foundation](#) [Mashup Authoring](#).
2. Create a new mashup by clicking the [+ Create](#) button.
3. Under the [General Information](#) section, enter a mashup name and description. Use a meaningful title. For example: Create Service Order.
4. Under the [Input Parameters](#) section, enter the service order create URL of your SAP S/4HANA or SAP S/4HANA Cloud tenant in the [URL](#) field.
 - Example URL for SAP S/4HANA: `https://my12345.org/sap/bc/ui5_ui5/ui2/ushell/shells/abap/FioriLaunchpad.html?sap-client=210&sap-ushell-config=headerless#ServiceOrder-create?sap-processtype=SRV0&sap-soldtoparty=171000030`
 - Example URL for SAP S/4HANA Cloud: `https://my123456.s4hana.ondemand.com/ui?sap-ushell-config=headerless#ServiceOrder-create?sap-processtype=SV01&sap-soldtoparty=17100004`
5. Click [Extract Parameters](#).
6. Under the [Parameter, Constant and API Key](#) section, ensure that the [sap-processtype](#) is set to SV01 and [sap-soldtoparty](#) is blank. You configure the parameter binding of the sap-soldtoparty by using the extensibility feature in Agent Desktop.

Post extraction of the parameters from the URL, the section should appear as:

Input Parameters

URL *

https://n.../ui?sap-ushell-config=headerless#ServiceOrder-create

Parameter, Constant and API Key

sap-processtype	•	SV01	•	☐ Mandatory	•	No System Parameter a...
sap-soldtoparty	•	No Constant added	•	☐ Mandatory	•	No System Parameter a...

7. [Save](#) and activate your mashup.

- Ensure to create a new mashup to use with agent desktop integration.
- Set the screen binding parameters in Agent Desktop via Extensibility and not in Mashup authoring.

Related Information

[Add Business Catalogs to Business Users in SAP S/4HANA Cloud](#)

9.8 Configure Mashup: Display Scenario

Configure a display scenario mashup in SAP Service Cloud Version 2 for integration with SAP S/4HANA Cloud.

Prerequisites

- Assign the customer service manager role `SAP_BR_CUSTOMER_SERVICE_MGR` to the user in S/4HANA.
- Assign the business catalog `SAP_S4CRM_BC_SRVC_DSP_PC` to the user in S/4HANA Cloud Service Management.

For further details on assigning business catalogs to S/4HANA Cloud users, see the related links section following the steps.

Procedure

1. Go to your user profile menu and choose [System Settings](#) [Foundation](#) [Mashup Authoring](#).
2. Create a new mashup by clicking the [+ Create](#) button.

- Under the [General Information](#) section, enter a mashup name and description. Use a meaningful title. For example: Display Service Order.
- Under the [Input Parameters](#) section, enter the service order display URL of your SAP S/4HANA or SAP S/4HANA Cloud tenant in the [URL](#) field.
 - Example URL for SAP S/4HANA: `https://my12345.org/sap/bc/ui5_ui5/ui2/ushell/shells/abap/FioriLaunchpad.html?sap-client=210&sap-ushell-config=headerless#ServiceOrder-display?ServiceOrder=ID`
 - Example URL for SAP S/4HANA Cloud: `https://my123456.s4hana.ondemand.com/ui?sap-ushell-config=headerless#ServiceOrder-display&/C_SrvOrdDocListRptAndObjPg('{ID1}')/?FCLLayout=MidColumnFullScreen`
- Click [Extract Parameters](#).
- Under the [Parameter, Constant and API Key](#) section, ensure that the [FCLLayout](#) is set to `MidColumnFullScreen` and `{ID}` is blank. You configure the parameter binding of the `{ID}` by using the extensibility feature in Agent Desktop.

Post extraction of the parameters from the URL, the section should appear as:

Input Parameters

URL *

`https://my12345.org/sap/bc/ui5_ui5/ui2/ushell/shells/abap/FioriLaunchpad.html?sap-client=210&sap-ushell-config=headerless#ServiceOrder-display?ServiceOrder=ID`

Parameter, Constant and API Key

FCLLayout	• MidColumnFullScreen	• ☐ Mandatory	• No System Parameter a...
{ID1}	• No Constant added	• ☐ Mandatory	• No System Parameter a...

- [Save](#) and activate your mashup.
 - Ensure to create a new mashup to use with agent desktop integration.
 - Set the screen binding parameters in Agent Desktop via Extensibility and not in Mashup authoring.

Related Information

[Add Business Catalogs to Business Users in SAP S/4HANA Cloud](#)

9.9 Adapt Agent Desktop for Mashup

9.9.1 Service Order

9.9.1.1 Create Service Order

Adapt the agent desktop screen to include the create scenario mashup in SAP Service Cloud Version 2 for integration with SAP S/4HANA Cloud.

Procedure

1. Go to [Agent Desktop](#).
2. Under the [Search](#) tab, [Confirm](#) an account and navigate to the [Customer Hub](#) tab.
3. Go to your user profile menu and choose [Start Adaptation](#)
4. Select the [What would you like to do?](#) section and select the **Edit Section** icon (✎) to open the [Section Items](#) pop-up box.
5. Select [Create](#) and click the **Create Button** icon (+).
6. In the [Add Buttons](#) dialog box, enter the [Button Text](#) as **Service Order** and enter an [Event Name](#).
7. In the Parameters section, ensure the parameter binding of sap-soldtoparty is set to [Fields/External Business Partner - Default ID](#) and click [Apply](#).

This links the service order to a customer account.

8. Select the tabs section and click the icon (✎) to open the [Edit Tabs](#) pop-up box.
9. Click the **Create Tab** icon (+) and provide a new tab name, for example: Create Service Order and press Enter.
10. Click the icon (➤) to launch Edit Tabs pop-up dialog again. Select the event you provided for service order create and click [Apply](#).
11. Click the icon (✎) under the Create Service Order screen area.
12. Click [Add Mashup](#) and select the mashup that you created for create service order.
13. In the Edit Properties window, set the value of [Height\(%\)](#) to 100 and ensure that sap-soldtoparty is set to [accountInfo/externalReferenceBusinessPartnerId](#) and click [Apply](#).
14. Click [End Session](#) to exit the adaptation mode and to view your adaptations on the screen.

9.9.1.2 View Service Order

Adapt the agent desktop screen to include the View Service Order Details mashup in SAP Service Cloud Version 2 for integration with SAP S/4HANA Cloud.

Procedure

1. Go to [Agent Desktop](#).
2. Under the [Search](#) tab, [Confirm](#) an account and navigate to the [Customer Hub](#) tab.
3. Go to your user profile menu and choose [Start Adaptation](#).
4. Select the tabs section and click the icon (✎) to open the [Edit Tabs](#) pop-up box.
5. Click the **Create Tab** icon (+) and provide a new tab name, for example: Display Service Order and press Enter.
6. Click the icon (➤) to launch Edit Tabs pop-up dialog again. Select the event you provided for service order display and click [Apply](#).
7. Click the icon (✎) under the Display Service Order screen area.
8. Click [Add Mashup](#) and select the mashup that you created for display service order.
9. In the Edit Properties window, set the value of [Height\(%\)](#) to 100 and ensure that {ID} is set to [s4ServiceOrderInfo/serviceOrderId](#) and click [Apply](#).
10. Click [End Session](#) to exit the adaptation mode and to view your adaptations on the screen.

9.9.1.3 Create an SAP S/4HANA Service Order from Agent Desktop Mashup

Create an SAP S/4HANA Service Order from Agent Desktop mashup by connecting to SAP Service Cloud Version 2.

Context

Now that the mashup is configured, and the Agent Desktop screen is adapted to show the mashup, service agents can create a service order directly in the back-office SAP S/4HANA Cloud Service Management application from Agent Desktop screen:

Procedure

1. Log in to your SAP Service Cloud Version 2 solution.

2. Choose [Service](#) [Agent Desktop](#).
3. [Search](#) for an account and click [Confirm](#).
4. Under the [Customer Hub](#) tab, choose [Create](#) [Service Order](#).

The [Service Order](#) tab loads the SAP S/4HANA Cloud system's [Create Service Order](#) screen using the mashup.

5. Enter a description and fill-in other details in the service order and click [Save](#).

The [Timeline](#) tab opens with a new [Service Order](#) created event on the timeline.

6. To see a quick view of service order, click on the [Service Order ID](#) in the timeline event.

To see a detailed view of the service order in new tab in the [Customer Hub](#), click the [🔗](#) icon.

Note

Only users with a role that has access to the service `sap.crm.service.s4hcServiceOrderService` can view the Service Order quick view. You can assign business roles under [User](#) [Settings](#) [Users and Control](#).

9.9.2 Service Quotation

9.9.2.1 Create Service Quotation

Adapt the agent desktop screen to include the create scenario mashup in SAP Service Cloud Version 2 for integration with SAP S/4HANA Cloud.

Procedure

1. Go to [Agent Desktop](#).
2. Under the [Search](#) tab, [Confirm](#) an account and navigate to the [Customer Hub](#) tab.
3. Go to your user profile menu and choose [Start Adaptation](#).
4. Select the [What would you like to do?](#) section and select the **Edit Section** icon ([🔗](#)) to open the [Section Items](#) pop-up box.
5. Select [Create](#) and click the **Create Button** icon ([+](#)).
6. In the [Add Buttons](#) dialog box, enter the [Button Text](#) as **Service Quotation** and enter an [Event Name](#).
7. In the Parameters section, ensure the parameter binding of `sap-soldtoparty` is set to [Fields/External Business Partner - Default ID](#) and click [Apply](#).

This links the service order to a customer account.

8. Select the tabs section and click the icon (✎) to open the [Edit Tabs](#) pop-up box.
9. Click the **Create Tab** icon (+) and provide a new tab name, for example: Create Service Quotation and press Enter.
10. Click the icon (➤) to launch Edit Tabs pop-up dialog again. Select the event you provided for service quotation create and click [Apply](#).
11. Click the icon (✎) under the Create Service Quotation screen area.
12. Click [Add Mashup](#) and select the mashup that you created for create service quotation.
13. In the Edit Properties window, set the value of [Height\(%\)](#) to 100 and ensure that `sap-soldtoparty` is set to [accountInfo/externalReferenceBusinessPartnerId](#) and click [Apply](#).
14. Click [End Session](#) to exit the adaptation mode and to view your adaptations on the screen.

9.9.2.2 View Service Quotation

Adapt the agent desktop screen to include the View Service Quotation Details mashup in SAP Service Cloud Version 2 for integration with SAP S/4HANA Cloud.

Context

Procedure

1. Go to [Agent Desktop](#).
2. Under the [Search](#) tab, [Confirm](#) an account and navigate to the [Customer Hub](#) tab.
3. Go to your user profile menu and choose [Start Adaptation](#).
4. Select the tabs section and click the icon (✎) to open the [Edit Tabs](#) pop-up box.
5. Click the **Create Tab** icon (+) and provide a new tab name, for example: Display Service Quotation and press Enter.
6. Click the icon (➤) to launch Edit Tabs pop-up dialog again. Select the event you provided for service quotation display and click [Apply](#).
7. Click the icon (✎) under the Display Service Quotation screen area.
8. Click [Add Mashup](#) and select the mashup that you created for display service quotation.
9. In the Edit Properties window, set the value of [Height\(%\)](#) to 100 and ensure that `{ID}` is set to [s4ServiceOrderInfo/serviceOrderId](#) and click [Apply](#).
10. Click [End Session](#) to exit the adaptation mode and to view your adaptations on the screen.

9.9.2.3 Create an SAP S/4HANA Service Quotation from Agent Desktop Mashup

Create an SAP S/4HANA Service Quotation from Agent Desktop mashup by connecting to SAP Service Cloud Version 2.

Context

Now that the mashup is configured, and the Agent Desktop screen is adapted to show the mashup, service agents can create a service quotation directly in the back-office SAP S/4HANA Cloud Service Management application from Agent Desktop screen:

Procedure

1. Log in to your SAP Service Cloud Version 2 solution.
2. Choose **Service** > **Agent Desktop**.
3. **Search** for an account and click **Confirm**.
4. Under the **Customer Hub** tab, choose **Create** > **Service Quotation**.

The **Service Quotation** tab loads the SAP S/4HANA Cloud system's **Create Service Quotation** screen using the mashup.

5. Enter a description and fill-in other details in the service quotation and click **Save**.

The **Timeline** tab opens with a new **Service Quotation** created event on the timeline.

6. To see a quick view of service quotation, click on the **Service Quotation ID** in the timeline event.

To see a detailed view of the service quotation in new tab in the **Customer Hub**, click the  icon.

Note

Only users with a role that has access to the service `sap.crm.service.s4hcServiceOrderService` can view the Service Quotation quick view. You can assign business roles under **User** > **Settings** > **Users and Control**.

10 Configure Cases

Administrators can configure cases in [Settings](#).

Find [Settings](#) in the user profile menu at the top of the screen.

→ Tip

Expand a collapsible section heading to view the configuration steps.

10.1 Configure Status Dictionary Entries

You, as an administrator, can use your own case statuses in your processes and schemas. Create the [Status Dictionary Entries](#) and then assign them a lifecycle and assignment status.

Context

The lifecycle statuses are used for service level agreement (SLA) calculations. Assignment statuses are used to identify the assignee of the case (the processor and requestor) to help case duration determination.

The following table lists the predelivered status values that you can select in the case and the next status you can reach from there. When you create your statuses, keep these possibilities in mind.

Current Status	Reachable Statuses	Non-Reachable Statuses	Description
Open	• In Process • Customer Action • Completed	• Closed	This status isn't reachable from any other value.
In Process	• Customer Action • Completed	• Open • Closed	If the status changes from In Process to Closed the system completes and closes the case.
Customer Action	• In Process • Completed	• Open • Closed	
Completed	• In Process • Customer Action • Closed	• Open	

Current Status	Reachable Statuses	Non-Reachable Statuses	Description
<i>Closed</i>	None	<i>Open</i> <i>In Process</i> <i>Customer Action</i> <i>Completed</i>	If a case is closed, it isn't possible to open it again. This status can only be used for cases still in the system. New cases aren't allowed to use <i>Closed</i>. Note Once a case is set to closed, you can't change the case status.

Procedure

- Go to [User Menu](#) > [System Settings](#) > [All Settings](#) > [Cases](#) > [Status Dictionary Entries](#).
- Select the add icon (+) at the top of the table to add a new status code.
- Enter a code and a description for your new status.

Custom status codes have to start with **Z** or **Y**.

- Select the *Lifecycle Status* and *Assignment Status* for your new code.

Allowed Combinations

Lifecycle Status	Assignment Status
Open	Processor Action
In Process	Processor Action
In Process	Requestor Action
Completed	Requestor Action
Closed	Not Assigned

- Select the confirm icon (✓) to save changes.

You can delete a status code by selecting the delete icon (🗑) in the *Action* column.

Note

You can change the description of the predelivered statuses, but you can't delete them.

Related Information

[Configure Status Schemas \[page 187\]](#)

You, as an administrator, can edit the standard status schemas, and create new schemas for cases.

10.2 Configure Status Schemas

You, as an administrator, can edit the standard status schemas, and create new schemas for cases.

Context

Change the visibility, and sequence of the standard statuses. Use status schemas to realize different processes in your company. For example: Create a status schema for use with your internal employee support, and a different status schema valid for interactions with partners.

Procedure

1. Go to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Cases](#) ► [Status Schemas](#) ►.
2. Select the add icon (+) at the top of the table to add a new status schema.
3. Enter an ID for your new schema in the [Status Schema](#) field. The ID consists of two characters and starts with either **Z** or **Y**.
4. Enter a description for your new schema.
5. Select the add icon (+) at the top of the table to [Add Status to Schema](#).

This step adds a row to the table where you can select from the status dictionary entries you've previously defined. Select the option if this is the [Initial Status](#) for the schema, and if you want the [Status Visible](#) for the case.

- Only the [Open Life Cycle Status](#) can be selected as initial status.
 - You can select the [Closed Life Cycle Status](#) but you can't set it to visible.
6. Save your changes.

You can [Hide](#) a status, or [Set as Visible](#) from the [Actions](#) menu (⋮).

ⓘ Note

Only statuses not currently in use can be deleted.

Related Information

[Configure Status Dictionary Entries \[page 185\]](#)

You, as an administrator, can use your own case statuses in your processes and schemas. Create the [Status Dictionary Entries](#) and then assign them a lifecycle and assignment status.

10.3 Configure Priority Color

Assign a new color to any of the four priority codes.

Procedure

1. Go to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Cases](#) ► [Priorities](#) ►.
2. Select the edit (✎) icon to change the color assigned to a priority code.
3. Select a new color from the menu.

Your changes are saved automatically.

10.4 Configure Service Levels

You, as an administrator, can create new service levels in your solution.

Context

Set up service level based on target milestones with response times, and specifying working hours and timezone.

Procedure

1. Go to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Cases](#) ► [Service Levels](#) ►.
2. Enter a name for the new service level.
3. Add target milestones.
4. Select a milestone type.
 - [Due Date for Completion](#)
 - [Due Date for Initial Review](#)
 - [Due Date for Resolution](#)
 - [Due Date for Response](#)

→ Tip

If you enter the duration in hours, the system only considers operating hours. If you enter the duration in days, the system uses the [Working Configuration](#), but not the time ranges. You enter this information under [Operating Hours](#) in the next step.

❖ Example

Suppose that your working days are from Monday to Friday, from 8:00 to 18:00 and the reporting date is Wednesday 06 April 2022, 15:00.

- If you enter a duration of 2 days the system calculates a due date on Friday, 08 April 2022, 15:00.
- If you enter a duration of 48 hours the system calculates a due date on Wednesday, 13 April 2022, 13:00.

📘 Note

Attribute specific service level definition is not supported. You need to create multiple service level definitions for different attributes with different durations for milestones. For example, Priority.

5. Set the *Working Day Calendar*, *Time Zone*, and *Working Configuration* daily time ranges.

Working Configuration allows you to set the hours when your service staff is available. If you don't enter operating hours, the solution assumes that service is available 24x7.

→ Tip

If you want to intentionally set up 24x7 availability, create a calendar with all 7 days as working days and 12 AM to 12 AM as working hours.

You can use customer working hours for milestone calculations by maintaining a working day calendar, time zone, and working configuration consistent with the customer location.

6. Save the new service level. The service level now appears in the *Service Levels* list.
7. Activate the new service level.

Related Information

[Configure Service Level Determination \[page 190\]](#)

You, as an administrator, can set up determination rules that assign service levels to cases automatically.

10.5 Configure Service Level Determination

You, as an administrator, can set up determination rules that assign service levels to cases automatically.

Context

When you create or edit a case, the system uses determination rules to evaluate which service level to apply. When a rule is found that matches the case, the system applies the appropriate service level. The service level pushes details about reaction times and due date calculations to the case.

Procedure

1. Go to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Cases](#) ► [Service Level Determination](#) ►.
2. Set up the rules table with [Adapt Columns](#).

The first time you set up business rules you must configure the rules table. You can have up to 15 columns in the table, including the results column.

→ Tip

Use only the columns required for assigning objects. Extra columns can impact performance. You can add additional columns and rules at any time.

3. Select the add icon (+) at the top of the table to add a new rule. A new row with an asterisk in each field of the row is added to the table.
4. Click the asterisk symbol (*) in a cell to enter a value for that field. The rules table offers a variety of logical operators to define your values.

You can add values to more than one field in a row. The fields in one row evaluate with a logical **AND** relationship. The field values must **all** evaluate to true in order for the rule to evaluate to true.

5. Continue to add rows for each desired result.

The system compares rows using the logical **OR** operator. The system compares each rule, top to bottom, until **ONE** evaluates to **true**.

Rearrange rows by selecting the row and selecting the sort icon (⬆⬆).

6. Select the confirm icon (✓) to save changes to the row when finished.
7. Click [Activate](#).

Enable your changes by activating the new rules.

→ Tip

Once you've set up your columns and you're familiar with the operators and table properties, you can export the table to a Microsoft Excel file. If you have a large number of service level rules, you can edit the file off line and then import the rules file back into your solution.

Related Information

[Configure Service Levels \[page 188\]](#)

You, as an administrator, can create new service levels in your solution.

10.6 Configure Case Types

Customize case types to match the workflow used by your organization.

Context

Create case types to model any type of service workflow based on your business processes. You can add phases, steps, approvals, automated flows, mashups, and feedback to your case type. Phases and steps guide your agents visually through processing the case.



Standard Case Statuses

Status Code	Status
01	Open
02	In Process
04	Customer Action
05	Completed
06	Closed

Procedure

1. Go to [User Menu](#) > [System Settings](#) > [All Settings](#) > [Cases](#) > [Case Types](#).

→ Tip

Select the pin icon (📌) to add [Case Types](#) to the [Dashboard](#) for quick access.

2. Select a case type in status *Active* that you want to use as a starting point to customize and create your own case type. You can use the *Sample Case Type* provided by SAP to get started.
3. Select *Copy to New Case Type* from the *Action* menu (⋮) for an active version of the case type template.
4. Enter a case type code starting with the letter **Z**. The type code can be up to four characters in length, with no spaces or special characters.
5. Enter a description for your new case type.
6. Choose the relevant business units and select *Create*.
7. The Case Template screen opens where you can edit the *General Settings*, *Case Flow*, *Relationship Settings*, *Additional Features*, and *Feature Settings*.
8. Go to the *Case Flow* tab to create Phases and Steps. Phases contain one or more specific steps. Steps are discrete actions that can be:
 - **To-Do:** Tasks or actions that can include explanatory text. If a step is mandatory, the agent must complete the task or action before the case can proceed to the next step. The agent must confirm manually once they complete the step.
 - **Autoflows:** Triggers an autoflow action. For example: update a field in an object, create a time line event, or send an email message. You can use autoflow steps in addition to autoflows triggered at the case level.
 - **Approvals:** Use approval steps to require that the agent request approval before continuing to process the case. Agents can include notes before submitting the approval request. You can use approval steps in addition to approvals triggered at the case level.
 - **Mashup:** A step consisting of a mashup screen from an external system. Integrate external systems directly into your case workflow.
 - **Feedback:** Set up a feedback step with channel, template, customer response survey, and optional preconditions. Once the agent reaches this step and conditions are met, the solution sends an email message containing a link to the survey and collects survey feedback.
 - **Input:** Use Input to embed case fields and attributes into the case runtime flow. The data entered is persisted onto relevant case fields across tabs and sections. This helps the service agent to focus on processing on the same screen.
 - **Assignment:** Use Assignment to enable automatic updates to field values based on the defined actions.
9. Add, edit, reorder, or remove phases and steps to set up the case template to reflect your desired business process.

Drag the move icon (≡) to reorder phases and steps.

10. (Optional) Add step preconditions. Edit the desired step to include conditions that evaluate if the step is ready to be processed.

Preconditions determine if a particular step can be processed or not. Agents can only set the step as done when the preconditions are met.

When preconditions are **not** met:

- Mandatory step: agents can proceed with other steps, but can't complete the current phase and continue to the next phase.
 - Optional step: agents can proceed with next steps as well as continue to next phase.
 - First step of the first phase: the solution takes agents to the next step that can be processed.
11. (Optional) Select *View/edit number range table* to select an existing number range for the case type, or create a new number range.
 12. (Optional) Select a service catalog for the case type.

Note

You can only select catalogs set up for use with cases. If you want to associate a service catalog to your case type, you must select a catalog when the case type is in draft status. You cannot add a catalog later.

13. Go to [Relationship Settings](#) and select a completion mechanism for main and subcases.
14. Select a business document from the [Additional Features](#) tab.
15. Enable the required features under the [Features Setting](#) tab.
16. Save your changes.
 - [Activate](#) sets your case type to active status.
 - [Close](#) saves your case type as a draft. You can activate it later.

Related Information

[Autoflow](#)
[Approval](#)

10.7 Configure Party Schemas

You, as an administrator can configure [Parties](#) for cases.

Context

You can determine automatically all involved parties for business transactions and their related documents using party roles and determination rules. These rules allow you to streamline account team assignments, and ensure that business partners are correctly assigned to business objects in a way that matches your company processes. You can decide which party roles to use in your organization, however some party roles are mandatory in some standard party schemas, and can't be deactivated.

Procedure

1. Go to [User Menu](#) > [System Settings](#) > [All Settings](#) > [Cases](#) > [Party Schema](#).
2. Add or edit party roles under [Party Role Assignments](#). Select the add icon (+) at the top of the table to add a new party role assignment.

You can also select the edit () icon to edit the existing party role assignments.

3. Select a *Party Role* from the menu.
4. Select the appropriate toggles to set the role to: active, mandatory, or unique.
5. Click  against any party role to open the *Party Role Determination Steps* screen where you can select the determination steps you want to keep active.

Note

When you check the *Event: Enforce consistency with source party during all subsequent determination steps* box for both *40 - Processor* and *28 - Service and Support Team* Party Role, it enables the system to maintain consistency between the assigned agent and the support team by automatically clearing or updating fields based on routing rules whenever one is manually changed, ensuring agents are only assigned within valid team structures.

10.8 Configure Case Routing to Employee

You, as an administrator, can set up rules that route cases to employees automatically.

Procedure

1. Go to  *User Menu*  *System Settings*  *All Settings*  *Cases*  *Case Routing to Employee* .
2. Set up the rules table with *Adapt Columns*.

The first time you set up business rules you must configure the rules table. You can have up to 15 columns in the table, including the results column.

→ Tip

Use only the columns required for assigning objects. Extra columns can impact performance. You can add additional columns and rules at any time.

3. Select the add icon (+) at the top of the table to add a new rule. A new row with an asterisk in each field of the row is added to the table.
4. Click the asterisk symbol (*) in a cell to enter a value for that field. The rules table offers a variety of logical operators to define your values.

You can add values to more than one field in a row. The fields in one row evaluate with a logical **AND** relationship. The field values must **all** evaluate to true in order for the rule to evaluate to true.

5. Continue to add rows for each desired result.

The system compares rows using the logical **OR** operator. The system compares each rule, top to bottom, until **ONE** evaluates to **true**.

Rearrange rows by selecting the row and selecting the sort icon (⬆).

6. Select the confirm icon (✓) to save changes to the row when finished.

7. Click [Activate](#).

Enable your changes by activating the new rules.

→ Tip

Once you've set up your columns and you're familiar with the operators and table properties, you can export the table to a Microsoft Excel file. If you have a large number of service level rules, you can edit the file off line and then import the rules file back into your solution.

Related Information

[Configure Case Routing to Team \[page 195\]](#)

You, as an administrator, can set up rules that route cases to service teams automatically.

10.9 Configure Case Routing to Team

You, as an administrator, can set up rules that route cases to service teams automatically.

Procedure

1. Go to [User Menu](#) > [System Settings](#) > [All Settings](#) > [Cases](#) > [Case Routing to Team](#).
2. Set up the rules table with [Adapt Columns](#).

The first time you set up business rules you must configure the rules table. You can have up to 15 columns in the table, including the results column.

→ Tip

Use only the columns required for assigning objects. Extra columns can impact performance. You can add additional columns and rules at any time.

3. Select the add icon (+) at the top of the table to add a new rule. A new row with an asterisk in each field of the row is added to the table.
4. Click the asterisk symbol (*) in a cell to enter a value for that field. The rules table offers a variety of logical operators to define your values.

You can add values to more than one field in a row. The fields in one row evaluate with a logical **AND** relationship. The field values must **all** evaluate to true in order for the rule to evaluate to true.

5. Continue to add rows for each desired result.

The system compares rows using the logical **OR** operator. The system compares each rule, top to bottom, until **ONE** evaluates to **true**.

Rearrange rows by selecting the row and selecting the sort icon (↕).

6. Select the confirm icon (✓) to save changes to the row when finished.
7. Click [Activate](#).

Enable your changes by activating the new rules.

→ Tip

Once you've set up your columns and you're familiar with the operators and table properties, you can export the table to a Microsoft Excel file. If you have a large number of service level rules, you can edit the file off line and then import the rules file back into your solution.

Related Information

[Configure Case Routing to Employee \[page 194\]](#)

You, as an administrator, can set up rules that route cases to employees automatically.

11 Configure Registered Products

As an administrator, you can add new categories and custom party types in addition to the default values.

Procedure

1. Navigate to ► *User Menu* ► *System Settings* ► *All Settings* ► *Categories* ► *Registered Products Category* .
2. Select  *(Create)*.
3. Enter the *Code* and *Description*.
4. Save your changes.

Select  *(Delete)* to delete an existing value.

12 Create Installation Points

From the [Installation Point](#) tab, you can create a new installation point only with the category of functional location.

In addition to the common features, this category of installation point supports warranty, sales and distribution data, and replication from SAP S/4 HANA. The customer information and installed base sections are prepopulated when you create an installation point from an installed base.

Installation points that are categorized as registered products are only displayed in the [Registered Product](#) tab. You can assign a registered product to an installation point. You can also assign an existing registered product to a current installation point.

To assign an installation point, with product or text category, to an installed base, create the installation point from the [Installed Base](#).

Note

You can add a functional location to an installed base using a migration template.

13 Configure Installed Bases

As an administrator, you can add new custom part types in addition to the default values.

Procedure

1. Navigate to ► *User Menu* ► *System Settings* ► *All Settings* ► *Installed Base* ► *Custom Party Type* ►.
2. Select **+** *(Create)*.
3. Enter the *Code* and *Description*.
4. Save your changes.

Select  *(Delete)* to delete an existing value.

14 Output Management

Output management comprises all activities related to the output of documents. Form-based documents can be output on an ad hoc basis or as an integrated part of a business process.

14.1 Main Templates

As an administrator, you can create company-specific main templates for use in form templates.

Form main templates are used to define the logo, header, sender address, and language-dependent footer that are used in form-based business documents.

SAP provides one default form main template for your solution. If you require a different logo or sender address for different companies or org units in your organization, you can create additional form main templates. These can then be assigned to the relevant company/org unit.

If you do not want to use any form main template for a business document, you can deactivate the use of main templates in the form template. If you specify that a main template must not be used for a form template, then you can alternatively create a unique header, logo, and footer for the form template using Adobe® LiveCycle Designer.

When a business document is output, the system uses the following logic to determine which form main template to use:

- The system searches for a form main template assigned to the company/org unit in which the business document was created. For example, if an invoice is created in the invoicing unit *Sales US West Coast* and is to be sent by email, the system tries to find a main template assigned to the org unit *Sales US West Coast*.
- If no match is found, the system searches upwards in the organizational structure for a main template assigned to a higher-level org unit.
- If no match is found, the system uses the default main template.

14.1.1 Create Main Templates

If you use templates, creating company-specific main templates makes all templates consistent, saving you time. You can define company-specific main templates to use as a basis for other forms.

Procedure

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Output Management](#) ► [Maintain Main Template](#) ►.

2. Choose **+** (*Create*)..
3. Enter a name and assign org unit (if any) for your new main template.

Use the *Is Default* toggle button to mark the template as the default template.

4. In the *Design Main Template* tab, enter the header and footer information.
 - a. In the *Header* tab, upload a logo.

Adjust the size and alignment of the logo, as required. The file formats .gif, .bmp, .jpg, .png, and .tif are supported. The logo must be lesser than 40 mm in width and 20 mm in height. If you upload a graphic that is larger than 40 mm x 20 mm, it is resized automatically.

- b. Use the *Header Divider* toggle button to add a graphical divider on your form.
 - c. Enter your company's address as the sender address. The sender address is shown above the recipient's address in a letter.
 - d. In the *Footer* tab, choose the number of footer columns you want to display.

Enter the footer text exactly as you would like it to appear on the form. You can maintain up to four footer blocks with a maximum of 10 lines per footer.

Note that the width of each footer block is equally divided over the available space. If a line of text in the footer is too long for the current font size, the system automatically enters a line break. If this automatic line break does not meet your needs, you can enter the text after the line break as a separate line.

5. Preview your changes by refreshing the screen.
6. Save your changes.

You can now assign the main template to one or more companies, org units, or output channels.

14.1.2 Edit Form Main Templates

As an administrator, you can update the main template design or assign organizational units to the template.

Procedure

1. Navigate to **User Menu** > **System Settings** > **All Settings** > **Output Management** > **Maintain Main Template**.
2. Select any main template that you want to edit and choose **Edit**.
3. Update the name and org unit details in the *Create Main Template View* tab or the header and footer details in the *Design Main Template* tab.
4. Save your changes.

14.2 Form Templates

Use form templates to define the content and layout of documents that can be output from the system.

Form templates are in the form of print forms that are read-only portable document format (PDF) documents that are generated from data stored in the system. Print forms can be printed, sent as an email attachment, or faxed to business partners.

Variants

Variants of a form template are copies of the content displayed in other languages, or with information specific to that country/region. Language variants are used to provide translations of form templates into other languages.

When a document is output from the system, the system automatically determines which language version to use. Most form templates are intended for internal review. Therefore the system uses the form template matching the language selected by the user on the log on screen. Certain objects are intended for review by a business partner or customer, and the system selects a template based on the recipient language.

Language variants are determined by the user log-on language, or in certain cases, by the preferred language of the receiving business partner. For example, if a customer in a German company sends product information to a business partner in France, the system searches for a French language variant of the form template and uses this to generate the document. If a language variant is not available in the recipient's language, then the form template in the fallback language English is used.

You can use the appropriate buttons to delete form templates or variants.

Restriction

It is not possible to delete a form template that has already been used to output a document because that template is referenced in a document's output history.

14.2.1 Create Form Templates

You can create custom form templates, but we recommend editing templates provided by SAP, which saves you setup time.

Context

→ Recommendation

While it's possible to create new form templates using this procedure, we recommend that you edit form templates provided by SAP or partners instead. Editing provided templates means that you don't have

to adapt any output settings. It is not possible to copy or edit print forms that contain legally-sensitive content.

Procedure

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Output Management](#) ► [Maintain Form Template](#) ►.
2. Choose a predelivered template, choose ⋮ and select [Copy Template](#).
3. Enter a name and description and select a language.
4. If you are creating a form template specific to a language, select the language. If the template is not specific to language, leave this field blank.
5. Turn on the [Form Main Template](#) switch button to use master templates to determine the layout of the template.
6. Save and publish your changes.

When a new version of a predelivered template is released, the new version of the predelivered template is automatically set to active. To avoid any changes to the existing version, create a copy of the predelivered template and use the custom template while selecting the form template. If you want to make changes in the newer predelivered templates, create a copy of the same and start using it. Else, continue using the existing custom template.

14.2.2 Edit Form Templates

As an administrator, you can update the name and description of a form template, upload a form template variant, or download a template and make changes using Adobe LiveCycle Designer.

Prerequisites

You have installed Adobe LiveCycle Designer and installed SAP CNSADDINS (SUPPORT PACKAGES AND PATCHES). To download and install or update the Adobe LiveCycle Designer, follow the instructions in the SAP note [2187332](#) (download and installation of version 11.0).

Context

→ Recommendation

We recommend that you edit form templates provided by SAP or partners instead of creating new form templates. Editing provided templates means that you do not have to adapt any output settings and

you can also manage all versions of the template. Form template versions that you customize are not overwritten by SAP or partner upgrades.

Procedure

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Output Management](#) ► [Maintain Form Template](#) ►.
2. Click ⋮ and select [Edit](#).
3. Update the name and description of a form template, upload a form template variant, or download a template and make changes using Adobe LiveCycle Designer.

By default, the [Form Main Template](#) switch is on. If you do not want to use company-specific main templates, turn off the switch. Note that only .xdp files can be uploaded.

4. Save and publish your changes.

In the Adobe LiveCycle Designer, you can make changes such as the following:

- Change position of title, logo, and fields
- Change alignment of title and logo (unless you define these with main templates)
- Change title and template texts
- Change sender address, number, and content of footers (unless you define these with main templates)
- Change style and size of font
- Change line style and add borders
- Adjust visibility of fields, change field labels, and the sequence of fields
- Adjust visibility of columns, change column order, and column headers
- Add standard fields available in the form data structure
If you want to add new fields that are not available in the form data structure, you can add extension fields. Other fields require support from the SAP Cloud Service Center.
- Add placeholders to free text blocks
- Add new texts or captions
- Show and hide page numbers or folder marks

Note

If you require complex changes to a print form, you can also contact the SAP Cloud Service Center. They can assist you with complex scripting such as dynamically displaying or hiding fields, sorting table rows, adding new or extension fields to a form, or creating a completely new form. For more information, see the [Business Center for Cloud Solutions from SAP](#) 📄.

For more information about using Adobe LiveCycle Designer, see the Adobe product documentation.

14.2.3 Create Language Variants

Variants are copies of an existing form template that are defined in other languages. You can use variants to create translations of forms in other languages.

Procedure

1. Navigate to [User Menu](#) > [System Settings](#) > [All Settings](#) > [Output Management](#) > [Maintain Form Template](#).
2. Click  and select [Copy Variant](#).
3. In the dialog box that appears, select a language.
4. By default, the [Form Main Template](#) switch is on. If you do not want to use company-specific main templates, turn off the switch.
 - If you use a main template, some of the elements are predefined by the main template, which creates consistency among documents.
 - If you do not use a main template, then you can define the template's header, footers, logo, and sender address individually for this template.
5. Save and publish your changes.

14.2.4 Download Template Variants

Download a template variant either to edit it offline or to use it as sample data to preview a form template.

Procedure

1. Navigate to [User Menu](#) > [Settings](#) > [All Settings](#) > [Output Management](#) > [Maintain Form Template](#).
2. Select the template you want and choose [Download](#).
3. Save the archive file using `.zip` as the file extension.

The ZIP file contains the following files:

- An XML data package (XDP) file containing the selected form template variant.
- An XML schema definition (XSD) file that defines the structure of the XML document. This is required for offline editing.
- If available, XML files containing sample data for previewing your changes.

Note

If no sample data files are available, you can also create your own sample data files using the [View XML Data](#) button on a document's output history.

14.2.5 Create Form Template Rule

Administrators can create rules to determine which form template is used for the output of a business document. If you create new templates, you must create a form template rule to define when the templates are used.

Context

You can either define a form template rule that applies to all instances of the selected business document, or you can use the parameters provided to create more complex rules. If you have created more than one rule for a document, you also need to define the sequence of the rules. The system processes the rules in the order in which they appear in the table until a rule is found for which the conditions are met. Therefore, enter more specific rules at the top of the list and generic rules at the bottom. If you create a rule and leave a parameter blank, then this is handled as a wildcard, and any value is considered as having fulfilled the condition.

Note

In some cases, a business document represents a group of related document types. For example, the business document Customer Invoice includes invoices, credit memos, correction invoices, and down payments.

Procedure

1. Navigate to **User Menu** > **System Settings** > **All Settings** > **Output Management** > **Select Form Template**.
2. Select the **Entity** and **Type**.
3. Click **Add**.
4. From the **Form Template** drop-down list, select a form template.
5. Under **Parameter Settings**, enter the parameters for which you want this rule to apply.
6. Use the **Rearrange** icon to adjust the sequence in which the form template rules are applied in order to ensure that the required rule is applied by the system.
7. Save your changes.

15 Adaptation

As an administrator, you can change the look and feel of the solution for all users by changing layout settings, adding mashups and fields, as well as defining extension fields.

This document describes the functions that you can access as an administrator using the [Start Adaptation](#) button that you can find under the [User Menu](#) dropdown list.

For an overview, watch the following video:

15.1 Customize Your Solution

As an administrator, you can customize the UI look and feel for all users.

Procedure

1. Navigate to ► [User Menu](#) ► [Start Adaptation](#) ►.

The system displays a thin border in the header to indicate that you are in adaptation mode.

2. Make your changes. Note that any change you make applies to only the main layout.
3. When your changes are complete, select [End Session](#) in the header area.

15.2 Manage Fields

Add, hide, and reorder fields.

Procedure

1. Navigate to ► [User Menu](#) ► [Start Adaptation](#) ►.

The system displays a thin border in the header to indicate that you are in adaptation mode.

2. Hover over the desired section to view the editing icons.
3. Select  **(Edit Section)** to open the [Section Items](#) pop-up box.
 - a. To change the sequence of fields, drag  **(Move)** next to a field and move the fields around.

- b. To hide fields, select  (Hide Field).
 - c. To add fields, select  (Add from list) to bring up the fields available in the entity. You can select one or multiple fields at a time.
4. Select [Apply](#).

15.3 Set Default Sorting

As an administrator, you can set a default sort column and direction for tables in adaptation mode. You can choose one sortable column to be the default.

Procedure

1. Navigate to an entity and open the list view.
2. From the [User Menu](#) select [Start Adaptation](#).
3. Choose  ([Table](#)) to switch to the table view.
4. Choose  ([Edit](#)) and then choose  ([Sort](#)).
5. In the pop-up box, select a column from the dropdown to define the sort criteria.
6. Select if you want the column to be sorted in the ascending or descending order.
7. Choose [Apply](#).

The system displays the active sort on the UI. For example, Sorted By Account.

15.4 Create and Organize Filters

As an administrator, you can define filters that end users often use. This makes it easy for them to find the information they need quickly.

Context

Customers use our platform to manage tickets or sales modules such as leads, but currently have to manually enter their most common queries each time. This feature allows them to save their own queries saving their time making the user experience better.

There are two types of custom filters:

- **End User or Personalized Filters:** Personalized filters applicable only for that end user.

- **Key User Filters:** Filters created by the administrator and propagated to all end users or certain group of end users using page layouts and assignment rules.

❁ Example

A customer service manager wants their team regularly to monitor high priority tickets or platinum customer incidents. With this feature, they can create a predefined query that filters for high priority tickets, saving them from having to manually enter this query each time.

Procedure

1. Navigate to [User Menu](#) > [Start Adaptation](#).

The system displays a thin border in the header to indicate that you are in adaptation mode.

2. Navigate to the list view for which you want to create a filter.
3. Select [▼ \(Advanced Filter\)](#)
4. Select an existing filter from the drop down.
5. In the advanced search fields, fill the values as per your requirements and select [Search](#). The system shows you all the items related to the filters.
6. Depending on whether you have selected a standard or custom filter, the system offers the following options:
 - The [Save](#) and [Save As](#) buttons are enabled if you have chosen a custom filter. Select [Save](#) to override the values and edit the existing filter. Select [Save As](#) to create a copy of the selected filter.
 - Only the [Save As](#) button is enabled if you have chosen a standard filter. Select [Save As](#) to create a copy of the selected filter. The system stores the references of standard filter and builds additional parameters. This means that in in future if the application changes the conditions for the standard filter, it will reflect in the custom filter. The [Save](#) button is grayed out because you are not allowed to edit a standard filter.
7. In the [Save Filters](#) pop-up box, enter a name for your filter and save your changes.

📘 Note

- The system considers only Advanced Filter fields for custom filter. Quick Filter field values are not considered. However, if Quick Filter values are connected to the Advanced Filter, the system uses the values from the Advanced Filter section.
- You can switch between saved filters in the dropdown. The system resets the values saved as part that specific saved filter.
- If you try to remove or hide a field that's being used as a parameter in any saved filter, the system displays an error message indicating that the field is currently in use. It will request that you eliminate its usage before proceeding with removal.

The system displays the list of all saved filters in the [Saved Filters](#) pop-up window. You can perform the following actions:

- Delete: Select [🗑 \(Delete\)](#) next to the filter you wish to remove, and then select [Save](#). Note that you cannot delete standard filters.

- Rearrange: To change the sequence of filters, drag  (Move) next to the filters and move them around.
 - Show and Hide: Use the switch next to the filters.
 - Set as default: Under the *Default Filter* header, select a filter from the available list.
 - Rename custom filters: Hover over the filter name and select  (Edit) to edit the name.
8. Save your changes.
 9. End the adaptation session.

15.4.1 Customize Quick Filters

As an administrator, you can customize quick filters according to your specific preferences or needs. This customization allows you to adjust which filters are readily available and how they are displayed or accessed within the application.

Context

It helps in improving efficiency and usability by ensuring that the most relevant filters are easily accessible for quick navigation or sorting of information.

Procedure

1. Navigate to  *User Menu*  *Start Adaptation* .

The system displays a thin border in the header to indicate that you are in adaptation mode.

2. Select an entity from the left navigation menu.
3. Select a saved filter.

The system displays the quick filter list associated with the saved filter.

4. Hover over the quick filter bar to view the edit option.

Choose  (Edit) to view the list of available fields. You can perform the following actions:

- Add From List: To add filters, select  (Add from list) to bring up the fields available in the entity.

Note

You cannot add more than six filters.

- Delete: To delete filters from the list, select  (Delete)
 - Hide: To hide the filter from the screen, choose  (Mark as Hidden)
 - Rearrange: To change the sequence of filters, drag  (Move) next to the filters and move them around.
5. Apply your changes.

The system reflects the new filters immediately. It also displays the filters in the order in which you placed them during editing.

15.5 Add Extension Fields

Add extension fields for your objects.

Procedure

1. Navigate to ► [User Menu](#) ► [Start Adaptation](#) ►.

The system displays a thin border in the header to indicate that you are in adaptation mode.

2. Hover over the desired section to view the editing icons.
3. Select  **(Edit Section)** to open the [Section Items](#) pop-up box.
4. To add fields, select  **(Add from list)** to bring up the fields available in the entity.
5. Find the [Extensions](#) node in the list and drill down inside the structure to view the list of available extension fields.
6. Select **+** **(Add Field)** to add the desired field.

Note: You can add only part of structure fields, for example, you can add only the amount value and not the currency.

7. Select [Apply](#).

Related Information

[Field Attributes \[page 221\]](#)

Field attributes, also known as column attributes or field properties, are characteristics or settings associated with a database field (column) that define various aspects of how data is stored, displayed, and processed within a database. These attributes help ensure data integrity, enforce constraints, and control how the data is used.

15.6 Create and Manage Custom Tabs

As an administrator, you can create custom tabs and embed mashups.

Procedure

1. Navigate to ► [User Menu](#) ► [Start Adaptation](#) ►.

The system displays a thin border in the header to indicate that you are in adaptation mode.

2. Hover over the tab header section to view the editing icons.
3. Select  ([Edit](#)) to open the [Edit Tabs](#) pop-up box.
4. Select  ([Add](#)) to create a custom tab.
5. Add a name for your tab and press the **Return** key.
6. Select [Apply](#).

A custom tab is now created with the same title.

7. Options:
 - To hide the tab and launch it based on a button, select  ([Drill Down](#)) next to the tab and select the [Show on Event Trigger](#) checkbox. The system now hides the tab by default and makes it visible only after you select the button you associate with an event. Ensure that the desired parameters of the event are associated to the mashup present in custom tab.
 - To hide a custom tab, select  ([Hide](#)).
 - To delete a custom tab, select  ([Delete](#)).
 - To change the sequence of tabs, drag  ([Move](#)) next to the tabs and move them around.
8. Select [Apply](#) after performing all the optional steps for the changes to reflect.
9. End your adaptation session.

15.7 Add HTML Mashups to Custom Sections or Tabs

As an administrator, you can add HTML mashups to custom sections or tabs.

Procedure

1. Navigate to ► [User Menu](#) ► [Start Adaptation](#) ►.

The system displays a thin border in the header to indicate that you are in adaptation mode.

2. Open a custom section, or a custom tab.

3. Select > *(Drill Down)* next to *Add Mashup* to drill down to the list of available mashups.
4. Search for the desired mashup and select > *(Drill Down) to view the mashup properties*.
5. If necessary, you can adjust the appearance of the mashup by editing the following options in the *Edit Properties* pop-up box.

Note: The options available to you'll vary depending on the mashup.

- Select the *Show Header* checkbox if you want to see the mashup title on the screen.
- Define the height of HTML Mashups to be displayed in the available screen. For example, if you set the value of *Height (%)* to 100, the newly added mashup occupies the full height of the screen.
- If the mashup has a parameter, you can choose the fields that need to be bound to the parameter value.
- Select *Event Parameters* to choose from the available parameters configured for this mashup. This helps you trigger certain events from the mashup. For example, if you want to use a button in the mashup to refresh the entire screen or specific sections where it's embedded, your admin creates a mashup with the required events and parameters in Mashup Authoring.

📌 Note

The embedded mashup instantly updates when you change the mapped parameter.

6. Select *Apply*.

15.8 Add URL Mashups as Button or Link

As an administrator, you can add URL mashups to a screen as a button or a link.

Context

You can place URL mashups in supported locations on the screen, depending on how the application supports adaptation. Implementation may vary by application and screen layout.

Procedure

1. Navigate to ► *User Menu* ► *Start Adaptation* ►.

The system displays a thin border in the header to indicate that you are in adaptation mode.

2. Open a custom section, or a custom tab.
3. Select  (*Edit*).
4. Select  (*Add URL Mashup*).
5. Select *Add Mashup as Link* or *Add Button* to view the list of available URL mashups.

6. Search for the desired mashup and select  (*Drill Down*) to view its properties.
7. Enter the following values:
 - a. *Button Text* to add the mashup as button.
 - b. *Link Text* to add the mashup as link.
8. Select *Apply*.

15.9 Add and Manage Sections

Add, hide, show, and reorder sections.

Procedure

1. Navigate to  *User Menu*  *Start Adaptation* .

The system displays a thin border in the header to indicate that you are in adaptation mode.

2. Hover over the desired section to view the editing icons.
 - a. To move the section up or down, select  (**Sort**).
 - b. To hide the section from the screen, select  (**Hide Section**).
 - c. To add sections, follow these steps:
 1. Select  (**Add Section**) on the orange/yellow outline.
 2. Select *Create Section*. The system creates an empty section above the selected spot and a *Create Section* pop-up box opens up.
 3. Enter the section name.
 4. Optional: Select the *Show Section Header* checkbox to make the section name visible as header.
 5. Select *Apply*.
 - d. To manage multiple sections together, follow these steps:
 1. Select .
 2. Select *Manage Sections*. The system displays the list of visible sections in the *All Sections* pop-up box.
 3. You can perform the following actions:
 - To change the sequence of sections, drag  (**Move Section**) next to sections and move them around.
 - To hide sections from the screen, Select .
 - To make hidden sections visible, select  (**Add from list**) to bring up the *Add Section* pop-up box. Search and select the desired sections and select *Apply* to make those sections visible again in the UI.
 - To delete custom sections from the screen, select  (**Delete Section**).

ⓘ Note

You can't delete standard sections; you can only hide them from the screen.

4. Select [Apply](#) to make changes to all these explicit actions at one go.

15.10 Create Custom Buttons

As an administrator, you can create custom buttons to trigger end user action such as release sales orders, print invoices, and so on. You can create them at designated sections on the screen, for example, near an action group.

Procedure

1. Navigate to [User Menu](#) [Start Adaptation](#).

The system displays a thin border in the header to indicate that you are in adaptation mode.

2. Hover over the desired section to view the editing icons.
3. Select [Edit](#) to open the [Section Items](#) pop-up box.
4. Select [Add](#) to open the [Add Buttons](#) pop-up box.

ⓘ Note

You can create a custom button under a standard button group or create a stand-alone button. To create a custom button under a standard button group, drill down inside the group.

5. Add a button name.
6. Add an event name. We recommend choosing a name that clearly describes what the button will do. For example, Create a Sales Order.

User actions, such as select of a button, or clicking a link triggers certain events. These events are associated to a button.

ⓘ Note

In this context, an **event** is the action that the system performs when the user interacts with the UI. Whenever a user selects a button or clicks a link, it triggers an event, which then tells the system what to do next.

For example, if you create a custom button called **Create Sales Order**, you then assign it to an event such as `OnCreateSalesOrder`. This event contains the logic that runs when the user clicks the button, such as opening a new form or creating a new record.

The button is what the user clicks, and the event is the operation that gets executed behind the scenes. Linking the button to an event ensures the system knows exactly what should happen when the user interacts with it.

7. Optional: Add parameters.

Parameters are data that is passed when an event is triggered. You can use these data points as input for mashups.

For example, when you capture the current address and send it to a third-party system through a button click, the system uses the address specified in the parameter section during button creation.

8. Select *Apply*.

16 Extensibility

Extensibility allows you and your partners to extend the software platform without having to modify the original codebase. So, you can add to the base functionality, thereby offering new capabilities and outputs.

Extensions range from customizing through to the development of add-ons or composites. Extensions can either be made by customers, partners, or SAP. Extensibility allows customers and partners to support specific use cases and requirements that are not covered by standard software.

16.1 Create Extension Fields

Extension fields are additional fields that you, as an administrator, can add to a cloud solution from SAP.

Context

Extension fields provide the means to store additional data for specific use cases beyond what is natively present in the solution. You can create these fields for a screen that has been enabled for extension fields.

Procedure

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Extensibility](#) ► [Extensibility Administration](#) ►.

The system opens a new screen and displays a list of all business entities in the left panel. These entities are grouped by services.

2. Select the [Extendable](#) checkbox to view the entities that can be extended.
3. Drill down to the entity that you want to add an extension field to, and select it. The system opens the corresponding worklist.
4. In the [Custom Fields](#) tab, select + [\(Add\)](#) to open the [Create Fields](#) quick create screen.
5. Enter the [Label](#). The system automatically generates the [Technical Name](#). You can edit the technical name if the system displays a duplication error.

Note

Technical names must be unique and are not case-sensitive.

6. Select the [Data Type](#) and the corresponding [Data Format](#).
7. (Optional) If the [Data Type](#) is [String](#) and the [Data Format](#) is [UUID](#), you can make this field a standard value help option. To do so, select the [Standard Value Help](#) checkbox, and choose the desired value help from the dropdown list to associate it with the extension field.

Note

Standard Value Help has restricted availability and is available only for Sales Quotes and Sales Orders. To request access to this feature, you must create a case with the component CEC-CRM-CZM-EXT.

8. (Optional) If you want to map this field with an existing field in a different entity, turn on the *Default Mapping* switch and select .

Note

To request this feature for other applications, create a case for the relevant application.

9. Select the field either from the direct or indirect references.

Note

- In the direct references, when the value of the mapped entity changes, the value of custom fields update automatically. There are associated entity references.
- Indirect references are follow-up or create with reference type entity. The value of the custom field remains unchanged until manually updated.

10. (Optional) If you want to mark the custom field that you are creating as a custom reference, enable the *Custom Reference Attribute* switch. Based on the data type and data format you enter, you must select the attributes. Any changes made to this standard field is automatically updated in the reference attributes across the referenced entities.
11. Select *Apply* and save your changes.

The extension field is now created and is listed under the *Custom Fields* tab.

Note

You can create up to 150 extension fields for root or main entity and up to 30 extensions fields for a child entity.

Related Information

[Extension Field Data Types \[page 219\]](#)

Extension Field Data Types refer to the various formats or structures of data that can be used when creating custom fields (extension fields) in an application. These data types define the kind of data the field can store, such as text, numbers, dates, boolean values, amounts, ensuring that the data entered aligns with the intended use and system requirements.

[Field Attributes \[page 221\]](#)

Field attributes, also known as column attributes or field properties, are characteristics or settings associated with a database field (column) that define various aspects of how data is stored, displayed, and processed within a database. These attributes help ensure data integrity, enforce constraints, and control how the data is used.

[Delete Extension Fields \[page 225\]](#)

As an administrator, you might delete an extension field to eliminate redundancy, improve system performance, and maintain data accuracy.

16.1.1 Extension Field Data Types

Extension Field Data Types refer to the various formats or structures of data that can be used when creating custom fields (extension fields) in an application. These data types define the kind of data the field can store, such as text, numbers, dates, boolean values, amounts, ensuring that the data entered aligns with the intended use and system requirements.

Extension Field Data Types

Data Type	Data Format	Description
STRING	STRING (Default)	This data type is used for storing alphanumeric characters. It can be a single line or a multiline text field.
	UUID	A UUID (Universally Unique Identifier) is a 128-bit identifier used for unique resource identification, typically represented as a 36-character string.
	EMAIL	Email data types are designed for storing email addresses and often include validation checks.
	CODE	CODE data type is used for code list. Fixed code lists are defined via Enum options. Dynamic code lists are defined via reference to the corresponding entity.
	TOKEN	A TOKEN represents a small piece of data, often a single word, number, or symbol used as a fundamental unit in programming or language processing.
	URI	A URI (Uniform Resource Identifier) is a text string used to identify a resource, typically consisting of a scheme, authority, path, query, and fragment components. Example: https://www.example.com/resource?param=value#section .
NUMBER	FLOAT (Default)	Number data types can include integers, decimals, or floats. They are used for storing numeric values. Use FLOAT when memory conservation is critical, and you can tolerate lower precision
	DOUBLE	The DOUBLE data type is used to represent real numbers with a high level of precision. Double provides a higher level of precision compared to the FLOAT data type. It can represent approximately 15-16 decimal digits accurately.

Note

Do not store ID fields as the number data type. Use the string data type for ID values. Even if the IDs consist only of numbers, we recommend that you store them as string type, especially when number formatting is not required.

Data Type	Data Format	Description
BOOLEAN		Boolean fields can have only two values - true or false. They are used for binary data or simple yes/no choices.
OBJECT		OBJECT is a versatile and often generic data type that can represent a wide range of values, data structures, or entities.
	AMOUNT	AMOUNT stores numerical values that represent quantities, prices, costs, or any other kind of monetary or non-monetary amounts.
	QUANTITY	QUANTITY typically refers to a numerical value that represents a count, measurement, or amount. Such values are generally represented using standard numeric data types, which can be either integer or floating-point types, depending on the nature of the quantities being measured.
INTEGER	INT32 (Default)	The INT32 data type can store a range of integer values, both positive and negative, within the constraints of a 32-bit binary representation. Int-32 is more memory-efficient than larger integer types like int-64, making it a good choice for applications where memory optimization is a concern.
	INT64	The INT64 data type is capable of storing a wide range of integer values, both positive and negative, within the constraints of a 64-bit binary representation. The range of values for a 64-bit integer is extensive, making it suitable for applications where very large or very small integer values need to be handled.
DATETIME	DATETIME (Default)	DATETIME data types store information about both the date (day, month, and year) and time (hours, minutes, seconds, and sometimes fractions of a second).
	DATE	Date fields store date values, including options for time and time zone settings.
	TIME	Time fields store time values without date information.

Data Type	Data Format	Description
	DURATION	Duration fields are used to represent a specific length of time, typically measured in terms of hours, minutes, seconds, and sometimes milliseconds. It allows you to work with time intervals or durations, as opposed to specific points in time, which are handled by date and time data types.

Note

- You can add up to 100 codes in a code list field. For existing static code lists having more than 100 codes, further code addition is restricted. However, other modifications such as activation / deactivation of the codes are allowed.

16.1.2 Field Attributes

Field attributes, also known as column attributes or field properties, are characteristics or settings associated with a database field (column) that define various aspects of how data is stored, displayed, and processed within a database. These attributes help ensure data integrity, enforce constraints, and control how the data is used.

Available Field Attributes

Field Attribute	Description
creatable	Createable fields indicate whether a field can be populated with data when inserting a new record (i.e., during the creation of a new record in a database table). If a field is createable, you can provide a value for it when adding a new record to the database. If it's not createable, the field must remain empty or be set to a default value during insertion.
updateable	Updateable fields determine whether a field's value can be modified after the record has been created. If a field is updateable, you can change its value when updating existing records. Fields marked as non-updateable typically remain unchanged after the record's creation, but this behavior can be controlled using field attributes.

Field Attribute	Description
nullable	A field is nullable if it can contain null values, meaning that it's not required to have a value in every record. If a field is nullable, you can leave it empty (null) when creating or updating a record. If a field is not nullable, it must always have a valid value, and it can't be null. Non-nullable fields are typically used for data that's required and must always have a value.
required	Required fields are those that must have a value in every record, and null values are not allowed. When defining a field as required, you must ensure that it is populated with a valid value whenever a new record is created or an existing record is updated. These fields are essential for data integrity and accuracy.
hidden	Hidden fields are not visible to the user by default, but they may contain data that is used for various purposes, such as calculations or internal processing. Hidden fields are often used in forms or data structures where the data is needed for processing but doesn't need to be displayed to the user.

Note

- You cannot change the `searchable` property for the `OBJECT`, `ARRAY`, `INTEGER`, `NUMBER`, and `DATETIME` datatypes.
- You cannot change the `filterable` and `sortable` properties for the `OBJECT`, `ARRAY`, `INTEGER`, and `NUMBER` datatypes.
- The `required` and `nullable` properties are not applicable for the `BOOLEAN` datatype.
- `OBJECT` `dataType` can not have `dataFormat` as `DEFAULT` or not defined. `dataFormat` `AMOUNT` or `QUANTITY` or `TEXT` are only supported for custom/extension attributes.
- You cannot update `dataType` and `dataFormat`.

16.1.3 Field Query Properties

As an administrator, you can define whether a field is filterable, searchable, or sortable to control how it behaves in queries and list views.

The following table gives an overview of the available properties.

Field Query Properties

Field Query Property	Description
sortable	Sortable fields allow data to be arranged in a particular order, typically in ascending or descending fashion. Sorting is based on the values within these fields. You can click the column headers in a table or a list to sort the data by that column. Sorting helps with data organization and simplifies data analysis.
filterable	Filterable fields permit data to be filtered based on specific criteria or conditions. Users can apply filters to display only the records that meet their chosen criteria. Filtering is employed to reduce a dataset's size, making it easier to focus on specific information or patterns within the data.
searchable	Searchable fields enable data to be searched or queried using keywords, phrases, or search terms. Searching allows users to find specific records or information within a larger dataset. You can enter search queries to locate records that contain the desired information, which is crucial for efficient information retrieval.

Procedure

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Extensibility](#) ► [Extensibility Administration](#) ►.
2. In the [Field Query Properties](#) tab, set the filterable, searchable, or sortable properties by turning on the switch next to them.
Turn on the [Show Limits](#) switch to display the total number of fields that can be marked as searchable, or search-relevant, along with how many have already been configured, for example, 5/10.
The system displays the limits for the following fields:
 - **Searchable Fields:** Searchable fields allow you to find data using keywords, phrases, or search terms.
 - **Search-Relevant Fields:** Search-relevant fields are those marked as filterable, searchable, or sortable. These attributes determine whether a field can be used in search, filtering, or sorting functionality.

Behavior of Search Relevant Properties (searchable, sortable, filterable)

Standard Fields

- **Sortable & Filterable:** You can mark a maximum of 50 standard attributes for an entity in a tenant as sortable or filterable. This limit is aggregated.

- **Searchable:** You can mark a maximum of 10 standard attributes for an entity in a tenant as searchable.

Extension Fields

- Search properties are counted row-wise for each attribute in an entity.
- If any one of the properties (searchable, sortable, or filterable) is enabled for an extension field, that field is counted for all three properties.
- The row-wise limit for extension fields is 10.

Note

When marking search-relevant fields (filterable, searchable, or sortable) it's important to stay within the defined limits. Note that you can mark up to 10 custom fields as searchable. However, the overall limit for search-relevant fields is also 10. This means that if you have marked 10 fields as searchable, you can no longer mark any other field as sortable or filterable within a particular entity.

In some cases, you may see totals like **73/50** for standard fields due to inherited or legacy configurations. Although this appears incorrect, the system still enforces the cap and prevents further changes. To resolve this, you must deselect some fields using adaptation before adding new ones. The system prompts you if the field count exceeds the limit.

Note

You can also define up to 100 search-relevant fields, including 30 searchable fields, without needing to differentiate between standard and custom fields using the [Feature Activation Hub](#). Once the feature is activated, the aggregated limits are applied across all entities in a service. You can continue to add or modify fields as long as the defined limits are not exceeded. If you are unable to customize a standard field for a particular service, create a case for the relevant application.

For more information regarding the enhanced search field limits feature, check the **Related Information** section.

Data Types and Data Formats for Search Relevant Properties

Data Type	Data Format	Searchable	Filterable	Sortable
STRING	STRING (Default)	Yes	Yes	Yes
	UUID	Not Applicable	Yes	Yes
	EMAIL	Yes	Yes	Yes
	CODE	Yes	Yes	Yes
	TOKEN	Yes	Yes	Yes
	URI	Yes	Yes	Yes
	ALPHANUMERIC	Yes	Yes	Yes

Data Type	Data Format	Searchable	Filterable	Sortable
NUMBER	FLOAT (Default)	Not Applicable	Yes	Yes
	DOUBLE	Not Applicable	Yes	Yes
BOOLEAN		Not Applicable	Yes	Yes
ARRAY		Not Applicable	Not Applicable	Not Applicable
OBJECT	AMOUNT	Not Applicable	Yes	Yes
	AMOUNT (VALUE)	Not Applicable	Yes	Yes
	AMOUNT (UOM)	Not Applicable	Yes	Yes
	QUANTITY	Not Applicable	Yes	Yes
	QUANTITY (VALUE)	Not Applicable	Yes	Yes
	QUANTITY (UOM)	Not Applicable	Yes	Yes
	TEXT	Yes	Yes	Yes
	TEXT (CONTENT)	Yes	Yes	Yes
	TEXT (LANG CODE)	Not Applicable	Yes	Yes
INTEGER	INT32 (Default)	Not Applicable	Yes	Yes
	INT64	Not Applicable	Yes	Yes
DATETIME	DATETIME (Default)	Not Applicable	Yes	Yes
	DATE	Not Applicable	Yes	Yes
	TIME	Not Applicable	Yes	Yes
	DURATION	Not Applicable	Yes	Yes

Related Information

[Enhanced Search Field Limits](#)

16.1.4 Delete Extension Fields

As an administrator, you might delete an extension field to eliminate redundancy, improve system performance, and maintain data accuracy..

To delete an extension field, navigate to  [Extensibility Administration](#)  [Custom Fields](#)  where all the fields are listed and select  ([Delete](#)).

Note that deleted fields cannot be recovered, and all data will be lost.

You are not allowed to delete an extension field if it is already in use in other business processes such as Validations, Determinations, Autoflows, and so on. We recommend that you use the where-used function to identify and detect all usages. Make sure to delete these usages before deleting the extension field.

Select



(View Usage) to check the services where the field has been used. If the usage list is blank or shows an error, ensure that the where-used service `sap.crm.service.whereUsedService` is assigned to your corresponding user role.

16.2 Custom Logic

Custom logic enables key users to create their own implementations to customize application functionalities with predelivered logic.

Two major types of custom logic implementation are supported in the system:

- Business expression-based custom logic: These validations and determinations are based on simple expressions and configured using in-build tools. These custom logic expressions are simple logic and limited by the capabilities of business expressions and custom logic tools.
- External API hooks: These validations and determinations are based on external API calls. The external APIs can be implemented in another external platform such as SAP Build Apps but must support the message structure defined by the system.

Note

The pre hook and post hook execution flow is as follows:

- For Pre Hook, the flow follows:
 1. In-App Determinations
 2. External Hook
- For Post Hook, the flow follows:
 1. In-App Determinations
 2. External Hook
 3. In-App Validations

Source Tenant Copy for Custom Logic

Custom Logic supports both *Complete Copy of Source Tenant* and *Copy of Configuration* scenarios.

In *Complete Copy of Source Tenant*, the custom logic runtime database and all its collections such as determinations, validations, and dynamic properties are copied except external hooks (copying data connector service isn't supported).

In *Copy of Configuration*, custom logic runtime database and all its collections such as determinations, validations, and dynamic properties are copied except for external hooks (copying data connector service isn't supported), however, in this scenario, only customization is copied and data isn't copied.

Related Information

[Requesting a New Tenant](#)

16.2.1 Create Validations

Validations help you, as an administrator, to implement a custom logic where you can prevent a save or issue a warning message to users during save based on certain conditions.

Procedure

1. Navigate to [User Menu](#) > [System Settings](#) > [All Settings](#) > [Extensibility](#) and choose [Extensibility Administration](#).

The system opens a new screen and displays a list of all business entities in the left panel. These entities are grouped by services.

2. Select a service and choose [Drill Down](#) to open and select an entity.
3. Navigate to the [Validations](#) tab and choose [Create](#) > [Using Forms](#) to launch the [New Validation](#) editor tab.

The editor has three main blocks:

- Header: Specify the validation name and description here.
 - Condition: Set the desired conditions using a flow-based visual condition editor.
 - Action: Specify an error or warning message based on the conditions defined in the condition block.
4. In the header block, specify the validation name and description.
 5. You can enable the [Show Inactive Codelist](#) switch to display all the codelist values including the inactive ones in the [Condition Block](#) dropdown.

Note

- By default, only active codelist values are shown in the dropdown. If a service does not support filtering active and inactive values using the filter parameter, all codelist values are shown in the dropdown.
- When you edit an existing condition, the switch is off by default. If an inactive code value was previously selected, it appears (in condition block) without description. You cannot save when the switch is disabled and an inactive value is selected.
- When you activate or deactivate the switch, all conditions using codelist fields are reloaded that takes some time and impacts performance.

6. In the [Condition Block](#) set the desired conditions using a flow-based visual condition editor.
 - a. Select a function from the dropdown.
 - b. Select the standard or extension field from the dropdown.

To create a condition based on the previous value of the field, slide the [Previous Value](#) switch.

- c. Select the comparison operator.
- d. Select the target comparison. You can compare either by a fixed value or with another field.
- e. If you choose *Value*, enter a corresponding target value. If you choose *Field*, select a target field.

Add multiple conditions by choosing **+** (*Add Condition*). To remove a condition, choose **X** (*Remove Condition*). To clear all conditions in a block, choose **🗑** (*Delete Condition Block*).

Note that when you create multiple conditions, the condition group is considered fulfilled only if all the conditions within a group are met.

7. Optional: Create an OR condition by choosing the **+ Or** button. The system displays additional condition blocks. Follow the same steps as earlier to fill one or more OR condition blocks.

Note

When you create multiple OR conditions, the condition group is considered fulfilled if any one of the block conditions are met.

8. In the third block, under *Action* choose to show either an error or a warning message.
9. Add the message text.
10. Save and activate your changes.

Note

You can create a maximum of 20 validations per entity.

In the *Validations* tab, the system lists all validations for a specified entity. Select  or  to edit or delete the corresponding validation.

If you need to temporarily turn off certain rules for testing or troubleshooting, you can do so by using the switch under *Actions* to disable the validation.

Note

- If you've used any IDs or code values in your determination or validation logic in the source system ensure the same IDs and code values are also present in your other tenants (test and production). This verification is important to ensure consistent behavior of custom logic across all tenants.
- Country/Region codes aren't unique. When setting up validation or determination conditions involving country/region codes, go to *Language, Region, and Currency* settings to check the codes listed under each country/region.

Note

To create a validation using a block-based editor, watch the following video:

Related Information

[Disable Custom Logic \[page 246\]](#)

As an administrator you can disable custom logic that you have previously configured or added to meet specific requirements. This includes validation and determination rules.

[Functions in Custom Logic \[page 247\]](#)

This document lists all the functions available for creating validation and determination rules in *Extensibility Administration*.

[Supported Entities for Custom Logic \[page 250\]](#)

Custom logic feature is available only for specific entities and sub-entities. All these entities have restricted availability. To access this feature, you must submit a support ticket in the relevant application component.

16.2.2 Create Determinations

Use determinations to calculate specific field value based on certain conditions. You can either default or propose the value or mandate a certain value using determinations.

Procedure

1. Navigate to **User Menu** > **System Settings** > **All Settings** > **Extensibility** and choose *Extensibility Administration*.

The system opens a new screen and displays a list of all business entities in the left panel. These entities are grouped by services.

2. Select a service and choose **Drill Down** to open and select an entity.
3. Navigate to the *Determinations* tab, and choose **Create** > **Using Forms** to launch the *New Determination* editor tab.

The editor has three main blocks:

- Header: Specify the determination name, event, and description here.
 - Condition: Set the desired conditions using a flow-based visual condition editor.
 - Assignment: Assign a value to specific fields. You can use each row to assign a single field value. For example, `total_amount = item_count * item_price`.
4. In the header block, perform the following actions:
 - a. Enter a name.
 - b. Choose one of the following events:
 - Pre Hook: You can use it to default or suggest standard and custom field values. The user can change the field values.
 - Post Hook: You can use it to mandate custom field values. The user can't change the standard field values.
 - c. Add a description.
 5. You can enable the *Show Inactive Codelist* switch to display all the codelist values including the inactive ones in the *Condition Block* dropdown.

Note

- By default, only active codelist values are shown in the dropdown. If a service doesn't support filtering active and inactive values using the filter parameter, all codelist values are shown in the dropdown.
- When you edit an existing condition or assignment, the switch is disabled by default. If an inactive code value was previously selected, it appears (in condition/assignment block) without a description. You can't save when the switch is disabled and an inactive value is selected.
- When you activate or deactivate the switch, all conditions using codelist fields are reloaded that takes some time and impacts performance.

6. In the *Condition Block*, set the desired conditions using a flow-based visual condition editor.

- Select a function from the dropdown.
- Select the standard or extension field from the dropdown.

To create a condition based on the previous value of the field, slide the *Previous Value* switch.

- Select the target comparison. You can compare either by a fixed value or with another field.
- If you choose *Value*, enter a corresponding target value. If you choose *Field*, select a target field.

Add multiple conditions by choosing + (*Add Condition*). To remove a condition, choose ✕ (*Remove Condition*). To clear all conditions in a block, choose 🗑️ (*Delete Condition Block*).

Note that when you create multiple conditions, the condition group is considered fulfilled only if all the conditions within a group are met.

7. Optional: You can also create an OR condition by choosing the + Or button. The system displays additional condition blocks. Follow the same steps as earlier to fill one or more OR condition blocks.

Note

When you create multiple OR conditions, the condition group is considered fulfilled if any one of the block conditions are met.

Under the *Summary* header, the system shows you a summary of all the conditions that you've entered.

8. In the *Assignment Block*, perform the following actions:

- Use a row to assign or calculate a value to a specific field.
- Write the expressions and calculations in free-style format.
- Mention the target field first, followed by "=" and then specify the expression. For example, {status} = ('Open')
- Enclose all entity fields in curly braces: For example, {status}, {ID}, and so on.
- Ensure that you add the extension/custom fields under the extensions structure. For example, {extensions/customfield}
- Get the technical name of the fields using the dropdown in the *Condition Block*.
- Use expressions consisting of operators and certain pre-defined functions specified in the *Assignment Block*.
- Ensure that you check the feedback provided by the system in case of invalid expressions. The error messages are configured to give exact feedback so that you can correct the expression.

- Create additional assignment rows by choosing **+** ([Add Assignment](#)). To remove a row, choose **X** ([Remove Assignment](#)). Choose [Clear All](#) to remove all assignment rows.
9. Optional: Select [Else](#) or [Else If](#) to create one more set of condition blocks that are triggered if the initial sets of conditions aren't satisfied. Follow the same steps to create **Else If/Else** conditions and assignments inside the respective blocks.

Note

- You can create an **Else If/Else** block only when you have at least one condition in the initial IF block.
- You can create up to 20 **Else If** conditions.
- If you select an **Else** condition, the system doesn't allow you to create additional blocks indicating that your entire logic is complete.

10. Save and activate your changes.

Note

You can create a maximum of 20 determinations per entity.

In the [Determinations](#) tab, the system lists all determinations for a specified entity. Choose  or  to edit or delete the corresponding determination.

To control the sequence in which the determination rules must be implemented, drag and drop the rules in order of precedence by choosing  ([Move](#)).

If you need to temporarily turn off certain rules for testing or troubleshooting, you can do so by using the switch under [Actions](#) to disable the determination.

Note

- If you've used any IDs or code values in your determination or validation logic in the source system, ensure the same IDs and code values are also present in your other tenants (test and production). This verification is important to ensure consistent behavior of custom logic across all tenants.
- Country/Region codes aren't unique. Go to [Language, Region, and Currency](#) settings, and check for country/region codes under a particular country/region while setting conditions for validations and determinations that involve country/region codes.

Note

To create a determination using a block-based editor, watch the following video:

Related Information

[Disable Custom Logic \[page 246\]](#)

As an administrator you can disable custom logic that you have previously configured or added to meet specific requirements. This includes validation and determination rules.

[Functions in Custom Logic \[page 247\]](#)

This document lists all the functions available for creating validation and determination rules in [Extensibility Administration](#).

[Supported Entities for Custom Logic \[page 250\]](#)

Custom logic feature is available only for specific entities and sub-entities. All these entities have restricted availability. To access this feature, you must submit a support ticket in the relevant application component.

16.2.3 Create External Hooks

As an administrator, you can create and use external hooks to activate external functions when you save an application. This means you can customize or add extra functionalities to the saving process according to your needs using side-by-side tools such as Kyma, BTP, and so on.

Prerequisites

You have already configured the communication arrangement between the solution and a communication partner.

Context

External hooks refer to mechanisms or points in a system or software where external processes or custom code can be integrated or triggered to extend or modify the behavior of the system. These hooks provide a way for external components to interact with the main system at specific events or actions, allowing for customization and flexibility in the overall functionality.

Note

When you send data to external systems, you're responsible for managing or deleting personal data from those systems. SAP or its affiliated companies shall not be liable for errors or omissions with respect to the data that is transmitted outside of the SAP solution.

Procedure

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Extensibility](#) ► and choose [Extensibility Administration](#).

The system opens a new screen and displays a list of all business entities in the left panel. These entities are grouped by services.

2. Select a service and choose  ([Drill Down](#)) to open and select an entity.

3. Navigate to the [External Hooks](#) tab and choose + [\(Create External Hook\)](#) to launch the [Create External Hook](#) quick create window.

Note

You can use [Generate Schema](#) to download the schema JSON files for both request and response. You can also use [Generate Sample Payload](#) to download sample request and response payloads for both the Pre Hook and Post Hook.

The values in the generated sample payloads are illustrative only. Ensure that you update them according to your specific requirements before sending the payload.

These JSON files can be used to implement parsing logic in an external system. The downloaded schema file always reflects the latest metadata information.

In the schema, some fields originate from a referenced entity. To use these fields and receive their data in custom logic or hooks, you must first enable them through the Reference Fields Usage feature. This requirement applies only to supported entities.

The schema displayed in the Generate Schema feature represents an example of the possible structure. It doesn't guarantee that all fields are available at runtime. The actual data returned depends on the specific instance and the reference fields that are enabled for the entity.

The External Hook payload contains only fields that have been populated in the instance at the time of the change or save. Fields left empty on the UI aren't sent. The schema shown in the [Generate Schema](#) feature is only an example of possible structures and doesn't guarantee that all fields are present.

4. Enter a name and description.
5. Select one of the following events:
 - Pre Hook: You can use it to default or suggest standard and custom field values. The user can change the field values.
 - Post Hook: You can use it to mandate custom field values. The user can't overwrite the field values.

The system displays the [Service Full Name](#) based on your selection in Extensibility Administration. It's a read-only field.

6. Enter the [API Path](#) suffix that must be applied. In the [Communication System](#) field, define the system URL but this API path can be different for different logic. For example, your system URL `my123131.ondemand.com` is defined in the communication system but you may call `my123131.ondemand.com/case_validation` for each case save. This step is achieved by mentioning the API path, which is added to the system instance URL.

The [HTTP Method](#) field is already auto-populated with [POST](#), since this method is the only method allowed currently. All the data exchange happens via a POST message.

7. Select the [Communication System](#) from the dropdown. Note that you have already created the communication system.
8. Save your changes.

Note

- In the pre hook, you have the flexibility to update both standard and custom fields. However, in post hook, you can only update custom fields. Updating standard fields isn't permitted.
- You can't add or delete entries from arrays of objects in either pre hook or post hook.

For instance, in the opportunity entity (`sap.ssc.opportunityservice.entity.opportunity`), you have items which are an array of objects referring to the `sap.ssc.opportunityservice.entity.opportunityItems` entity. Here, you can't add another object in the items array, nor can you remove the object (with `"id"= "6cc451bc-2969-4cf4-a303-92607abc7b5c"`).

Sample Code

```
{
  "items": [
    {
      "id": "6cc451bc-2969-4cf4-a303-92607abc7b5c",
      "productId": "c14c84f8-2f86-4890-b0a4-b6a956c3d622",
      "productDisplayId": "0000000000,020,255",
      "netAmount": {
        "content": 0,
        "currencyCode": "USD"
      },
      "revenueStartDate": "2024-03-01",
      "revenueEndDate": "2024-03-01"
    }
  ]
}
```

However, you can add or remove items from arrays of primitive types (String, Number, and so on.) For example, a case entity (`sap.ssc.caseservice.entity.case`) has an attribute called 'toRecipients' which is an array of strings. You can add or remove entries here.

Sample Code

```
{
  "emailInteraction": {
    "id": "fd63de17-e971-11ed-a33e-a7a619a69637",
    "senderEmailAddress": lorem.ipsu@gmail.com,
    "toRecipients": [
      icanaddandremovehere@gmail.com
    ]
  }
}
```

- In the post hook, you can perform validations and respond with error/warning/info messages. Responding with an error message results in a transaction save failure.
- Metadata validations are conducted on modified data, and errors from these checks are returned to the application service.
These validation errors must be visible in the user interface, and the save operation fails accordingly.
- If there's an error in connecting to an external system implementation (due to incorrect configuration or unavailability of the service), appropriate errors are returned to the application service.
These errors must be visible in the user interface, and the save operation fails.
- The response returned in hooks must adhere to the published and supported response structure. Schema validations are performed on the response data, and any validation errors are returned to the application service.
These errors must be visible in the user interface, and the save operation fails.
- Only one external pre hook and one external post hook per root entity are allowed.
Creating hooks on sub-entities isn't permitted.

- The external hook implementation must be lean, efficient, and performant as these implementations are called synchronously within the transaction. Sub-optimal implementations hamper the performance of your application.
- In unforeseen cases, to protect the system's stability and performance, requests taking more than 10 seconds are canceled. An error message indicating the cancellation appear in the application.

⚠ Caution

- Data can only be modified by sending updated data back to the pre- and post-hook response **data** object.
- Don't use APIs to create, modify, or delete the same entity (for which the hook is called), another entity in SAP Sales and Service Cloud Version 2, or an entity in an external system (SAP or non-SAP). At this stage, the transaction is still in progress and not yet complete. It can be aborted after the hook returns for various reasons, which can lead to inconsistencies in the solution. For such scenarios, use the entity state-changed events that are triggered after the entity is saved.

Related Information

[External Hook Request and Response Structure \[page 235\]](#)

Provides Request and Response structure for both preHook and postHook to external system.

[Create Communication Systems](#)

[Reference Fields Usage \[page 244\]](#)

Enable administrators to reference attributes from other entities when defining custom logic on an entity.

16.2.3.1 External Hook Request and Response Structure

Provides Request and Response structure for both preHook and postHook to external system.

ExternalHook PreHook

Request format for preHook to external system:

↗ Sample Code

```
{
  "entity": "",
  "beforeImage": { },
  "currentImage": { },
  "context": { }
}
```

Element	Element Of	Mandatory	Type	Description
entity		Yes	String	Full name of the aggregated entity.
beforeImage		No	Object	Before image of the aggregate entity data. This is empty in the create scenario.
currentImage		Yes	String	Current image of the aggregate entity data.
userId	context	Yes	String	ID of the user for the current transaction.
employeeId	context	No	String	Employee ID of the user for the current transaction. This isn't provided for technical users.
skipValidations	context	Yes	Boolean	A flag to bypass custom logic validation. When passed as true , external hooks can bypass validations, and the validation results are ignored.
isDryRun	context	Yes	Boolean	A flag to simulate custom logic without saving any changes. When set to true , the transaction runs in dry run mode, letting users preview the outcome before committing.
language	context	Yes	String	The current login language in which the user has logged in.
operation	context	Yes	String	This field provides information about the aggregate entity. Supported values include CREATE_DRAFT , CREATE , UPDATE , or DELETE .

Operation

The `operation` value is determined based on the action performed on the aggregate entity.

- **CREATE_DRAFT**: The operation will have this value when the aggregate entity instance is a **Draft Document** in the **Creation** stage.

Note

This operation value is only for entities that support draft documents/draft mode, otherwise CREATE is sent.

- **CREATE**: The operation will have this value when an aggregate entity instance is in the Creation stage.
- **UPDATE**: The operation will have this value when the aggregate entity instance is being updated. This could be the ROOT being updated or Items being created/updated/deleted etc.
- **DELETE**: Operation will have this value when the aggregate entity instance is being deleted.

Example

Incoming Externalhook PreHook request for case entity

Sample Code

```
{
  "entity": "sap.ssc.caseservice.entity.case",
  "currentImage": {
    "subject": "Very Important Case",
    "priority": "03",
    "priorityDescription": "Normal",
    "origin": "MANUAL_DATA_ENTRY",
    "caseType": "ZSRK",
    "caseTypeDescription": "SRK Case Type",
    "timePoints": {
      "reportedOn": "2023-10-30T05:28:00Z"
    },
    "categoryLevel1": {},
    "extensions": {
      "Price": 12345
    }
  },
  "context": {
    "employeeId": "11ee5843-e4c2-d70e-afdb-81a24fa8c000",
    "userId": "f5b4e4ad-5843-11ee-829c-21688db8d25e",
    "skipValidations": true,
    "isDryRun": true,
    "language": "de",
    "operation": "CREATE_DRAFT"
  }
}
```

Response format for preHook

Sample Code

```
{
  "data": { },
  "noChanges": true/false
}
```

Element	Mandatory	Type	Description
data	Yes (if noChanges = false or if noChanges not sent in response)	Object	Modified entity aggregates current image data returned by the external system. Standard and extension fields can be modified.
noChanges	No	Boolean	Indicates that the external system hasn't modified the data. If set to 'true', you don't need to return the 'data' attribute in the response. This helps reduce unnecessary data transfer over the wire.

❖ Example

Externalhook PreHook response for case entity where subject and extensions/price are changed.

↔ Sample Code

```
{
  "data": {
    "subject": "Not important",
    "priority": "03",
    "priorityDescription": "Normal",
    "origin": "MANUAL_DATA_ENTRY",
    "caseType": "ZSRK",
    "caseTypeDescription": "SRK Case Type",
    "timePoints": {
      "reportedOn": "2023-10-30T05:28:00Z"
    },
    "categoryLevel1": {},
    "extensions": {
      "Price": 999
    }
  }
}
```

❖ Example

Externalhook PreHook response for case entity where no modifications are made on the data.

↔ Sample Code

```
{
  "noChanges": true
}
```

ExternalHook PostHook

Request format for postHook to external system

Sample Code

```
{
  "entity": "",
  "beforeImage": { },
  "currentImage": { },
  "context": { }
}
```

Element	Element Of	Mandatory	Type	Description
entity		Yes	String	Full name of the aggregated entity
beforeImage		No	Object	Before image of the aggregate entity data. This is empty in the create scenario.
currentImage		Yes	Object	Current image of the aggregate entity data
userId	context	Yes	String	ID of the user for the current transaction.
employeeId	context	No	String	Employee ID of the user for the current transaction. This isn't provided for technical users.
skipValidations	context	Yes	Boolean	A flag to bypass custom logic validation. When passed as true , external hooks can bypass validations, and the validation results are ignored.
isDryRun	context	Yes	Boolean	A flag to simulate custom logic without saving any changes. When set to true , the transaction runs in dry run mode, letting users preview the outcome before committing.
language	context	Yes	String	The current login language in which the user has logged in.

Element	Element Of	Mandatory	Type	Description
operation	context	Yes	String	This field provides information about the aggregate entity. Supported values include CREATE_DRAFT, CREATE, UPDATE, or DELETE.

Operation

The operation value is determined based on the action performed on the aggregate entity.

- **CREATE_DRAFT**: The operation will have this value when the aggregate entity instance is a **Draft Document** in the **Creation** stage.

Note

This operation value is only for entities that support draft documents/draft mode, otherwise CREATE is sent.

- **CREATE**: The operation will have this value when an aggregate entity instance is in the Creation stage.
- **UPDATE**: The operation will have this value when the aggregate entity instance is being updated. This could be the ROOT being updated or Items being created/updated/deleted etc.
- **DELETE**: Operation will have this value when the aggregate entity instance is being deleted.

Example

Incoming Externalhook PostHook request for case entity

Sample Code

```
{
  "entity": "sap.ssc.caseservice.entity.case",
  "currentImage": {
    "subject": "Very Important Case",
    "priority": "03",
    "priorityDescription": "Normal",
    "origin": "MANUAL_DATA_ENTRY",
    "caseType": "ZSRK",
    "caseTypeDescription": "SRK Case Type",
    "timePoints": {
      "reportedOn": "2023-10-30T05:28:00Z"
    },
    "categoryLevel1": {},
    "registeredProducts": [
      {
        "serialId": "11",
        "referenceProduct": {
          "id": "00163e03-552b-1ee2-8daa-9827ce027ec0"
        },
        "referenceDate": "2023-10-16T00:00:00.000Z",
        "id": "3c448270-67fd-00d2-0000-726b8bb0f5cb",
        "displayId": "6",
        "status": "ACTIVE",

```



```

        "description": "Test Product",
      },
      {
        "serialId": "123",
        "referenceDate": "2023-10-16T00:00:00.000Z",
        "displayId": "Test",
        "status": "ACTIVE",
        "description": "Test Product2",
      }
    ],
    "extensions": {
      "Price": 12345
    }
  },
  "context": {
    "employeeId": "11ee5843-e4c2-d70e-afdb-81a24fa8c000",
    "userId": "f5b4e4ad-5843-11ee-829c-21688db8d25e",
    "skipValidations": true,
    "isDryRun": true,
    "language": "de",
    "operation": "CREATE_DRAFT"
  }
}

```

Example

Response format for postHook

Sample Code

```

{
  "data": { },
  "error": [
    {
      "code": "",
      "message": "",
      "target": ""
    }
  ],
  "info": [
    {
      "code": "",
      "message": "",
      "target": "",
      "severity": "WARNING/INFO"
    }
  ]
}

```

Element	Element Of	Mandatory	Type	Description
data		Yes (if noChanges = false or if no-Changes not sent in response)	Object	Modified entity aggregates current image data returned by the external system. Only extension fields are editable.

Element	Element Of	Mandatory	Type	Description
noChanges		No	Boolean	Indicates no modifications to data by an external system. If passed as 'true', data need not be provided in the response.
error		No	Array of objects	List of validation error messages
code	Error	Yes		Combination of the external service name and error code of integer (5 digits), with a dot (.) as the delimiter
message	Error	Yes	String	Description of the error message Messages can be up to 200 characters long. Anything longer is truncated.
target	Error	No	String	Object that is used to provide reference to the target element that caused the error. The target may either identify the resource directly using the URL notation (or) may identify an element in the payload using the binding "{ ... }" notation
info		No	Array of objects	List of validation info/warning messages.
code	Info	No	String	Combination of the external service name and code integer (5 digits), with a dot (.) as the delimiter
message	Info	Yes	String	Description of the info/warning message Messages can be up to 200 characters long. Anything longer is truncated.
target	Info	No	String	Object that is used to provide reference to the target element that caused the error. The target may either identify the resource directly using the URL notation (or) may identify an element in the payload using the binding "{ ... }" notation
severity	Info	Yes	Enum	Indicates the type of the information. Permitted values INFO WARNING

❧ Example

Outgoing Externalhook PostHook response for case entity with error, info, and warning messages

🔗 Sample Code

```
{
  "data": {
    "subject": "Very Important Case",
```

```

    "priority": "03",
    "priorityDescription": "Normal",
    "origin": "MANUAL_DATA_ENTRY",
    "caseType": "ZSRK",
    "caseTypeDescription": "SRK Case Type",
    "timePoints": {
      "reportedOn": "2023-10-30T05:28:00Z"
    },
    "categoryLevel1": {},
    "registeredProducts": [
      {
        "serialId": "11",
        "referenceProduct": {
          "id": "00163e03-552b-1ee2-8daa-9827ce027ec0"
        },
        "referenceDate": "2023-10-16T00:00:00.000Z",
        "id": "3c448270-67fd-46d2-ae80-726b8bb0f5cb",
        "displayId": "6",
        "status": "ACTIVE",
        "description": "Test Product",
      },
      {
        "serialId": "123",
        "referenceDate": "2023-10-16T00:00:00.000Z",
        "displayId": "Test",
        "status": "ACTIVE",
        "description": "Test Product2",
      }
    ],
    "extensions": {
      "price": 12345
    }
  },
  "error": [
    {
      "code": "external_CaseService.10000",
      "message": "Case type is not allowed",
      "target": "{caseType}"
    },
    {
      "code": "external_CaseService.10001",
      "message": "Reported on is older than required date.",
      "target": "{timePoints.reportedOn}"
    },
    {
      "code": " external_CaseService.10004",
      "message": "Error message raised from external hook. Registered
Products node does not contain id.",
      "target": "{registeredProducts[1].id}",
      "severity": "ERROR"
    }
  ],
  "info": [
    {
      "code": " external_CaseService.10002",
      "message": "The price is greater than 1000.",
      "target": "{extensions.price}",
      "severity": "WARNING"
    },
    {
      "code": " external_CaseService.10003",
      "message": "Case processed by external system.",
      "target": "",
      "severity": "INFO"
    },
    {
      "code": " external_CaseService.10004",
      "message": "Case processed by external system.",

```

```

        "target": "{registeredProducts[0].id}",
        "severity": "INFO"
      }
    ]
  }
}

```

ⓘ Note

The 'isDraftMode' attribute in currentImage and beforeImage attributes is for SAP internal usage only. Don't use this field for read or write in your external implementations.

16.2.4 Reference Fields Usage

Enable administrators to reference attributes from other entities when defining custom logic on an entity.

These referenced entities can either have standard or extension attributes. For example, A scenario where end users want to create a determination on a case entity and use standard and extension fields from the individual customer entity.

These reference fields are classified into four types:

- Standard Reference Field, for example, `case/id`
- Cross-Entity Reference Field, for example, `referenceUsages/case_priority`
- Cross-Entity Extension Reference Field, for example, `referenceUsages/caseExtensionsIsPremium`
- Extension Reference Field, for example, `Extensions/field1` for Case/Subject

ⓘ Note

To use a reference field in [Condition Configuration](#) of custom logic, ensure it is explicitly tagged using the [Reference Fields Usage](#) UI. However, to use a reference field in the left-hand side (LHS) of an assignment, it is not mandatory to be tagged. In such a case, the reference field is available as long as it is marked as creatable or updatable.

ⓘ Note

These reference fields usage objects are read-only and not intended for end-user modification. If there are any changes made by users via External Hooks, it is ignored by the system. These fields are a part of both pre-hook and post-hook data, and in both the scenarios the behavior is the same.

Structure:

```
referenceUsages: { caseStatus: 'ACTIVE', ownerEmployeeDisplayId: 'b96a8f70-01fb-4f36-866f-4dc8882b669a' } *referenceUsage object is present directly under the root entity.
```

16.2.4.1 Create Reference Fields

Create reference fields that are used in custom logic.

Prerequisites

Ensure you've chosen [Custom Logic](#) from the dropdown on the [Reference Fields Usage](#) screen.

Context

A reference field is an attribute that points to another attribute in a different entity to fetch the data. These attributes are called custom reference fields.

Procedure

1. Navigate to [User Menu](#) > [System Settings](#) > [All Settings](#) > [Extensibility Administration](#).
2. Select an entity from the left pane.
3. Navigate to the [Reference Fields Usage](#) tab, and choose + [\(New\)](#). In the [Select KeyGroups](#) quick create screen, choose entities, select the attributes for these entities, and then choose [OK](#).
You've now created a reference field for usage in custom logic.

Note

- For a selected KeyGroup, only level one attributes are displayed.
- If attributes are already selected for a custom logic, such attributes are grayed out for selection.
- While creating a reference field, you can select a maximum of 10 fields at once. You can create a maximum of 30 cross-entity reference fields for custom logic from a maximum of 10 KeyGroups.
- Cross-entity reference fields and extension fields can be accessed using the path `referenceUsages/generatedfield`, which consists of the keyGroup and fieldName details.
- For each referenced field, additional details such as [Path](#), [Mapped Reference Fields](#), [External Hook](#), and [Actions](#) are displayed under respective columns. You can select the



(View Usage) under the [Actions](#) column to view the usage of the reference field within custom logic (whether used in validations, determinations, or external hooks) and choose [\(Delete\)](#) to delete the reference field.

- [View Usage](#) shows all the platform application usage of the reference field across custom logic and autoflow.
- To delete a reference field from usage, all the occurrences must be removed and also must be deleted from validations, determinations, and external hooks across custom logic and autoflow.

4. If you want to enable the usage of the reference field in external hooks, in addition to validations and determinations, turn on the [External Hook](#) switch for a particular reference field.

16.2.5 Disable Custom Logic

As an administrator you can disable custom logic that you have previously configured or added to meet specific requirements. This includes validation and determination rules.

Context

For debugging purposes, if you wish to disable the implementation of custom logic for specific users in a test landscape, you can use the [Custom Logic Administration](#) workspace.

Here, you have the option to temporarily disable the custom logic that you created for debugging purposes. By doing so, you can differentiate between SAP-generated messages and those from your own logic, aiding in your analysis and resolution process. This feature is particularly useful during debugging, enabling you to mimic the behavior of a fresh SAP solution by turning off custom logic implementation for specific users or in a test landscape.

In the [Custom Logic Administration](#) workspace, you can specify the users for whom you want to disable these custom processes. This feature is helpful when you need to temporarily suspend all custom logic implementation for selected users during debugging or testing phases.

Procedure

1. Navigate to **User Menu** > **System Settings** > **All Settings** > **Extensibility** and choose [Custom Logic Administration](#).

The system opens the [Disable Custom Logic](#) screen and displays the list of all the users who created the custom logic.

2. Select the check box next to the user that you want to delete.
3. Choose  ([Delete](#)).

All the validation and determination rules connected to that user is disabled.

Note

This feature is available only in the test landscape and you cannot add more than five users.

16.2.6 Functions in Custom Logic

This document lists all the functions available for creating validation and determination rules in [Extensibility Administration](#).

The list of functions available are as follows:

- **IS NULL**

The function supports null check for both array and non-array attributes. The result data type is Boolean.

- **Non-Array Fields**

This function accepts an expression or an attribute as an input. This function is supported for all data types.

❖ Example

- `$IS NULL ( {closeDate} )` returns true if closeDate is null otherwise false.
- `$IS NULL ( {description} == 'desc' )` returns true if description is null otherwise false.
- `( $IS NULL ( {account/name} ) == false ) && ( {account/name} == 'acc' )`

- **Array Root Fields**

This function only operates on the top-level array attribute, and doesn't support nested arrays.

The expression returns true only if the array is either not present in the input JSON or is explicitly set to null.

In case of empty arrays or arrays with values in it, the result of the expression would be false.

❖ Example

`$IS NULL ( {salesData} )` returns true if salesData array is null or not part of the input json.

- **MATCH_ALL**

Returns true if every occurrence of an attribute in the array matches with a value. Otherwise, returns false.

ⓘ Note

If a condition using `$MATCH_ALL` is evaluated on an empty array or an attribute belonging to an empty array, it shall evaluate to false.

- **MATCH_ANY**

Returns true if there's at least one occurrence of an attribute in the array that matches a value. Otherwise, returns false.

ⓘ Note

If a condition using `$MATCH_ANY` is evaluated on an empty array or an attribute belonging to an empty array, it shall evaluate to false.

- **SUM**

Returns the sum of attributes in the array. This function is applicable only for Integer and Number array operands.

- **IS_EMPTY**

Returns true if the array is empty, else returns false.

❖ Example

salesData is an array, \"salesData\": [] will return true. But \"salesData\": [{}] would return false.
When salesData is missing or \"salesData\": null the result is UNKNOWN

- **IS_CHANGED**

The function compares before image and current image of a specific attribute. The result data type is Boolean.

It isn't supported for arrays, objects, or multi-value codelist.

❖ Example

`$IS_CHANGED ( {priority} )` returns true if value of priority field has changed from its before image.

- **\$CURRENTDATETIME()**

Returns current date time value, supported as Date Time data type. Usage (`$CURRENT_DATETIME()` > '2019-08-27T00:00:00Z').

When used alone, for example (`$CURRENT_DATETIME()`), it returns the current date time value in EPOC format on evaluation.

The consuming application can then convert it in the format that is required.

- **\$CURRENT_DATE_IN_UTC**

Returns current date value, supported as Date data type. Usage (`$CURRENT_DATE_IN_UTC()` > '2019-08-27').

When used alone, for example (`$CURRENT_DATE_IN_UTC()`), it returns the current date value in UTC format on evaluation.

The consuming application can then convert it in the format that is required.

- **\$LOGGED_IN_EMPLOYEE_ID**

Returns logged in user's employee ID as a string. Usage ({ownerId} = (`$GET_LOGGED_IN_EMPLOYEE_ID()`)).

ⓘ Note

When this function is called for a technical user (for example, in a data impex scenario), it returns **null** because no value is assigned to the LHS field since no employee is associated with a technical user.

- **\$CONCATENATE (list of comma-separated string arguments)**

Returns concatenated string value. It accepts arguments of the data type String.

Usage `$CONCATENATE ( 'String1' , {account/name} , 'String2' )`

ⓘ Note

Condition configuration UI supports the following functions: **IS Null, Is Empty, Match All, Match Any, and Sum** based on the source field datatype. The assignment editor (in determination) supports **Current date time, Current date in UTC, Concatenate and Sum** functions.

- **\$COUNT**

Returns the number of attributes in an array that evaluate to true for a given condition.

Usage: `$COUNT({arrayExpression})`

Parameters

Parameter	Type	Required	Description
arrayExpression	Array(Boolean)	Yes	Array expression that evaluates to boolean values

Return Value

- **Type:** Integer (Number)
- **Value:** Number of elements that evaluate to true

Supported Data Types: Array expressions

Behavior:

- **Boolean Evaluation:** Each element is evaluated as true or false.
- **True Values Counted:** Only elements evaluating to true are counted.
- **False Ignored:** Elements evaluating to false are ignored.
- **Array-Based Operation:** Intended for use with array attributes and conditions.
- **Missing Attribute Handling:** If an attribute referenced in the condition isn't present in one or more array entries, the function doesn't throw an error. Those entries are skipped in evaluation and execution continues.
- **Zero Safe:** Returns 0 if no elements match the condition.

Null Handling

Input Array	Output	Additional Information
null	0	Null array returns zero
[]	0	Empty array returns zero

Examples:

-  **Example**
Count Matching Items

```
$COUNT({items/description} == 'Test')  
// Returns the number of items whose `description` equals Test.
```
-  **Example**
Counting Open Items

```
$COUNT({items/lifeCycleStatus} == 'OPEN')  
// Returns the number of items where `lifeCycleStatus` is OPEN.
```
-  **Example**
No Matching Entries

```
$COUNT({items/description} == XYZ)  
// Returns : 0
```
-  **Example**
Numeric Condition

```
$COUNT({items/price} > 100)  
// Returns the number of items where the price is greater than 100.
```

16.2.7 Supported Entities for Custom Logic

Custom logic feature is available only for specific entities and sub-entities. All these entities have restricted availability. To access this feature, you must submit a support ticket in the relevant application component.

List of supported entities in custom logic

- Account
 - Account Address
 - Account Domain
 - Account Contact Person
 - Account Identification
 - Account Relationship
 - Account Sales Arrangement
 - Account Sales Territory
 - Account Team Member
 - Assigned Product List
 - Call List
 - Case
 - Chat Interaction
 - Competitor
 - Competitor Product
 - Contact Person
 - Contract Account
 - Email Interaction
 - Employee
 - Functional Location
 - Installed Base
 - Individual Customer
 - Individual Customer Address
 - Individual Customer Identification
 - Individual Customer Relationship
 - Individual Customer Sales Arrangement
 - Individual Customer Team Member
 - Lead
 - Move Integration Service
 - Partner
 - Phone Call
 - Premise
 - Product
 - Product Group
 - Product Install Point
 - Product List
 - Registered Product
 - Return Order
 - Sales Quote
 - Service Order
 - Service Quotation
-

16.2.8 View Application Logs

View a summary of all errors that occurred during the processing of custom logic external hooks. This enables customers to review and address issues with their external system calls.

Procedure

1. Navigate to **User Menu** > **System Settings** > **All Settings** > **Extensibility** and choose **Extensibility Administration**.

The system opens a new screen and displays a list of all business entities in the left panel. These entities are grouped by services.

2. Select a service and choose **Drill Down** to open and select an entity.
3. Navigate to the **Logs** tab.

4. Select the [Log ID](#) or choose  in the [Actions](#) column to open the screen with the details of that log.

You can view details such as timestamps, process, user names, description and so on.

Note

These logs are visible for a period of seven days, after which they're automatically deleted from the database.

16.3 Create Dynamic Property Rules

Create and apply dynamic rules to manage property behaviors for entities and sub-entities based on specific conditions.

Context

Dynamic property rules are essential for providing flexibility and adaptability in managing entity behaviors. In complex systems, entities and sub-entities often require different property configurations based on varying business rules, conditions, or contexts.

Note

Dynamic property rules can be created only for specific entities and sub-entities, some of which have restricted availability. To access this feature, you must submit a support ticket in the relevant application component.

Account	Individual Customer Address	Product
Account Address	Individual Customer Identification	Appointment
Account Domain	Individual Customer Relationship	Sales Order
Account Identification	Individual Customer Sales Arrangement	Visit (unrestricted)
Account Contact Person	Individual Customer Team Member	Phone Call
Account Relationship	Lead	Chat Interaction
Account Sales Arrangement	Opportunity	Service Order
Account Sales Territory	Sales Quote	In-Person Interaction
Account Team Member	Key Account Plan (unrestricted)	Registered Product
Functional Location	Product Install Point	Installed Base
Individual Customer	Case	

Restriction

Dynamic property rules apply to screens and controls explicitly designed for one single type and instance of an entity. For example: a lead, an opportunity, or a case. Dynamic property rules **cannot** be applied to

screens or controls that display tables of entities or multiple types of entities. For example: the interaction log, which shows multiple phone calls, chats, in-person interactions, and more.

Procedure

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Extensibility](#) and choose [Extensibility Administration](#).

The system opens a new screen and displays a list of all business entities in the left panel. These entities are grouped by services.

2. Select a service and choose  ([Drill Down](#)) to open and select an entity.
3. Navigate to the [Dynamic Properties](#) tab and choose [+](#) ([Create](#)).
4. Select either [Field Attributes](#) or [Layout Conditions](#) to launch the [Create Dynamic Properties](#) editor tab.

Note

[Field Attributes](#) define the properties of fields (entity attributes) such as creatable, updatable, required, or hidden based on the application's context. In contrast, [Layout Conditions](#) control the visibility of sections, tabs, and buttons based on the application's context.

Field Attributes are configured directly on the entity, making them applicable across all layouts. Layout Conditions are specific to individual page layouts.

5. In the header block, specify the dynamic property name and description.
6. Turn on the [Allow Override by Business Role](#) switch if you want field settings defined in this rule to be overridden by role-based rules. This gives priority to rules created in the [Role-Based Attributes](#) editor for users with specific business roles. For more information, see the **Create Role-Based Attributes** guide in the **Related Information** section.

Note

When this switch is enabled, the system applies the role-based rule instead of the current rule for users with the specified business role.

7. In the [Condition Block](#) set the desired conditions using a flow-based visual condition editor.
 - a. Select a function from the dropdown.
 - b. Select the standard or extension field from the dropdown.

When you add a condition in a child entity, the dropdown normally shows only child fields. But in some cases, where a child field property needs to be defined based on a root field, such as, case status, you need access to root fields. To create a rule based on the root field, turn on the **Root Fields** switch.

- c. Select the comparison operator.
- d. Select the target comparison. You can compare either by a fixed value or with another field.
- e. If you choose [Value](#), enter a corresponding target value. If you choose [Field](#), select a target field.

Add multiple conditions by choosing [+](#) ([Add Condition](#)). To remove a condition, choose  ([Remove Condition](#)). To clear all conditions in a block, choose  ([Delete Condition Block](#)).

Note that when you create multiple conditions, the condition group is considered fulfilled only if all the conditions within a group are met.

- Optional: Create an OR condition by choosing the **+ Or** button. The system displays additional condition blocks. Follow the same steps as earlier to fill one or more OR condition blocks.

Note

When you create multiple OR conditions, the condition group is considered fulfilled if any one of the block conditions are met.

- In the *Properties* section, choose **+ (Create)** to add a new property.

Note

The *Properties* tab is available only for *Field Attributes*.

- Select a field from the dropdown and use the check boxes to define the properties as required.
- Save and activate your changes.

Note

You can create up to 30 dynamic property rules per entity.

Related Information

[Create Role-Based Attributes \[page 253\]](#)

As an administrator, you can create and assign role-based business rules in dynamic properties.

16.3.1 Create Role-Based Attributes

As an administrator, you can create and assign role-based business rules in dynamic properties.

Procedure

- Navigate to **User Menu** > **System Settings** > **All Settings** > **Extensibility** and select *Extensibility Administration*.

The system opens a new screen and displays a list of all business entities in the left panel. These entities are grouped by services.

- Select a service and choose  **(Drill Down)** to open and select an entity.
- Navigate to the *Dynamic Properties* tab and choose **+ (Create)**.
- Select *Role-Based Attributes*.

The editor opens with two sections:

1. Header:

- Specify the dynamic property name and description.
- Use the [Default](#) switch to apply the rule to all business roles at once. Use this when all roles should follow the same restrictions.

Note

Only one rule can be marked as default. Creating a default rule is optional and depends on your use case.

- Use the [Business Role](#) dropdown to select a specific role and customize the property behavior for that role. This allows different field attributes for different roles.

2. Properties: Use the check boxes to define restricted or unrestricted properties as required.

5. Select [Save](#) to apply your changes.

Note

The scope of a role-based attributes rule is limited to the dynamic property where it's defined. If a restriction is set in a static property or another dynamic property, that restriction takes priority. To override it based on business roles, use the [Allow Override by Business Role](#) switch in the corresponding [Field Attribute](#) rule. For more information, refer to the [Create and Assign Dynamic Property Rules](#) topic in the **Related Information** section.

Related Information

[Create Dynamic Property Rules \[page 251\]](#)

Create and apply dynamic rules to manage property behaviors for entities and sub-entities based on specific conditions.

16.4 Create Page Layouts

As an administrator, you can create customized page layouts to cater to the diverse needs of different user groups. This allows you to provide an optimized and user-centric experience for each segment of users.

Context

While the main layout is the default layout that impacts all users and establishes consistency across the entire project, there are situations where specific user groups may require a different look and feel or user interface (UI) design. In such cases, page layouts play a crucial role in customizing the user experience for different groups of users. For example, users in Bangalore may have distinct preferences, cultural influences, or language requirements compared to users in New York. By using different page layouts for each user group, administrators can create a more personalized and relevant experience, enhancing user engagement and satisfaction.

By default, when you create a new page layout, the changes you make to that layout are not automatically applied to any of the existing users. You must explicitly assign the new page layout by using the assignment rules.

This approach allows administrators to have full control over which users or user groups will be using the customized page layout. You can selectively assign the new layout based on countries, languages, business roles, individual users, territories and their hierarchies, and so on, using the assignment rules. Once the assignment is made, the users who have been assigned the new page layout start experiencing the changes and enjoy the optimized user experience tailored to their requirements.

Procedure

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Extensibility](#) ► and choose [Page Layouts](#).
2. Choose + ([Add](#)) to open the [New Layout](#) quick create screen.
3. Add a name.
4. Save your changes.

The page layout is now created and is listed under the [Page Layouts](#) screen.

You can make changes for the specific page layout by using the adaptation mode and selecting the new page layout.

Choose  ([Edit](#)) or  ([Delete](#)) to edit or delete the corresponding page layout.

Note

Conflicts can occur when changes are made to the same control in both the main layout and the page layout. In such cases, the behavior defined in the page layout takes precedence over the main layout. For example, if a field is made visible in the main layout and then explicitly hidden in the page layout, the page layout will take precedence, and the field will be eventually hidden when the page is displayed to the user.

In case of non-conflicting changes, both the main layout and the page layout changes will get applied. In this scenario, any visibility configurations set at the main layout level will be applied universally across all pages where that field appears unless explicitly overridden in individual page layouts.

16.5 Create and Assign Rules to Page Layouts

As an administrator, you can create and assign rules to page layouts.

Context

Assignment rules allow users to implement diverse logic, customize the user interface, and create distinct appearances for various user groups based on territories, departments, and users.

Procedure

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Extensibility](#) and choose [Assignment Rules](#).
2. Choose + [\(Add\)](#) to open the [New Rule](#) screen.
3. Enter the rule name and description.
4. Add a condition following these steps:
 - a. Select a property from the dropdown.
 - b. Select the comparison operator.
 - c. Enter a value.

Add multiple conditions by choosing + [\(Add Condition\)](#). To remove a condition, choose ✕ [\(Remove Condition\)](#). To clear all conditions in a block, choose 🗑 [\(Delete Condition Block\)](#)

Note that when you create multiple conditions, the condition group is considered fulfilled only if all the conditions within a group are met.

5. Optional: You can create a new condition by choosing the + [New Condition Group](#) button. The system displays additional condition blocks. Follow the same steps as earlier to fill one or more OR condition blocks.

📘 Note

When you create multiple OR conditions, the condition group is considered fulfilled if any one of the block conditions are met.

6. Save your changes.

The rule is now created and is listed under the [Assignment Rules](#) screen.
7. Navigate to the [Page Layouts](#) screen.
8. Choose + [\(Assign Rules\)](#) next to the page layout. The system opens the [Assign Rules](#) quick create screen.
9. Select the rules you want to assign to the page layout and choose [Save](#).

The rule is now assigned to the page layout.

Note

You can assign multiple rules to a single page layout. The system applies the **OR** condition in case of multiple assignment rules per page layout. Even if one assignment rule is satisfied, the page layout is assigned to the user.

If there are multiple rules assigned, the page layout that appears on top of the list on the UI takes precedence and is applied for that particular user. One user can have only one page layout applied at a time and this is decided based on precedence. To change the sequence, you can drag and drop the page layout.

Example

You have created two different Page Layouts **Sales Manager** and **Sales Rep**. They appear on the UI as follows:

1. Sales Manager
2. Sales Rep

If you fall under both the conditions, i.e., a sales manager who can act like a sales rep as well, the Sales Manager layout is given more precedence since the Sales Manager Page Layout precedes the Sales Rep page layout on the UI.

16.6 Create and Add Custom Code Lists

Create customized code lists and use them in all supported entities.

Prerequisites

As an administrator, you must configure code lists in **Settings**. Add the business service `customCodeListService` to the required business roles to provide access to custom code list. Enable the app in the [App List](#) section.

Procedure

1. To create a custom code list:
 - a. Navigate to **User Menu** > **System Settings** > **All Settings**.
 - b. Under **Extensibility**, select **Custom Code Lists**.
 - c. Choose **+ (Create Code List)** from the code list section on the left pane.
The system opens a window.
 - d. Enter a **Code List ID** and **Name**.

- e. Choose your [Default Language](#).
- f. Select the [Sorting Options](#). You can choose the order in which you want to see the code list, based on the code or description.
- g. Save your changes.

The system displays the details of your newly created code list, along with an option to add more entries to it.

- h. Choose + [\(Create Code List Entry\)](#) from the detail section on the right pane.
- i. Enter a [Code](#) and [Description](#).
- j. Choose ✓ [\(Create\)](#)

Similarly, you can add more entries based on your requirement. You can search, delete, and view usage of code lists. You can also remove and modify notes.

Note

You can create a maximum of 100 code lists, and each code list can contain up to 2000 values.

2. To create an extension field with the code list in an entity:
 - a. Go to [Extensibility](#) in the [Settings](#) page, and then select [Extensibility Administration](#).
The system opens a new screen and displays a list of all business entities in the left panel. These entities are grouped by services.
 - b. Select the [Extendable](#) checkbox to view the entities that can be extended.
 - c. Drill down to the entity you want to add an extension field to, and select it. The system opens the corresponding worklist.
 - d. In the [Custom Fields](#) tab, choose + [\(Add\)](#) to open the [Create Fields](#) quick create screen.
 - e. Enter a [Label](#).
The system auto-populates the name.
 - f. In the [Data Type](#) field, choose [String](#).
 - g. In the [Data Format](#) field, choose [Code](#).
 - h. Select the [Custom](#) checkbox.
This displays a field for code list with a single-selection option. To enable multiple selections, select the [Multi Value](#) checkbox.
 - i. Search and add the code lists.
 - j. Save your changes.
3. To add the extension field to an entity, navigate to a supported entity and add the extension field via adaptation mode.

Related Information

[Adaptation \[page 207\]](#)

As an administrator, you can change the look and feel of the solution for all users by changing layout settings, adding mashups and fields, as well as defining extension fields.

16.7 Create Code List Restrictions

Code list restriction refers to the process of limiting the available code values based on other code fields or context. As an administrator, you can restrict the values available from a drop-down list by creating and maintaining code list restrictions for different business entities.

Prerequisites

You have assigned the [Code List Restrictions](#) service to the respective business roles.

Context

By default, the system allows all the code values for a business entity to be visible. You can restrict those values based on an instance field through a combination of the following values: Business entity, code field, and control field.

Procedure

1. Navigate to **User Menu** > **System Settings** > **All Settings** > **Extensibility** and select [Code List Restrictions](#).
2. Choose **+ (Create)** to open the [Create Code List Restrictions](#) guided activity.
3. In the [Define Code List Restriction](#) step, do the following:
 - a. Enter a name and description.
 - b. From the drop-down list, select a business entity for which you want to restrict certain code types.
 - c. From the [Code to Restrict](#) drop-down list, select a code that you want to restrict for the business entity.

Note

The drop-down list includes both active and inactive codes. Make sure you select an active code.

- d. Do one of the following:
 1. From the [Control Field](#) drop-down list, select a field.
 2. Choose [Next](#) to view all the allowed code values for that business entity.
 3. In the [Configure Restrictions](#) step, deselect the check boxes to restrict all or some of the values. Select [View Details](#) to view more than 20 values.
- e. Do one of the following:
 1. Choose [Next](#) to view all the allowed code values for that business entity.
 2. In the [Configure Restrictions](#) step, select one or multiple roles from the value help and mark them as restricted if you so require.
You can choose not to define the code list restrictions based on roles.
 3. Choose [Next](#).

4. In the [Set Default Value](#) step, select a role from the value help to mark it as default. Note that this feature has restricted availability and is supported only for [Case](#). To request for access to this feature, you must create a case with the component CEC-CRM-CLR.
4. Save your changes.

Note

Code list restrictions with circular dependencies are not supported, as they can result in infinite loops during defaulting. For example, if Priority controls Status, Status controls Source Code, and Source Code controls Priority, this creates a circular dependency and is therefore not allowed.

Result: You have restricted specific code values from appearing in certain drop-down menus in the applications. This restriction is based on a combination of business entity, code field, and control field.

To view the code list restrictions that you have created, select the business entity from the drop-down menu on your screen.

Note

The [Validation Relevant](#) switch determines whether Code List Restrictions (CLR) checks are applied across different data creation channels, such as the UI or back-end APIs.

When turned on, the CLR checks are enforced for both UI-based actions and back-end channels, including API, Data Replication, and Data Import Tools (e.g., Data Impex)

When turned off, the CLR checks are performed only for UI-based operations, skipping validation for bulk imports or API-based entries. This option is designed for performance optimization. For example, when creating many accounts at once using systems like S/4HANA, skipping validation can help avoid delays.

Example

In the application UI, you have made the following selections:

- [Business Entity](#): Sales Orders
- [Code to Restrict](#): Cash Discount Term Code
- [Control Field](#): Distribution Channel

The system displays all the available code list values for the code Cash Discount Term Code, for the Sales Orders business entity. You will have an option to restrict or allow the code values for the control field values [Direct Sales](#) and [Indirect Sales](#) individually.

16.7.1 Role-Based Code List Restrictions

As an administrator, you can define code list restrictions (CLR) based on user roles.

For each code field within a business entity, you can create up to two restrictions:

- With a Control Field: Restricts values based on another field's input.
- Without a Control Field: Allows you to apply role-based restrictions, specifying which values are available for users in different roles.

Rules for Restriction Computations at Runtime

When managing CLRs for a code field in a business entity, the system follows specific rules at runtime to determine access based on role-based or instance-based restrictions:

- A restricted code value without specified roles applies universally to all users, overriding any instance-based CLR.
- A restricted code value with specified roles applies only to those roles. Other roles are allowed by default, but instance-based CLRs can further restrict these values for other users.
- An allowed code value with specified roles explicitly permits users with those roles and implicitly restricts others, taking precedence over instance-based CLR.
- An allowed code value without specified roles is open to all users, with instance-based CLR potentially applying further restrictions.
- If only an instance-based CLR exists, its restrictions apply universally to all users.
- If a single CLR with role-based configurations exists for a code field, its restrictions will apply accordingly. Specifically, if a code value is permitted for one role, it will be restricted for all other roles. Conversely, if it is restricted for one role, it will be allowed for all others. If no roles are specified, the restrictions or allowed values will apply to all roles.

16.8 Mashups

Learn how to configure, use, and access mashups.

Mashups are used to integrate data from SAP's cloud solution with data provided by an online Web service or application. Users can access the content provided by these Web services and applications, and use it in their daily work. Mashups can include Web searches, company or industry business information, or online map searches.

16.8.1 Configure Mashups

Administrators can configure mashups in [Settings](#).

→ Tip

Expand a collapsible section heading to view the configuration steps.

16.8.1.1 Enable Mashups for Business Roles

As an administrator, you must add the mashup business service to your business users.

Procedure

1. Go to [User Menu](#) > [System Settings](#) > [All Settings](#) and under *Users and Control*, select [Business Roles](#).
2. Select a [Business Role ID](#) to open the details.
3. Under [Business Services](#), choose [+](#) to open the [Add Business Services](#) window.
4. Select the check box for [mashupService](#) and choose [Add](#).

Note

You can create HTML mashups based on a mashup category provided by SAP.

Results

You have added the mashup service to the selected business user.

Related Information

[Create Business Roles and Assign Business Users \[page 28\]](#)

Administrators can create business roles with appropriate authorizations and assign them to business users.

16.8.1.2 Configure Mashup URL in Content Security Policy

As an administrator, you must configure the mashup URL in the content security policy settings.

Procedure

1. Go to [User Menu](#) > [System Settings](#) > [All Settings](#) > [System Preferences](#) > [Content Security Policy](#)
2. Go to the [Frame Source](#) section and choose [+](#).
3. Enter the hostname URL and choose [✓](#).

If you're creating a mashup with the URL `https://my123456.s4hana.ondemand.com/ui?sap-ushell-config=headerless#ServiceOrder-create` then, add `*.ondemand.com` in the [Frame Source](#).

4. To add an image to the RSS Feed card, go to the [Image Source](#) section and choose **+**.
5. Enter the hostname URL and choose **✓**.

If you're creating a mashup with the URL `https://media-cldnry.s-nbcnews.com/image/upload/t_fit_1500w/mpx/2704722219/img.jpg` then, add `*.s-nbcnews.com` in the [Image Source](#).

16.8.2 Create HTML Mashups

Create HTML mashups to embed a HTML or JavaScript based web page into a screen of your SAP cloud solution.

Context

Some web services may pass your business data to a third-party organization, for example, account data is passed to a search engine when performing a reverse lookup in an online address book. We recommend that you check whether the mashup conforms to your company's data privacy policies before activating the mashup.

Procedure

1. As an administrator, navigate to **User Menu** > **System Settings** > **All Settings** and under **Extensibility**, select **Mashup Authoring**.
2. Choose **+** (**Create**) and then choose **Create HTML Mashup**.
3. Under **General Information**, enter a mashup name.

By default, the status of the mashup is **Active**. You can also change the status to **Inactive** if necessary.

4. To use this mashup on your mobile, turn on the **Mobile Usage** switch.
5. To allow aliasing of mashup parameters during navigation (for example, from a timeline card), you must turn on the **Object-Based Navigation** switch. Once enabled, you can alias your parameters using either the **Object ID** or the **Object Display ID**.
6. Choose **Show More** and update other required fields.

Note

You can adjust the screen height. The default height is 500 pixels.

7. From the **Manage Permissions** dropdown, select **Microphone**, **Camera**, **Notifications**, or **Clipboard** to allow mashups to access your system's microphone, notifications, and clipboard, respectively.

Note

Ensure that your system permissions are enabled.

8. Under [Input Parameters](#), enter the URL of the website in the field.
9. Select [Extract Parameters](#).

The system verifies the URL domain and shows  if the domain is already available under the trusted domain list.

Note

For URLs that don't use queries, you can manually add curly brackets around terms that should act as placeholders. For example, in the URL `https://mail.google.com/mail/#search/SAP`, you can replace the word SAP with a search term in curly brackets, for example, `{term}`. If you then enter `https://mail.google.com/mail/#search/{term}` in the [URL](#) field and click [Extract Parameters](#), the word in brackets is extracted as a parameter.

10. If the URL domain is not in the trusted domain list, hover over  and [Add to Trusted Domain List](#).
11. To add parameters to this mashup, select [Add Parameter](#) and enter the required details.

In the [Parameters and Value Binding](#) section, configure the required parameters for the mashup.

- Enter a [Parameter Name](#) to identify the mashup parameter.
- Choose [Constant](#) to assign a fixed value to the parameter, which cannot be changed at runtime.
- Turn on the [Mandatory](#) switch if the parameter must always have a value.
- Turn on the [Multi Value](#) switch to allow the mapped application field to hold multiple values in an array format.
- Select [System Parameter](#) to bind the parameter to a predefined system value.

Note

Field Parameter Binding allows you to bind a mashup parameter to an application field's value. The field must be exposed by the business object, and the binding is done in adaptation mode when adding the mashup to a screen.

Caution

Do not add any sensitive information.

12. Under [Event Configuration](#), enter an event name and add parameters according to your requirements.

Note

The admin must map this event to the application event or directly to a standard operation on the respective screens through adaptation.

13. Choose [Show Preview](#) to check the details.
14. Save your changes

16.8.2.1 Initiate Navigation from Mashups

You can initiate navigation to specific screens in SAP Sales Cloud and SAP Service Cloud Version 2 from any external web applications using a mashup.

Note

- The external web application must be added as an HTML mashup in SAP Sales Cloud and SAP Service Cloud Version 2.
- Navigation is supported only for the list view, detail view, quick view, and quick-create view.
- This feature is also supported for custom UIs.

Initiate navigation refers to the action of navigating to a specific screen or view within SAP Sales Cloud and SAP Service Cloud Version 2, such as the list, detail, quick, or quick create view, based on the parameters passed from an external web application.

To initiate navigation, the external application must use the `window.postMessage()` method to send data to SAP Sales Cloud and SAP Service Cloud Version 2. This data must include specific parameters required for the navigation to work.

The following details explain these parameters and help you create the required HTML code in the external application:

Parameter	Default Value	Description	Mandatory/Optional	Usage Detail
operation	navigation	Name of the operation.	Mandatory	Used by the Mashup web component for determining the action that needs to be triggered.
params	null	Parameters required for initiating navigation.	Mandatory	Used by the Mashup web component for passing the required parameters to trigger the mentioned action.
params.objectKey	null	UUID of the entities or things.	Mandatory	Used to navigate to the specific entity or thing.
params.routingKey	null	Name of the entities, retrieved from the <code>routingKey</code> property in the application UI view metadata in SAP Service Cloud Version 2.	Mandatory	Used for determining the entity to which user wants to navigate.

Parameter	Default Value	Description	Mandatory/Optional	Usage Detail
params.viewType	details	Name of the screen or view to navigate to.	Mandatory	Used for determining the view or screen to which the user wants to navigate. The following view types are supported: • "list" for list view • "details" for entity detail view • "quickcreate" for entity quick create screen • "quickview" for entity quick view.

You can use the following code snippet as a reference. Add a similar script to your external web application and integrate it using an HTML mashup to initiate the navigation.

Note

For custom UI navigation, you can obtain the `routingKey` information from the Design App.

```
<script>
function openCaseView() {
    var case={
        operation:"navigation",
        params: {
            objectKey: "00163ea6-f284-1eda-b895-42c904e08752",
            routingKey: "case",
            viewType:"details"
        }
    };
    window.parent.postMessage(case, '*')
}
</script>
<input type='button' value='Open Case Details' onclick='openCaseView();' />
```

16.8.2.2 Initiate Search Events from Mashups

As an administrator, you can initiate search events to specific screens in SAP Sales Cloud Version 2 and SAP Service Cloud Version 2 from any external web applications using a mashup.

Note

- The external web application must be added as an HTML mashup in SAP Sales Cloud and SAP Service Cloud Version 2.

To initiate a search event, the external application must use the `window.postMessage()` method to send data to SAP Sales Cloud and SAP Service Cloud Version 2. This data must include specific parameters required for the navigation to work.

The following details explain these parameters and help you create the required HTML code in the external application:

Parameter	Default Value	Description	Mandatory/Optional
operation	null	Name of the operation.	Mandatory
params	null	Parameters required for initiating navigation.	Mandatory if required by the specific event
event	null	Name of the event..	Mandatory

You can use the following code snippet as a reference. Add a similar script to your external web application and integrate it using an HTML mashup to initiate the search event.

```
<script>
function triggerSearchEvent() {
    var event = {event: "mashup100.magiccom.customExtendedSearch", operation:
"search", params: {$search: "alex"}};
    window.parent.postMessage(event, '*')
}
</script>
<input type='button' value='Search' onclick='triggerSearchEvent();' />
```

16.8.2.3 Initiate Custom Events from Mashups

Use the Mashup callback feature in SAP Sales Cloud and SAP Service Cloud Version 2 to initiate custom events from embedded web applications. These events must be defined and maintained by the respective application teams.

Note

The external web application must be added as an HTML mashup in SAP Sales Cloud and SAP Service Cloud Version 2.

Implementation Guidelines for External Web Applications

To initiate a custom event from a mashup, note the following:

- Use `window.parent.postMessage` within your embedded web application to communicate with the SAP Sales Cloud and SAP Service Cloud Version 2 mashups.
- A specific JSON format (described in the table below) must be sent from the embedded web application to the SAP Sales Cloud and SAP Service Cloud Version 2 mashups using `window.parent.postMessage`.
- Parameters passed to the event are sanitized using `encodeURIComponent` before processing.
- The **Name of event** and **parameter names** must exactly match those defined during mashup authoring.

Format of the JSON data

The following table outlines the structure of the JSON object that must be sent via `window.parent.postMessage` from the embedded web application:

Property Name	Data Type	Default Value	Mandatory/Optional	Description
operation	string	null	Mandatory	Name of the operation
event	string	null	Mandatory	Name of the event
params	object	null	Mandatory if required by specific event	Parameters required to trigger the event

Example

Scenario: A user wants to initiate a refresh event from a web application embedded in a mashup within the Account object detail screen. The event name is "accountRefreshEvent", and it does not require any parameters.

The following sample provides an idea of the logic that must be written using `window.parent.postMessage`:

```
<script>
function triggerRefreshEvent() {
    var event = { event: "accountRefreshEvent",
                  operation: "triggerCustomAction"
                };
    window.parent.postMessage(event, '*')
}
<input type='button' value='Refresh' onclick= triggerRefreshEvent();' />
```

16.8.3 Create URL Mashups

As an administrator, you can create URL mashups to send data from SAP Sales Cloud and SAP Service Cloud Version 2 to a web service provider's URL. The results are displayed on the provider's website in a new browser window.

Context

Some web services may pass your business data to a third-party organization, for example, account data is passed to a search engine when performing a reverse lookup in an online address book. We recommend that you check whether the mashup conforms to your company's data privacy policies before activating it.

Note

You can create URL Mashups for the following entities: Accounts, Individual Customers, Contacts, Competitors, Partner Contacts, Leads, Cases, Guided Selling, Sales Quotes, Sales Orders, Employee, Partner, Supplier, and Supplier Contact.

Procedure

1. Navigate to [User Menu](#) > [System Settings](#) > [All Settings](#) > [Extensibility](#) > [Mashup Authoring](#).
2. Choose [+](#) ([Create](#)) and then select [URL Mashup](#).
3. Under [General Information](#), enter a mashup name.

By default, the status of the mashup is [Active](#). You can also change the status to [Inactive](#) if necessary.

4. To use this mashup on your mobile, turn on the [Mobile Usage](#) switch.
5. To allow aliasing of mashup parameters during navigation (for example, from a timeline card), you must turn on the [Object-Based Navigation](#) switch. Once enabled, you can alias your parameters using either the [Object ID](#) or the [Object Display ID](#).
6. Choose [Show More](#) and update other required fields.
7. Under [Input Parameters](#), enter the URL of the website in the field.
8. Select [Extract Parameters](#).

The system verifies the URL domain and shows  if the domain is already available under the trusted domain list.

Note

For URLs without query parameters, you can manually insert curly brackets around terms you want to use as placeholders. For example, replace **SAP** in `https://mail.google.com/mail/#search/SAP` with `{term}` to create `https://mail.google.com/mail/#search/{term}`. When you enter this in the [URL](#) field and select [Extract Parameters](#), the word in brackets is extracted as a parameter.

9. If the URL domain is not in the trusted domain list, hover over  and choose [Add to Trusted Domain List](#). The system redirects you to the [Content Security Policy Setting](#), where you can add the URL to the trusted domains list.
10. To add parameters to this mashup, select [Add Parameter](#) and enter the required details.

In the [Parameters and Value Binding](#) section, configure the required parameters for the mashup.

- Enter a [Parameter Name](#) to identify the mashup parameter.
- Choose [Constant](#) to assign a fixed value to the parameter, which cannot be changed at runtime.
- Mark the [Mandatory](#) checkbox if the parameter must always have a value.
- Choose [System Parameter](#) to bind the parameter to a predefined system value.

Note

Field Parameter Binding allows you to bind a mashup parameter to an application field's value. The field must be exposed by the business object, and the binding is done in Adaptation mode when adding the mashup to a screen.

⚠ Caution

Do not add any sensitive information.

11. Select [Show Preview](#) to open the preview in a new browser tab.

12. Save your changes.

To use the mashup, perform the following steps:

1. Navigate to an entity and open a record in detailed view.
2. Choose [Start Adaptation](#) to add the URL mashup as a button or a link at any place where a mashup can be added.
3. Choose [End Adaptation](#) after adding the mashup.

Results

The URL is launched in a separate browser tab on choosing the button or the link.

16.8.4 Create Web Component Mashups

Create web component mashups to embed that into a screen of your SAP cloud solution.

Context

Some web services may pass your business data to a third-party organization, for example, account data is passed to a search engine when performing a reverse lookup in an online address book. We recommend that you check whether the mashup conforms to your company's data privacy policies before activating the mashup.

📘 Note

To enable this feature, create a case with the CEC-CRM-PM-PF component using the subject **Enable Creation of Web Component Mashups**.

Procedure

1. As an administrator, navigate to [User Menu](#) > [System Settings](#) > [All Settings](#) and under [Extensibility](#), select [Mashup Authoring](#).
2. Choose [+ \(Create\)](#) and then choose [Create Web Component Mashup](#).
3. Under [General Information](#), enter a mashup name.

By default, the status of the mashup is [Active](#). You can also change the status to [Inactive](#) if necessary.

4. Optional: To use this mashup on your mobile, turn on the [Mobile Usage](#) switch.
5. Select [Show More](#) and update the required fields.

Note

You can adjust the screen height. The default height is 500 pixels.

6. From the [Manage Permissions](#) dropdown, select [Microphone](#), [Notifications](#), or [Clipboard](#) to allow mashups to access your system's microphone, notifications, and clipboard, respectively.
7. Under [Input Parameters](#), enter the [Resource URL](#) of the website in the field.
8. Enter the [Web Component](#) name.
9. To add parameters to this mashup, select [Add Parameter](#) and enter the required details.

Caution

Do not add any sensitive information.

10. Choose [Show Preview](#) to check the details.
11. Save your changes.

Related Information

[Create Analytics Widget Mashups \[page 271\]](#)

Create an SAP Analytics Cloud mashup that is embedded on the home page card to show advanced charts via SAP Analytics Cloud.

16.8.5 Create Analytics Widget Mashups

Create an SAP Analytics Cloud mashup that is embedded on the home page card to show advanced charts via SAP Analytics Cloud.

Context

Some web services may pass your business data to a third-party organization, for example, account data is passed to a search engine when performing a reverse lookup in an online address book. We recommend that you check whether the mashup conforms to your company's data privacy policies before activating the mashup.

Note

To enable this feature, create a case with the CEC-CRM-PM-PF component using the subject **Enable Creation of Analytics Widget Mashups**.

Procedure

1. As an administrator, navigate to ► [User Menu](#) ► [Settings](#) ► [All Settings](#) ► and under [Extensibility](#), select [Mashup Authoring](#).
2. Select [+ \(Create\)](#) and then choose [Analytics Widget](#).
3. Under [General Information](#), enter a mashup name.

By default, the status of the mashup is [Active](#). You can also change the status to [Inactive](#) if necessary.

4. Select [Show More](#) and update the required fields.

Note

You can adjust the screen height. The default height is 500 pixels.

5. Under [Parameters and Value Binding](#), select [Add Parameter](#) and enter the required details.
6. Save your changes

16.8.6 Copy Existing Mashups

You can create a copy from the existing mashups.

1. Go to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Extensibility](#) ► [Mashup Authoring](#) ►.
2. From the new tab, select the mashup you want to copy by clicking on the mashup name.
3. From the [Mashup Details](#) screen, choose [Duplicate](#).
The system opens a copy of the mashup.
4. Enter a new name for the mashup and adapt the configuration settings, as required.
5. Save your changes.

16.8.7 Add Mashups as Tabs

As an administrator, you can add a mashup as a new tab.

Using page layouts and visibility property, you can control who is allowed to use the mashup.

To add a mashup as a tab, follow these steps:

1. **Create a tab on a screen.**
 1. Navigate to a screen where you want to add the mashup.
 2. Go to ► [User Menu](#) ► [Start Adaptation](#) ►. The system opens in the adaptation mode.
 3. Hover over the area where you see a plus icon (+).
The system highlights the area with a red border along with a tool tab
 4. From the tool tab, select the pencil icon to edit. A new tab opens.
 5. Click the plus icon (+) and enter a tab name.
 6. Click [Apply](#).
The new tab is added to the header of the screen.

2. Add a mashup to the tab.

1. Open the new tab and hover over the empty space to enable the red highlight with the tool tab.
2. From the tool tab, select the pencil icon to edit. A new tab opens.
3. Select the arrow available next to [Add Mashup](#) to open a new window.
4. Click the arrow available next to the required mashup to display the properties.
5. If necessary, you can adjust the appearance of the mashup by selecting one of the following options under [Edit Properties](#).

Note

The options available to you may vary depending on the mashup.

- [Show Header](#): Select the [Show Header](#) checkbox if you want to see the mashup title on the screen.
- [Height \(%\)](#): You can define the height of HTML Mashups to be displayed in the available screen. For example, if you set the value of Height (%) to 100, the newly added mashup occupies the full height of the screen.
If the selected mashup has only a parameter name added, without a constant or system parameter, the system displays an additional field to choose the value to be mapped.

Note

The embedded mashup instantly updates when you change the mapped parameter.

6. Click [Apply](#).
7. To save your settings, click [End Session](#) on the top of the screen.

Related Information

[Adaptation \[page 207\]](#)

As an administrator, you can change the look and feel of the solution for all users by changing layout settings, adding mashups and fields, as well as defining extension fields.

16.8.8 Delete or Deactivate Mashups

Delete or deactivate mashups that you and other users have created.

1. Go to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► and under [Extensibility](#), select [Mashup Authoring](#).
2. Select a mashup from the list.
3. Choose the [Delete](#) icon, or turn the [Active](#) switch off.
If you choose to deactivate, the mashup is no longer visible on screens. If you choose to delete, the mashup is removed from the [Mashup Authoring](#) view and is also deleted from all screens for which it had been made visible.

16.8.9 Edit URL and Parameter Value for Transported Mashups

As an administrator, you can manually enter the URL and parameter value for mashups in the target system.

Context

When you transport a mashup to the target system, the URL field and parameter value are not transported and remains empty. As a result, any screen where the mashup is used displays a warning message indicating that no URL is added. To ensure the mashup works correctly, you must manually enter the URL and parameter value in the target system.

Procedure

1. Navigate to **User Menu** > **System Settings** > **All Settings** and under **Extensibility**, select **Mashup Authoring**.
2. Select the transported mashup for which you need to enter a URL.
3. Choose **Edit**, located on the top-right corner of the screen.
4. Under **Input Parameters**, enter the URL of the website in the field.
5. Under **Parameters and Value Binding** edit the values for **Constant** and **System Parameter**.
6. Select **Save**.

See the related link section for information on transporting configuration changes from a source tenant to a target tenant.

Related Information

[Change Project](#)

16.8.10 Retain Mashup State

As an administrator, you can retain the mashup state when switching tabs.

Context

When you switched between tabs in the application, mashups would refresh, causing any data you had entered to be lost. This disrupts the workflow for users, especially when they need to copy information from different screens into the mashup.

With the [Mashup Retain State](#) feature, mashups built with HTML or Web Components will no longer reload when you switch tabs. Your data stays just the way you left it, making it easier to work without interruptions or repeated inputs.

Note

To enable this feature, create a case with the CEC-CRM-PM-PF component using the subject **Enable Mashup Retain State**.

Procedure

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Extensibility](#) ► [Mashup Authoring](#) ►
2. Select the mashup for which you want to retain the state while switching tabs.
3. Choose  [\(Edit\)](#).
4. Under [General Information](#), turn on the [Retain State](#) switch.
5. Save your changes.

Caution

Be careful when choosing which mashups to keep open. Keeping too many can impact performance, so use this feature wisely.

Note

- Browser support details:
 - Google Chrome: Supported from version 133 onwards
 - Microsoft Edge: Supported from version 133 onwards
 - Mozilla Firefox and Safari: Not supported yet
- This feature supports all shell tab switching but may not function consistently when switching between tabs within the entity details screen. The entities where it works are: Accounts, Contact, Individual Customer.

16.8.11 Mashups Where Used List

The *Where Used* feature tracks where mashups are used in other entities or as homepage cards.

It displays the usage in a hierarchical structure that shows the complete dependency from the top-level component down to the mashup.

The feature tracks the usages only for mashups created on or after January 15, 2026.

When you delete such mashups the system will display an error message. Use the Where Used feature to identify and remove all usages before deleting the mashup.

Note

Mashups created before January 15, 2026 may not have their usages tracked. If you delete such mashups, the system allows the deletion even if they are used in other entities, which may lead to errors. To track the usage of these mashups, go to the screen where the mashup is used, make a minor change in adaptation mode, and save it. This enables the system to register the mashup usage.

To view the where used list do the following:

1. Navigate to ► *User Menu* ► *System Settings* ► *All Settings* ► and under *Extensibility*, select *Mashup Authoring*.
2. From the list of mashups, navigate to the desired mashup and choose ≡ (*Where Used*).
A pop up opens listing the usages of the mashup.

16.9 Modify Work Center Sequence

As an administrator, you can rearrange the order in which work centers appear in the navigation menu. You can also set a default work center that opens each time you access the screen.

Procedure

1. Navigate to ► *User Menu* ► *System Settings* ► *All Settings* ► *Extensibility* ► and choose *Work Center Sequence*.
2. From the *Default Work Center* dropdown, select the work center that should open by default whenever you access the screen.
3. Choose ≡ (*Move*) next to a work center and drag it to adjust its position in the sequence.
4. Confirm your changes.

16.10 Language Adaptation

Language adaptation customizes communication for different users, considering business and linguistic preferences. It ensures content resonates effectively, improving comprehension and engagement.

This practice is vital for global communication, making information accessible and enhancing the user experience across diverse linguistic and cultural backgrounds.

16.10.1 Change User Interface Texts for Applications

As an administrator, you can translate or change field labels, codelist descriptions, custom messages and UI texts for applications.

Context

You can use this feature to override standard SAP texts and custom message texts, as well as customize field label terminology to fit your business needs.

Procedure

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Extensibility](#) ► [Language Adaptation](#) . The system opens the **Language Adaptation** tool in a new window.
2. Select an application, source language, and target language for maintaining custom translations. Keep the source language and target language same if you just want to change the field labels or texts.
3. Choose [Next](#).
4. Navigate to one of the following:
 - [Field Labels](#) tab if you want to change the field labels. The system displays the current target language text.
 - [Code Lists](#) tab if you want to change the code lists descriptions. If no code lists are selected, select one or multiple code lists from the drop down and click outside the box. The system displays the list of selected code lists and the current target language text.
 - [Messages](#) tab if you want to edit custom and standard messages (currently available only for [Opportunities](#)). You must create a case with the component CEC - CRM - TXA to enable standard message editing.
 - [Timeline Texts](#) tab if you want to translate the texts that appear in Timeline events. Note that parameters such as "{\$leadId}", "{\$closeDate}", and other placeholders starting with "{\$...}" cannot be added or removed; they must be included in the target language text. Only the surrounding static text can be modified.
5. Modify the text in one of the following ways:

- For simple texts, use the [Target Language Text](#) column and save your changes. To undo the changes, choose [↶ \(Reset\)](#).
You can also change the source text by selecting [Edit Source Text](#). To go back to the SAP delivered target language text, select [↶ \(Revert\)](#)
- To change texts enmasse, use the following steps:
 - Choose [↵ \(Export\)](#).
 - In the [Export](#) screen, select the file type as either [Excel](#) or [CSV](#) and select [Export](#). The system generates a zip file.
 - Select the zip file.
It gets download to your local system. The file contains two excels with the following column headers:
 - File 1: ID, Service, Source Language, Target Language
 - File 2: ID, Label ID, Default Target Language Text, Source Language Text, Target Language Text, Text Parent_Technical_Key and a new column header called **New Target Language Text**
 - Add a new text or update the existing text in the **New Target Language Text** column.
 - Save your changes and re-compress the Excel files into a ZIP file.
 - Navigate to the original tab and select [↶ \(Import\)](#).
 - In the [Import](#) screen, select the file type as either [Excel](#) or [CSV](#). You can either choose to import the complete entity by switching on the [Complete Entity](#) option, or select from [Field Labels](#), [Code Lists](#), and [Messages](#).
 - Browse for the *.zip prepared data file you want to upload.

→ Remember

Use that latest file for import. If an old file is used, existing data is overwritten considering update file data as the latest data. Data which is not to be updated can be left unchanged in the exported file. We recommend that you import only a maximum of 50,000 records in a single file.

- Select [Import](#) and save your changes.

Related Information

[Data Import and Export \[page 70\]](#)

This document describes how you can use Data Import and Export to import data into your solution and export data from your solution.

16.11 Extensibility Using SAP Build Apps

SAP Build Apps enables you to build apps for all form factors including mobile, desktop, browser, TV, and others without writing code.

You can extend your system using SAP Build Apps by designing, developing, and embedding these extensions on your system. You can access Build Apps by choosing [⋮ \(Switch Application\)](#) in the Shell bar.

Note

If you've 100 or more users, SAP Build Apps base package is included as a default with SAP Service Cloud Version 2. If you've fewer than 100 users, you need additional licensing for SAP Build Apps to build extensions. You must contact your account executive to purchase licenses.

Related Information

[Design and Develop Extensions on SAP Service and Sales Cloud Version 2 Using SAP Build Apps](#)
[Setup SAP Build Apps with SAP Sales Cloud and SAP Service Cloud Version 2](#)

16.12 Configure Inbound Authentication

Propagate user identity from BTP service to SAP Sales Cloud Version 2 and SAP Service Cloud Version 2 using OAuth2 SAML Bearer Assertion Flow BTP destinations.

Context

The following steps describe how to connect a SAP Sales Cloud Version 2 and SAP Service Cloud Version 2 system using the OAuth2 SAML Bearer Assertion Authentication flow. In this flow, the user identity of the logged-in user from source system can be propagated to SAP Sales and Service Cloud Version 2 system.

Procedure

1. Login to your SAP BTP cockpit, and navigate to **BTP Subaccount** > **Destinations** > **Subaccount Destinations** > **Download Trust**.
2. Copy contents of the certificate between the header and the footer.
3. Create an inbound configuration in your tenant.
 - a. Verify if your tenant is already configured.

Code Syntax

```
Method: GET
URL: <Your tenant host url>/sap/c4c/api/v1/iam-service/
oauth2InboundConfiguration/34f8b9d2-78c2-4617-bcf2-6eb36403c667
```

- b. If there exists a configuration, verify if it is the same certificate or different. If the certificate is same, you can use the existing configuration. Otherwise, override it.

Code Syntax

```
Method: PUT
URL: <Your tenant host url>/sap/c4c/api/v1/iam-service/
oauth2InboundConfiguration/34f8b9d2-78c2-4617-bcf2-6eb36403c667
Body:
{
  "id": "34f8b9d2-78c2-4617-bcf2-6eb36403c667",
  "clientAppName" : "<give your own name>",
  "certificate" : "<certificate extracted in above step>"
}
```

- c. If there is no inbound configuration, create a new inbound configuration.

Code Syntax

```
Method: POST
URL: <Your tenant host url>/sap/c4c/api/v1/iam-service/
oauth2InboundConfiguration
Body:
{
  "id": "34f8b9d2-78c2-4617-bcf2-6eb36403c667",
  "clientAppName" : "<give your own name>",
  "certificate" : "<certificate extracted in above step>"
}
```

Once you create or update the configuration, you get the following response.

Output Code

```
{
  "value": {
    "id": " ... ",
    "adminData": {
      "createdBy": " ... ",
      "createdOn": " ... ",
      "updatedBy": " ... ",
      "updatedOn": " ... ",
      "deletedBy": null,
      "deletedOn": null,
      "deleted": false
    },
    "description": " ... ",
    "grantType": "SAML2_BEARER",
    "scopes": [
      "<SCOPE_ID>"
    ],
    "nameIdType": "EMAIL",
    "spEntityId": " ... ",
    "audience": "<AUDIENCE>",
    "tokenServiceUrl": "<TOKEN_URL>",
    "clientId": "<CLIENT_ID>",
    "clientAppName": " ... ",
    "certificate": " ... "
  }
}
```


⚠ Restriction

Only one OAuth2 SAML Assertion Bearer Flow Inbound Configuration can be configured for a tenant.

- Now, go to your [BTP Subaccount](#) > [BTP Subaccount - Create New Destination](#) and create a new destination entering the following information:

Option	Description
Name	<As per your choice>
Type	HTTP
URL	<Your tenant host URL>
Authentication	OAuth2SAMLBearerAssertion
Audience	<AUDIENCE> from the inbound configured JSON response
AuthContextClassRef	urn: none
Client Key	<CLIENT KEY> from the inbound configured JSON response
Token Service URL	<TOKEN_URL> from the inbound configured JSON response
Scope	<SCOPE_ID> from the inbound configured JSON response
nameIdFormat	urn:oasis:names:tc:SAML:1.1:nameid-format:emailAddress

📌 Note

If this destination is also used for SAP Build Apps, select additional properties and enter the information.

AppGyverEnabled	true
HTML5.DynamicDestination	true
IsSAPSalesAndServiceCloudAPI	true

- Configure trusted domains for SAP authorization and trust management service: [Configure Trusted Domains for SAP Authorization and Trust Management Service](#).

Related Information

[Security Considerations for the SAP Authorization and Trust Management Service](#)
[OAuth SAML Bearer Assertion Authentication](#)

16.13 Custom Services

Custom Service refers to a service that is developed, implemented, and deployed outside of the SAP Sales Cloud and SAP Service Cloud Version 2 systems. However, its metadata, such as entity definitions and event structures, is imported into the metadata repository of the systems.

This is a key feature that enables add-ons, allowing external services to integrate seamlessly with the SAP Sales Cloud and SAP Service Cloud Version 2 processes.

Once integrated, the custom service can fully participate in the SAP Sales Cloud and SAP Service Cloud Version 2 ecosystem and leverage core platform capabilities, such as:

- **Built-in features** like autoflow, timeline, and so on.
- **Custom work centers** with auto-generated user interfaces, including:
 - List View
 - Detail View
 - Quick View
 - Quick Create
 - Timeline

This approach allows customers to extend the functionality without altering the core platform, ensuring flexibility and future-proofing custom developments.

There are two types of custom services: one is mashup based, and the other is entity based. The mashup-based version features a new UI flow and expanded configuration options .

Note

You can create up to 30 custom services in total, with the following limits:

- A maximum of **30 mashup-based** custom services.
- A maximum of **15 entity-based** custom services.

These limits are shared. For example: If you create **30 mashup-based** services, you cannot create any more entity-based services. If you create **15 entity-based** services, you can create up to **15 mashup-based** services, totaling 30 overall.

Related Information

[API Guidelines \[page 298\]](#)

This page outlines the API guidelines that external or custom services must follow to integrate with SAP Sales Cloud Version 2 and SAP Service Cloud Version 2. These guidelines are for REST based externally implemented APIs.

[Metadata Guidelines \[page 287\]](#)

When integrating external services, inconsistencies may arise due to differences in data types, attributes, associations, and operations. To address this, a JSON file is used to map the external service details to structure expected in the solution. By explicitly defining these mappings in the JSON file, you ensure seamless and accurate service integration.

16.13.1 Create a Mashup-Based Custom Service

As an administrator, you can create a mashup-based custom service.

Procedure

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Extensibility](#) ► [Custom Services](#) ►.
2. Choose **+** ([Create](#)) and then select [Mashup-Based Custom Service](#).
 - a. On the [Service Details](#) screen, enter a title and description.
 - b. Choose [Confirm](#).
 - c. Navigate to the [Custom Services](#) screen and choose **↻** ([Refresh](#)) to view the new service.

Note

You can create a maximum of 30 mashup-based custom services.

- d. To edit or delete a service, choose **✎** ([Edit](#)) or **🗑** ([Delete](#)) under the [Actions](#) column.
3. To assign a mashup into the new service, choose **📄** ([Design App](#)) under the [Actions](#) column.

Note

Ensure that the new service is activated. If it is not, wait until the status changes from [In Process](#) to [Activated](#).

- a. On the [UI View](#) screen, choose **+** ([Create](#)) available on the top-right corner of the screen.
The system opens a new screen.
- b. Enter a title.
- c. Choose an icon.
- d. Select a UI View Pattern from the available options:
 - Work List
If you intend to use the mashup solely as a work center feature, assign it to a worklist.
 - Detail View
 - Quick View
 - Quick Create
- e. Choose **📄** ([Add Mashup](#)) to open the Mashup pop-up window.
- f. Select a mashup and choose [Apply](#).

Note

The mashup name that you configured in Mashup Authoring will not be visible on the UI.

- g. Save your changes.
The [UI View](#) page shows the new design.
- h. Turn on the [Include in Navigation Menu](#) switch to include this screen to the navigation menu as a work center.

- i. Choose *Confirm*.
4. To manage the custom service and the interface you created, choose **⋮ (More)** available on the top-right corner of the screen and choose the following options:
 - *Assign Business Roles*: For more information, see [Create Business Users and Assign Business Roles \[page 23\]](#).
 - *Manage Work Center Sequence*: For more information, see [Modify Work Center Sequence \[page 276\]](#).

📘 Note

The previous **Mashup as a Work Center** flow experience has been deprecated on September 1, 2025 and is replaced by the new custom service flow. This change applies only to the admin creation flow and not to the underlying functionality. The existing mashups will continue to work as-is and no changes are required to your current configurations or data. The update introduces a revamped UI with enhanced features for a better authoring experience.

Results

You have added the mashup as a work center under the navigation menu.

16.13.2 Create an Entity-Based Custom Service

Create a custom service to integrate external functionality into the SAP Sales Cloud and SAP Service Cloud Version 2 systems without modifying the core application.

Procedure

1. Navigate to **► User Menu ► System Settings ► All Settings ► Extensibility ► Custom Services ►**.
2. Choose **⊕ (Create)** and then select *Entity-Based Custom Service*.
 - a. On the *Service Details* screen, enter a *Service Title* and update other required details.

📘 Note

For CAP-based custom services, you can automatically generate the metadata JSON file using the **CAP File Conversion** feature. This reduces the manual effort required to fill in details based on your CAP service. For more details, see the **Related Information** section.

- b. Upload a service JSON file that contains custom metadata using the *Drop or Browse Files* option.
You can select **↓ (Download JSON Files)** to download the following files based on your requirement:
 - *Template*: This downloads a JSON file template without any data
 - *Sample*: This downloads an example of JSON file with data
 - *Current Metadata*: This downloads a copy of your uploaded JSON file

SAP Sales Cloud and SAP Service Cloud Version 2 requires that both the service API and metadata conform to specific guidelines. For more details, refer to the **Metadata Guidelines** and **API Guidelines** in the **Related Information** section.

- c. Choose *Confirm*.

Navigate to the *Custom Services* screen and in the **Actions** column, choose  (*Refresh*) to view the new service.

- d. Optional: To edit or delete a service, choose the  (*Edit*) or  (*Delete*) icons in the *Actions* column.

3. To Configure the host for your custom service, Go to the *Actions* column for the service you want to configure the host for, and then select  (*Configure Host*).

Note

The service must contain a JSON file with the API metadata.

- a. In the *Configure Host* window, select one of the following connection types:
 - *Direct Connection*: If you select this option, you must provide the **Host URL** for Direct Connection. As the name suggests, a direct call is made from the SAP Sales and Service Cloud application to the external or custom service.

Note

- The external service must support **Single Sign-On (SSO)**.
 - **CORS (Cross-Origin Resource Sharing)** settings must be handled by the external service.
- *Backend Connection*: In this connection type, calls to the external service API are not made directly. Instead, they are routed through a standard backend service within SAP Sales and Service Cloud, essentially functioning as a server-to-server call.
To use the backend connection type, you must configure an outbound communication system with the domain URL of the external service. Once the communication system is set up, you can select the operations (as defined in the metadata) that should be supported.

Note

- The connection uses server-to-server communication.
- Authentication is based on the method configured in the communication system.
- Supported authentication methods:
 - OAuth 2.0 – Client Credentials
 - OAuth 2.0 – SAML Bearer Assertion (Recommended for user propagation)

- b. Save your changes.

4. To embed an interface into the new service, choose  (*Design App*) in the *Actions* column.

Note

Ensure that the new service is activated. If it is not, wait until the status changes from *In Process* to *Activated*.

- a. On the *UI View* screen, choose  (*Create*) available on the top-right corner of the screen.
- b. Enter a title.
- c. Choose an icon.

- d. Select a UI View Pattern from the available options:

- Work List
- Detail View
- Quick View
- Quick Create

Note

You can either generate a custom UI or assign a mashup. After saving, you can switch from a mashup to a custom UI, but not the other way around.

- e. To generate a custom timeline card for this external service, select [Timeline Card](#) under the [Select Embedded Component](#) section. To enable navigation from the timeline card, make sure a [Quick View](#) is assigned to the custom service under the [Select UI View Pattern](#) section. If a custom UI is selected for the Quick View, it will be used for navigation. If a URL mashup is selected, navigation will open the mashup in a new browser tab.
- f. Save your changes.
- The [UI View](#) page shows the new interface design.
- g. To include this screen to the navigation menu as a workspace, turn the [Navigation Launchpoint](#) switch on.
- h. Choose [Confirm](#).

Note

The system currently supports up to two child entities per root entity. Multi-level hierarchies are not yet supported.

5. To manage the service, choose [⋮ \(More\)](#) available on the top-right corner of the screen and select the following options:
- [Assign Business Roles](#): For more information, see [Create Business Users and Assign Business Roles \[page 23\]](#).
 - [Manage Workspace Sequence](#): For more information, see [Modify Work Center Sequence \[page 276\]](#).

Related Information

[Metadata Guidelines \[page 287\]](#)

When integrating external services, inconsistencies may arise due to differences in data types, attributes, associations, and operations. To address this, a JSON file is used to map the external service details to structure expected in the solution. By explicitly defining these mappings in the JSON file, you ensure seamless and accurate service integration.

[API Guidelines \[page 298\]](#)

This page outlines the API guidelines that external or custom services must follow to integrate with SAP Sales Cloud Version 2 and SAP Service Cloud Version 2. These guidelines are for REST based externally implemented APIs.

[CAP File Conversion \[page 315\]](#)

For CAP-based custom services, you can automatically generate the metadata JSON file using the [CAP File Conversion](#) feature. This reduces the manual effort required to fill in details based on your CAP service.

[Manage Communication Systems](#)

[Create Custom Timeline Events](#)

[Create HTML Mashups \[page 263\]](#)

Create HTML mashups to embed a HTML or JavaScript based web page into a screen of your SAP cloud solution.

16.13.2.1 Metadata Guidelines

When integrating external services, inconsistencies may arise due to differences in data types, attributes, associations, and operations. To address this, a JSON file is used to map the external service details to structure expected in the solution. By explicitly defining these mappings in the JSON file, you ensure seamless and accurate service integration.

Note

Custom services are primarily implemented outside the SAP Sales Cloud and SAP Service Cloud Version 2 infrastructure. Their metadata is imported into the SAP Sales Cloud and SAP Service Cloud Version 2 metadata repository, enabling seamless integration with SAP Sales Cloud and SAP Service Cloud Version 2 processes such as generating custom UIs to list, create, or update entities, adding timeline entries, or defining autoflow actions.

These metadata artifacts are described in the following section. You can download a sample metadata file or template from the **Custom Service Administration** screen.

Service

The **Service** metadata acts as a container for all other metadata artifacts and is required if you plan to define any of them.

name

A unique technical name of the service.

Must start with a letter and contain only alphanumeric characters. It should not be suffixed with the word **service**. (for example: "projectOrder").

Type: string

fullName

Auto-generated using the **name** and the default namespace (`customer.ssc`).

A "Service" suffix is added automatically (for example: "`customer.ssc.service.projectOrderService`"). This is why the name itself should not end in **service**.

Type: string

title

Display name of the service, readable name used on the UI.

Type: string

description

Description of the service. This property can be used to explain the purpose of the service.

authorizationRelevant

If this property is set to true, the service needs to be added to a business role assigned to the user who will access it via the custom UI (or SAP Sales Cloud and SAP Service Cloud Version 2 channels). If the user sets it to false, there will be no check from SAP Sales Cloud and SAP Service Cloud Version 2. To generate a custom UI, this property must be set to true (so that an admin can assign the corresponding UI app to a user role).

Note

This mechanism only provides filtering of UI apps and API calls to the custom service (from SAP Sales Cloud and SAP Service Cloud Version 2). Your external service must implement its own access control logic and not depend on this filtering.

```
"service": {
  "name": "ProjectOrder",
  "title": "ProjectOrder",
  "description": "Project Order Service For Demo",
  "authorizationRelevant": true
}
```

Entity

A Service is an aggregate of one or more entities. Entities can have references to other entities. An entity represents a resource that can be accessed from an API. In that sense, it represents the publicly accessible model of entity data

This is a core metadata to define most of the artifacts. For example, it is used to define the request and response structure of the (custom service) API, payload structure of the events, and used to generate the UI views. An entity can be composed of attributes and can refer other entities in the same or other service. Usually every service has one aggregate entity (marked as root). A service can have only one aggregate entity.

In addition to business entities, code list entities are also supported.

An entity metadata sample:

```
{
  "name": "projectOrder",
  "label": "Project Order",
  "entityType": "BUSINESS",
  "objectTypeCode": "CUS1271",
  "root": true,
  "attributes": [
    {
      "name": "id",
      "label": "Name",
      "dataType": "STRING",
      "dataFormat": "UUID",
      "nullable": false,
      "creatable": true,
      "updatable": false,
      "filterable": true,
      "searchable": true,
      "keyType": "PRIMARY"
    },
    {

```



```

        "name": "name",
        "label": "Name",
        "dataType": "STRING",
        "description": true,
        "nullable": false,
        "creatable": true,
        "updatable": true,
        "filterable": true,
        "searchable": true
    },
    {
        "name": "account",
        "dataType": "OBJECT",
        "label": "Account",
        "objectDefinition": [
            {
                "name": "id",
                "dataType": "STRING",
                "creatable": false,
                "updatable": false,
                "nullable": false,
                "sortable": false,
                "filterable": true,
                "searchable": false,
                "label": "Account ID",
                "dataFormat": "UUID",
                "descriptionAttribute": "formattedName",
                "keyType": "FOREIGN",
                "objectReference": {
                    "associationType": "ASSOCIATION",
                    "targetAttribute": "id",
                    "targetEntity":
"sap.ssc.md.accountservice.entity.account",
                    "keyGroup": "account",
                    "targetService":
"sap.ssc.md.service.accountService"
                }
            },
            {
                "name": "formattedName",
                "dataType": "STRING",
                "creatable": true,
                "updatable": true,
                "nullable": true,
                "sortable": true,
                "filterable": true,
                "searchable": true,
                "label": "Account Name",
                "objectReference": {
                    "associationType": "REFERENCE",
                    "targetAttribute": "formattedName",
                    "keyGroup": "account"
                }
            }
        ]
    },
    {
        "name": "estimatedRevenue",
        "label": "Estimated Revenue",
        "dataType": "OBJECT",
        "dataFormat": "AMOUNT",
        "nullable": false,
        "creatable": true,
        "updatable": true
    },
    {
        "name": "products",
        "dataType": "ARRAY",

```

```

        "creatable": true,
        "updatable": true,
        "nullable": true,
        "sortable": false,
        "filterable": true,
        "searchable": true,
        "label": "Products",
        "itemDataType": "OBJECT",
        "entityReference": "product"
    }
}

```

name

Technical name for the entity (for example: "projectOrder"). Every entity needs to have a unique technical name within a service. Only alphanumeric characters are supported and must start with a letter.

Type: string

label

Display name of the entity (for example: "Project Order"). The translation support of these labels is supported via the Language Adaptation Tool.

Type: string

entityType

It can have following values:

- BUSINESS - Indicates that the entity is a business entity.
- CODEVALUE - Indicates that the entity is used to stored code values.

Type: string

objectTypeCode

Object type code for the entity type. It is used in various object relationship. This is a 4 digit value prefixed with 'CUS'. This property is added for Root Entity only. If type code is not provided, it is generated by the system.

Type: string

root

Indicates if the entity is the root of the aggregate.

Type: boolean

Attributes

An attribute represents a field in the entity. Attributes can also be complex or be defined by an association to another entity, within the same service or across services. An attribute can have following properties in the metadata:

name

The technical name of the attribute (for example: "id", "name"). It should be alphanumeric starting with a letter. It can contain underscore. It is recommended to use lower camel case.

Type: string

label

Human-readable and translatable label for the attribute. Translation support of these labels is supported via the Language Adaptation Tool.

Type: string

dataType

Data type of the attribute (for example: string, number). The API implementation should validate the data type in the payload to ensure it matches the expected type. Custom UI uses this property to determine the type of input control (for example: text box, number field, date picker).

For supported data type and their characteristics, see the **Datatypes in SAP Sales Cloud and SAP Service Cloud Version 2** section.

Type: string

dataFormat

The data format enables the consumer to understand the way the data needs to be processed (or) formatted (for example: CODE, FLOAT). In custom UI, this property would help in rendering the field with specific formatting or validation (for example: dropdown for CODE, numeric input for FLOAT). Custom service API implementation should ensure the data adheres to the specified format.

For supported data format and their characteristics, see the **Datatypes in SAP Sales Cloud and SAP Service Cloud Version 2** section.

Type: string

nullable

Indicates if value null is allowed for the attribute. In the custom UI, an attribute marked nullable as false will be validated to ensure they are filled before submission. API implementation should also not allow null value if this property is set to false.

Type: boolean

creatable

Indicates if the attribute can be set during entity creation. Custom UI will mark corresponding UI field as read-only if the value of this property is false. API implementation should ignore any value passed in the payload for such cases. Note that these attributes can be set in the backend processing (calculation based on other attributes) even if creatable is false.

Type: boolean

updateable

Indicates if the attribute can be modified during update. If value of this property is true, the respective UI field will appear as read-only. API implementation should ignore the value of such attribute in the payload if updateable is false.

Type: boolean

filterable

Specifies if the entity instances can be filtered based on this attribute. This property ensures that only relevant attributes are available for filtering in the UI. If the value is true, the API implementation should respect the attribute value in a filter expressions, but if the value is false such attributes can be ignored or rejected (in filter expressions).

Type: boolean

searchable

Specifies if the entity instances can be searched using this attribute. If the value is true, the API implementation should respect the attribute value in a search expressions, but if the value is false such attributes can be ignored or rejected (in a search expression).

Type: boolean

keyType

This should be used only if the attribute is a key. Possible values are PRIMARY and FOREIGN.

- An entity should have an attribute named `id` of type `UUID` that is marked as a PRIMARY key.
Exception: In case it is not possible to have a key of type `UUID`, `displayId` attribute can be used as a key. However support for such entities would be limited to certain scenarios (time-line and related activities). An entity must have either `id` or `displayId` (or both).
- There should not be more than one attribute marked as PRIMARY in an entity
- FOREIGN is used for attribute that has reference to another entity and represents primary key for that entity. An attribute with reference to another entity cannot be marked as PRIMARY.

Type: string

objectDefinition

This is used to define object structure for object data type. For example see the "account" attribute in the above entity metadata example.

Type: Object

objectReference

An attribute can refer to an attribute of an associated entity. When there is an association to another entity, this setting specifies the referenced foreign attribute. Note that this is only an indicator of the source entity from which the data originates. It helps consumers (such as a custom UI) during edit operations by enabling value help (Object Value Selector) for these attributes. For read operations, the custom service API must include the corresponding data in the response payload.

Type: Object

An object reference has following properties:

- **"associationType":** "ASSOCIATION" or "REFERENCE". In the entity relationship, one attribute need to establish the relationship. It needs to point the primary-key of the target entity. For the source entity this attribute is marked as foreign key. For the foreign key attribute, the association type must be of type "ASSOCIATION". For other reference-attributes via same association it needs to be set as "REFERENCE".
- **"keyGroup":** The keyGroup represents the association name. The keyGroup is defined on the foreign key attribute and used for all other attributes that use the same association.
 - Use a readable name for the keyGroup as it is also used in the admin screen.
 - The keyGroup should be unique for an entity.
 - keyGroup must be alpha-numeric only, and should not start with a number.

- **"targetService"**: Target service name. This needs to be specified only on the foreign-key attribute.
- **"targetEntity"**: Specifies the target entity name in the association. This needs to be specified only on the foreign-key attribute.
- **"targetAttribute"**: Attribute name in the target entity.

description

Indicates if this attribute represents the description of the entity. The description-attribute is also used as description of the corresponding `id` attribute, specially in the case if the `id` is referenced in another entity. For example in the above entity metadata sample, `formattedName` is marked as `"descriptionAttribute"` of the `"account/id"`. There can be only one attribute marked as description-attribute (`description: true`) for an entity.

Type: boolean

displayID

display

Indicates if this attribute is display id attribute, which could be used as an alternate key. This property designates an attribute as a user-friendly identifier for the entity. In custom UI, attributes marked as `displayId` are often displayed prominently in tables, forms, or dropdowns to help users identify records.

itemDataType (for array data type)

This property required for the `array` data type to specify the data type of the items in the array.

- `itemDataType` could be a primitive data type (like `string`) or another entity. For example in the above sample it points to the `"product"` entity (to define array of products).
- An array of entity must have an `id` attribute that identifies an item uniquely.

Type: string

Events

Defines supported events for the service. Events provides asynchronous communication from custom service to SAP Sales Cloud and SAP Service Cloud Version 2. These events can be consumed in the SAP Sales Cloud and SAP Service Cloud Version 2 processes, such as, autoflow and timeline.

```
{
  "name": "projectOrderCreate",
  "entityReference": "ProjectOrder",
  "trigger": "CREATE"
}
```

name

Technical name for the event (for example: `"projectOrderCreate"`). Every event needs to have a unique technical name within the service. This name would be used to generate the event's fullname. The technical name needs to be alpha numeric with lower Camel Case.

Type: string

fullName

This is generated out of name. E.g.: `"customer.ssc.projectorderservice.event.projectOrderCreate"`. Here `"customer.ssc"` is name space, `"projectorderservice"` is the service name and `"projectOrderCreate"` is the name of the event.

Type: string

entityReference

The entity this event is associated with (for example: "projectOrder"). Entity metadata defines the structure of the event payload.

Type: string

trigger

Event trigger type, such as CREATE, UPDATE, or DELETE. It defines the condition under which the event is triggered, such as when an entity is created, updated, or deleted.

Type: string

API

Note

Refer to the **API Guidelines** in the **Related Information** section for more details on response headers and messages.

API metadata sample:

```
{
  "name": "projectOrder",
  "description": "API for interacting with the projectOrder service",
  "title": "Project Order Collection API",
  "apiPath": "/project-order-service/projectOrders",
  "entityReference": "projectOrder",
  "operations": [
    {
      "id": "readProjectOrder",
      "method": "QUERY",
      "request": {
        "pathVariables": []
      }
    },
    {
      "id": "readProjectOrder",
      "path":("/{key})",
      "method": "READ",
      "request": {
        "pathVariables": [
          {
            "name": "id",
            "dataType": "STRING",
            "dataFormat": "UUID"
          }
        ]
      }
    },
    {
      "id": "createProjectOrder",
      "path": "/",
      "method": "POST",
      "request": {
        "pathVariables": []
      }
    }
  ]
}
```

```

    },
    {
      "id": "updateProjectOrder",
      "path": "/{key}",
      "method": "PATCH",
      "request": {
        "pathVariables": [
          {
            "name": "id",
            "dataType": "STRING",
            "dataFormat": "UUID"
          }
        ]
      }
    }
  ],
}

```

name

Technical name for the API (for example: "projectOrder"). Every api needs to have a unique technical name within a service. This name would be used to generate the event's fullname. The technical name needs to be alpha numeric with lower Camel Case.

fullName

Like other metadata artifacts full name is generated based on the name space and name. E.g: "customer.ssc.projectorderservice.api.productOrder".

Type: string

entityReference

Name of the entity to which this API provides access to. The request/response schema is determined based on the entity metadata

Type: string

description

Brief description of the API's purpose.

Type: string

title

Display title for the API.

Type: string

apiPath

Relative base path for the API (for example: "/project-order-service/projectOrders").

Type: string

operations

It defines the supported operations, such as, READ, CREATE, UPDATE, or DELETE, along with binding to the respective HTTP methods, relative path and variables

Type: array

Each operation has the following properties:

- **"id"** : An unique id for the operation. Example: "readProjectOrder".
- **"path"** : The relative path (from API path) to perform operation. For example to read a particular entity ("readProjectOrder" operation), the path is "{key}". Here particular record will be fetched based on the **primary** key of the entity. See the following path variables. The actual path to call the API would be //.
- **"pathVariables"**: Path variables defines the place holders in the path that can vary based on the resource requested. In the above example the consumer needs to pass the primary key (usually **id** attribute) of the entity to be fetched.

Note: Placeholder name in path attribute should be unique for each entity per service. Means, if "{key}" is already defined as path for say root entity, then for child entity, path should be different, such as "{parentKey}"

-
- **"method"** : HTTP method like POST

Datatypes in SAP Sales Cloud and SAP Service Cloud Version 2

In SAP Sales Cloud and SAP Service Cloud Version 2, metadata attributes are defined using datatypes (aligned with OpenAPI/JSON Schema standards) and data formats (providing additional context). Please refer to the following table for more information on available datatypes.

Data Type	Data Format	Description
STRING	STRING (Default)	Default string format. If dataFormat is not set, default data format assumed.
	UUID	Universally Unique Identifier.
	EMAIL	Email address.
	CODE	Code list (fixed via enumOptions or dynamic via code entity reference).
	TOKEN	Token format.
	URI	Uniform Resource Identifier.
	ALPHANUMERIC	An alphanumeric string (0-1, A-Z, a-z)
NUMBER	FLOAT (Default)	Default number type.
	DOUBLE	Double-precision number.
BOOLEAN	No value	Represents true/false.
OBJECT	No value	For structured or complex types.
	AMOUNT	Amount with a structure (e.g., sap.crm.common.struct.amount).
	QUANTITY	Quantity with a structure (e.g., sap.crm.common.struct.quantity).

Data Type	Data Format	Description
	TEXT	Text with a structure (e.g., <code>sap.crm.common.struct.text</code>)
ARRAY	No value	Applicable data format specified via <code>itemDataType</code> .
INTEGER	INT32 (Default)	Default integer type.
	INT64	Long integer.
DATETIME	DATETIME (Default)	Default datetime type.
	DATE	Date only.
	TIME	Time only.
	DURATION	Duration type.

UI Metadata

Apart from the metadata uploaded there are also metadata that is generated and attached to the service. These metadata are generated as new user interface(UI) created. These include:

- UI View: A UI View represent an UI component such as list, detail and quick view.
- UI App: The UI Views logically belong to one or more UI Apps. UI Apps perform many roles, such as, filtering the UI components based on user roles and hold routing information.

Lifecycle and Versioning

All the artifacts are owned by the service. If the service is deleted, all the related artifacts are also deleted. Currently old versions of custom service metadata are not stored in the SAP Sales Cloud and SAP Service Cloud Version 2 repository. Since the custom services are primarily implemented outside the SAP Sales Cloud and SAP Service Cloud Version 2 infrastructure, versioning of the metadata also needs to be maintained externally.

Related Information

[API Guidelines \[page 298\]](#)

This page outlines the API guidelines that external or custom services must follow to integrate with SAP Sales Cloud Version 2 and SAP Service Cloud Version 2. These guidelines are for REST based externally implemented APIs.

16.13.2.2 API Guidelines

This page outlines the API guidelines that external or custom services must follow to integrate with SAP Sales Cloud Version 2 and SAP Service Cloud Version 2. These guidelines are for REST based externally implemented APIs.

The custom UIs, generated via extensibility tools in Sales & Service Cloud (SSC), use the [REST \(Representational State Transfer\)](#) APIs provided by the custom service. The custom service APIs are implemented outside the SSC infrastructure but needs to follow the guidelines outlined in this page.

REST APIs (also called RESTful Web Services) are cross-platform APIs used to perform QCRUD (Query, Create, Read, Update, Delete) operations on data resources using the HTTP protocol. It uses HTTP methods to interact with resources that are identified using a Uniform Resource Identifier (URI). Requests and responses are sent in a structured format.

Operations

The following table maps CRUD operations to HTTP methods:

CRUD Operation	Semantics	HTTP Method	Example
Query	Query a collection of resources.	GET	GET /project-order-service/projectOrders
Read	Read a single resource.	GET	GET /project-order-service/projectOrders/1234
Create	Create a resource in collection.	POST	POST /project-order-service/projectOrders`
Update (partial)	Partial update of the resource	PATCH	PATCH /project-order-service/projectOrders/1234
Delete	Deletes a resource.	DELETE	DELETE /project-order-service/projectOrders/1234

📌 Note

The PUT verb is not currently used by the custom UI.

URL

The complete API URL is constructed as : `https://<application base url>/<api-path>`

- **application-base-url**: The application's base URL, specified when creating a custom service in the admin settings.
- **api-path** : The relative API path provided in the metadata JSON.

Further **application-base-url** should be constructed as following: `///`. Especially **version** is required to be present. Currently this must be "v1".

🔗 Example

In `https://XXX.ondemand.com/customer/ssc/api/v1//project-order-service/projectOrders,`

- "https://XXX.ondemand.com" is **host**.
- "customer/ssc" is **namespace**.
- "api" is **prefix**. It can be any other suitable prefix.
- "v1" is version.
- "/project-order-service/projectOrders" is (**api-path**)

Refer to the **Metadata Guidelines** topic for more details on API path. Various CRUD operations will be performed as per API information uploaded in the metadata.

JSON as a Content Type

All the APIs must accept and provide response in JSON format unless specifically mentioned otherwise.

The content type must be manifested in the API response by using the HTTP header Content-Type.

Content-Type: application/json

JSON payload for a single resource as well as a collection should be wrapped within a root object with **value** property as below.

Example: Single resource response

```
{
  "value": {
    "id": "192399f2-00ef-4024-80bf-9940c8a1e320",
    "name": "Meow"
  }
}
```

Example: Collection resource response

```
{
  "value": [
    {
      "id": "192399f2-00ef-4024-80bf-9940c8a1e320",
      "name": "Meow"
    },
    {
      "id": "192399f2-00ef-4024-80bf-9940c8a1e321",
      "name": "Woof"
    }
  ]
}
```

Query a Collection

For query on a collection, GET (HTTP) method is used. The query should support query parameters \$top, \$skip, \$search, \$filter, \$orderBy and \$count as defined below.

\$top, \$skip

For query, the pagination is primarily client-driven, for example, the client (Custom UI) would request for the pages from the API and the API responds accordingly.

Query Parameter	Description
\$top	Indicates page size.
\$skip	Indicates the offset (number of items). For example, if the page size is 100 and the user is navigating to the 10th page, the offset (\$skip) would be 900 (9 * 100).

Example: Get 3rd page with page size of 50
GET /project-order-service/projectOrders?\$skip=100&\$top=50"

\$search

Provides the basic search (for example, free text search) expression in the query to search in a collection.

Example: Search for `custom service` within project order collection.
GET /project-order-service/projectOrders?\$search="sample project"

- The search expression required to be enclosed in double quotes ("..").
- Phrases are required to be enclosed in double quotes for search("This is a value").
- If the search value is a single word, quotes can also be skipped though not recommended. API providers need to support this.

\$filter

This is used to provide filtering on the individual attributes, with the ability to use complex conditions.

Example: Get all the project orders with estimated revenue
GET /project-order-service/projectOrders?\$filter=account/id eq '0002a5fa-4e10-11ed-be55-6b0cc10b11d2' and estimatedRevenue/content eq 200 and estimatedRevenue/currencyCode eq 'USD'

- The filter expressions are not enclosed in single or double quotes.
- Only the **equals** (eq) and **starts with** operators are supported in the custom UI. The **starts with** operator appears as **startswith** for Data services and **sw** for REST services.

Literal Values

Literal values follow the [OData URL conventions](#) .

Type	Description	Example
Boolean	Boolean literals are represented as low-ercase names	true, false
Date	RFC 3339 full-date without quotes	2024-01-19
DateTimeStamp	RFC 3339 date-time without quotes	2024-01-19T13:43:21Z
Duration	dayTimeDuration in single quotes	'P21DT15H42M3.1415927S', '-PT5M'
Null	The null literal is represented as a low-ercase name	null

Type	Description	Example
Number	Numeric values may contain a decimal point and use exponential notation	42, 3.14, 6.371e3
String	String literals are surrounded by single quotes, single quotes within the string are doubled	'l' 'amour'
Time	RFC 3339 partial-time or full-time without quotes	23:59:59.999999, 13:43:21Z
UUID	RFC 4122 representation without quotes	192399f2-00ef-4024-80bf-9940c8a1e320

\$orderby

Used for ordering the result. It is comma separated list of attribute names, attributed with asc or desc.

Example: Get Products ordered by release data (Ascending) and Price (Descending)
GET /project-order-service/projectOrders?\$orderby=account/name asc

\$select

The \$select is used as query parameter to select specific attributes (or) nodes from the aggregate. When attributes are specified via \$select, the rest of the attributes and nodes are excluded automatically from the payload in the response. The following are some of the other characteristics of \$select.

- For a list, only the attributes in value node can be specified.
- For deep structures, the path can be navigated by using forward slash /.
- Comma , is used as the separator.
- Primary key (entity primary key) is always returned.
- The \$select does not determine the order of fields.
- Wildcards (*) or regex are not supported.
- If an attribute is specified that does not exist in entity, ignore the attribute.
- The entire string or parts must not be enclosed in single or double quotes.

❖ Example

GET /project-order-service/projectOrders?\$select=name&\$top=1

RESPONSE

```
{
  "value": [{
    "id": "192399f2-00ef-4024-80bf-9940c8a1e320",
    "name": "Sample project "
  }]
}
```

\$count

The \$count is used in the path of the api to get only the count of records. While retrieving the list of records, \$count is used as a query parameter to indicate if the count should be returned along with list of records.

GET /project-order-service/projectOrders?\$search="sample"

Response:

```
{
  "count" : 11
}
```

```
GET /project-order-service/projectOrders?$search="sample"&$select=name&$count=true
Response
{
  "count": 11,
  "value": [{
    "id": "192399f2-00ef-4024-80bf-9940c8a1e320",
    "name": "sample"
  },
  {...}]
}
```

Read a Single Resource

To read a single resource, GET (HTTP) method is used.

Example read project order with the id value '0002a5fa-4e10-11ed-be55-6b0cc10b11d2`
 GET /project-order-service/projectOrders/0002a5fa-4e10-11ed-be55-6b0cc10b11d2

Create

To create a resource, POST (HTTP) method is used.

❄ Example

```
POST /project-order-service/projectOrders
{
  "name": "Sample Project",
  "account": {
    "id": "111399f2-00ef-4024-80bf-9940c8a1e311"
  }
}
```

📌 Note

In POST, PATCH and PUT methods, the content is enclosed directly without a wrapper root object.

Request Content

- The content is of JSON type and represents the entity.
- Unless specifically mentioned, id attribute is not passed in the request payload and it must be generated by the service.
- If request payload contains an attribute that is marked as creatable=false (in entity metadata), ignore the attribute.
- Unknown attributes should be ignored.

Response

Headers

ETag: Return the ETag of the created resource in the response header. Usually, for the child entity create, the state of the root is also updated and ETag of root is returned.

Status

On successful creation, API should return 201 response code with the relative path of the resource created.

HTTP/1.1 201 Created

Location: /project-order-service/projectOrders/0002a5fa-4e10-11ed-be55-6b0cc10b11d2

Response Content

- The created resource is returned on successful execution of the create request.
- The content returned is dependent on the entity being created. For e.g. if POST is on /projectOrders , the response should contain the entire Project Order aggregate. If the POST is on /projectOrders/4dde9b8e-6fe9-11ea-bc55-0242ac130003/products then the created Product content is returned.
- For error cases, an error payload may be returned as described in the error responses
- If additional messages have to be sent as part of response, these are handled as part of the info message. Refer message handling

Update (Partial)

To update a resource (partial), PATCH (HTTP) method must be supported.

PATCH /project-order-service/projectOrders/444399f2-00ef-4024-80bf-9940c8a1e222

Content-Type: application/merge-patch+json

If-Match: "2022-06-29T14:45:42.272Z"

```
{
  "name": "Changed Sample Project ",
}
```

Request Content

- The content is of JSON type and represents the entity.
- id is passed as part of the URL.
- If request payload contains an attribute that is marked as updatable=false (in entity metadata), ignore the attribute.
- Unknown attributes should be ignored.

Request Header

If-Match PATCH requests are not idempotent and to avoid conflicting updates, this header must be supported. The value represents the state of the resource available with the client (ETag obtained from API is used as value for If-Match)

When a consumer tries to update a resource, it informs the server to perform the update if and only if the state of resource known to the client (ETag) matches the state of the resource in server.

For more details see the [HTTP Headers](#) guide.

Response

Headers

ETag: Return the ETag of the updated resource in the response header. Usually, for the child entity update, the state of the root is also updated and ETag of root is returned.

Status

- Return HTTP code:
 - 200, if the request is successful and content is being sent as part of the response payload.

- 204, if the request is successful and no content is being sent back as part of the response payload.
- 412, if the passed **If-Match** value does not match against the current state.
- 428, if the precondition (If-Match) is missing.
- For other response code see the **HTTP Response Code and Message Handling Guidelines** document.

Response Content

- The updated resource content may (http code 200) or may not (HTTP code 204) be returned on successful execution of the update request.
- The content returned must conform to the metadata of entity updated.
- For error cases, an error payload may be returned as described in the error responses.
- If additional messages have to be sent as part of response, these are handled as part of the info message.

Delete

To delete a resource, DELETE (HTTP) method must be supported. Similar to PATCH, the resource to be deleted is specified in the URL.

DELETE /project-order-service/projectOrders/444399f2-00ef-4024-80bf-9940c8a1e222

Request Content

- Entity id is passed as part of the URL.
- Content is not provided in the response body.

Request Header

If-Match may or may not be specified in the request header. But if passed in the request header it should be treated similar to PATCH. It means if the state on the server does not match with the state passed in this header, the service should return HTTP error code 412.

Response

Headers

ETag: For aggregate entity deletion, no ETag is returned. For the child entity deletion, the state of the root is also updated and ETag of root is returned.

Status

- Returns HTTP code 204, if the request is successful.
- Returns 412, if the passed **If-Match** value does not match against the current state of the resource.

Response Content

- No content is send back in the response payload.
- For error cases, an error payload may be returned as described in the **HTTP Response Code and Message Handling Guidelines**.

📌 Note

For more details on error cases, response codes, and response headers, see the **Related Information** section.

Related Information

[HTTP Headers \[page 305\]](#)

The following list contains common HTTP headers that custom services should support.

[HTTP Response Code and Message Handling Guidelines \[page 307\]](#)

This documents explains the guidelines to be followed with respect to response codes and messages (Error, Info, Warnings) raised by external custom service.

[Metadata Guidelines \[page 287\]](#)

When integrating external services, inconsistencies may arise due to differences in data types, attributes, associations, and operations. To address this, a JSON file is used to map the external service details to structure expected in the solution. By explicitly defining these mappings in the JSON file, you ensure seamless and accurate service integration.

16.13.2.2.1 HTTP Headers

The following list contains common HTTP headers that custom services should support.

Service Headers

Request Header

HTTP Request Header	Description
Accept - Language	Used to provide the response language expected by consumer. Refer to Accept-Language Header  for more information about the header and examples- - APIs must support both 2-character language codes (e.g., Accept - Language: de) and 2-2 character codes (e.g., Accept - Language: en-US). - APIs must return a Content - Language header indicating the language of the response.

HTTP Request Header	Description
If-Match	This header is used as part of request (PUT and PATCH) in cases where conflict management is required. Its value represents the state of the resource as known to the client (using the ETag previously obtained from the API as the If-Match value). When a consumer tries to update a resource, it informs the server to perform the update if and only if the state of resource known to the client (ETag) matches the state of the resource in server. Services shall implement conflict handling for PUT and PATCH depending on the use case. The value of If-Match is enclosed in double quotes ("2022-06-29T14:45:42.272Z"). If double quotes are missing, the extraction of the tag may fail and pre-condition checks may be unsuccessful (412 Precondition Failed).
content-type	Content

Response Header

HTTP Response Header	Description
Content-Language	Content-Language is used for indicating the language intended for the users. It doesn't indicate that the document is written in that language. - Services must respond using Content-Language header to indicate the language intended for audience. - For services/APIs that are language-neutral, the Content-Language header may be skipped to indicate the content is intended for all users.
Allow	When the service responds with 405 Not Allowed (request HTTP verb not allowed), the service must also indicate the verbs that are allowed for the requested resource. Example: Allow: GET, POST, PUT
Location	For a POST request, services must include the URL of the created resource in this response header. Relative URL of the API is sent as the value. The relative URL should start from the service-URL

HTTP Response Header	Description
ETag	The ETag is used to represent the state of a resource. APIs use strong ETags with strong comparison. APIs must include the ETag in responses for GET, PUT, and PATCH requests. The ETag typically applies to the entire aggregate; creating or updating a child entity uses the same aggregate ETag.

16.13.2.2.2 HTTP Response Code and Message Handling Guidelines

This documents explains the guidelines to be followed with respect to response codes and messages (Error, Info, Warnings) raised by external custom service.

HTTP Response Codes

Follow standard codes for HTTP status

- 2XX - Success codes
- 4XX - Error Due to client
- 5XX- Error due to server

2XX Response Code

Status	Verb	Response	Description
200 OK	GET	Query: Requested resources for GET	Represents successful operation
200 OK	GET	Read: Requested resource for GET	Represents successful operation
200 OK	PATCH	Updated resource (in full)	Represents successful operation
200 OK	DELETE	Representation of status (for example, minimal payload with identifier, status)	Represents successful operation

Status	Verb	Response	Description
201 Created	POST	Created resource after the successful creation	Only used for successful create. The location header in response should also contain the URL to the resource.
202 Accepted	POST		This needs to be supported only where asynchronous operations are supported, for example, request was received and will be processed asynchronously.
202 Accepted	PATCH		Represents successful operation
202 Accepted	DELETE		Represents successful operation

Example for 2XX response code.

Consider an example where the entity metadata is as following.

```
{
  "value": {
    "name": "authors",
    "label": "Authors",
    "root": true,
    "attributes": [
      {
        "name": "id",
        "label": "ID",
        "dataType": "STRING",
        "dataFormat": "UUID",
        "nullable": false,
        "creatable": true,
        "updatable": false,
        "filterable": true,
        "searchable": true,
        "keyType": "PRIMARY"
      },
      {
        "name": "Name",
        "label": "Name",
        "dataType": "STRING",
        "description": true,
        "nullable": false,
        "creatable": true,
        "updatable": false,
        "filterable": true,
        "searchable": true
      },
      {
        "name": "books",
        "dataType": "ARRAY",
        "creatable": true,
        "updatable": true,
        "nullable": true,
        "sortable": false,
        "filterable": true,
        "searchable": true,
        "label": "Books",
        "itemDataType": "OBJECT",
        "objectDefinition": [
```

```

{
  "name": "id",
  "dataType": "STRING",
  "creatable": false,
  "updatable": false,
  "nullable": true,
  "sortable": false,
  "filterable": true,
  "searchable": true,
  "label": "Technical ID",
  "dataFormat": "UUID",
  "keyType": "PRIMARY"
},
{
  "name": "title",
  "label": "Title",
  "dataType": "STRING",
  "nullable": true,
  "creatable": true,
  "updatable": true,
  "filterable": true,
  "searchable": true,
  "description": true
},
{
  "name": "description",
  "label": "Description",
  "dataType": "STRING",
  "nullable": true,
  "creatable": true,
  "updatable": true,
  "filterable": true,
  "searchable": true
},
{
  "name": "inStock",
  "dataType": "BOOLEAN",
  "creatable": true,
  "updatable": true,
  "nullable": true,
  "sortable": true,
  "filterable": true,
  "searchable": true,
  "label": "In Stock"
}
],
"entityReference": "books"
}
]
}
}

```

A successful response would look as following:

```

{
  "value": [
    {
      "id": "0f8cbc4c-0417-45a1-8a32-75563d8ef3bc",
      "name": "John Doe",
      "email": "john.doe@example.com",
      "books": [
        {
          "id": "1",
          "title": "Book 1",
          "inStock": true,
          "description": "A great book"
        },
        {
          "id": "2",

```

```

        "title": "Book 2",
        "inStock": false,
        "description": "Another great book"
    },
    ],
    "createdOn": "2025-02-01T05:00:02.669Z",
    "updatedOn": "2025-05-06T17:00:02.689Z"
}
]
}

```

Status	Verb	Response Body	Description
400 Bad Request	All	Error Message	Request cannot be processed as it (URL or request body) does not match what was defined for the API.
401 Unauthorized	All		User is not authenticated. This error code would be sent at the API gateway. At the service implementation, if the JWT is not received or cannot be decoded, then this status SHOULD be raised.
403 Forbidden	All		When the user is authenticated but does not have the authorizations to access the resource, then this error code is used.
404 Not Found	All		When a user has authorizations to access a resource type, but the resource does not exist in the system then this error code is used.
405 Not Allowed	All		If a specific HTTP verb is not supported by an API, then this status code should be returned when the consumer attempts to use that HTTP verb.
412 Precondition Failed	PATCH	Error Message	When the client requests for an update of a resource based on the known state of the object (ETag / If-Match), the server checks for state conflict, and if a conflict exists, this response is sent.
415 Unsupported Media Type	All		When the client sends the payload in a content type that is not supported by the server, then this status code is returned.

Status	Verb	Response Body	Description
422 Unprocessable Content	POST, PATCH	Error Message	This should also be used when the request body is syntactically correct but it was not able to be processed. For example, the request body fails business logic validation because of an invalid code value.
428 Precondition Required	PATCH		If the precondition (If-Match) is missing in the header, this status is returned. For PATCH , this precondition (If-Match) is mandatory.

Status	Verb	Response Body	Description
500 Internal Server Error	All	Error Message	This error code is used in general for any exception or error that may be raised by the server. For exceptions that are raised by the service, the error has to be returned in a standard error format.
501 Not Implemented	GET	Error Message	When a system query option is specified that is planned to be supported, but not yet implemented by the service. The details of the options not supported are returned in the error message.
523 Upstream Server Error	All	Error Message	This error code is reserved for failed calls made to an external service. The specific reason for the failure (e.g. service unavailable) will be included in the response payload.

Errors and Info Messages

When an error is raised and a response has to be sent back, the service must send only the error response in the payload.

Warnings and Info messages must not be part of the error response.

Where required, warnings and info messages **MAY** be sent as part of the HTTP header.

Error Response

When an error occurs, the response is sent as a single object `error` as shown in the following example. When the service sends an error response, no other payload is sent back to consumer.

```
{
  "error": {
    "code": "external.12345.ValidationsMessages",
    "message": "Invalid email address",
    "target": "{emailAddress}"
  }
}
```

If the error object has to return multiple errors, then a list of error details are sent. The primary error is defined in the root. The details, if present, contain the inner/secondary errors and do not include the primary error. The detailed error instances cannot have further child-level details.

```
{
  "error": {
    "code": "projects.234567.Auth",
    "message": "insufficient authorizations for creating a projects",
    "details": [
      {
        "code": "projects.22345.Auth",
        "message": "insufficient authorizations for ..."
      }
    ]
  }
}
```

Error Response Attributes

Element	Element of	Mandatory	Description
error		Yes	Single instance object that represents the primary error. An error can contain other errors as a flattened array (refer to details).
code	error	Yes	Represents an error code. The error code is a combination of the group (service prefix) and error code integer (5 digits), with a dot (.) as the delimiter.
message	error	Yes	Description of the error message. The error message must be localized using the standard language rules.

Element	Element of	Mandatory	Description
details	error	No	Array of error objects excluding details. The child errors should not contain further children (for example, the element "details" is not permitted for child error objects). The primary error is not repeated in details.
target	error	No	Object that is used to provide reference to the target element that caused the error. The target may either identify the resource directly using the URL notation (or) may identify an element in the payload using the binding { ... } notation.

Note

The error response must not include warnings and info messages.

Target Element Specification

The URL notation is used to define the target resource and must provide the complete URL path to the resource (for example, `https://<application-url>/external-projectorder-service/projectorders/124/products`).

The binding notation is used to identify the element within the payload. The binding is enclosed in curly braces `{}` and the value is the attribute target path in the payload (for example, `{id}`, `{amount/currency}`, `{products/1/name}`).

Information Message Response

There could be cases where the service needs to send additional messages, like a warning or info. This may be required even when the service is responding with the expected payload — for example, sending info with 200 OK or warnings along with error responses. A generic message response payload can be used in such situations to send the message via the HTTP header.

The info messages are sent as part of the response payload. These are included in the `m` (message) element of the payload.

The info messages do not stop the execution flow. These messages may or may not be displayed (or) acted upon, and hence a follow-up action should not be expected.

```
{
  "info": {
```

```

"message": "lead warning message",
"target": "/leads/90299929",
"severity": "WARNING",
"details": [
  {
    "message": "phone number may be invalid",
    "target": "{phoneNumber}"
  },
  {
    "message": "another warning",
    "target": "{anotherAttribute}",
    "severity": "WARNING"
  }
]
}
}
}

```

A code is not mandatory in info or warning messages.

Info Response Attributes

Element	Element of	Mandatory	Description
info		Yes	Single instance object that represents the primary message.
message	info	Yes	Description of the info message.
details	info	No	Array of info objects excluding details. The child messages should not contain further children (for example, the element "details" is not permitted for child message objects).
target	info	No	Object that is used to provide reference to the target element that needs to be mapped to the message. The target may either identify the resource directly using the URL notation (or) may identify an element in the payload using the binding { ... } notation.
severity	info	No	Indicates the type of the information. Permitted values: INFO , WARNING , and ERROR . INFO is the default, and a message is considered as INFO if severity is not specified.

Note

The severity **ERROR** should be used only if the create or update is successful, but there are errors that need to be conveyed as messages. These are not transactional / server errors.

Do not use severity **ERROR** for server errors. Server errors must only be returned as error objects, and the HTTP status should indicate an error. The severity **ERROR** is only for communicating a message to the user with an error icon!

The info message is only applicable for HTTP status 200 and 201

16.13.2.3 CAP File Conversion

For CAP-based custom services, you can automatically generate the metadata JSON file using the [CAP File Conversion](#) feature. This reduces the manual effort required to fill in details based on your CAP service.

The feature helps you convert CAP-CDS-based metadata into the JSON format used by SAP Sales and Service Cloud, allowing you to upload the final metadata with minimal effort and create custom services with the corresponding UI.

Current Scope

- Supports conversion to create service, entities, events, APIs based on available metadata.
- Partial automation: Some manual intervention is required post the conversion.
- Limited to JSON format support for conversion.

Pre-Requisites

- A CAP service must be developed and deployed.
- If the CDS service is available locally, run the following command in the CLI to convert it to JSON: `cds -2 json <service.cds file path>`
- For services deployed to BTP, use `cds pull` to fetch the complete service, then convert it locally using the same command: `cds -2 json <service.cds file path>`
- Follow guidelines to avoid inconsistency in metadata conversion and reduced manual effort to upload final metadata to SAP Sales Cloud and SAP Service Cloud Version 2 while creating custom service in [Extensibility Administration](#) [Custom Service](#).

Guidelines for Conversion Supportability

You must follow these guidelines in `schema.cds` while defining the entity and events:

- Use the custom annotation `@isRoot` to mark the root entity. If not specified, the first entity will be treated as the root by default. Only one root entity is allowed.
- The `source` property in the service definition must not be modified and should retain the ".cds" extension. All required properties must remain intact when converting `serviceSchema.cds` to JSON and uploading it for metadata conversion. For example:

```
"ProjectOrderService": {
  "@source": "srv/projectorder-service.cds", // .cds extension
  "kind": "service",
  "@path": "/project-order-service"
}
```

Note: Do not use the word `service` while defining the service name in your .cds file.

- Use `@odata.containment: false` for entities in `schema.cds` that can exist independently and are not contained within another entity. This is important for maintaining the entity's `creatable` and `updatable` properties.
- Each entity must have at least one attribute (element) with `"key": true`, preferably a UUID field. For example:

```
key ID : UUID;
```

- Use the `@description` custom annotation to mark an attribute as the entity's description field. Only attributes annotated this way can be used as `descriptionAttribute` in associations. For Example:

```
"Name": {
  "@description": true, // Custom annotation.
  "type": "cds.String",
  "notNull": true
},
```

- Use `@Core.Immutable: true` to make an attribute non-updatable under all conditions. For such attributes, or attributes of type `UUID`, the system will set `creatable: true` and `updatable: false`. For all other attributes, `updatable` will be set to `true` by default. For example to maintain either true or false for `Name` attribute:

```
"@Core.Immutable": true/false,
Name : String not null;
```

- Use `@readonly` to make an attribute neither creatable nor updatable. In the converted metadata, this results in: `creatable: false` and `updatable: false`.

```
@readonly
opportunityId : UUID;
```

- If none of the above annotations are applied, `creatable` and `updatable` will default to `true`, except for attributes of type `UUID`, where `updatable` will be `false`.
- Use `@UI.HiddenFilter: true` to mark an attribute as `filterable: false`. If `@UI.HiddenFilter: false` is explicitly set, the attribute will be marked as `filterable: true`. If this annotation is not maintained, the default behavior is `filterable: true`.

```
@UI.HiddenFilter: false/true
displayId : String;
```

- Use `not null` to mark an attribute as non-nullable. If not specified, the attribute is considered `nullable: true` by default.

```
Name : String not null;
```

- For attributes that are associated with another entity, the custom annotation `@dataFormat` should be maintained on the attribute in the parent entity.

For example:

```
@dataFormat: 'AMOUNT'
estimatedRevenue : Composition of EstimatedRevenue;
```

In this case, the `estimatedRevenue` attribute in the `ProjectOrder` entity is associated with another entity, `EstimatedRevenue`, which contains an amount field structure. To indicate that the `estimatedRevenue` association follows the **AMOUNT** structure, it is mandatory to maintain the `@dataFormat` custom annotation as described above.

EstimatedRevenue entity:

```
entity EstimatedRevenue {
  key ID                      : UUID;
    currencyCode              : String;
    content                    : Decimal(10, 3);
}
```

Similarly, for attributes associated with an entity that represents a **quantity structure**, the custom annotation `@dataFormat: 'QUANTITY'` must be maintained.

For example:

```
@dataFormat: 'QUANTITY'
quantity : Composition of ProductQuantity;
```

ProductQuantity entity:

```
entity ProductQuantity {
  key ID          : UUID;
    content        : Integer;
    unitOfMeasure  : String;
}
```

For entities defined with `@dataFormat: 'QUANTITY'` or `'AMOUNT'`, be sure to mark them with the annotation `@isCnsEntity: true`.

- ```
type Etype : String @assert.range enum {
 BUSINESS = 'Business';
 CODEVALUE = 'Code Value';
}
```
- Events can be defined in the `service.cds` file of a CAP project using the `event` keyword. To ensure proper event metadata generation during conversion, the event name should end with the corresponding action: `CREATE`, `UPDATE`, or `DELETE`.

For example:

```
EVENT
ProjectOrderService.customer.ssc.projectorderservice.event.projectOrderCreate;
```

- The CDS import functionality can be used to convert an OpenAPI schema into CSN (Core Schema Notation). You can retrieve the OpenAPI schema from the **SAP Business API Hub** under the SAP Sales Cloud and SAP Service Cloud Version 2 space. Then, run the following command to generate the CSN:

The conversion involves the following steps:

1. Upload CAP Schema: Provide CAP-CDS based metadata in JSON format generated as mentioned in pre-requisites section. This schema should adhere to guidelines mentioned in below section.
2. Download Converted Metadata: Retrieve the converted metadata in JSON format. Please note that the conversion has limitations. For more details, refer to the **Guidelines for Conversion Supportability** section
3. Validate and Modify: Review the output and update the JSON file if any details are missing or inconsistencies are found.
4. Re-upload Metadata: Use the existing upload feature in the custom service UI to submit the revised metadata.

### Manual Steps Before Uploading Metadata to Sales and Service Cloud

After downloading the converted metadata file through the automated process, complete the following manual steps before using the upload feature in Sales and Service Cloud:

1. Validate the downloaded metadata to ensure it is correct and aligns with the current scope. The conversion might introduce inconsistencies or discrepancies. Make any necessary corrections based on the guidelines for uploading custom service metadata to the Sales and Service Cloud.
2. Set the correct root entity. The first entity defined in the CAP service metadata is automatically considered the root. If a different entity should be the root, update the root: true property accordingly.
3. Update names of entities, services, and other full names with namespace, as the converted metadata may only reflect CAP-specific names. These should be modified to match the naming conventions required by the Sales and Service Cloud. Example: For associations to CNS entities, target entity and target service needs to be updated. Adding entityReference for ASSOCIATION depends on customer's use case (If the customer wants all the attributes from the associated entity, then entity reference can be maintained. If not then only attributes defined under the objectDefinition will be taken into considerations)
4. For entity attributes which are marked for description, update the descriptionAttribute property accordingly. Example: If @description: true is set upon an element in CDS, then please make use of it as a descriptionAttribute for the appropriate entity attribute, for which it serves as the description field. Refer to the sample json payload for examples of attributes which are marked under the descriptionAttribute property.
5. For APIs, update the API metadata, with the relevant API paths corresponding to all the child entities. Since at the CDS side, @path is maintained at the service level, the metadata conversion takes care of only one api metadata i.e. the root entity. For other child entities, the api metadata has to be manually maintained.
6. Once all validations and modifications are complete and aligned with the custom service metadata guidelines(compare it with the sample metadata that is available), use the upload feature to submit the final metadata.

#### Metadata Conversion Support and Limitations:

##### Service

- No discrepancy in service metadata converted with metadata defined in template data.json.
- **Supported:** name, title, description, authorizationRelevant
- **Unsupported:** fullName (not present in template.json)

##### Entities

- **Supported:**
  - name
  - label (human-readable format of name)
  - entityType (enum)
  - root (first entity in service)
  - creatable, updatable
  - attributes
- **Unsupported:** objectTypeCode
- **Notes:**
  - entityType is based on provided guidelines.
  - objectTypeCode not supported yet.
  - creatable, updatable of entity: default is false.
  - As per guidelines: @odata.containment: false → creatable, updatable = true.  
If annotation exists but is true, then creatable, updatable = false (default behavior anyway).

##### Entity Attributes

- **Supported:**
  - name
  - label (human-readable format)
  - dataType (CAP ↔ Sales & Service Cloud): STRING, BOOLEAN, NUMBER, DATETIME, OBJECT, ARRAY, INTEGER
  - dataFormat: STRING, UUID, EMAIL, CODE, URI, ALPHANUMERIC, FLOAT, DOUBLE, AMOUNT, QUANTITY, INT32, INT64, DATETIME, DATE, TIME, DURATION
  - nullable: derived from not null property
  - creatable, updatable: default true;
    - If @Core.Immutable: true or UUID field → updatable: false
    - If @readonly: true → creatable, updatable: false
  - filterable: determined by @UI.HiddenFilter: true
  - enumOptions: populated per guideline
- **Unsupported:**
  - description (not available due to CODELIST association not supported)
  - dataFormat: TEXT, DEFAULT
- **Associations Supported:**
  - AMOUNT: If custom annotation is defined. Also, for the AMOUNT field, if the elements will have a attribute of type - "cds.string" then it will be considered as a CODE field (currencyCode etc.), and the remaining field will be added as it is. For eg: if content field is present of type cds.integer or cds.decimal it will be handled accordingly.
  - QUANTITY: If a custom annotation is defined, it will be handled in the same way as an Amount field.
  - CODE: Enum-based.
- **Associations Not Supported:**
  - Dynamic code list (CODEVALUE entity type associations)
- **Other Notes:**
  - searchable, sortable, searchResult, langDependent: always false
  - searchWeightage: always 1
  - keyType: primary for key: true attributes
  - filterable: false for OBJECT/ARRAY dataType or when @UI.HiddenFilter: true
  - sourceEntity, targetEntity, targetService: namespaces not included, to be maintained manually

## Events

- **Supported:**
  - technicalName
  - label
  - entityReference
  - trigger (should follow naming convention ending in CREATE/UPDATE/DELETE)

## APIs

- **Supported:**
  - technicalName
  - title

- description
- apiPath
- operations:
  - id
  - path
  - method: GET, POST, PATCH, DELETE
  - request:
    - pathVariable: name, dataType, dataFormat
  - responses: description, responseCode (200, 400, 401, 403, 404, 500)
  - entityReference
- **Unsupported:** method: query

**Note:** Currently only API metadata for root entity is supported. APIs for child entities are not yet generated. Need to be set manually. Api methods - GET POST PATCH DELETE

### Limitations

- First entity in CDS schema is assumed as the root.
- @create, @update required on attributes. Defaults:
  - creatable, updatable = true for non-key fields
  - For key: creatable = true, updatable = false
- @Core.Immutable: true or UUID → creatable: true, updatable: false
- @readonly: true → creatable, updatable = false
- @UI.HiddenFilter: true → filterable = false.
  - If not present → filterable = true by default
- Attribute composition:
  - to many → ARRAY
  - 1:1 → OBJECT
- @dataFormat: AMOUNT or QUANTITY must be set for correct mapping
- Enums via type keyword: type Etype : String @assert.range enum { BUSINESS = 'Business'; CODEVALUE = 'Code Value'; }
- Multiple service .cds files supported by combining into single JSON via cds commands. Currently only one service-based JSON is supported.
- Event must be defined in service .cds file. It can be defined using the EVENT keyword. Also, the trigger (CREATE, UPDATE, DELETE) should be maintained at the end of the event name, like projectOrderCreate.

## Mapping DataType and Data formats of CAP and SAP Sales Cloud and SAP Service Cloud Version 2

| Cloud Data Type | Data Format | CAP Data Type | Description    | Supported | Comments |
|-----------------|-------------|---------------|----------------|-----------|----------|
| STRING          | STRING      | String        | Default string | YES       |          |



| Cloud Data Type | Data Format     | CAP Data Type | Description        | Supported | Comments                                          |
|-----------------|-----------------|---------------|--------------------|-----------|---------------------------------------------------|
|                 | UUID            | UUID          | UUID               | YES       |                                                   |
|                 | EMAIL           | String        | Email              | YES       |                                                   |
|                 | CODE            | String        | Code List          | YES       |                                                   |
|                 | TOKEN           | String        | Token format       | YES       | Requires<br><code>@dataFormat:<br/>'TOKEN'</code> |
|                 | URI             | String        | URI format         | YES       | Requires<br><code>@dataFormat:<br/>'URI'</code>   |
| NUMBER          | FLOAT           | Decimal       | Default number     | YES       |                                                   |
|                 | DOUBLE          | Double        | Double-precision   | YES       |                                                   |
| BOOLEAN         | -               | Boolean       | Boolean type       | YES       |                                                   |
| OBJECT          | AMOUNT/QUANTITY | Aspect types  | Complex structures | YES       | TEXT not supported/tested                         |
| ARRAY           | -               | Array         | Array type         | YES       |                                                   |
| INTEGER         | INT32           | Integer       | Default integer    | YES       |                                                   |
|                 | INT64           | Int64         | Long integer       | YES       |                                                   |
| DATETIME        | DATETIME        | DateTime      | DateTime type      | YES       |                                                   |
|                 | DATE            | Date          | Date               | YES       |                                                   |
|                 | TIME            | Time          | Time               | YES       |                                                   |

**Note:** Database specific data types like `real_vector` or `blob` are not supported in CNS and there are no alternatives.

# 17 Mashups

Learn how to configure, use, and access mashups.

Mashups are used to integrate data from SAP's cloud solution with data provided by an online Web service or application. Users can access the content provided by these Web services and applications, and use it in their daily work. Mashups can include Web searches, company or industry business information, or online map searches.

## 17.1 Configure Mashups

Administrators can configure mashups in [Settings](#).

### → Tip

Expand a collapsible section heading to view the configuration steps.

### 17.1.1 Enable Mashups for Business Roles

As an administrator, you must add the mashup business service to your business users.

#### Procedure

1. Go to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► and under [Users and Control](#), select [Business Roles](#).
2. Select a [Business Role ID](#) to open the details.
3. Under [Business Services](#), choose + to open the [Add Business Services](#) window.
4. Select the check box for [mashupService](#) and choose [Add](#).

#### ⓘ Note

You can create HTML mashups based on a mashup category provided by SAP.

#### Results

You have added the mashup service to the selected business user.

## Related Information

[Create Business Roles and Assign Business Users \[page 28\]](#)

Administrators can create business roles with appropriate authorizations and assign them to business users.

## 17.1.2 Configure Mashup URL in Content Security Policy

As an administrator, you must configure the mashup URL in the content security policy settings.

### Procedure

1. Go to **User Menu** > **System Settings** > **All Settings** > **System Preferences** > **Content Security Policy** .
2. Go to the **Frame Source** section and choose **+** .
3. Enter the hostname URL and choose **✓** .

If you're creating a mashup with the URL `https://my123456.s4hana.ondemand.com/ui?sap-ushell-config=headerless#ServiceOrder-create` then, add `*.onedemand.com` in the **Frame Source**.

4. To add an image to the RSS Feed card, go to the **Image Source** section and choose **+** .
5. Enter the hostname URL and choose **✓** .

If you're creating a mashup with the URL `https://media-cldnry.s-nbcnews.com/image/upload/t_fit_1500w/mpx/2704722219/img.jpg` then, add `*.s-nbcnews.com` in the **Image Source**.

## 17.2 Create HTML Mashups

Create HTML mashups to embed a HTML or JavaScript based web page into a screen of your SAP cloud solution.

### Context

Some web services may pass your business data to a third-party organization, for example, account data is passed to a search engine when performing a reverse lookup in an online address book. We recommend that you check whether the mashup conforms to your company's data privacy policies before activating the mashup.

## Procedure

1. As an administrator, navigate to **User Menu** > **System Settings** > **All Settings** and under **Extensibility**, select **Mashup Authoring**.
2. Choose **+ (Create)** and then choose **Create HTML Mashup**.
3. Under **General Information**, enter a mashup name.  
By default, the status of the mashup is **Active**. You can also change the status to **Inactive** if necessary.
4. To use this mashup on your mobile, turn on the **Mobile Usage** switch.
5. To allow aliasing of mashup parameters during navigation (for example, from a timeline card), you must turn on the **Object-Based Navigation** switch. Once enabled, you can alias your parameters using either the **Object ID** or the **Object Display ID**.
6. Choose **Show More** and update other required fields.

### Note

You can adjust the screen height. The default height is 500 pixels.

7. From the **Manage Permissions** dropdown, select **Microphone**, **Camera**, **Notifications**, or **Clipboard** to allow mashups to access your system's microphone, notifications, and clipboard, respectively.

### Note

Ensure that your system permissions are enabled.

8. Under **Input Parameters**, enter the URL of the website in the field.
9. Select **Extract Parameters**.

The system verifies the URL domain and shows  if the domain is already available under the trusted domain list.

### Note

For URLs that don't use queries, you can manually add curly brackets around terms that should act as placeholders. For example, in the URL `https://mail.google.com/mail/#search/SAP`, you can replace the word `SAP` with a search term in curly brackets, for example, `{term}`. If you then enter `https://mail.google.com/mail/#search/{term}` in the **URL** field and click **Extract Parameters**, the word in brackets is extracted as a parameter.

10. If the URL domain is not in the trusted domain list, hover over  and **Add to Trusted Domain List**.
11. To add parameters to this mashup, select **Add Parameter** and enter the required details.

In the **Parameters and Value Binding** section, configure the required parameters for the mashup.

- Enter a **Parameter Name** to identify the mashup parameter.
- Choose **Constant** to assign a fixed value to the parameter, which cannot be changed at runtime.
- Turn on the **Mandatory** switch if the parameter must always have a value.
- Turn on the **Multi Value** switch to allow the mapped application field to hold multiple values in an array format.
- Select **System Parameter** to bind the parameter to a predefined system value.

#### Note

Field Parameter Binding allows you to bind a mashup parameter to an application field's value. The field must be exposed by the business object, and the binding is done in adaptation mode when adding the mashup to a screen.

#### Caution

Do not add any sensitive information.

12. Under *Event Configuration*, enter an event name and add parameters according to your requirements.

#### Note

The admin must map this event to the application event or directly to a standard operation on the respective screens through adaptation.

13. Choose *Show Preview* to check the details.
14. Save your changes

## 17.2.1 Initiate Navigation from Mashups

You can initiate navigation to specific screens in SAP Sales Cloud and SAP Service Cloud Version 2 from any external web applications using a mashup.

#### Note

- The external web application must be added as an HTML mashup in SAP Sales Cloud and SAP Service Cloud Version 2.
- Navigation is supported only for the list view, detail view, quick view, and quick-create view.
- This feature is also supported for custom UIs.

Initiate navigation refers to the action of navigating to a specific screen or view within SAP Sales Cloud and SAP Service Cloud Version 2, such as the list, detail, quick, or quick create view, based on the parameters passed from an external web application.

To initiate navigation, the external application must use the `window.postMessage()` method to send data to SAP Sales Cloud and SAP Service Cloud Version 2. This data must include specific parameters required for the navigation to work.

The following details explain these parameters and help you create the required HTML code in the external application:

| Parameter         | Default Value | Description                                                                                                                             | Mandatory/Optional | Usage Detail                                                                                                                                                                                                                                                                                                                                    |
|-------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| operation         | navigation    | Name of the operation.                                                                                                                  | Mandatory          | Used by the Mashup web component for determining the action that needs to be triggered.                                                                                                                                                                                                                                                         |
| params            | null          | Parameters required for initiating navigation.                                                                                          | Mandatory          | Used by the Mashup web component for passing the required parameters to trigger the mentioned action.                                                                                                                                                                                                                                           |
| params.objectKey  | null          | UUID of the entities or things.                                                                                                         | Mandatory          | Used to navigate to the specific entity or thing.                                                                                                                                                                                                                                                                                               |
| params.routingKey | null          | Name of the entities, retrieved from the <b>routingKey</b> property in the application UI view metadata in SAP Service Cloud Version 2. | Mandatory          | Used for determining the entity to which user wants to navigate.                                                                                                                                                                                                                                                                                |
| params.viewType   | details       | Name of the screen or view to navigate to.                                                                                              | Mandatory          | Used for determining the view or screen to which the user wants to navigate. The following view types are supported: <ul style="list-style-type: none"> <li>• "list" for list view</li> <li>• "details" for entity detail view</li> <li>• "quickcreate" for entity quick create screen</li> <li>• "quickview" for entity quick view.</li> </ul> |

You can use the following code snippet as a reference. Add a similar script to your external web application and integrate it using an HTML mashup to initiate the navigation.

#### Note

For custom UI navigation, you can obtain the **routingKey** information from the Design App.

```
<script>
function openCaseView() {
 var case={
 operation:"navigation",
 params: {
 objectKey: "00163ea6-f284-1eda-b895-42c904e08752",
 routingKey: "case",
 viewType:"details"
 }
 }
}
```

```

 }
 };
 window.parent.postMessage(case, '*')
}
</script>
<input type='button' value='Open Case Details' onclick='openCaseView();' />

```

## 17.2.2 Initiate Search Events from Mashups

As an administrator, you can initiate search events to specific screens in SAP Sales Cloud Version 2 and SAP Service Cloud Version 2 from any external web applications using a mashup.

### Note

- The external web application must be added as an HTML mashup in SAP Sales Cloud and SAP Service Cloud Version 2.

To initiate a search event, the external application must use the `window.postMessage()` method to send data to SAP Sales Cloud and SAP Service Cloud Version 2. This data must include specific parameters required for the navigation to work.

The following details explain these parameters and help you create the required HTML code in the external application:

| Parameter | Default Value | Description                                    | Mandatory/Optional                          |
|-----------|---------------|------------------------------------------------|---------------------------------------------|
| operation | null          | Name of the operation.                         | Mandatory                                   |
| params    | null          | Parameters required for initiating navigation. | Mandatory if required by the specific event |
| event     | null          | Name of the event..                            | Mandatory                                   |

You can use the following code snippet as a reference. Add a similar script to your external web application and integrate it using an HTML mashup to initiate the search event.

```

<script>
function triggerSearchEvent() {
 var event = {event: "mashup100.magiccom.customExtendedSearch", operation:
"search", params: {$search: "alex"}};
 window.parent.postMessage(event, '*')
}
</script>
<input type='button' value='Search' onclick='triggerSearchEvent();' />

```

## 17.2.3 Initiate Custom Events from Mashups

Use the Mashup callback feature in SAP Sales Cloud and SAP Service Cloud Version 2 to initiate custom events from embedded web applications. These events must be defined and maintained by the respective application teams.

### Note

The external web application must be added as an HTML mashup in SAP Sales Cloud and SAP Service Cloud Version 2.

## Implementation Guidelines for External Web Applications

To initiate a custom event from a mashup, note the following:

- Use `window.parent.postMessage` within your embedded web application to communicate with the SAP Sales Cloud and SAP Service Cloud Version 2 mashups.
- A specific JSON format (described in the table below) must be sent from the embedded web application to the SAP Sales Cloud and SAP Service Cloud Version 2 mashups using `window.parent.postMessage`.
- Parameters passed to the event are sanitized using `encodeURIComponent` before processing.
- The **Name of event** and **parameter names** must exactly match those defined during mashup authoring.

## Format of the JSON data

The following table outlines the structure of the JSON object that must be sent via `window.parent.postMessage` from the embedded web application:

| Property Name | Data Type | Default Value | Mandatory/Optional                      | Description                              |
|---------------|-----------|---------------|-----------------------------------------|------------------------------------------|
| operation     | string    | null          | Mandatory                               | Name of the operation                    |
| event         | string    | null          | Mandatory                               | Name of the event                        |
| params        | object    | null          | Mandatory if required by specific event | Parameters required to trigger the event |

## Example

Scenario: A user wants to initiate a refresh event from a web application embedded in a mashup within the Account object detail screen. The event name is "accountRefreshEvent", and it does not require any parameters.

The following sample provides an idea of the logic that must be written using `window.parent.postMessage`:

```
<script>
```



```
function triggerRefreshEvent() {
 var event = { event: "accountRefreshEvent",
 operation: "triggerCustomAction"
 };
 window.parent.postMessage(event, '*')
}
<input type='button' value='Refresh' onclick= triggerRefreshEvent();' />
```

## 17.3 Create URL Mashups

As an administrator, you can create URL mashups to send data from SAP Sales Cloud and SAP Service Cloud Version 2 to a web service provider's URL. The results are displayed on the provider's website in a new browser window.

### Context

Some web services may pass your business data to a third-party organization, for example, account data is passed to a search engine when performing a reverse lookup in an online address book. We recommend that you check whether the mashup conforms to your company's data privacy policies before activating it.

#### Note

You can create URL Mashups for the following entities: Accounts, Individual Customers, Contacts, Competitors, Partner Contacts, Leads, Cases, Guided Selling, Sales Quotes, Sales Orders, Employee, Partner, Supplier, and Supplier Contact.

### Procedure

1. Navigate to **User Menu** > **System Settings** > **All Settings** > **Extensibility** > **Mashup Authoring**.
2. Choose **+ (Create)** and then select **URL Mashup**.
3. Under **General Information**, enter a mashup name.

By default, the status of the mashup is **Active**. You can also change the status to **Inactive** if necessary.

4. To use this mashup on your mobile, turn on the **Mobile Usage** switch.
5. To allow aliasing of mashup parameters during navigation (for example, from a timeline card), you must turn on the **Object-Based Navigation** switch. Once enabled, you can alias your parameters using either the **Object ID** or the **Object Display ID**.
6. Choose **Show More** and update other required fields.
7. Under **Input Parameters**, enter the URL of the website in the field.
8. Select **Extract Parameters**.

The system verifies the URL domain and shows  if the domain is already available under the trusted domain list.

#### Note

For URLs without query parameters, you can manually insert curly brackets around terms you want to use as placeholders. For example, replace **SAP** in `https://mail.google.com/mail/#search/SAP` with `{term}` to create `https://mail.google.com/mail/#search/{term}`. When you enter this in the *URL* field and select *Extract Parameters*, the word in brackets is extracted as a parameter.

9. If the URL domain is not in the trusted domain list, hover over  and choose *Add to Trusted Domain List*. The system redirects you to the *Content Security Policy Setting*, where you can add the URL to the trusted domains list.
10. To add parameters to this mashup, select *Add Parameter* and enter the required details.

In the *Parameters and Value Binding* section, configure the required parameters for the mashup.

- Enter a *Parameter Name* to identify the mashup parameter.
- Choose *Constant* to assign a fixed value to the parameter, which cannot be changed at runtime.
- Mark the *Mandatory* checkbox if the parameter must always have a value.
- Choose *System Parameter* to bind the parameter to a predefined system value.

#### Note

Field Parameter Binding allows you to bind a mashup parameter to an application field's value. The field must be exposed by the business object, and the binding is done in Adaptation mode when adding the mashup to a screen.

#### Caution

Do not add any sensitive information.

11. Select *Show Preview* to open the preview in a new browser tab.
12. Save your changes.

To use the mashup, perform the following steps:

1. Navigate to an entity and open a record in detailed view.
2. Choose *Start Adaptation* to add the URL mashup as a button or a link at any place where a mashup can be added.
3. Choose *End Adaptation* after adding the mashup.

## Results

The URL is launched in a separate browser tab on choosing the button or the link.

## 17.4 Create Web Component Mashups

Create web component mashups to embed that into a screen of your SAP cloud solution.

### Context

Some web services may pass your business data to a third-party organization, for example, account data is passed to a search engine when performing a reverse lookup in an online address book. We recommend that you check whether the mashup conforms to your company's data privacy policies before activating the mashup.

#### Note

To enable this feature, create a case with the CEC-CRM-PM-PF component using the subject **Enable Creation of Web Component Mashups**.

### Procedure

1. As an administrator, navigate to  [User Menu](#)  [System Settings](#)  [All Settings](#)  and under [Extensibility](#), select [Mashup Authoring](#).
2. Choose  ([Create](#)) and then choose [Create Web Component Mashup](#).
3. Under [General Information](#), enter a mashup name.

By default, the status of the mashup is [Active](#). You can also change the status to [Inactive](#) if necessary.

4. Optional: To use this mashup on your mobile, turn on the [Mobile Usage](#) switch.
5. Select [Show More](#) and update the required fields.

#### Note

You can adjust the screen height. The default height is 500 pixels.

6. From the [Manage Permissions](#) dropdown, select [Microphone](#), [Notifications](#), or [Clipboard](#) to allow mashups to access your system's microphone, notifications, and clipboard, respectively.
7. Under [Input Parameters](#), enter the [Resource URL](#) of the website in the field.
8. Enter the [Web Component](#) name.
9. To add parameters to this mashup, select [Add Parameter](#) and enter the required details.

#### Caution

Do not add any sensitive information.

10. Choose [Show Preview](#) to check the details.
11. Save your changes.

## Related Information

[Create Analytics Widget Mashups \[page 271\]](#)

Create an SAP Analytics Cloud mashup that is embedded on the home page card to show advanced charts via SAP Analytics Cloud.

## 17.5 Create Analytics Widget Mashups

Create an SAP Analytics Cloud mashup that is embedded on the home page card to show advanced charts via SAP Analytics Cloud.

### Context

Some web services may pass your business data to a third-party organization, for example, account data is passed to a search engine when performing a reverse lookup in an online address book. We recommend that you check whether the mashup conforms to your company's data privacy policies before activating the mashup.

#### Note

To enable this feature, create a case with the CEC-CRM-PM-PF component using the subject **Enable Creation of Analytics Widget Mashups**.

### Procedure

1. As an administrator, navigate to  [User Menu](#)  [Settings](#)  [All Settings](#)  and under [Extensibility](#), select [Mashup Authoring](#).
2. Select  ([Create](#)) and then choose [Analytics Widget](#).
3. Under [General Information](#), enter a mashup name.

By default, the status of the mashup is [Active](#). You can also change the status to [Inactive](#) if necessary.

4. Select [Show More](#) and update the required fields.

#### Note

You can adjust the screen height. The default height is 500 pixels.

5. Under [Parameters and Value Binding](#), select [Add Parameter](#) and enter the required details.
6. Save your changes

## 17.6 Copy Existing Mashups

You can create a copy from the existing mashups.

1. Go to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Extensibility](#) ► [Mashup Authoring](#) .
2. From the new tab, select the mashup you want to copy by clicking on the mashup name.
3. From the [Mashup Details](#) screen, choose [Duplicate](#).  
The system opens a copy of the mashup.
4. Enter a new name for the mashup and adapt the configuration settings, as required.
5. Save your changes.

## 17.7 Add Mashups as Tabs

As an administrator, you can add a mashup as a new tab.

Using page layouts and visibility property, you can control who is allowed to use the mashup.

To add a mashup as a tab, follow these steps:

1. **Create a tab on a screen.**
  1. Navigate to a screen where you want to add the mashup.
  2. Go to ► [User Menu](#) ► [Start Adaptation](#) . The system opens in the adaptation mode.
  3. Hover over the area where you see a plus icon (+).  
The system highlights the area with a red border along with a tool tab
  4. From the tool tab, select the pencil icon to edit. A new tab opens.
  5. Click the plus icon (+) and enter a tab name.
  6. Click [Apply](#).  
The new tab is added to the header of the screen.
2. **Add a mashup to the tab.**
  1. Open the new tab and hover over the empty space to enable the red highlight with the tool tab.
  2. From the tool tab, select the pencil icon to edit. A new tab opens.
  3. Select the arrow available next to [Add Mashup](#) to open a new window.
  4. Click the arrow available next to the required mashup to display the properties.
  5. If necessary, you can adjust the appearance of the mashup by selecting one of the following options under [Edit Properties](#).

### Note

The options available to you may vary depending on the mashup.

- [Show Header](#): Select the [Show Header](#) checkbox if you want to see the mashup title on the screen.
- [Height \(%\)](#): You can define the height of HTML Mashups to be displayed in the available screen. For example, if you set the value of Height (%) to 100, the newly added mashup occupies the full height of the screen.  
If the selected mashup has only a parameter name added, without a constant or system parameter, the system displays an additional field to choose the value to be mapped.

#### Note

The embedded mashup instantly updates when you change the mapped parameter.

6. Click [Apply](#).
7. To save your settings, click [End Session](#) on the top of the screen.

## Related Information

### [Adaptation \[page 207\]](#)

As an administrator, you can change the look and feel of the solution for all users by changing layout settings, adding mashups and fields, as well as defining extension fields.

## 17.8 Delete or Deactivate Mashups

Delete or deactivate mashups that you and other users have created.

1. Go to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► and under [Extensibility](#), select [Mashup Authoring](#).
2. Select a mashup from the list.
3. Choose the [Delete](#) icon, or turn the [Active](#) switch off.  
If you choose to deactivate, the mashup is no longer visible on screens. If you choose to delete, the mashup is removed from the [Mashup Authoring](#) view and is also deleted from all screens for which it had been made visible.

## 17.9 Edit URL and Parameter Value for Transported Mashups

As an administrator, you can manually enter the URL and parameter value for mashups in the target system.

## Context

When you transport a mashup to the target system, the URL field and parameter value are not transported and remains empty. As a result, any screen where the mashup is used displays a warning message indicating that no URL is added. To ensure the mashup works correctly, you must manually enter the URL and parameter value in the target system.

## Procedure

1. Navigate to **User Menu** > **System Settings** > **All Settings** and under **Extensibility**, select **Mashup Authoring**.
2. Select the transported mashup for which you need to enter a URL.
3. Choose **Edit**, located on the top-right corner of the screen.
4. Under **Input Parameters**, enter the URL of the website in the field.
5. Under **Parameters and Value Binding** edit the values for **Constant** and **System Parameter**.
6. Select **Save**.

See the related link section for information on transporting configuration changes from a source tenant to a target tenant.

## Related Information

[Change Project](#)

## 17.10 Retain Mashup State

As an administrator, you can retain the mashup state when switching tabs.

### Context

When you switched between tabs in the application, mashups would refresh, causing any data you had entered to be lost. This disrupts the workflow for users, especially when they need to copy information from different screens into the mashup.

With the **Mashup Retain State** feature, mashups built with HTML or Web Components will no longer reload when you switch tabs. Your data stays just the way you left it, making it easier to work without interruptions or repeated inputs.

#### Note

To enable this feature, create a case with the CEC-CRM-PM-PF component using the subject **Enable Mashup Retain State**.

## Procedure

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Extensibility](#) ► [Mashup Authoring](#) ►
2. Select the mashup for which you want to retain the state while switching tabs.
3. Choose [✎ \(Edit\)](#).
4. Under [General Information](#), turn on the [Retain State](#) switch.
5. Save your changes.

### ⚠ Caution

Be careful when choosing which mashups to keep open. Keeping too many can impact performance, so use this feature wisely.

### ℹ Note

- Browser support details:
  - Google Chrome: Supported from version 133 onwards
  - Microsoft Edge: Supported from version 133 onwards
  - Mozilla Firefox and Safari: Not supported yet
- This feature supports all shell tab switching but may not function consistently when switching between tabs within the entity details screen. The entities where it works are: Accounts, Contact, Individual Customer.

## 17.11 Mashups Where Used List

The [Where Used](#) feature tracks where mashups are used in other entities or as homepage cards.

It displays the usage in a hierarchical structure that shows the complete dependency from the top-level component down to the mashup.

The feature tracks the usages only for mashups created on or after January 15, 2026.

When you delete such mashups the system will display an error message. Use the Where Used feature to identify and remove all usages before deleting the mashup.

### ℹ Note

Mashups created before January 15, 2026 may not have their usages tracked. If you delete such mashups, the system allows the deletion even if they are used in other entities, which may lead to errors. To track the usage of these mashups, go to the screen where the mashup is used, make a minor change in adaptation mode, and save it. This enables the system to register the mashup usage.

To view the where used list do the following:

1. Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► and under [Extensibility](#), select [Mashup Authoring](#).



2. From the list of mashups, navigate to the desired mashup and choose  (*Where Used*).  
A pop up opens listing the usages of the mashup.

# 18 Machine Learning

SAP applies machine learning intelligence to automate business processes.

Machine learning improves the productivity and quality of these processes by making time-efficient decisions and reducing redundant tasks.

## Related Information

[SAP Business AI Features for SAP Service Cloud Version 2](#) 

## 18.1 Intelligent Services

Use case data to create and train machine learning models to recognize similar cases, predict case categories, and sentiment.

Intelligent Services categorize your cases and improves accuracy. Intelligent Services predict the case category and recommends similar cases that can provide you with potential answers for the customer issue. The solution can scan the case text to identify sentiment, translate case text, and identify profane language. The model constantly captures feedback to be retrained for improved accuracy over time.

### 18.1.1 Case Categorization

Use machine learning to automate the case categorization process.

Save agents' time by automatically categorizing cases for your most commonly used case categories. Machine learning analyzes your past cases and category catalog to learn how cases are categorized in your organization. Once you train and activate the model and if the confidence thresholds are met, the system automatically populates the proposals into the case category.

Case categorization increases agent productivity, provides better prioritization of incoming cases and automatic classification based on model accuracy.

Case categorization is currently supported in the following languages:

- Chinese
- English
- French
- German
- Japanese

- Portuguese
- Russian
- Spanish

## Prerequisites

- Data volume: Includes minimum 1000 categorized cases and an additional 500 instances of each category or subcategory values
- Data quality: Includes subject, case type, description, and proper category

### 18.1.1.1 Case Intelligence Readiness Report

Use the readiness report to analyze your case data to determine if you can get useful predictions for case categorization.

Before you start creating and training the *Case Categorization* model, you must run the readiness report.

Find the readiness report under ► *System Settings* ► *All Settings* ► *Machine Learning* ► *Intelligent Service* ► *Case Categorization* . Select *Readiness Report* at the top right.

When the check is complete, the readiness report displays one of the following status:

- Ready to use machine learning
- Data improvements required
- Requirements not fulfilled

Each readiness factor shows you the minimum required value, the recommended value, and the actual value in your system. You can adjust the data in your system to meet the minimum and recommended thresholds and run the readiness check again. Each check factor should meet the recommended value to ensure reliable prediction results. If the check factor value falls between the minimum value and the recommended value, the report shows a caution. If the value is less than the minimum value, the report shows an error message. You can still proceed with the training, however the prediction result will be of lower accuracy.

#### Note

The data volume, or total number of objects (cases) shown is an **approximate** number of case records from the previous 12 months that could potentially be used to train the machine learning model.

## Example

Let's look at a simplified example for case categorization readiness.

Suppose you have 10 categories in your service category catalog. The readiness report shows that only 33% (3) of these categories are assigned to the required number of cases to meet the qualification threshold.

The percentage is below the recommended minimum threshold. However, 80% of your qualifying cases are assigned to these three categories. If you train a model based on this data you can potentially automate up to 80% of your incoming cases, despite using only 33% of your category catalog. In this case, machine learning could still yield significant agent productivity gains.

## 18.1.1.2 Configure Case Categorization

As an administrator, you must add, train, test, and activate the Case Categorization model.

### Procedure

1. Navigate to the *User Menu*, and select ► *System Settings* ► *All Settings* ► *Machine Learning* ► *Intelligent Service* ► *Case Categorization* ►.
2. To create a model, click +.
3. Enter a name for your model.
4. Use up to four levels of categories to categorize cases.
5. Expand the *Data Selection Criteria* section. From the *Training Data Selection* dropdown, select one of the training datasets. From the *Inference Eligibility Check* section, define a condition for inference check.

#### Note

Data Selection can be performed in two ways, code-based selection and filtering and UI-based selection and filtering methods. If you're enabling the *Training Data Selection* and *Inference Eligibility Check* switches, ensure that the code method of training data selection and inference eligibility check isn't active. At a time, only one of the methods can be active in the system, either the code-based selection and filtering method or the UI-based filtering and selection method.

#### Note

- The training dataset is the query list, available on the cases UI. Only the training dataset that is chosen from the dropdown is selected for training the model.
- Ensure that the training dataset and the defined inference eligibility check are functionally compatible. If you've chosen the case type dataset under training data selection, ensure that the inference eligibility check also follows the same.

6. In the *Custom Processing Communication Configuration* subsection, activate both *Training Data Selection* and *Inference Eligibility Check* switches.
7. Click *Save*.

A new model is added with status **Created**.

8. Locate your model and click *Train* from *Actions* to start training the model.

#### Note

If you update your service category catalog, you must map new category IDs to existing categories and retrain your model.

- a. Select the model for which you want to update the category mapping.
- b. Click [Download Mapping](#) to download the existing category mapping of the model.

#### Note

If the model doesn't have a mapping document, click [Prepare Mapping](#) to create one and then download it.

- c. Edit the category listing as required and upload the updated mapping document.
- d. Click [Save](#).
- e. Select [Train](#) to retrain the model.

The training takes time to complete depending on the volume of the historical data. The mode goes through the following statuses during the training: [Created](#) > [Training in Preparation](#) > [Data Extraction is Pending](#) > [Data Extraction is in Progress](#) > [Data Preprocessing is Pending](#) > [Data Preprocessing is in Progress](#) > [Training Triggered](#) > [Training is Pending](#) > [Training in Progress](#) > [Training Completed or Training Failed](#) > [Active or Inactive](#).

The system stops the training if data extraction fails and updates the status as [Data Extraction Failed](#).

You can refresh and update the training status of the model by selecting the  (refresh) icon.

9. Choose the  (activate) icon to activate the model.

You can deactivate a model by selecting the  (deactivate) icon.

#### Note

You can enable automated recommendations using the [Enable](#) switch in the [Settings](#) tab of your model. Adjust the confidence level for automated prediction. The default value of the confidence level slider is 60 in the solution.

#### Restriction

You can't add new categories from the settings dialog. You can only deselect, or select the categories you included during model creation.

## 18.1.1.3 View Case Categorization Prediction

Find category prediction results in cases.

### Procedure

1. Go to [Cases](#).
2. Create a new case and enter the subject and description fields.
3. Select [Save](#).

Case categorization prediction works for all case creation channels such as manual, e-mail, social media, chat, messaging, and so on. The main usage for case category prediction is for cases generated from

incoming e-mail messages. Cases are categorized during case creation. You will only see predictions for cases created after the model is set to active status.

If a model is created with multiple category levels, and is based on the threshold set, the prediction is bottom up, wherever a threshold set. For example, If the threshold is set at 70% for a three-level categorization model, and for a certain case text if the category level prediction is as follows:

- Category Level 1 is predicted with 90%
- Category Level 2 is predicted with 80%
- Category Level 3 is predicted with 65%

In such a scenario, auto categorization predicts results up to two levels. In this scenario, both category level 1 and category level 2 predictions are auto populated.

## 18.1.2 Similar Case Recommendation

Find solutions from similar cases from the past that you can apply to the current case.

### Note

To ensure accurate similar case recommendations, the model uses a sample dataset of active case types to predict results in real time.

The machine learning algorithm scans the subject and description of categorized cases and returns the top six most similar cases to the current case. The model learns the semantic relationship of words in the description and subject text and uses this context to find cases similar to the current case. This semantic, sentence-level representation of case text yields more accurate results than a simple keyword search. You can check similar cases for possible solutions to the current case.

Similar case recommendation is currently supported in the following languages:

- Dutch
- English
- Finnish
- French
- German
- Greek
- Italian
- Polish
- Slovak
- Spanish

## 18.1.2.1 Configure Similar Case Recommendation

As an administrator, you must add, train, and activate the Similar Case Recommendation model.

### Procedure

1. Navigate to your user profile, and select [System Settings](#) > [All Settings](#) > [Machine Learning](#) > [Intelligent Service](#) > [Similar Case Recommendation](#).
2. To add a mode, click [+](#).
3. Enter a name for your model.
4. Click [Save and Train](#) to save your model and start the training.

The training takes time to complete depending on the volume of the historical data. The mode goes through the following statuses during the training: [Created](#) > [Training in Preparation](#) > [Data Extraction is Pending](#) > [Data Extraction is in Progress](#) > [Data Preprocessing is Pending](#) > [Data Preprocessing is in Progress](#) > [Training Triggered](#) > [Training is Pending](#) > [Training in Progress](#) > [Training Completed or Training Failed](#) > [Active or Inactive](#).

The system stops the training if data extraction fails and updates the status as [Data Extraction Failed](#).

You can refresh and update the training status of the model by selecting the [refresh](#) (refresh) icon.

#### Note

You can also save your model and train your model later.  
Locate your model and click [Train](#) from the [Actions](#) to start training the model.

5. After the training is complete, click [Activate](#) from [Actions](#) to activate the model.  
You can deactivate a model by selecting [Deactivate](#).

#### Note

You can use the test console to test if the model is functioning correctly. Select your model, enter a description in the [Input](#) field and click [Test](#).

You can enable similar case recommendation for incoming cases by turning on the [Enable Similar Case Recommendation](#) switch. Testing via console is possible with an active model even when automation is disabled.

## 18.1.2.2 View Similar Case Recommendation

View similar case recommendations in cases.

### Procedure

1. Go to the [Case](#) worklist.
2. Create a new case and enter the subject and description fields.

Similar case recommendation happens at case creation. Hence, you'll only see predictions for cases created after the model is set to active status.

3. Select [Save](#).

Find similar cases in the the side pane by clicking the [Similar Cases](#) (🔍) icon. A confidence score of 0–100 shows the level of accuracy of the match.

Case summary displays on hover over the card view or list view of cases in the [Similar Cases](#) section, if case summary is activated and a summary exists for the case.

### 18.1.2.2.1 Similar Cases - Show More

View all similar cases to check for more solutions that apply to your current cases.

To view more similar cases, select [Show More](#) at the top right of the [Recommendations](#) side pane. The [Solution Center](#) opens and displays up to ten similar cases in the top table.

## 18.1.3 NLP Classification

Use Natural language processing (NLP) to identify sentiment from email.

The NLP Classification scenario includes the **Sentiment** output field that determines if the vocabulary used in the email from which a case is created, is positive or negative and the degree of positivity or negativity.

#### ⚠ Restriction

Sentiment analysis only identifies sentiment for cases created from email channels.

NLP sentiment classification returns the following possible values:

- Not Available
- Strong Positive
- Weak Positive
- Neutral
- Weak Negative



- Strong Negative

The sentiment value from an email from which a case is created, indicates the overall emotion.

The magnitude of sentiment indicates how much emotional content is present within the email, and is often proportional to the length of the text.

The natural language algorithm differentiates between positive and negative emotion in a case, but doesn't identify **specific** positive and negative emotions. For example, **angry** and **sad** are both negative emotions. However, the sentiment classification only indicates that the text is **negative**, not sad, or angry.

An email with a neutral value can indicate a low-emotion, or email text expressing mixed emotions; containing both high positive and high negative values that cancel out each other.

### 18.1.3.1 Configure NLP Classification

As an administrator, you must add, train, and activate the NLP Classification model.

#### Procedure

1. Navigate to your user profile, and select [Settings](#) [All Settings](#) [Machine Learning](#) [Intelligent Service](#) [NLP Classification](#).
2. To create a new model, click [+](#).
3. Enter a name for your model.
4. Turn on the [Sentiment](#) switch to enable sentiment detection.
5. Click [Save and Train](#) to save your model and start the training.

The model goes through the following statuses during the training: [Created](#) [Training in Preparation](#) [Data Extraction is Pending](#) [Data Extraction is in Progress](#) [Data Preprocessing is Pending](#) [Data Preprocessing is in Progress](#) [Training Triggered](#) [Training is Pending](#) [Training in Progress](#) [Training Completed or Training Failed](#) [Active or Inactive](#).

You can refresh and update the training status of the model by selecting the [refresh](#) (refresh) icon.

#### Note

You can also save your model and train your model later.

6. Locate your model and click [Train](#) from [Actions](#) to start training the model.
7. After the training is completed, click the [activate](#) (activate) icon to activate the model.

You can deactivate a model by selecting the [deactivate](#) (deactivate) icon.

#### Note

You can adjust the threshold confidence level. The default value of the confidence level is 60.

### Note

You can use the test console to test if the model is functioning correctly. Select your model, enter a description in the [Input](#) field and click [Classify](#). The test console invokes the prediction model API using the sample case data you provide.

## 18.1.4 Machine Translation

Use machine translation to translate emails.

Machine translation is available for the languages in the following table.

Machine Translation Supported Languages

| Source Language                          | Code | Target Language         | Code |
|------------------------------------------|------|-------------------------|------|
| Arabic                                   | ar   | English (United States) | en   |
| Catalan                                  | ca   | English (United States) | en   |
| Chinese                                  | zh   | English (United States) | en   |
| Danish                                   | da   | English (United States) | en   |
| Dutch                                    | nl   | English (United States) | en   |
| English (US – applicable for below rows) | en   | Arabic                  | ar   |
| English                                  | en   | Bulgarian               | bg   |
| English                                  | en   | Catalan                 | ca   |
| English                                  | en   | Chinese (Simplified)    | zh   |
| English                                  | en   | Chinese (Traditional)   | zf   |
| English                                  | en   | Croatian                | hr   |
| English                                  | en   | Czech                   | cs   |
| English                                  | en   | Danish                  | da   |
| English                                  | en   | Dutch                   | nl   |
| English                                  | en   | Estonian                | et   |
| English                                  | en   | Finnish                 | fi   |
| English                                  | en   | French (France)         | fr   |
| English                                  | en   | German                  | de   |
| English                                  | en   | Greek                   | el   |
| English                                  | en   | Hebrew                  | he   |
| English                                  | en   | Hindi                   | hi   |
| English                                  | en   | Hungarian               | hu   |
| English                                  | en   | Italian                 | it   |

| Source Language | Code | Target Language      | Code |
|-----------------|------|----------------------|------|
| English         | en   | Japanese             | ja   |
| English         | en   | Kazakh               | kk   |
| English         | en   | Korean               | ko   |
| English         | en   | Latvian              | lv   |
| English         | en   | Lithuanian           | lt   |
| English         | en   | Malay                | ms   |
| English         | en   | Norwegian            | no   |
| English         | en   | Polish               | pl   |
| English         | en   | Portuguese (Brazil)  | pt   |
| English         | en   | Romanian             | ro   |
| English         | en   | Russian              | ru   |
| English         | en   | Serbian (Latin)      | sh   |
| English         | en   | Slovak               | sk   |
| English         | en   | Slovenian            | sl   |
| English         | en   | Spanish (Spain)      | es   |
| English         | en   | Swedish              | sv   |
| English         | en   | Thai                 | th   |
| English         | en   | Turkish              | tr   |
| English         | en   | Ukrainian            | uk   |
| English         | en   | Vietnamese           | vi   |
| Finnish         | fi   | English              | en   |
| French (France) | fr   | English              | en   |
| German          | de   | Bulgarian            | bg   |
| German          | de   | Chinese (Simplified) | zh   |
| German          | de   | Croatian             | hr   |
| German          | de   | Czech                | cs   |
| German          | de   | English              | en   |
| German          | de   | French (France)      | fr   |
| German          | de   | Hungarian            | hu   |
| German          | de   | Italian              | it   |
| German          | de   | Polish               | pl   |
| German          | de   | Romanian             | ro   |
| German          | de   | Russian              | ru   |
| German          | de   | Serbian (Latin)      | sh   |

| Source Language     | Code | Target Language | Code |
|---------------------|------|-----------------|------|
| German              | de   | Slovakian       | sk   |
| German              | de   | Slovenian       | sl   |
| German              | de   | Spanish (Spain) | es   |
| Hindi               | hi   | English         | en   |
| Italian             | it   | English         | en   |
| Japanese            | ja   | English         | en   |
| Korean              | ko   | English         | en   |
| Norwegian           | no   | English         | en   |
| Portuguese (Brazil) | pt   | English         | en   |
| Romanian            | ro   | English         | en   |
| Russian             | ru   | English         | en   |
| Spanish (Spain)     | es   | English         | en   |
| Swedish             | sv   | English         | en   |

## 18.1.4.1 Configure Machine Translation

As an administrator, you must add, train, test, and activate the Machine Translation model.

### Procedure

1. Navigate to your user profile, and select [Settings](#) > [All Settings](#) > [Machine Learning](#) > [Intelligent Service](#) > [Machine Translation](#).
2. To create a new model, click [+](#).
3. Enter a name for your model.
4. Click [Save and Train](#) to save your model and start the training.

The model goes through the following statuses during the training: [Created](#) > [Training in Preparation](#) > [Training Completed or Training Failed](#).

You can refresh and update the training status of the model by selecting the [refresh](#) icon.

#### Note

You can also save your model and train your model later.

5. Locate your model and click [Train](#) from [Actions](#) to start training the model.
6. After the training is complete, click [Activate](#) from [Actions](#) to activate the model.  
You can deactivate a model by selecting [Deactivate](#).

#### Note

You can use the test console to test if the model is functioning correctly. Select your model, enter a description in the [Input](#) field and click [Translate](#). The test console invokes the prediction model API using the sample case data you provide.

## 18.1.4.2 View Machine Translated Text

View the translated description for the email in the user logon language.

### Procedure

1. Navigate to the [Emails](#) worklist and select an email.

The same feature is available in Case timeline.

2. Select (Machine Translation) to view the [Machine Translation](#) popup.

The [Machine Translation](#) popup has two sections, [Original Text](#) and [Translated Text](#). You can remove some parts of the text in the [Original Text](#) before translation. This is helpful when you have a mix of languages in an email. For example, when email footer signature or disclaimer text is in a language other than the email content language and is much more in volume as compared to the email content which could be a shorter one or two line which need to be translated. In this scenario, you can remove a part of text which is in different language and translate.

3. To translate the required text, select [Translate](#) in the [Translated Text](#) section.

## 18.1.5 Case Summarization

Use Natural Language Processing (NLP) to summarize the text in the subject field and email interactions.

#### Restriction

- Text summarization is tested only for English language texts.
- Text summarization is applicable only for email based interactions between a customer and a service agent.
- Text summarization is available only for email interactions generated from Outlook and Gmail.
- For existing cases, a summary is generated only when a new interaction is added (after model activation).
- There's a delay between sending or receiving interaction and summary updates. Text summarization takes place asynchronously from the email communication. The delay depends on the number of interactions in the queue and the number and size of the interactions.

## 18.1.5.1 Configure Case Summarization

As an administrator, you must add, train, test, and activate the Case Summarization model.

### Procedure

1. Navigate to your user profile, and select ► [Settings](#) ► [All Settings](#) ► [Machine Learning](#) ► [Intelligent Service](#) ► [Case Summarization](#) ►.
2. To create a new model, click **+**.
3. Enter a name for your model.
4. Click [Save and Train](#) to save your model and start the training.

The model goes through the following statuses during the training: ► [Created](#) ► [Training in Preparation](#) ► [Data Extraction is Pending](#) ► [Data Extraction is in Progress](#) ► [Data Preprocessing is Pending](#) ► [Data Preprocessing is in Progress](#) ► [Training Triggered](#) ► [Training is Pending](#) ► [Training in Progress](#) ► [Training Completed or Training Failed](#) ► [Active or Inactive](#) ►.

You can also save your model and train your model later.

#### Note

You can refresh and update the training status of the model by selecting the  (refresh) icon.

5. After the training is complete, click [Activate](#) from [Actions](#) to activate the model.

You can deactivate a model by selecting the  (deactivate) icon.

#### Note

You can use the test console to test if the model is functioning correctly. Select your model, enter a Case ID in the [Input](#) field and click [Summarize](#). The test console invokes the prediction model API using the sample case data you provide.

You can enable case summarization for incoming cases by turning on the [Enable text summarization for case](#) switch. When you enable the automation settings, then the system starts summarizing email interactions at case level. Testing via console is possible with an active model even when automation is disabled.

#### Restriction

There's an upper limit on the number of interactions summarized for test purposes. Hence, you may not see all interactions in the summary.

## 18.1.5.2 View Case Summarization

View the case summary results.

### Procedure

1. Log into your solution and go to [Cases](#).
2. Open a case in detailed view using the  icon.
3. Go to the [General](#) tab to view the summary of each interaction.

You can see an overall summary, a summary of agent-generated interactions, and a summary of customer interactions.

If the email interactions and the corresponding case summaries were created in English, and later when you login in a different language, then you can translate the summary to the logged in language by selecting [Show Translation](#).

#### Note

You cannot keep both Generative AI based case summary and classical Machine Learning based case summary (under Intelligent Services) in active state together. If you are activating Generative AI based case summary, then you must deactivate the other one.

Also, while switching from classical machine learning to Generative AI based summary (or vice-versa), case summary generated by the previous approach does not get updated until summary is re-generated.

## 18.1.6 Case Topic Analyzer

Use the [Case Topic Analyzer](#) machine learning model to view trending topics from cases.

The [Case Topic Analyzer](#) machine learning model analyzes keywords from case descriptions and groups cases based on labels assigned to a model. A label is a topic which acts as a heading for a list of similar keywords.

### Prerequisites

To ensure accurate results from the model, your system must contain historical data (cases) in order to train the model. Ideally around 3000 cases (minimum) with cases having meaningful text content in the case description (multiple sentences which conveys the issue).

#### Note

When setting up a demo or test environment, you need to consider the following points:

- If most cases are talking about totally different distinct problems, topic detection will not happen properly.
- Cases must have meaningful text which conveys the issue correctly.

### 18.1.6.1 Configure Case Topic Analyzer

As an administrator, you must add, train, test, and activate the Case Topic Analyzer model.

#### Procedure

1. Navigate to your user profile, and select [Settings](#) > [All Settings](#) > [Machine Learning](#) > [Intelligent Service](#) > [Case Topic Analyzer](#).
2. Under [Case Topic Analyzer](#), click the create icon (+) to add a model.
3. Enter a name for your model.
4. Click [Save and Train](#) to start training the model.

The model goes through the following statuses during the training: [Created](#) > [Training in Progress](#) > [Training Completed or Training Failed](#) > [Active or Inactive](#).

You can refresh and update the training status of the model by selecting the  (refresh) icon.

5. After the training is completed, select your model to assign a label to the list of keywords.

#### Note

A label is a topic which acts as a heading for a list of similar keywords.

You must set appropriate topic label which matches your business.

6. Select [Settings](#) to see a number of topics with keywords associated with each one and three sample case IDs for each topic.
7. Enter a topic label.
8. Click [Save](#).
9. Click the  (activate) icon to activate the model.

#### Note

- You can deactivate a model by selecting the  (deactivate) icon.
- You can use the test console to test if the model is functioning correctly. Select your model, enter a description in the [Input](#) field, and click [Test](#).
- You can enable the case topic analyzer for incoming cases by turning on the [Enable topic analyzer for case](#) switch.
- Only English language is supported currently for the Case Topic Analyzer models.



## 18.1.6.2 View Topic Analyzer

View trending topics from case descriptions.

After you train and activate the case topic analyzer model, you can view trending topics in [Topic Analyzer](#). New cases created in the system are grouped according to the topics they belong to. Existing cases up to a maximum of 12 months are grouped into topics.

You can choose to view the trending topics in the form of cards or lists that display the topic name, the rank, number of cases, and percentage trend. The topics are ranked according to the number of cases for the topics for the calendar year. The percentage trends are calculated on monthly basis.

You can filter trending topics using a date filter with values such as [This week](#), [This Month](#), and [This Year](#). By default, the date filter is set to filter trending topics for the current week. For a selected filter, topics will appear only if it has cases from that selected time period. However, the overall topic rank for the calendar year and percentage trend (month to month) does not change.

The trending topics display information for all topics in the following forms:

- **Bar charts** - displays the number of cases for topics
- **Graphs** - displays topic trends across months.
- **Keywords clusters** - displays list of trending keywords as word clouds and as lists.

When you select a trending topic, the list, graph and keywords display information specific to the selected topic only. You can view the list of associated cases and navigate to specific cases.

## 18.1.7 Case Type Determination

The [Case Type Determination](#) machine learning model determines the case type during the conversion of email to case.

During the conversion of emails to cases, the case type determination model analyzes subject and description. It then establishes the [Case Type](#) for the case creation.

The system inputs the determined case type into the case, initiating the case creation process.

Case type determination is currently supported in the following languages:

- Chinese
- English
- French
- German
- Japanese
- Portuguese
- Russian
- Spanish

## Prerequisites

- Data volume: Includes minimum 1000 cases with case types assigned and an additional 500 instances of each case type.
- Data quality: Includes subject, description, and case type.

### 18.1.7.1 Configure Case Type Determination

As an administrator, you must add, train, test, and activate the Case Type Determination model.

1. Navigate to your user profile, and select ► [Settings](#) ► [All Settings](#) ► [Machine Learning](#) ► [Intelligent Service](#) ► [Case Type Determination](#) ►.
2. To add a new model, click +.
3. In the [Create Model](#) popup, enter a name for the model.
4. Go to [Settings](#).
5. Turn on the [Confidence Level](#) and select the confidence value.
6. Save your changes.
7. Click [Save and Train](#) to save your model and start the training.  
You can also save your model and train your model later.  
The model goes through the following statuses during the training: ► [Created](#) ► [Training in Progress](#) ► or [Training Failed](#).  
You can refresh and update the training status of the model by selecting the ↻ (refresh) icon.
8. Click the ⚙️ (activate) icon to activate the model.

#### Note

- After you activate the model, ensure that you have enabled the [Case Type Determination From AI](#) switch from the [Determination Rules](#) section in the email channels. For more information, see the Related information section.
- You can deactivate a model by selecting the ⏻ (deactivate) icon.
- You can use the test console to test if the model is functioning correctly. Select your model, enter a description in the [Input](#) field, and click [Test](#).

## Related Information

[Configure Case Types \[page 191\]](#)

Customize case types to match the workflow used by your organization.

## 18.1.8 Custom Code Extensions for Machine Learning Scenarios

You can now control the training data set and select which instances must be predicted using the [Custom Code Extensions for ML Scenarios](#) for case categorization. This scenario helps train the model with relevant data and exclude irrelevant ones, making the training process more efficient.

### Note

This scenario is supported explicitly for case categorization only. One year's data is consumed for model training. It's ideal that the data filtering criteria for both training and inferencing are aligned such that the predictions are accurate, to avoid inferencing data which has been excluded from model training as it can yield incorrect predictions

Use the predelivered [Custom Code Extensions for ML Scenarios](#) communication configuration template, along with the synchronous HTTP-data outbound communication configuration [Machine Learning: Custom Processing for Setting Training Data Selection Criteria](#) and [Machine Learning: Custom Processing for Inference Eligibility Check](#) extension points to enable this capability.

## Machine Learning: Custom Processing for Setting Training Data Selection Criteria

The [Machine Learning: Custom Processing for Setting Training Data Selection Criteria](#) extension point enables you to influence the data set used for ML model training. It uses historical case data of one year. As a user, you can adjust the filtering criteria, that is, limit the case data usage to six months or limit the case data usage of specific case types only using the filterable case attributes.

### Request format

The request payload is sent through a POST call in JSON in the following format.

#### Sample Code

```
request payload
{
 "entities" : ["sap.crm.caseservice.entity.case"],
 "scenarioName": "CASE_CATEGORIZATION"
}
```

### Response format

The application expects a response back in JSON in the following format:

#### Sample Code

```
[{ "entity": "sap.crm.caseservice.entity.case", "filter": "adminData/createdOn
ge 2025-04-03T04:23:39.582Z" }]
```

## Machine Learning: Custom Processing for Inference Eligibility Check

The [Machine Learning: Custom Processing for Inference Eligibility Check](#) extension point determines if an incoming case instance qualifies for ML inference. It uses user-defined filters (case type, priority, and so on) compatible with API specifications to decide whether a particular case instance must be categorized using machine learning or must be skipped. Any invalid or incompatible filters result in processing failure.

### Request format

The request payload is sent through a POST call in JSON in the following format.

#### Sample Code

```
request payload
{
 "entity" : "sap.crm.caseservice.entity.case", "entityId": "755724b5-9334-4c81-be78-e8f644aa8603", "entityDisplayId": "12345",
 "scenarioName": "CASE_CATEGORIZATION"
}
```

### Response format

The application expects a response back in JSON in the following format:

#### Sample Code

```
{
 "eligible": true
}
```

## Configure Custom Processing Exit

1. Create a communication system with outbound authentication details for the application endpoint, following the steps in the [Create Communication Systems](#) topic in this guide. You can ignore the inbound configuration. This application endpoint holds the custom logic to determine enabling of Machine Learning scenario.
2. Create a communication configuration by copying the [Machine Learning: Custom Processing for Setting Training Data Selection Criteria](#) and [Machine Learning: Custom Processing for Inference Eligibility Check](#) predelivered templates by following the steps in the [Create Communication Configurations](#) topic in this guide. You can ignore steps related to inbound and outbound configuration.
3. Change the status of the communication arrangement to [Active](#) by clicking the status and selecting the [Active](#) radio button.

#### Note

Currently, the exit processing is supported only for ML scenario Case Categorization only.

The endpoint configured in the outbound configuration gets called for the processing the Machine Learning Scenario Case Categorization and Inference Checks.

## Related Information

[Create Communication Systems](#)  
[Create Communication Configurations](#)

### 18.1.9 Profanity Check

Use profanity check to define a set of pejorative vocabulary.

Profanity check is currently available only in English.

#### 18.1.9.1 Configure Profanity Check

As an administrator, you must add, train, test, and activate the Profanity Check model.

#### Procedure

1. Navigate to your user profile, and select ► [Settings](#) ► [All Settings](#) ► [Machine Learning](#) ► [Intelligent Service](#) ► [Profanity Check](#) .
2. To create a new model, click [+](#) .
3. Enter a name for your model.
4. Click [Save and Train](#) to save your model and start the training.

The model goes through the following statuses during the training: ► [Created](#) ► [Training in Preparation](#) ► [Data Extraction is Pending](#) ► [Data Extraction is in Progress](#) ► [Data Preprocessing is Pending](#) ► [Data Preprocessing is in Progress](#) ► [Training Triggered](#) ► [Training is Pending](#) ► [Training in Progress](#) ► [Training Completed or Training Failed](#) ► [Active or Inactive](#) .

You can refresh and update the training status of the model by selecting the  (refresh) icon.

#### Note

You can also save your model and train your model later.

Locate your model and click [Train](#) from [Actions](#) to start training the model.

5. After the training is completed, click the  (activate) icon to activate the model.  
You can deactivate a model by selecting the  (deactivate) icon.

#### Note

You can use the test console to test if the model is functioning correctly. Select your model, enter a description in the [Input](#) field and click [Check Profanity](#). The test console invokes the prediction model API using the sample case data you provide.

## 18.1.9.2 View Profanity Check

View the results of profanity check.

### Procedure

1. Navigate to the [Email](#) worklist.
2. Reply to an email.
3. Draft a reply with profane language.

The system opens a pop up with the profane text.

## 18.1.10 Intelligent Services FAQ

Here are some frequently asked questions (FAQ) and corresponding answers about Machine Learning.

### What are Intelligent Services?

Intelligent Services is a collection of machine learning models and scenarios embedded in SAP Service Cloud CNS. Intelligent Services is not a separate product. You don't require any additional licensing, deployment, or upgrades. You can manage everything through the SAP Service Cloud CNS administrator interface.

### Is there any impact on the system performance as the data is cached?

There's no impact on day-to-day system performance as the data is cached and stored in the solution. All compute-intensive operations, such as training the model, happen outside of your SAP Service Cloud CNS tenant.

### How many models are active at any time?

Only one model can be active in production. However, you can create multiple models.

### Are there extension fields to the model?

Extension fields aren't available right now.

### How many category levels does Case Categorization support?

Case Categorization supports upto four levels.

## Related Information

[Service Ticket Intelligence](#) [Service Ticket Intelligence](#)

## 18.2 Generative AI

Use Generative AI to generate account synopsis, case summary, draft email response for cases, and backend entity extraction.

Generative AI features are available only when you have a valid license for usage of SAP AI Units. To enable Generative AI features, you must raise an incident with the **CEC-CRM-ML** component, mentioning that you have a valid license for SAP AI Units. After you activate the Generative AI features, you must assign the [sap.crm.service.mlScenarioManagementGenAIService](#) service and app to your role.

### Note

- For details on how the metrics consume the credits, see the [SAP AI Services List](#).
  - For SAP Service Cloud Version 2, one transaction as per AI services list for each Generative AI capability translates to the following:
    - one account synopsis
    - one email draft
    - one case summaryEvery regeneration for these capabilities also counts as a transaction.
- Generative AI features are tested only for English, German, French, and Spanish language texts.

### 18.2.1 Case Summary

Use Generative AI to summarize text in email interactions for cases.

Case summary is displayed in the [General](#) tab of cases. Case summary is displayed on hover over the card view or list view of cases in the [Similar Cases](#) section, if case summary is activated and a summary exists for the case. Case summary is applicable only for email-based interactions between a customer and a service agent. Case summary is generated only for new interactions that occur after activating [Case Summary](#).

### Note

You can't keep both Generative AI-based case summary and classical Machine Learning based case summary (under Intelligent Services) in an active state together. If you're activating a Generative AI-based case summary, then you must deactivate the other one.

Also, while switching from classical machine learning to Generative AI-based summary (or conversely), case summary generated by the previous approach doesn't get updated until the summary is re-generated.

A complete case summary consists of two different types of summaries, interaction and resolution summary. Resolution summary is specific to cases that are in either [Closed](#) or [Completed](#) status, while interaction summary is a summary of ongoing interactions within a case.

- **Interaction Summary:** An interaction summary is generated based on email interactions between a customer and the service agent. It includes an [Interaction Summary](#), [Products](#), [Key Dates](#), [Case Sentiment](#) or [Case Sentiment Graph](#). If an interaction summary isn't available for a case, you can choose [Generate Summary](#) on the [General](#) tab of the case to generate it.
- **Resolution Summary:** A resolution summary is generated for cases that are in [Completed](#) or [Closed](#) status only. A resolution summary consists of an [Overview](#), [Resolution](#), [Case Metrics](#), and [Resolution Satisfaction](#) sections based on the resolution provided in the case. You can generate a resolution summary by choosing the [Generate Resolution Summary](#) on the [General](#) tab of completed or closed cases.

## Configure Case Summary

1. As an administrator, navigate to your user profile, and select ► [System Settings](#) ► [All Settings](#) ► [Machine Learning](#) ► [Generative AI](#) ► [Case Summary](#) ►.
2. To activate summary for cases, enable [Activate Case Summary](#), and accept [Acknowledgment](#).
3. Use the [Test Console](#) to test the case summary.
4. To update case summary for your business users, enable [Update Case Summary](#) under [Case Summary Automation](#).

### Note

The administrator can use the test console even if this step isn't performed, however, the summary of a case is available to business users only if [Update Case Summary](#) is enabled.

5. Select one of the following:
  1. **Manual** - to manually generate the summary and update the case summary by regenerating the summary when required. Business users must click [Generate Summary](#) to generate the case summary for the first time, and use [Regenerate](#) to update the case summary with the latest changes.
  2. **Automatic** - to automatically update the case summary for every interaction between the agent and customer.

### Note

Summary generation is an asynchronous process and takes some time for a summary update to be visible in the UI.

6. Go to [Interaction](#). The [Email Interaction](#) and [Description](#) fields are included in the interaction summary by default. Optionally, you can also enable the [Priority Code](#) and [External Case Notes](#) fields by selecting note types to include in the interaction summary.
7. Go to [Resolution](#). The [Email Interaction](#) and [Description](#) fields are included in the case summary by default. Optionally, you can also enable [Internal Case Notes](#) and [External Case Notes](#) fields to include details from these fields in the resolution summary overview.
8. Save your changes.



## View Interaction Summary

### Prerequisite:

- Ensure you've enabled the [Natural Language Processing Classification](#) service from Intelligent Services to display either a [Sentiment Trend](#) for a single inbound interaction or the [Sentiment Graph](#) in cases of multiple inbound interactions. This graph is displayed within the Interaction Summary.
  - Ensure you've created and enabled a [Business Information Extraction](#) scenario from Intelligent Services to display the [Products](#) information and the associated [Key Dates](#) in the interaction summary. For the Products information to be extracted from a case, a business information extraction scenario must be configured and activated including [Product Name](#) [Product ID](#), [Serial ID](#), and linked to a case type. The product information is extracted once the case is logged and also when the case is updated based on subsequent interactions.
1. Open a case in detailed view using the  icon.
  2. Go to .
  3. Select [Generate Summary](#) to generate an interaction summary manually. Once the interaction summary is generated, select [Regenerate](#) to update the summary with latest interactions. This step is applicable if you've selected the [Manual](#) option for [Update Case Summary](#) during configuration. The generated summary includes [Products](#), [Key Dates](#), [Case Sentiment](#) or a [Case Sentiment Graph](#), either of which appear based on single or multiple interactions respectively.

#### Note

- Case summary is generated by Generative AI using case-specific data. You must verify it before making decisions.
- If the email interactions and the corresponding case summaries are created in English, and later when you log on in a different language, you can translate the summary to the logged in language.

## View Resolution Summary

1. Open a closed or a completed case in detailed view using the  icon.
2. Go to .
3. Select [Generate Resolution Summary](#) to generate resolution summary manually. You can view resolution summary only for those cases that are in [Completed](#) or [Closed](#) status only. This step is applicable if you've selected the [Manual](#) option for [Update Case Summary](#) during configuration.

## Related Information

### [NLP Classification \[page 344\]](#)

Use Natural language processing (NLP) to identify sentiment from email.

### [Business Information Extraction \[page 370\]](#)

Business information extraction is the process of identifying, extracting, and organizing relevant data from business-related content such as text descriptions to create, update, or identify case-related entities.

## 18.2.2 Registered Product Summary

Use Generative AI to generate a registered products summary for the registered products.

Registered product summary provides a summary of registered products and cases created for the registered products in the last three months.

A registered product summary consists of an overview and aggregated cases summary.

- Overview: A summary of when the product was registered, the type of warranty, the warranty coverage, and the location of the product is provided.
- Aggregated Case Insights: Provides consolidated insights of all cases registered in the last three months, their status, SLA for cases, and so on. It also provides a quick information of the acumen of the cases in the last three months.

### Configure Registered Product Summary

1. As an administrator, navigate to your user profile, and select ► [System Settings](#) ► [All Settings](#) ► [Machine Learning](#) ► [Generative AI](#) ► [Registered Product Summary](#) .
2. To activate case summarization, enable [Activate Registered Product Summary](#), and accept [Acknowledgment](#). By default, the [Registered Products Overview](#) and [Related Case Insights](#) is included in the configuration.
3. Save your changes.

## 18.2.3 Email Drafter

Use email drafter from Generative AI to draft email replies or interactions for cases (currently tested in English, German, Spanish, and French).

### Configure Email Drafter

1. Navigate to your user profile and select ► [System Settings](#) ► [All Settings](#) ► [Machine Learning](#) ► [Generative AI](#) ► [Email Drafter](#) .
2. To activate email draft recommendation for business users, enable [Activate Email Drafter](#), and accept the [Acknowledgment](#).
3. Use [Test Console](#) to test email draft recommendation.
4. Go to [Configure](#) and enable the required entities (currently supported only for Cases).
5. To include prompts in the email draft recommendation, enable the required prompts from the following list in [Available Prompts](#):
  - Case Status Update email
  - Update the Service Level timelines

- Share relevant KB articles
6. To create a custom prompt, select **+** (Create Custom Prompts) icon, enter prompt name, assign business roles from the list, and enter instructions.

#### **Note**

- You can assign business roles choosing the  (edit) icon under the **Actions** column for both standard and custom prompts.
- Up to three business roles can be assigned to a prompt. You can select more than three roles from the selector, however, a maximum of three roles can be assigned to a prompt and is also displayed in the **Assigned Roles** section. If a prompt isn't assigned to a particular role, it's available for all business roles in the system.
- Only users assigned to these business roles can access the prompts.
- While writing instructions, you can use the field names from **Dynamic Values** as placeholders for providing input context to Generative AI. This action can enhance prompt generation. The maximum length allowed for instructions is 500 characters.

7. Save your changes.

## **View Email Drafter Recommendation**

1. To view the email drafter recommendation, open a case in detailed view using the  icon and select **Email**.
2. Select **Email Drafter**. The **Email Drafter** popup opens.
3. Select a prompt from **Available Prompts**.
4. Select the style, tone, and length.
5. Select **Generate Draft**. The recommended email draft appears in the **Draft** section.
6. Click **Apply** to copy the draft content to the email editor.

#### **Note**

Draft is generated by Generative AI using the application context data. You must verify the draft thoroughly before using it in the response email. If you want to change the text, style, tone, or purpose, then you must make the changes and click **Generate Draft** and **Apply**.

## **18.2.4 Account Synopsis**

Account Synopsis provides all the information related to an account that helps the service agents.

### **Configure Account Summary**

1. As an administrator, navigate to **User Menu** > **System Settings** > **All Settings** > **Machine Learning** > **Generative AI** > **Account Summary**.

2. To activate account summary, enable [Activate Account Summary](#), and accept [Acknowledgment](#).
3. In the [Configure](#) section, you can enable the following aspects of accounts that you want to display in the [AI Synopsis](#) tab within accounts:
  - Overall Information
  - Business Overview
  - Typical Customers
  - Competitive Landscape
  - Culture and Background
  - Business Strategies
4. Save the changes.

## View Account Summary

1. From the Navigation Menu, Search for [Accounts](#).
2. In the worklist of the accounts view, select the account name and then click  ([Open in Detail View](#)).
3. Select the [AI Synopsis](#) tab to view the AI-generated summary.

## 18.2.5 Duplicate Check

Duplicate Check leverages Generative AI to check and display the duplicate check results with a confidence score, before creating the new respective entities.

### Note

[Duplicate Check](#) feature is also available in the system. At once, only one of the versions can be enabled in the system. Ensure that you've enabled only one of these features in your system.

### 18.2.5.1 Account Duplicate Check

Account Duplicate Check leverages Generative AI to check and display the duplicate check results with a confidence score, before creating an account.

Duplicate check for accounts performs a check comparing the configured fields with the list of accounts existing within the system. As an administrator, you can configure the fields for duplicate check. Once duplicate check is configured for accounts, any new account that is to be created is checked against configured fields and the result is displayed, containing a list of the existing accounts with a confidence score. This confidence score enables you to decide whether or not to proceed with the creation.

### Note

[Duplicate Check](#) (non-AI) feature is also available in the system. At once, only one of the features can be enabled in the system. Ensure that you've enabled only one of these features in your system.

## Configure Account Duplicate Check

1. As an administrator, navigate to your [User Menu](#), and select [System Settings](#) > [All Settings](#) > [Machine Learning](#) > [Generative AI](#) > [Duplicate Check](#) > [Account](#).
2. Enable the [Activate Account Duplicate Check](#) switch to activate duplicate check.
3. Under [Configure](#), include the fields such as [Full Name](#), [Role](#), [Industry](#), [Email](#), and so on, that you wish to configure accounts for checking duplicates. You can remove the fields clicking on the delete icon. You can also add more fields by selecting fields from the [Add Fields](#) option.

### Note

The [Add Fields](#) section contains additional configurable account fields that you can include in your check configuration. These fields can include custom fields as well, and these custom fields must be set to filterable in [Field Query Properties](#) of extension fields properties.

4. Save your changes.
5. Use the [Test Console](#) to check for duplicate accounts. Enter the [Account ID](#) to check for duplicate accounts. If duplicates exist, a list of duplicates is displayed along with the confidence score.

### Note

Confidence indicates how closely the attributes of the new account match with the duplicates.

## Related Information

[Duplicate Check](#)

### 18.2.5.2 Individual Customer Duplicate Check

Individual Customer Duplicate Check leverages Generative AI to check and display the duplicate check results with a confidence score, before creating an individual customer.

Duplicate check for individual customers performs a check comparing the configured fields with the list of individual customers existing within the system. As an administrator, you can configure the fields for duplicate check. Once duplicate check is configured for individual customers, any new individual customer that is to be created is checked against configured fields and the result is displayed, containing a list of the existing individual customers with a confidence score. This confidence score enables you to decide whether or not to proceed with the creation.

### Note

[Duplicate Check](#) (non-AI) feature is also available in the system. At once, only one of the features can be enabled in the system. Ensure that you've enabled only one of these features in your system.

## Configure Individual Customer Duplicate Check

1. As an administrator, navigate to your [User Menu](#), and select [System Settings](#) > [All Settings](#) > [Machine Learning](#) > [Generative AI](#) > [Duplicate Check](#) > [Individual Customer](#).
2. Enable the [Activate Individual Customer Duplicate Check](#) switch to activate the duplicate check.
3. Under [Configure](#), include the fields such as [Name](#), [Website](#), [Street](#), [Role](#), [Postal Code](#), and so on, that you wish to configure individual customers for checking duplicates. You can remove the fields clicking on the delete icon. You can also add more fields by selecting fields from the [Add Fields](#) option.

### Note

The [Add Fields](#) section contains additional configurable individual customer fields that you can include in your check configuration. These fields can include custom fields as well, and these custom fields must be set to filterable in [Field Query Properties](#) of extension fields properties.

4. Save your changes.
5. Use the [Test Console](#) to check for duplicate individual customers. Enter the [Customer ID](#) to check for duplicate individual customers. If duplicates exist, a list of duplicates is displayed along with the confidence score.

### Note

Confidence indicates how closely the attributes of the new individual customer match with the duplicates.

## Related Information

[Duplicate Check](#)

### 18.2.5.3 Contact Duplicate Check

Contact Duplicate Check leverages Generative AI to check and display the duplicate check results with a confidence score, before creating a contact.

Duplicate check for contacts performs a check comparing the configured fields with the list of contacts existing within the system. As an administrator, you can configure the fields for duplicate check. Once duplicate check is configured for contacts, any new contact that is to be created is checked against configured fields and the result is displayed, containing a list of the existing contacts with a confidence score. This confidence score enables you to decide whether or not to proceed with the creation.

### Note

[Duplicate Check](#) (non-AI) feature is also available in the system. At once, only one of the features can be enabled in the system. Ensure that you've enabled only one of these features in your system.

## Configure Contact Duplicate Check

1. As an administrator, navigate to your [User Menu](#), and select ► [System Settings](#) ► [All Settings](#) ► [Machine Learning](#) ► [Generative AI](#) ► [Duplicate Check](#) ► [Contact](#) .
2. Enable the [Activate Contact Duplicate Check](#) switch to activate duplicate check.
3. Under [Configure](#), include the fields such as [Name](#), [Department Code](#), [Function Code](#), [Email](#), and so on, that you wish to configure contacts for checking duplicates. You can remove the fields clicking on the delete icon. You can also add more fields by selecting fields from the [Add Fields](#) option.

### Note

The [Add Fields](#) section contains additional configurable contact fields that you can include in your check configuration. These fields can include custom fields as well, and these custom fields must be set to filterable in [Field Query Properties](#) of extension fields properties.

4. Save your changes.
5. Use the [Test Console](#) to check for duplicate contacts. Enter the [Contact ID](#) to check for duplicate contacts. If duplicates exist, a list of duplicates is displayed along with the confidence score.

### Note

Confidence indicates how closely the attributes of the new contact match with the duplicates.

## Related Information

[Duplicate Check](#)

## 18.2.6 Control Enablement of Generative AI features

You can make context data based Generative AI feature disablement or enablement decision using [Generative AI: Custom Processing Decision Maker](#).

As an administrator, you can introduce your own logic to enable or disable Generative AI scenarios, allowing the use of Generative AI based on document type or business context. Use the pre-delivered [Custom Code Extensions for Generative AI Scenarios](#) communication configuration template, along with the synchronous outbound communication configuration [Generative AI: Custom Processing Decision Maker](#) to enable this capability.

## Configure Enablement of Generative AI features

1. Create a communication system with outbound authentication details for the application endpoint which holds the custom logic to determine enablement of Generative AI scenario by following the

steps mentioned in the *Create Communication Systems* topic in this guide. You can ignore the inbound configuration.

2. Create a communication configuration by copying the pre-delivered template configuration [Custom Code Extensions for Generative AI Scenarios](#) by following the steps mentioned in the *Create Communication Configurations* topic in this guide. You can ignore steps related to inbound and outbound configuration.
3. Change the status of the communication arrangement to [Active](#) by clicking the status and selecting the [Active](#) radio button.

The endpoint configured in the outbound configuration gets called before the Generative AI scenario is enabled for a specific document.

#### Request format

The request payload will be sent through a POST call in JSON in the following format.

##### { } Sample Code

```
{
 "scenarioName": "<Name of Gen AI scenario(English and Caps) in Admin UI
with space replaced by underscore>"
 "id": "<entity uuid>",
 "objectTypeCode": "<context object type code Example: 2886 for case, 72
for Opportunity >"
}
```

#### Response format

The application expects a response back in JSON in the following format:

##### { } Sample Code

```
{
 "isAllowed": "true/false"
}
```

The Generative AI scenario is allowed for the document only if the `isAllowed` value is returned as true and also if there is no exit configured or the exit endpoint is unavailable.

## Related Information

[Create Communication Systems](#)

[Create Communication Configurations](#)

## 18.2.7 Remove Personally Identifiable Information

You can check and remove Personally Identifiable Information(PII) in Generative AI scenarios by using the [Generative AI: Custom Processing Personally Identifiable Information](#) template.

Use the pre-delivered [Custom Code Extensions for Generative AI Scenarios](#) communication configuration template, along with the synchronous HTTP data outbound communication configuration [Generative AI: Custom Processing Personally Identifiable Information](#) to enable this capability.



The configuration is expected to produce response text with PII data replaced by unique tags and generate a dictionary mapping unique identifier tags to PII data values. This mapping can then be used to associate the data back in the model response, providing users with responses containing PII contextual data.

## Configure Custom Processing Exit

1. Create a communication system with outbound authentication details for the application endpoint which holds the custom logic to determine enablement of Generative AI scenario by following the steps mentioned in the *Create Communication Systems* topic in this guide. You can ignore the inbound configuration.
2. Create a communication configuration by copying the pre-delivered template configuration [Custom Code Extensions for Generative AI Scenarios](#) of category All HTTP Data by following the steps mentioned in the *Create Communication Configurations* topic in this guide. You can ignore steps related to inbound and outbound configuration.
3. Change the status of the communication arrangement to [Active](#) by clicking the status and selecting the [Active](#) radio button.

The endpoint configured in the outbound configuration gets called before the Generative AI scenario is enabled for a specific document.

### Request format

The request payload will be sent through a POST call in JSON in the following format.

#### { } Sample Code

```
request payload
{
 "text": <contextual data with PII>,
 "scenarioName": "SCENARIO_NAME"
}
```

### Response format

The application expects a response back in JSON in the following format:

#### { } Sample Code

```
{
 "text": <Anonymized contextual data>,
 "piiDataValueMappingDictionary": [
 {
 "identifier": <unique tag used for anonymizing PII>,
 "value": <PII value>
 }
]
}
```

## 18.2.8 Business Information Extraction

Business information extraction is the process of identifying, extracting, and organizing relevant data from business-related content such as text descriptions to create, update, or identify case-related entities.

Use the generative AI-based information extraction capability to extract business information from the subject or description of the case and add that information to the case-related entities.

### Related Information

[Generative AI \[page 359\]](#)

Use Generative AI to generate account synopsis, case summary, draft email response for cases, and backend entity extraction.

[Business Documents](#)

[Add Business Document to Case Type](#)

### 18.2.8.1 Configure Business Information Extraction

As an administrator, activate and configure business information extraction for your users.

#### Context

To configure business information extraction:

1. Activate Business Information Extraction.
2. Create or include standard elements.
3. Create or include standard classes.
4. Create a scenario, and link the scenario to a case type.

#### Note

You can't keep both Generative AI-based *Business Information Extraction* scenario with first level case category element and classical Machine Learning based case categorization (under Intelligent Services) in an active state together.

Business information extraction can be consumed in the following scenarios:

- Case: For cases, custom fields (case header levels) are supported in addition to the standard fields such as *Contract Account External ID*, *Product ID*, *Product Name*, and *Serial ID*.
- Business Documents: For business documents, custom elements and custom classes are supported.
- Joule: For Joule, the case type of standard class is supported.

## Procedure

1. Go to your *User Menu* and select ► *System Settings* ► *All Settings* ► *Machine Learning* ► *Generative AI* ► *Business Information Extraction* ►.
2. Activate business information extraction by enabling the *Activate Business Information Extraction* switch, and accept the *Acknowledgment*. The *Activation Log* button displays when the capability is enabled.
3. **If you intend to choose Standard Elements:** Select *Elements* from the side panel, activate standard elements such as *Product Name*, *Product ID*, and *Serial ID*, and save your configuration. Enter a *Short Sample*. This short sample is a sample text that helps the system identify your elements. Enter at least one short sample.
4. **If you intend to choose Custom Elements:** Select *Elements* from the pane, and click the + icon. Enter a name, the same name of the custom field technical name so that extracted value can be populated in an extension field in case. Enter a label and description for your element.
5. **If you intend to choose Standard Classes:** Select *Classes*, and activate the *Category Level 1* and case type standard class. Enter *Short Sample*. This short sample is a sample text and the corresponding value that helps the system identify the classes. The short sample text has a limit of 150 characters. The short sample value must be of the **Catalog ID-Category ID** format. Save your configuration.

### Note

For a particular catalog-category combination, multiple samples can be defined.

6. **If you intend to choose Custom Classes:** Select *Classes* from the side panel, and click the + icon. Enter a name, the same name of the custom field technical name so that classified value can be populated in an extension field in case. Enter a label and a description for your class to define a custom class.

### Note

For custom fields, only case header fields of type string are currently supported.

7. Select *Scenario* from the side panel, and click the + icon.
8. Enter a scenario name and scenario description. Choose a *Usage Context: Business Document, Joule, or Case* based on your need. Include elements and classes by selecting from *Element Selection* and *Class Selection* respectively for your scenario.

### Note

- Case: Custom fields (case header levels of type string) are supported in addition to standard fields such as *Contract Account External ID*, *Product ID*, *Product Name*, and *Serial ID*.
- Business Documents: Only custom elements and custom classes are supported.
- Joule: Case type of standard class is supported.

9. Perform a test extraction by entering sample text in the text box, and click *Test*. Once information is extracted, you get the extracted elements and class derivations with confidence percentage values. Choose *Save* to save you configuration.

You've now successfully configured *Business Information Extraction* capability on your system.

## Related Information

[Business Documents](#)

[Generative AI \[page 359\]](#)

Use Generative AI to generate account synopsis, case summary, draft email response for cases, and backend entity extraction.

[Business Information Extraction \[page 370\]](#)

Business information extraction is the process of identifying, extracting, and organizing relevant data from business-related content such as text descriptions to create, update, or identify case-related entities.

[Configure Business Information Extraction in Cases](#)

## 18.2.9 Utilities Insights

Use Utilities Insights from Generative AI to generate a summary for billed consumption details

1. As an administrator, navigate to your user profile, and select ► [System Settings](#) ► [All Settings](#) ► [Machine Learning](#) ► [Generative AI](#) ► [Utilities Insights](#) .
2. To activate Premise Billed Consumption Summary, enable [Activate Utilities Insights](#), and accept the [Acknowledgment](#). The [Activation Log](#) button displays when the capability is enabled.
3. Save your changes.

## Related Information

[Billed Consumption Details](#)

## 18.3 Joule

Joule configuration.

### → Remember

[Joule enablement status](#) is enabled only if Joule is already onboarded on to your tenant via BTP configuration.

## Onboarding Prerequisites

The following information is applicable if you intend to enable Joule on your tenant.

- SKUs for Joule: 8019544 or 8019164
- Sign up for EAC on the following page:  
[https://influence.sap.com/go/joule\\_servicecloudv2](https://influence.sap.com/go/joule_servicecloudv2) for SAP Service Cloud Version 2
- Once you have opted for the SKU and the EAC request is accepted, raise an incident against the component **CEC-CRM-JOU**.
- Post activation, follow the onboarding steps as guided by the EAC program team.

## Business Information Extraction Scenario for Case Processing

When you're using Joule for Case Creation, if you wish that case type is determined based on the case content instead of being selected by user manually, select a *Business Information Extraction* scenario, which includes case type as a class in the scenario.

### Note

You can define a business information extraction scenario for selection.

## Related Information

[Configure Business Information Extraction \[page 370\]](#)

As an administrator, activate and configure business information extraction for your users.

# 19 SAP CX AI Toolkit

Use SAP CX AI Toolkit to analyze enterprise data with trusted AI that helps you enhance customer service.

SAP CX AI Toolkit uses the unique SAP data to provide you with output relevant to your solution. You can use SAP CX AI Toolkit to answer any business question you have, for example, to obtain a generated summary of your last conversation with a customer. You can also access tools to assist you with tasks, for example scheduling a meeting, generating a summary, or generating a product description. The answers and texts that the SAP CX AI Toolkit generates are based on the data sources that you connect it to, like PDF documents, Outlook, Excel, OneDrive, and Microsoft Teams.

## → Remember

- Ensure that you have purchased the SAP CX AI Toolkit and have a valid license. For more information, see [Onboarding Guide](#) and [Set Up Guide](#).
- To enable Generative AI features, you must enable SSO.
- SAP CX AI toolkit is available only in limited data center regions. For more information, see [Supported Data Centers](#).

Once you have integrated the toolkit with the solution, the system shows an icon on the shell bar. By clicking the



(CX AI Toolkit) icon, an embedded side panel opens from the right side of the solution, allowing you to adjust its width using drag and drop.

SAP CX AI Toolkit is supported for the following entities:

- Accounts
- Contacts
- Cases

The following are the contextual behaviors of the feature:

- **Contextual Smart Actions:** Currently respond only to a Contact entity in focus. If you are viewing a Contact, you can slide the toggle to [Show Related Insights](#), which will update the Smart Actions list to show any interaction with that person over the past 90 days. The lookup is performed using the Contact email address.
- **Contextual AI Tools:** Work by matching the tool to the entity in focus. If they match, the current record will be pre-populated in the lookup field.

For example:

- Viewing an Account and opening the Account Summary tool.
- Viewing a Contact and opening the Write Personalized Prospecting Email tool.
- Viewing a Case and opening the Write a KB Article tool.
- **AI Tools Fuzzy Search:** The AI tools in the embedded panel also support a fuzzy search that suggests recently viewed entities.

For example, if you have multiple accounts open (but not on the active page), you can type a few letters in the Account Summary lookup, and it will suggest any of the recently viewed accounts containing those letters.

## Related Information

[User Guide for SAP CX AI Toolkit](#)

[Important Notes](#)

## 19.1 Configure SAP CX AI Toolkit

As an administrator, you can configure SAP CX AI Toolkit.

### → Tip

Expand a collapsible section heading to view the mandatory configuration steps.

### 19.1.1 Enable CX AI Toolkit for Business Roles

As an administrator, you must add the CX AI Toolkit business service to your business users.

## Procedure

1. Go to **User Menu** > **System Settings** > **All Settings**, under **Users and Control**, select **Business Roles**.
2. Select a **Business Role ID** to open the details.
3. Under **Business Services**, click **+** to open the **Add Business Services** window.
4. Search for **digitalAssistant** using **Q** and select the following checkboxes:
  - **sap.crm.service.digitalAssistantCxaitkAdminService**: For admins to enable the CX AI Toolkit option on the settings page
  - **sap.crm.service.digitalAssistantCxaitkService**: For users to enable the CX AI Toolkit plugin on the shell bar
5. Select **Add**.

## Results

You've added the service to the selected business user.

## Related Information

[Create Business Roles and Assign Business Users \[page 28\]](#)

Administrators can create business roles with appropriate authorizations and assign them to business users.

## 19.1.2 Integrate SAP CX AI Toolkit

As an administrator, you must integrate the SAP CX AI Toolkit with the solution to start using the features.

## Procedure

1. Go to ► *User Menu* ► *System Settings* ► *All Settings* ►.
2. Under *Machine Learning*, select *CX AI Toolkit*.
3. Enter the *Organization ID* to the field from the settings page of SAP CX AI Toolkit.
4. Save your changes.

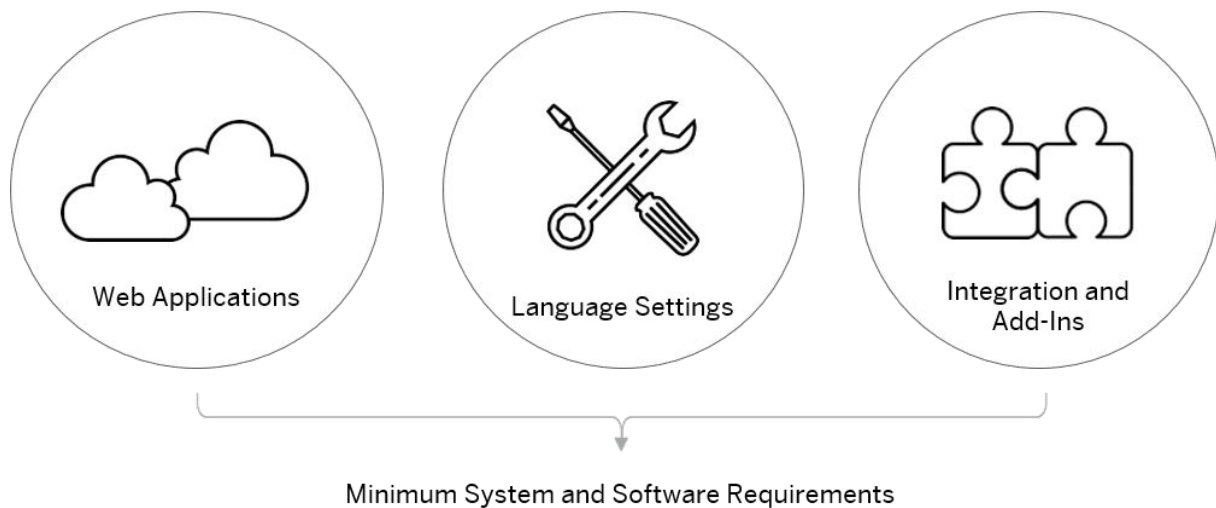
## Related Information

[Integrating SAP Sales and Service Cloud Version 2](#)



## 20 Minimum System and Software Requirements

This section provides an overview of minimum system and software requirements.



### 20.1 Web Applications

You will learn about the minimum hardware and network requirements, display resolution, additional software, browser settings and versions and domain settings for web applications.

#### 20.1.1 Minimum Hardware Requirements

Minimum desktop and laptop hardware requirements.

- Processor: Intel Core 2 Duo (2.4 GHz with a 1066 megahertz {MHz} front-side bus) or equivalent
- Memory: 6 gigabytes (GB)

## 20.1.2 Minimum Network Requirements

Minimum bandwidth and latency requirements for

| Number of Users           |         | 20   | 100  | 500 | 1000 |
|---------------------------|---------|------|------|-----|------|
| Bandwidth Required (Mbps) | Sales   | 0.25 | 1.24 | 6.2 | 12.4 |
|                           | Service | 0.6  | 3    | 15  | 30   |

Network latency should be 200ms.

### Note

The calculations consider only bandwidth used by the solution and don't include any other internet or internal network traffic.

SAP Support may request wired networking during investigation of incidents.

## 20.1.3 Display Resolution

The solution is best displayed at a screen resolution of 1920 x 1080 and 100% scale.

## 20.1.4 Additional Software

You will require Adobe Reader 8.1.3 or higher as an additional software to work on certain functions.

## 20.1.5 Browser Settings

Download settings and formats, pop-up settings and advanced settings.

### Downloads

Downloading files should be enabled in your browser, since the solution can export files to your local device. If your browser supports restricting downloads by file type, you should enable at least the XML and CSV file types.

If your browser supports trusted sites, we recommend maintaining the download settings only for the Trusted Sites security zone and adding the following URLs to this zone:

- Your system URL, or alternatively the general pattern to which the URL adheres, such as `http://*.crm.cloud.sap`
- The pattern `https://*.crm.cloud.sap`
- The pattern `https://*.cxm-salescloud.com`

## Pop-ups

The pop-up blocker should either be disabled, or the system URL should be registered as an exceptional site for the pop-up blocker. The Trusted Sites security zone settings listed in the **Downloads** section applies to pop-ups as well.

## 20.1.6 Browser Versions

Versions tested and handling browser updates.

SAP performs manual and automated testing on the supported and recommended browsers. Recommended browsers provide better performance and usability. Later versions of the recommended browser should be compatible, but have not yet been tested.

Use the latest, stable version of the recommended browsers. Please be aware that some browsers download and apply updates automatically. SAP makes every effort to test and support the most recent, stable versions of all recommended browsers.

## 20.1.7 Domain Settings

Allow applications to access resources from the **\*.crm.cloud.sap** domains.

If you use a URL filter to blocklist a category of websites, you must allow **\*.crm.cloud.sap** domains as an exception. Static resources based on the Content Delivery Network (CDN) should be in allow list for the domain **\*.cxm-salescloud.com**.

## 20.1.8 Desktops and Laptops

Browser support by operating system for desktops and laptops.

| Platform          | OS                 | Recommended Browser | Supported                 |
|-------------------|--------------------|---------------------|---------------------------|
| Microsoft Windows | Windows 10 & above | Google Chrome       | Google Chrome             |
|                   |                    |                     | Microsoft Edge (Chromium) |

| Platform       | OS               | Recommended Browser | Supported                 |
|----------------|------------------|---------------------|---------------------------|
| Apple Mac OS X | 10.10 "Yosemite" | Google Chrome       | Google Chrome             |
|                |                  |                     | Microsoft Edge (Chromium) |

## 20.2 Language Settings

Browser settings for language support.

To ensure that texts appear consistently in the same login language, review the following three language settings:

- Before you log on to the solution, choose the preferred language from the dropdown list.
- Make sure that the browser locale is correct, the locale is used to determine the dates, time zone, or amount formats. For add-ons, dates and times are not updated based on your browser settings.
- In the operating system itself, ensure that the preferred language is defined as the display language.

### 20.2.1 Languages Supported

Supported languages and restrictions for specific languages.

The user interface is available in the following languages.

- AR — Arabic (Functional limitations for this language are described in the note following this list)
- BG — Bulgarian
- CS — Czech
- DA — Danish
- DE — German
- EL — Greek
- EN — English
- ES — Spanish
- FI — Finnish
- FR — French
- HE — Hebrew (Functional limitations for this language are described in the note following this list)
- HR — Croatian
- HU — Hungarian
- IT — Italian
- JA — Japanese
- KK — Kazakh
- KO — Korean
- MS — Malay

- NL — Dutch
- NO — Norwegian
- PL — Polish
- PT — Portuguese
- RO — Romanian
- RU — Russian
- SH — Serbian
- SK — Slovak
- SL — Slovenian
- SV — Swedish
- TH — Thai
- TR — Turkish
- UK — Ukrainian
- VI — Vietnamese
- ZH-CN — Simplified Chinese
- ZF — Traditional Chinese

#### Note

Right-to-left languages (Arabic script and the Hebrew alphabet are written from right to left) are **NOT SUPPORTED** for email integration, analytics, or maps.

## 20.3 Integration and Add-Ins

Additional software requirements for integration scenarios and add-ins.

You can enable additional functionalities by integrating with external systems or software that can be installed, via the [Download](#) screen, for specific business purposes and users.

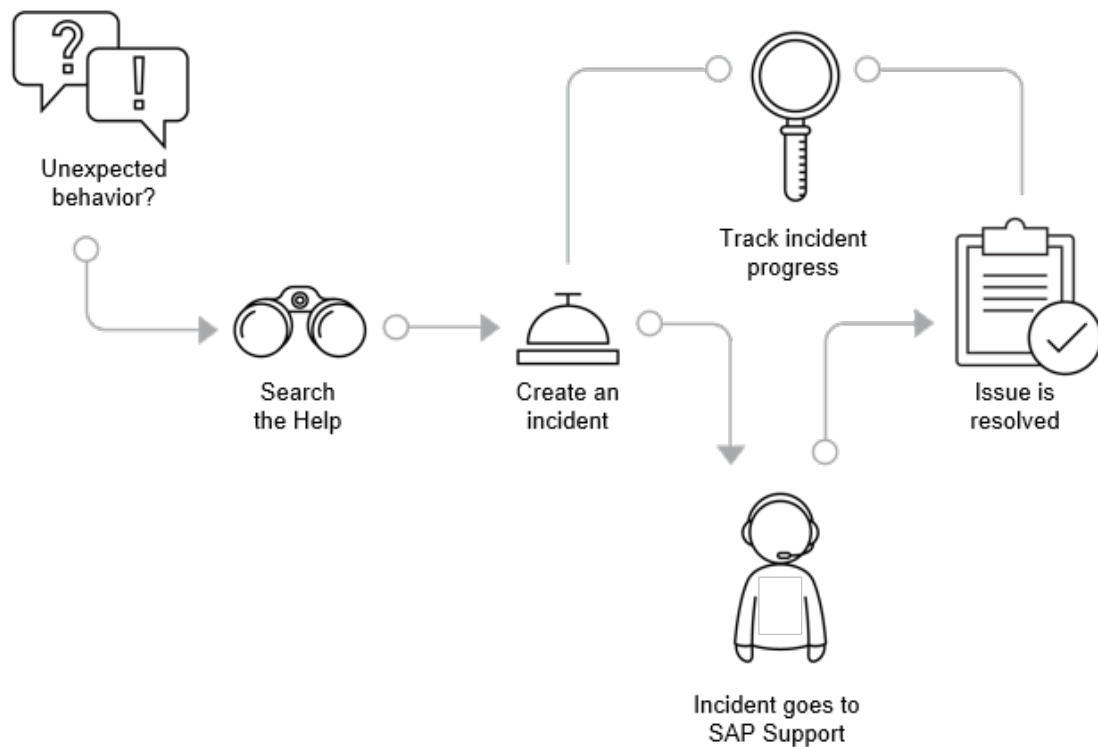
### 20.3.1 Requirements for Software Integration

To increase customer service, large enterprises can integrate existing SAP S/4HANA systems with the solution.

If you integrate the cloud service with licensed SAP on-premise software, you can access your on-premise solutions through the cloud service without additional licenses, providing the access is solely to perform functions in conjunction with the cloud service.

## 21 Help and Support

Learn how to find answers to your questions about the solution and how to report issues and problems.



As you're working, if you encounter unexpected behavior in the solution that interrupts your work or reduces the quality of the service, you can search the [Built-In Support](#) (🔍) for more information or create an incident that is sent to SAP Support. You can search the entire SAP knowledge database, as well as additional resources, for information that can solve your problem.

If you want to review the product documentation that can help you solve the issue yourself, you can find links to the SAP Help Center and this solution guide in the [Built-In Support](#) panel.

Your administrator is the first line of support but if they can't solve the issue for you, they can send the incident to SAP for further investigation.

**Prerequisite:** One time registration is required to create an SAP Universal ID and use the real-time chat and submit an incident. We recommend you register for your SAP Universal ID with your S-User ID. Once registered, incident reporting and expert chat are available in the [Built-In Support](#) panel.

When creating an incident, be sure to create it from the screen where you're experiencing the issue. The system identifies the screen automatically and includes this information in the incident. To help support process your incident more quickly, always include step-by-step instructions on how to re-create the behavior you're reporting.

As an administrator, after you submit an incident, you can track it for updates.

You can chat with product support specialists as well as with our customer interaction teams for updates on existing incidents. Each chat generates an incident that you can view and edit. The incident contains the chat transcript and can be used to follow up on the reported issue.

Get help within the *Built-In Support* tool by selecting the ( ⓘ ) icon at the top of the panel.

#### Restriction

Built-in support is available only in English.

## Related Information

[Full documentation for Built-In Support](#)

[About SAP Universal ID](#) 

## 22 Configure Approvals

As an administrator, you can configure approvals. You define the entity, event, and condition based on which a notification is sent to the approvers.

### Steps

1. [Configure Conditions \[page 384\]](#)

As an administrator, you define the conditions based on which a notification is sent to the approvers.

2. [Configure Approvers \[page 386\]](#)

Set up approval process with direct or responsible employees as approvers.

### 22.1 Configure Conditions

As an administrator, you define the conditions based on which a notification is sent to the approvers.

### Context

Conditions are structured in groups and include criteria that must be fulfilled if approval action must be triggered.

#### Note

- You can add multiple approval steps using the [Add](#) button and configure conditions and approval for each step.
- You can define more than one conditions using logical operators [AND/OR](#). Use the [AND](#) operator to define conditions within the same group.
- To add more than one condition, use the [OR](#) operator. Add a new group to define the [OR](#) operator amongst the approval conditions such that if conditions in at least one group are met, then the overall condition is considered fulfilled and the approval process is triggered.
- The maximum number of steps in an approval process is 10.
- Conditions can be based on standard fields or extension fields.

### Procedure

1. Go to your user menu and select  [Settings](#)  [All Settings](#)  [Business Flow](#)  [Approval](#) .

You can see a list of approvals that are available.



2. Select **+** and select [Create Approval Process](#).
3. Enter a name in the [Process Name](#) field.
4. From the [Entities](#) dropdown, select [Cases](#)

#### Note

Approvals are available only for [Case](#).

5. To include all steps while checking the conditions, enable [Run All Steps](#). This is applicable in a multi-step approval process only.

When [Run All Steps](#) is enabled, all the steps in the approval process would be evaluated and only those would be triggered for which conditions are met. The steps for which conditions are not met are marked [Not Applicable](#). However, the status of the approval as a whole will be [Not Applicable](#) only if condition for all the steps were not met.

If [Run All Steps](#) is not enabled, then approval process ends if any of the step conditions are not met and the status of the step and the approval is marked as [Not Applicable](#).

6. Select your language and enter subject and description in the [Task subject](#) and [Task Description](#) fields.
7. Enable [Email Notifications](#).

#### Note

Your administrator must define email templates for [Approval Request](#) and [Approval Response](#) template types for [Case](#) object type. For more details on defining templates, see Related information.

8. Optional: Define conditions to specify criteria that must be fulfilled to trigger approvals.

#### Note

- You can add multiple approval steps using the [Add](#) button and configure conditions and approval for each step.
- You can define more than one conditions using logical operators [AND/OR](#). Use the [AND](#) operator to define conditions within the same group.
- When one of the conditions is not met, the approval process ends and the requestor is notified about the same.
- To add more than one condition, use the [OR](#) operator. Add a new group to define the [OR](#) operator amongst the approval conditions such that if conditions in at least one group are met, then the overall condition is considered fulfilled and the approval process is triggered.
- The maximum number of steps in an approval process is 10.
- Conditions can be based on standard fields or extension fields.

- a. Choose [+OR Condition](#). By default, the trigger is set to [Conditions are met](#). While using operands, the first condition can only be defined using the [AND](#) operand.
- b. Select the field, comparator and the value or field. When previous value and current value influence the action, choose [Previous value](#) to consider the value before the event occurred.
- c. Specify an appropriate operator, and a value.

The following operators are available for defining conditions, based on the data type of the field:

- Equal to
- Not Equal to

- Greater than
  - Greater than or equal to
  - Less than
  - Less than or equal to
  - Contains
  - Does not contain
  - Is Null
  - Is not Null
- d. If desired, specify conditions with the following options:
- Click the **Add** icon (+) to add an **AND** expression.  
Conditions within a group are logical **AND** expressions. If all the conditions within a group are met, the condition group is considered fulfilled.
  - Click **+ OR Condition** to add an **OR** expression.  
If all conditions in at least one group are met, then the overall condition for the rule is considered fulfilled.
  - Click the **Delete** icon (🗑) to delete a condition group.
  - Click the **Dismiss** icon (✕) to remove a condition.

#### Note

- The maximum number of OR conditions per approval rule is 10.
- The maximum number of AND conditions per approval rule is 10.
- The maximum number of steps in an approval process is 10

**Task overview:** [Configure Approvals \[page 384\]](#)

**Next task:** [Configure Approvers \[page 386\]](#)

## 22.2 Configure Approvers

Set up approval process with direct or responsible employees as approvers.

### Procedure

1. Select the *Approval Request* and *Approval Response* templates.
2. In the *Define Work Distribution* section, define the approvers. You can choose from the following options:
  - **Direct Approvers:** You can select specific employees responsible as direct approvers, provided that they possess the corresponding access rights to the task type and business transaction. Such approvers also require read access to the relevant business transaction data and write access to the notes of the business transaction.

### Note

You can enable approvers to trigger actions directly from the details of the item to be approved. To do so, from the [Approval](#) tab of any related item, make the [Actions](#) column visible on the user interface

The maximum number of direct approvers per approval rule is 10.

- **Responsibles:** You can select any responsible employees associated with the Approval. It can include [Processor of case](#), [Manager of processor for case](#), [Suggested recipients](#) and so on.
- 3. To include multiple approvers, enable the [All Approvals Required](#) switch. However, sequential approval is not supported.
- 4. To enable automatic approval of your business process (only when there are no approvers determined for a step), turn on the [Automatic Approval](#) switch.
- 5. Choose [Save](#) and [Activate](#).

You can select the approvals that you define here in the [Approval](#) step type of [Case](#) designer.

When the entire approval process is completed, the system sends an in-app notification of type [Info Notification](#) to the approval requester with one of the following statuses so that requestor can proceed with subsequent actions.

- Approved -
  - Conditions are met and approvers have approved the approval request.
  - (Auto Approved) Conditions are met, however the approver can't be determined (approver is not added at the case step), and the [Automatic Approval](#) switch is enabled while setting up the approval process.
- Rejected - Conditions are met, however the approvers have rejected the approval request.
- Not Applicable - Conditions are not met and hence the approval process ends.

**Task overview:** [Configure Approvals \[page 384\]](#)

**Previous task:** [Configure Conditions \[page 384\]](#)

## 23 Configure Autoflow

An autoflow rule executes the actions immediately when the conditions are met.

### Steps

1. [Configure Conditions \(with Arrays\) \[page 388\]](#)  
As an administrator, you define the conditions that trigger actions in an autoflow.
2. [Configure Conditions \[page 391\]](#)  
As an administrator, you define the conditions based on which the actions are triggered for an autoflow.
3. [Configure Actions \[page 393\]](#)  
After you have defined conditions for an autoflow, you must configure actions.

### 23.1 Configure Conditions (with Arrays)

As an administrator, you define the conditions that trigger actions in an autoflow.

### Context

Conditions are structured in groups and include the criteria that must be fulfilled for an Autoflow rule to execute. You can configure multiple conditions using logical operators such as [AND](#) and [OR](#):

- Use the **AND (+)** operator to evaluate multiple conditions within the same group. If all the conditions within a group are met, the condition group is considered fulfilled.
- Use **+ OR Condition** to add a more than one condition, where if at least one group evaluates to True, the Autoflow is triggered.

#### ⓘ Note

- The maximum number of OR conditions allowed per rule is 10.
- The maximum number of AND conditions allowed per rule is 10.

### Procedure

1. Go to the [User Menu](#) and select [System Settings](#) > [All Settings](#) > [Business Flow](#) > [Autoflow](#) .
2. Choose **+ (Add)** and select [Rule](#).
3. From the [Choose an Entity](#) dropdown, select an entity.

4. Select the [Event Type](#) to define when the Autoflow rule is evaluated.

Common event types include:

- On create - Rule runs when a new record is created.
- On update - Rule runs when a record is updated.
- Explicit Trigger - Rule runs when you choose [Submit](#) in the [Autoflow](#) case step. You must select [Autoflow](#) for step type in case type configuration.

#### → Recommendation

- Autoflow rules that are set to **on create** and **on update** are run asynchronously after the transaction object has been saved and the corresponding event is triggered. Defining too many rules can impact overall system performance. Therefore, it is recommended that you consolidate autoflow rules wherever possible. Based on the entity you have selected, there are various event types for managing autoflows.
- Autoflow rules affect all users using the entity within a process. Plan and define the rules accordingly.
- While configuring conditions, It is recommended to not use language-dependent attributes (and are not available in conditions lists). Also, do not configure the attribute [createdByName](#), if you are not a technical user.

5. Select the Field type from the [Select Field](#) dropdown.

When previous value and current value influence the action, choose [Previous Value](#) to consider the value before the event occurred. For example, when the field - [Escalation Status](#) is changed from the previous value of [Not Escalated](#) to [Escalated](#).

6. Define conditions to specify criteria that must be fulfilled to trigger an autoflow.

You can define Autoflow conditions on array (1:N) fields such as Notes, Attachments, Approval ID, and so on. This enables Autoflow sequences to respond to changes in the related child objects, such as any addition of a note or attachment.

#### ⓘ Note

This condition modeling logic is currently supported only for the Case entity and non-scheduled Autoflows.

When selecting the array fields:

- a. Specify how the condition must be evaluated against the array elements using the [Match Type](#) dropdown:
    - [Match Any](#) → Returns True if at least one occurrence of the attribute in the array matches the value. Otherwise, returns False.
    - [Match All](#) → Returns True if every occurrence of the attribute in the array matches the value. Otherwise, returns False.
  - b. (Optional) Enable the [Add as Subset](#) toggle. This groups multiple conditions related to array fields into a subset within a larger condition block. This is useful when multiple notes or attachments are being evaluated as a group.
  - c. Add any additional logic or nesting using [AND / OR](#) as needed.
7. The available operators for defining conditions vary depending on whether the selected field is an array, non-array, technical ID, or based on its data type. The following list shows the applicable operators:

- Equal to
- Not equal to
- Greater than
- Greater than or equal to
- Less than
- Less than or equal to
- Contains
- Does not contain
- Is Null
- Is not Null
- Is Changed
- In
- Not In

8. Select the target type from the following options:

- Field
- Comparator
- Function (applicable only for date time and date field types). The current date and time displayed are based on UTC.

#### Note

Not all target types can be selected for all operators. Some are restricted based on the selected operator.

9. Specify an appropriate field, value, or function for comparison.

#### Note

You can also create conditions using custom fields. Set up custom fields before you create autoflow rules.

Choose  (*Delete*) to delete a condition group.

Choose  (*Dismiss*) to remove a condition.

10. Choose *Save* to save your condition set.

**Task overview:** [Configure Autoflow \[page 388\]](#)

**Next task:** [Configure Conditions \[page 391\]](#)

## 23.2 Configure Conditions

As an administrator, you define the conditions based on which the actions are triggered for an autoflow.

### Context

Conditions are structured in groups and include criteria that must be fulfilled if a workflow action must be triggered. More than one condition can be defined using logical operators [AND/OR](#). The [AND](#) operator can be defined using condition within the same group. To add more than one condition, use the [OR](#) operator. Add a new group to define the OR operator among the workflow conditions such that if conditions in at least one group are met, then the overall condition is considered fulfilled and the workflow rule has been triggered.

#### Note

- The maximum number of OR conditions allowed per rule is 15.
- The maximum number of AND conditions allowed per rule is 15.

### Procedure

1. Go to the [User Menu](#) and select [System Settings](#) > [All Settings](#) > [Business Flow](#) > [Autoflow](#) .
2. Choose [+ \(Add\)](#) and select [Rule](#).
3. From the [Choose an Entity](#) dropdown, select an entity.
4. Select the [Event Type](#).

This determines when an autoflow rule must be evaluated and triggered. Timing is relative to an event that occurs to an entity. For example,

- On create - The rule is applied at every create of an entity.
- On update - The rule is applied at every update of an entity.
- Explicit Trigger -The rule is applied when you choose [Submit](#) in the [Autoflow](#) case step. You must select [Autoflow](#) for step type in case type configuration.

#### → Recommendation

- Autoflow rules that are set to **on create** and **on update** are run asynchronously after the transaction object has been saved and the corresponding event is triggered. Defining too many rules can impact overall system performance. Therefore, it is recommended that you consolidate autoflow rules wherever possible. Based on the entity you have selected, there are various event types for managing autoflows.
- Autoflow rules affect all users using the entity within a process. Plan and define the rules accordingly.
- While configuring conditions, It is recommended to not use language-dependent attributes (and are not available in conditions lists). Also, do not configure the attribute [createdByName](#), if you are not a technical user.

5. Select the Field type from the [Select Field](#) dropdown.

6. Define conditions to specify criteria that must be fulfilled to trigger an autoflow.

The following operators are available for defining conditions, based on the data type of the field:

- Equal to
- Not Equal to
- Greater than
- Greater than or equal to
- Less than
- Less than or equal to
- Contains
- Does not contain
- Is Null
- Is not Null
- Is Changed

Conditions can be based on standard fields or extension fields.

- Use the [AND](#) operator to define conditions within the same group. If all the conditions within a group are met, the condition group is considered fulfilled. Choose [+OR Condition](#) to add more than one condition. If all conditions in at least one group are met, then the overall condition for the rule is considered fulfilled. By default, the trigger is set to [Conditions are met](#).
- Select the target type from the following options:
  - Field
  - Comparator
  - Value or Field
  - Function (applicable only for date time and date field types). The current date and time displayed are based on UTC.
- When previous value and current value influence the action, choose [Previous value](#) to consider the value before the event occurred. The value - [Before Entity Change](#) can be used to define conditions based on field value changes. For example, an email is sent when the field - [Escalation Status](#) is changed from [Not Escalated](#) to [Escalated](#).  
The [Previous Value](#) checkbox is not available for scheduled autoflow rules.
- Specify an appropriate operator, and a value.

#### Note

You can also create conditions using custom fields. Set up custom fields before you create autoflow rules.

Choose  ([Delete](#)) to delete a condition group.

Choose  ([Dismiss](#)) to remove a condition.

7. Choose [Save](#).

**Task overview:** [Configure Autoflow \[page 388\]](#)

**Previous task:** [Configure Conditions \(with Arrays\) \[page 388\]](#)



Next: [Configure Actions \[page 393\]](#)

## 23.3 Configure Actions

After you have defined conditions for an autoflow, you must configure actions.

### Note

The maximum number of action cards allowed per action type is 5.

Parent topic: [Configure Autoflow \[page 388\]](#)

Previous task: [Configure Conditions \[page 391\]](#)

## Related Information

[Configure Conditions](#)

### 23.3.1 Send Emails

You configure an autoflow to send emails when the conditions are met for business entities. Currently, the email action is supported only for entities such as Cases, Leads, and Tasks.

## Procedure

1. Select *Send Email* from the *Action* dropdown, to send an email.
  - a. Enter the following details:

| Field Name            | Description                                                                                                           |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------|
| Subject               | Enter the subject of your email.                                                                                      |
| Template              | Choose a template.                                                                                                    |
| Sender Name and Email | Select the name and email ID of the sender. This depends on the email channels defined for the entity selected.       |
| Role Based Recipients | Select the recipients of this email based on their role. These roles vary based on the entity that you have selected. |

| Field Name        | Description                                                                                                                                                                                                        |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | <div>  <b>Note</b><br/>           If there are no roles associated with the entity, you cannot see any recipients.         </div> |
| <b>Recipients</b> | Select the recipients of this email irrespective of their roles. You can select from a list of accounts, individual customers, and contacts that are maintained in the master data.                                |

2. Select **+** (*Add*) to add more actions.

The *Action* dropdown is displayed.

- Select *Send Emails* to send an email.
- Select *Send Event Notifications* to send event notification to a third-party communication system.
- Select *Send Info Notifications* to send in-app notifications to selected users.

3. You can disable or delete actions from the *Action* menu:

- a. Select  (*Delete*) to delete an action.
- b. Select the *Disable* icon to disable an action.

4. Select *Save*.

5. Select *Activate*.

You can select the autoflows that you define here in the *Autoflow* step type of the *Case* designer.

## Related Information

[Configure Conditions](#)

[Send Event Notifications \[page 394\]](#)

You configure an autoflow to send event notifications to third-party systems, when the conditions are met for business objects.

[Send Info Notifications \[page 395\]](#)

You configure an autoflow to send in-app notifications to selected users, when the conditions are met for business entities.

## 23.3.2 Send Event Notifications

You configure an autoflow to send event notifications to third-party systems, when the conditions are met for business objects.

### Procedure

1. Select *Send Event Notification* from the *Action* dropdown.

- a. In the panel that appears, enter the following details:

| Field Name           | Description                                                                                             |
|----------------------|---------------------------------------------------------------------------------------------------------|
| Event Name           | Enter a name.                                                                                           |
| Communication System | Choose a communication system. For more information, see <a href="#">Create Communication Systems</a> . |
| Subscribe Path       | Enter the URL of the third-party system.                                                                |

2. Select **+** (*Add*) to add more actions.

The *Action* dropdown displays.

- Select *Send Emails* to send email.
- Select *Send Event Notifications* to send event notification to a third-party communication system.
- Select *Send Info Notifications* to send in-app notifications to selected users.

3. You can disable or delete actions from the *Action* menu:

- a. Select  (*Delete*) to delete an action.
- b. Select the *Disable* to disable an action.

4. Select *Save*.

5. Select *Activate*.

You can select the autoflows that you define here in the *Autoflow* step type of *Case* designer.

## Related Information

[Configure Conditions](#)

[Send Info Notifications \[page 395\]](#)

You configure an autoflow to send in-app notifications to selected users, when the conditions are met for business entities.

[Send Emails \[page 393\]](#)

You configure an autoflow to send emails when the conditions are met for business entities. Currently, the email action is supported only for entities such as Cases, Leads, and Tasks.

## 23.3.3 Send Info Notifications

You configure an autoflow to send in-app notifications to selected users, when the conditions are met for business entities.

### Context

The info notifications appear in the Notifications area.

After you configure conditions, define sending info notification as the action of the autoflow.

## Procedure

1. Select [Send Info Notification](#) from the [Action](#) dropdown.
  - a. In the panel that appears, enter the following details:

| Field Name  | Description                               |
|-------------|-------------------------------------------|
| Language    | Select a language from the dropdown.      |
| Subject     | Enter a subject.                          |
| Description | Enter a description for the notification. |

 **Note**

Placeholders are supported for info notification action (subject and description). You can use a placeholder by typing a { symbol to see the list of placeholder attributes.

|            |                                                                                     |
|------------|-------------------------------------------------------------------------------------|
| Role based | Select roles from the dropdown, to send notifications to users with specific roles. |
| Employee   | Select employees from the list, to send notification to specific users.             |

2. Select [+](#) ([Add](#)) to add more actions.

The [Action](#) dropdown displays.

    - Select [Send Emails](#) to send email.
    - Select [Send Event Notifications](#) to send event notification to a third-party communication system.
    - Select [Send Info Notifications](#) to send in-app notifications to selected users.
  3. You can disable or delete actions from the [Action](#) menu:
    - a. Select  ([Delete](#)) to delete an action.
    - b. Select the [Disable](#) icon to disable an action.
  4. Select [Save](#).
  5. Select [Activate](#).
- You can select the autoflows that you define here in the [Autoflow](#) step type of [Case](#) designer.

## Related Information

### [Configure Conditions](#)

#### [Send Event Notifications \[page 394\]](#)

You configure an autoflow to send event notifications to third-party systems, when the conditions are met for business objects.

#### [Send Emails \[page 393\]](#)

You configure an autoflow to send emails when the conditions are met for business entities. Currently, the email action is supported only for entities such as Cases, Leads, and Tasks.

## 24 Configure Microsoft Teams Integration

As an administrator, configure Microsoft Teams integration.

To enable the Microsoft Teams integration, administrators must complete the following configuration steps:

- Add the business service `sap.crm.service.msteamsService` to activate Microsoft Teams features.
- Ensure that you have a Microsoft Office 365 Exchange Online license.
- In the [Cross-Site Request Forgery Settings](#), the *Strict* mode is enabled by default for third-party integrations. Change this setting to *None*. For details, see [Cross Site Request Forgery Settings](#).

Administrators must add the domain `teams.cloud.microsoft` to the [Frame Source](#) under the [Content Security Policy](#) section. This configuration is required for Microsoft Teams to host the application tabs within our interface.

Navigate to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [System Preferences](#) ► [Content Security Policy](#) ►. Under [Frame Source](#), add `teams.cloud.microsoft`.

### 24.1 Add SAP Sales and Service Cloud App in Teams

You can add SAP Sales and Service Cloud app in the Microsoft Teams app.

#### Procedure

1. Log in to Microsoft Teams admin center.
2. From the home page, expand the navigation menu and go to ► [Teams apps](#) ► [Manage apps](#) ►.
3. Search and select the [SAP Sales and Service Cloud](#) app from the list.
4. Choose [Allow](#). A new window opens.
5. Choose [Allow](#) again.

## 24.2 Enable and Consent to Teams Integrated Features

You can enable Microsoft Teams integrated features by providing consent.

### Procedure

1. Navigate to the [User Menu](#) and choose [System Settings](#) > [All Settings](#).
2. Under [Communication & Information Exchange](#), select [Microsoft Teams Integration](#).
3. Enable the following features based on your requirements:
  - [Share Workspace and Collaboration Room](#)  
You can enable [Collaboration Room](#) and [Share Workspace](#) separately.  
If you enable [Create Teams](#), the system allows you to create a new team while using the [Share Workspace](#) feature.  
If you enable [Post Sync](#), all the posts in the associated teams channel from the [Posts](#) tab are synced to SAP Sales Cloud Version 2.

#### Note

The [Post Sync](#) feature is applicable only for [Collaboration Rooms](#).

There are two permission types that you can choose from:

- **Application Permission:** Your administrator must consent on behalf of all the users.
- **Delegated Permission:** Your administrator must consent on behalf of your organization. To know more, see [Application Permissions vs. Delegated Permissions](#).  
If you're using the features such as Share Workspace, and Collaboration Rooms with delegated permission, the system shows a window where you must allow the permissions and login to your Microsoft Teams account. The validity of this permission is 30 days. Hence, you must allow the permissions and login to Microsoft Teams again when the permission expires.  
In the case of delegation permission type, your administrator has to select the check box [Consent on behalf of your organization](#) available at the end of all the permissions listed.

#### Note

The application and user delegation consent provided by the administrator must be renewed once in three years.

- [Teams Meeting](#): Only application permission is supported.
  - [Outbound Calls](#): No consent is required.
4. Choose [Provide Consent](#) for the feature that you want to use.

#### Note

- You can map multiple SAP Sales and Service Cloud V2 tenants to one Azure tenant.
- The domain of the logged in user (xxx@abc.com) and admin user (yyy@abc.com) providing consent must be the same.

5. Sign in with your Microsoft 365 username and password.

A new window opens for consent.

6. Review the permissions and click [Accept](#).

As an admin, you must grant consent to any feature before using it.

#### Note

There is no consent required for [Outbound Calls](#).

You can see the details of the user who provided consent, and the time of consent on the [Microsoft Teams Integration](#) page.

# 25 Configure Library

Administrators can create file type filter categories and manage access restrictions for [Library](#).

## 25.1 Create File Type Filter Category

Administrators must create file type categories for users to filter files.

### Procedure

1. Go to ► [User Menu](#) ► [System Settings](#) ► [All Settings](#) ► [Library](#) ► [File Type Filter Configuration](#) ►.
2. Choose [+](#) ([Create](#)).

#### ⓘ Note

The standard SAP system delivers file type filter categories without file types added to them. You can modify them with the required file types or delete them.

To add file types to a standard filter category, allow the file types at ► [System Settings](#) ► [Data Administration](#) ► [MIME Types for Attachments](#) ► if they are not allowed.

3. In the [Create File Type Filter Category](#) pane, enter [Title](#).
4. Choose



([Search and Add](#)) to add the supported file types.

5. Choose [Save](#).

The new file type filter category gets added in [File Type Filter Categories](#).

#### ⓘ Note

- To delete a file type configuration, click ([Delete](#)) in [File Type Filter Categories](#).
- You can also modify file types added to an existing file type filter category.
- Turn off the file type filter categories to disable them from the filter values in the [File Type](#) filter in [Library](#).



## Related Information

[Define MIME Types for Attachments](#)

## 25.2 Manage Access Restrictions

Administrators can provide users with restricted access to the library.

### Context

With this access context, read and write access to folders and files can be restricted based on sales area and employee.

Sub folders inherit access restrictions of their parent folder. You can also set sub folders' access to *Private* and set restrictions that apply to only sub folders.

These access restrictions also apply to attachments added from the library. Files added to the library and referenced in a business object, such as account, are visible to all under the *Attachments* section. However, unauthorized users cannot navigate to further details. Restricted documents are not listed in *Attachments*. Attachments from outside of the library are not subject to access restrictions.

### Procedure

1. Create a business role.
2. Add the *Library* (*sap.crm.service.libraryService*) business service.
3. Enable the *sap.crm.libraryservice.uiapp.libraryAdminApp* and *sap.crm.libraryservice.uiapp.libraryApp* apps.

## Related Information

[Create Business Roles and Assign Business Users \[page 28\]](#)

Administrators can create business roles with appropriate authorizations and assign them to business users.

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